

BEAM CLI Guide

Used in Containerized Deployment

Installation Instructions

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1 Prerequisites for BEAM CLI

The prerequisites listed in this section need to be in place to be able to use BEAM CLI.

- A running Kubernetes environment with helm support.
- Knowledge about the Kubernetes environment, including the networking details.
- Access rights to deploy and manage workloads.
- Availability of the kubectl CLI tool with correct authentication details.
- Availability of the helm package.
- Availability of Helm charts and Docker images for the service and setup BEAM CLI.
 - Ingress Controller
 - For ingress controller, the ingress configured over OAM interface is needed.
 - IAM (Identity and Access Management) and its database.
 - Rbac Proxies for services need to be accessed from BEAM CLI

Note: Applications listed are not applicable to every solution.



2 Download and Setup BEAM CLI

To download and set up the BEAM CLI, do the following steps. If BEAM CLI already installed and FQDN configured, follow steps 1-4 to download the latest version.

Steps

1. Run the following command to get the FQDN of `eric-bss-beam-cli`:

```
$ run_tool kubectl get httpproxy eric-bss-beam-cli --namespace <NAMESPACE>
```

Example

```
$ run_tool kubectl get httpproxy eric-bss-beam-cli --namespace <NAMESPACE>
NAME                                FQD                                     TLS SECRE
T
eric-bss-beam-cli                  eric-bss-beam-cli.beam-cluster01.operator.com  eric-bss
-beam-oam-external-server-cert    valid    Valid HTTPProxy
```

2. Download BEAM CLI using one of the following options:

- a. Run the following command to download the BEAM CLI binary from a Linux terminal:

```
$ curl -kO https://<fqdn>/images/linux/beamctl
```

Where:

`<fqdn>` is the FQDN of `eric-bss-beam-cli` HTTP proxy fetched from the previous command.

Example

```
$ curl -kO https://eric-bss-beam-cli.beam-cluster01.operator.com/images/linux/beamctl
```

- b. To download the BEAM CLI binary securely, the corresponding CA certificate must be passed.

```
$ curl -O https://eric-bss-beam-cli.beam-cluster01.operator.com/images/linux/beamctl --cacert ca-cert.crt
```

Note: The DNS name set for the BEAM CLI (default: `eric-bss-beam-cli`) must be resolvable to the external IP assigned to the corresponding Ingress.

- c. If a browser is used, download the BEAM CLI from the link available in the web page.



3. After downloading the BEAM CLI, the user must have executable rights on the BEAM CLI binary to be able to continue. Change the permissions of the updated beamctl by running the following command:

It must have at least the following access:

```
$ chmod 750 ./beamctl
```

4. Check the version of the updated beamctl by running the following command:

```
$ ./beamctl --version
```

Example

```
$ ./beamctl --version
beamctl version 1.26.0
```

5. Use the existing user in IAM which is created during the initial setup to access different services.

For more information, see section RBAC OIDC Setup.

6. If the Cluster FQDN is not configured earlier, configure it by running the following command:

```
./beamctl config set-config --key fqdn --value
<oamSiteDomainName>
```

Where:

<oamSiteDomainName> can be found in *<base directory>config/env/prod/*environ.yaml

Note: To connect another site (in a multi-site environment), always run the preceding command with another site cluster FQDN information. After successful beam login to another site, use `--server` parameter for beamctl commands only for the first time to change the respective server URL.

7. Log on to beamctl using one of the current users.

```
$ ./beamctl login -u <user> -p <path>/test_user.passphrase
-t https://eric-sec-access-mgmt.<oamSiteDomainName>/auth/realms/
master/protocol/openid-connect/token --verbose
```



3 BEAM CLI Commands

The following sections provide details of the BEAM CLI commands in the application components.

- Note:**
- After successfully logging in, user can optionally use the `--server` parameter with CLI commands to connect a different server URL.
 - The `--server` values would be used temporarily for that session and would not be saved to the `beamctl` configuration file.
 - Create User and assign group in IAM to access different services. For more information, see Installation Guide.

3.1 BAE Configuration Commands

3.1.1 BAE REST Client Configuration

The REST Client Configuration command offers functionality to interact with BAE REST Client ConfigMap. It allows to retrieve the ConfigMap and modify its values as needed.

Syntax

```
beamctl eric-bss-ael-bae-oam
```

Available Sub Commands

Table 1 Sub Commands

Command	Description
get-rest-config on page 4	Enables to retrieve the BAE REST client ConfigMap.
update-rest-config on page 8	Updates the BAE REST client ConfigMap.

3.1.1.1 get-rest-config

The `get-rest-config` command enables to retrieve the BAE REST client ConfigMap.

Syntax

```
beamctl eric-bss-ael-bae-oam get-rest-config
```




Arguments

The command has no arguments.

Parameters

Table 2 Parameters

Parameter	Parameter Type	Description
connectionTimeout	Optional	The duration, in milliseconds, to wait when establishing a connection before timing out.
numberOfRetries	Optional	The number of times the client must retry a request upon failure before considering it failed.
requestTimeout	Optional	The duration, in milliseconds, to wait for a request to complete before timing out.
restClientName	Mandatory	The name identifier of the REST client to be updated. This is mandatory for identifying the target REST client. For the BAE REST client, only this attribute is required.
restServiceName	Mandatory or Optional	The name of the REST service associated with the client. This attribute is mandatory when updating clients other than the BAE REST client to specify the target service.
sleepTimeBetweenRetries	Optional	The wait time, in milliseconds, between retry attempts when a request fails.

Note: As mentioned in the previous table, the parameters `restClientName` and `restServiceName` must be informed to identify the REST client that is updated. Except for BAE REST client, where only `restClientName` must be informed.

Example 1

```
beamctl eric-bss-ael-bae-oam get-rest-config
```

Sample Response:

```
{
  "restClientConfiguration": [
    {
      "connectionTimeout": 5000,
      "numberOfRetries": 2,
      "requestTimeout": 5000,
      "restClientName": "bae-rest",
      "sleepTimeBetweenRetries": 2000
    },
    {
      "connectionTimeout": 5000,
      "numberOfRetries": 2,
      "requestTimeout": 5000,
      "restClientName": "cha-rest",
      "restServiceName": "CHA_BalanceAdjustment",

```



```
"sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cha-rest",
  "restServiceName": "CHA_BalanceEnquiry",
  "sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cha-rest",
  "restServiceName": "CHA_BalanceManagement",
  "sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cha-rest",
  "restServiceName": "CHA_PurchaseCharge",
  "sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cha-rest",
  "restServiceName": "CHA_ReccurrenceManagement",
  "sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cha-rest",
  "restServiceName": "CHA_Refill",
  "sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cha-rest",
  "restServiceName": "CHA_TestManagement",
  "sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cha-rest",
  "restServiceName": "CHA_TransferProductData",
  "sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cpm-rest",
  "restServiceName": "CPM_BusinessLogic",
  "sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cpm-rest",
  "restServiceName": "CPM_OrganizationManagement",
  "sleepTimeBetweenRetries": 2000
},
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "cpm-rest",
  "restServiceName": "CPM_TestManagement",
```



```

    "sleepTimeBetweenRetries": 2000
  },
  {
    "connectionTimeout": 5000,
    "numberOfRetries": 2,
    "requestTimeout": 5000,
    "restClientName": "edmpoc-rest",
    "restServiceName": "EDM_MoveParty",
    "sleepTimeBetweenRetries": 2000
  },
  {
    "connectionTimeout": 5000,
    "numberOfRetries": 2,
    "requestTimeout": 5000,
    "restClientName": "ntf-rest-client",
    "restServiceName": "NTF_BssfPartyCommunicationManagement",
    "sleepTimeBetweenRetries": 2000
  },
  {
    "connectionTimeout": 5000,
    "numberOfRetries": 2,
    "requestTimeout": 5000,
    "restClientName": "ntf-rest-client",
    "restServiceName": "NTF_MoveParty",
    "sleepTimeBetweenRetries": 2000
  },
  {
    "connectionTimeout": 5000,
    "numberOfRetries": 2,
    "requestTimeout": 5000,
    "restClientName": "rmca-rest",
    "restServiceName": "RMCA_CatalogManagement",
    "sleepTimeBetweenRetries": 2000
  },
  {
    "connectionTimeout": 5000,
    "numberOfRetries": 2,
    "requestTimeout": 5000,
    "restClientName": "invoicing-rest",
    "restServiceName": "INV_InvoiceEnquiry",
    "sleepTimeBetweenRetries": 2000
  },
  {
    "connectionTimeout": 5000,
    "numberOfRetries": 2,
    "requestTimeout": 5000,
    "restClientName": "invoicing-rest",
    "restServiceName": "INV_OffCycleBillManagementService",
    "sleepTimeBetweenRetries": 2000
  },
  {
    "connectionTimeout": 5000,
    "numberOfRetries": 2,
    "requestTimeout": 5000,
    "restClientName": "fin-rest",
    "restServiceName": "FIN_Business",
    "sleepTimeBetweenRetries": 2000
  },
  {
    "connectionTimeout": 5000,
    "numberOfRetries": 2,
    "requestTimeout": 5000,
    "restClientName": "fin-rest",
    "restServiceName": "FIN_Customer_Account",
    "sleepTimeBetweenRetries": 2000
  },
  {
    "connectionTimeout": 5000,
    "numberOfRetries": 2,
    "requestTimeout": 5000,
    "restClientName": "fin-rest",
    "restServiceName": "FIN_Specification",
    "sleepTimeBetweenRetries": 2000
  }
]
}

```



3.1.1.2 update-rest-config

The `update-rest-config` command updates the BAE REST client ConfigMap.

Syntax

```
beamctl eric-bss-ael-bae-oam update-rest-config < <inputFile.json>
```

Arguments

The command has no arguments.

Parameters

Table 3 JSON Parameters

Parameter	Parameter Type	Description
<code>connectionTimeout</code>	Optional	The duration, in milliseconds, to wait when establishing a connection before timing out.
<code>numberOfRetries</code>	Optional	The number of times the client must retry a request upon failure before considering it failed.
<code>requestTimeout</code>	Optional	The duration, in milliseconds, to wait for a request to complete before timing out.
<code>restClientName</code>	Mandatory	The name identifier of the REST client to be updated. This is mandatory for identifying the target REST client. For the BAE REST client, only this attribute is required.
<code>restServiceName</code>	Mandatory or Optional	The name of the REST service associated with the client. This attribute is mandatory when updating clients other than the BAE REST client to specify the target service.
<code>sleepTimeBetweenRetries</code>	Optional	The wait time, in milliseconds, between retry attempts when a request fails.

Note: As mentioned in the previous table, the parameters `restClientName` and `restServiceName` must be informed to identify the REST client that is updated. Except for BAE REST client, where only `restClientName` must be informed.

Example 2 Example of an `<inputFile.json>`

```
{
  "connectionTimeout": 5000,
  "numberOfRetries": 2,
  "requestTimeout": 5000,
  "restClientName": "bae-rest",
```



```
    "sleepTimeBetweenRetries": 2000  
  }
```

Example 3

Updating the CPM REST client with `restClientName` and `restServiceName`:

```
beamctl eric-bss-ael-bae-oam update-rest-config < inputFile.json
```

Sample Response:

```
{  
  "restClientName": "cpm-rest",  
  "restServiceName": "CPM_BusinessLogic",  
  "numberOfRetries": 3,  
  "sleepTimeBetweenRetries": 3000,  
  "requestTimeout": 3000,  
  "connectionTimeout": 3000  
}
```

Example 4

Updating the BAE REST client with `restClientName`:

```
beamctl eric-bss-ael-bae-oam update-rest-config < inputFile.json
```

Sample Response:

```
{  
  "restClientName": "bae-rest",  
  "numberOfRetries": 3,  
  "sleepTimeBetweenRetries": 3000,  
  "requestTimeout": 3000,  
  "connectionTimeout": 3000  
}
```

3.1.2 BAE Supervisor Configuration

The Supervisor Configuration command offers functionality to interact with BAE Supervisor ConfigMap. It allows to retrieve the ConfigMap and to modify its values as needed.

Syntax

```
beamctl eric-bss-ael-bae-oam
```



Available Sub Commands

Table 4 Sub Commands

Command	Description
get-supervisor-config on page 10	Enables to retrieve the BAE Supervisor ConfigMap.
update-supervisor-config on page 11	Allows to update the BAE Supervisor ConfigMap.

3.1.2.1 get-supervisor-config

The `get-supervisor-config` command enables to retrieve the BAE Supervisor ConfigMap.

Syntax

```
beamctl eric-bss-ael-bae-oam get-supervisor-config
```

Arguments

The command has no arguments.

Parameters

Table 5 Parameters

Parameter	Parameter Type	Description
<code>minAvailable</code>	Mandatory	Minimum capacity that must be available.
<code>monitoringDelay</code>	Mandatory	Delay, in seconds, between successive monitoring attempts.
<code>monitoringFailureCount</code>	Mandatory	Number of successive monitoring failures that qualify the Application Group for failover.
<code>monitoringStatus</code>	Mandatory	Indicates whether pod health monitoring is enabled or disabled.
<code>overloadThreshold</code>	Mandatory	It is the overload percentage threshold: equal or above this value is considered overloaded.

Example 5

```
beamctl eric-bss-ael-bae-oam get-supervisor-config
```

Sample Response:

```
{
  "schemaVersion": "1.0.0",
  "supervisor": {
    "appMonitoringConfig": [
```



```
{
  "podHealthChecks": {
    "minAvailable": "30%",
    "monitoringDelay": 2,
    "monitoringFailureCount": 3,
    "monitoringStatus": "enabled",
    "overloadThreshold": 90
  }
},
"version": "1.0.0"
}
```

3.1.2.2 update-supervisor-config

The `update-supervisor-config` command allows to update the BAE Supervisor ConfigMap.

Syntax

```
beamctl eric-bss-ael-bae-oam update-supervisor-config <


```

Arguments

The command has no arguments.

Parameters

Table 6 JSON Parameters

Parameter	Parameter Type	Description
minAvailable	Mandatory	Minimum capacity that must be available.
monitoringDelay	Mandatory	Delay, in seconds, between successive monitoring attempts.
monitoringFailureCount	Mandatory	Number of successive monitoring failures that qualify the Application Group for failover.
monitoringStatus	Mandatory	Indicates whether pod health monitoring is enabled or disabled.
overloadThreshold	Mandatory	It is the overload percentage threshold: equal or above this value is considered overloaded.

Example 6 Example of an `<inputFile.json>`

```
{
  "minAvailable": "30%",
  "overloadThreshold": 90,
  "monitoringDelay": 2,
  "monitoringFailureCount": 3,
}
```



```
"monitoringStatus": "enabled"
}
```

Example 7

```
beamctl eric-bss-ael-bae-oam update-supervisor-config <
inputFile.json
```

Sample Response:

```
{
  "minAvailable": "40%",
  "monitoringDelay": 2,
  "monitoringFailureCount": 3,
  "monitoringStatus": "enabled",
  "overloadThreshold": 90
}
```

3.1.3 BAE Data Encryption Configuration

The Data Encryption Configuration command offers functionality to interact with BAE Data Encryption ConfigMap. It allows to retrieve the ConfigMap and modify its values as needed.

Syntax

```
beamctl eric-bss-ael-bae-oam
```

Available Sub Commands

Table 7 Sub Commands

Command	Description
get-data-encryption-config on page 12	Enables to retrieve a BAE Data Encryption ConfigMap.
update-data-encryption-config on page 14	Allows to update a BAE Data Encryption ConfigMap.

3.1.3.1 get-data-encryption-config

The `get-data-encryption-config` command enables to retrieve a BAE Data Encryption ConfigMap.



Syntax

```
beamctl eric-bss-ael-bae-oam get-data-encryption-config <apiName>
```

Arguments

Table 8 Mandatory Arguments

Argument	Description
apiName	It is the name of the API to retrieve the ConfigMap. The apiName can be: — bssf — customermanagement — partymanagement — tmf637 — tmf654 — tmf666 — tmf669 — tmf676 — tmf678

Parameters

Table 9 Parameters

Parameter	Parameter Type	Description
baeTransientDataEncryptionEnabled	Mandatory	Indicates whether transient data encryption is enabled for BAE.
frequency	Mandatory	Frequency for periodic tasks related to encryption. For example: rotation or monitoring.
initialDelay	Mandatory	Initial delay before the first execution of scheduled encryption-related tasks.
serverPort	Mandatory	Port on which the KMS gateway service listens.
serviceName	Mandatory	Name of the KMS gateway service.
tlsVersion	Mandatory	TLS protocol version used for secure communication with the KMS gateway.



Example 8

```
beamctl eric-bss-ael-bae-oam get-data-encryption-config bssf
```

Sample Response:

```
{
  "baeTransientDataEncryptionEnabled": "",
  "frequency": 5,
  "initialDelay": 0,
  "serverPort": 5000,
  "serviceName": "eric-bss-bam-kms-gateway",
  "tlsVersion": "TLSv1.3"
}
```

Example 9

```
beamctl eric-bss-ael-bae-oam get-data-encryption-config tmf637
```

Sample Response:

```
{
  "baeTransientDataEncryptionEnabled": "",
  "frequency": 5,
  "initialDelay": 0,
  "serverPort": 5000,
  "serviceName": "eric-bss-bam-kms-gateway",
  "tlsVersion": "TLSv1.3"
}
```

3.1.3.2 update-data-encryption-config

The `update-data-encryption-config` command allows to update a BAE Data Encryption ConfigMap.

Syntax

```
beamctl eric-bss-ael-bae-oam update-data-encryption-config apiName
< <inputFile.json>
```

Arguments

Table 10 Mandatory Arguments

Argument	Description
apiName	It is the name of the API to retrieve the ConfigMap. The apiName can be:



Argument	Description
	— bssf — customermanagement — partymanagement — tmf637 — tmf654 — tmf666 — tmf669 — tmf676 — tmf678

Parameters

Table 11 JSON Parameters

Parameter	Parameter Type	Description
baeTransientDataEncryptionEnabled	Mandatory	Indicates whether transient data encryption is enabled for BAE.
frequency	Mandatory	Frequency for periodic tasks related to encryption. For example: rotation or monitoring.
initialDelay	Mandatory	Initial delay before the first execution of scheduled encryption-related tasks.
serverPort	Mandatory	Port on which the KMS gateway service listens.
serviceName	Mandatory	Name of the KMS gateway service.
tlsVersion	Mandatory	TLS protocol version used for secure communication with the KMS gateway.

Example 10 Example of an *<inputFile.json>*

```
{
  "baeTransientDataEncryptionEnabled": "",
  "frequency": 5,
  "initialDelay": 0,
  "serverPort": 5000,
  "serviceName": "eric-bss-bam-kms-gateway",
  "tlsVersion": "TLSv1.3"
}
```



Example 11

```
beamctl eric-bss-ael-bae-oam update-data-encryption-config bssf <
inputFile.json
```

Sample Response:

```
{
  "baeTransientDataEncryptionEnabled": "false",
  "serviceName": "encrypt-data",
  "serverPort": 4444,
  "frequency": 5,
  "initialDelay": 10,
  "tlsVersion": "TLSv1.2"
}
```

3.1.4

BAE Hops Allowed During Translation Configuration

The Hops Allowed during External ID to ID Translation Configuration command offers functionality to interact with the BAE Maximum Number of Hops property.

This configuration controls the maximum number of translation message routing within the BAE translation service.

Syntax

```
beamctl eric-bss-ael-bae-oam
```

Available Sub Commands

Table 12 Sub Commands

Command	Description
get-max-number-of-hops on page 16	This command is capable of get BAE maximum number of hops properties.
update-max-number-of-hops on page 18	This command is capable of update BAE maximum number of hops properties.

3.1.4.1

get-max-number-of-hops

The `get-max-number-of-hops` command is capable of get BAE maximum number of hops properties.

Syntax

```
beamctl eric-bss-ael-bae-oam get-max-number-of-hops
```



Arguments

The command has no arguments.

Parameters

Table 13 Parameters

Parameter	Parameter Type	Description
eric-bss-ael-bae	Mandatory	It is the maximum number of hops (remote reads) allowed for the BAE microservice.
eric-bss-ael-bae-customermanagement	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF629 Customer Management</i> microservice.
eric-bss-ael-bae-partymanagement	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF 632 Party Management</i> microservice.
eric-bss-ael-bae-tmf637	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF637 Product Inventory Management</i> microservice.
eric-bss-ael-bae-tmf654	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF654 Prepay Balance Management</i> microservice.
eric-bss-ael-bae-tmf666	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF666 Account Management</i> microservice.
eric-bss-ael-bae-tmf669	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF669 Party Role</i> microservice.
eric-bss-ael-bae-tmf676	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF676 Payment Management</i> microservice.
eric-bss-ael-bae-tmf678	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF678 Customer Bill Management</i> microservice.

Example 12

```
beamctl eric-bss-ael-bae-oam get-max-number-of-hops
```

Sample Response:

```
{
  "eric-bss-ael-bae": "1",
  "eric-bss-ael-bae-customermanagement": "1",
  "eric-bss-ael-bae-partymanagement": "1",
  "eric-bss-ael-bae-tmf637": "1",
  "eric-bss-ael-bae-tmf654": "1",
  "eric-bss-ael-bae-tmf666": "1",
  "eric-bss-ael-bae-tmf669": "1",
  "eric-bss-ael-bae-tmf676": "1",
  "eric-bss-ael-bae-tmf678": "1"
}
```



Note: If the response is "Fail. See logs", refer to the logs from the OAM application.

3.1.4.2 update-max-number-of-hops

The `update-max-number-of-hops` command is capable of update BAE maximum number of hops properties.

Syntax

```
beamctl eric-bss-ael-bae-oam update-max-number-of-hops <

```

Arguments

The command has no arguments.

Parameters

Table 14 Parameters

Parameter	Parameter Type	Description
eric-bss-ael-bae	Mandatory	It is the maximum number of hops (remote reads) allowed for the BAE microservice.
eric-bss-ael-bae-customermanagement	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF629 Customer Management</i> microservice.
eric-bss-ael-bae-partymanagement	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF632 Party Management</i> microservice.
eric-bss-ael-bae-tmf637	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF637 Product Inventory Management</i> microservice.
eric-bss-ael-bae-tmf654	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF654 Prepay Balance Management</i> microservice.
eric-bss-ael-bae-tmf666	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF666 Account Management</i> microservice.
eric-bss-ael-bae-tmf669	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF669 Party Role</i> microservice.
eric-bss-ael-bae-tmf676	Mandatory	It is the maximum number of hops (remote reads) allowed for the <i>TMF676 Payment Management</i> microservice.
eric-bss-ael-bae-tmf678	Mandatory	It is the maximum number of hops (remote reads) allowed



Parameter	Parameter Type	Description
		for the <i>TMF678 Customer Bill Management</i> microservice.

Example 13 Example of an `<inputFile.json>`

```
{
  "maxNumberOfHops": "1"
}
```

Example 14

beamctl eric-bss-ael-bae-oam update-max-number-of-hops <inputFile.json>

Sample Response:

```
{
  "eric-bss-ael-bae": "1",
  "eric-bss-ael-bae-customermanagement": "1",
  "eric-bss-ael-bae-partymanagement": "1",
  "eric-bss-ael-bae-tmf637": "1",
  "eric-bss-ael-bae-tmf654": "1",
  "eric-bss-ael-bae-tmf666": "1",
  "eric-bss-ael-bae-tmf669": "1",
  "eric-bss-ael-bae-tmf676": "1",
  "eric-bss-ael-bae-tmf678": "1"
}
```

Note: If the response is "Fail. See logs", refer to the logs from the OAM application.

3.2 CHA Configuration Commands

Note: The commands are intended to be run against primary site unless its mentioned as site specific.

3.2.1 Diameter Interface Configuration

The Diameter interface commands provide the options to manage Diameter interface configurations. The following command gives the specification for managing the Diameter interface configuration:

beamctl diameter-interface -h

The available commands are described in the following subsections.



3.2.1.1 Add Diameter Peer

The command `add-diameter-peer` adds a diameter peer to the list of known peers for the specified AppGroup.

Syntax

```
beamctl diameter-interface add-diameter-peer <appgroupname>  
<diameterhost> [flags]
```

Arguments

Table 15 Mandatory Arguments

Argument	Description
appgroupname	Name of the app group.
diameterhost	FQDN, IPv4, or IPv6 of the Diameter Host.

Flags

Table 16 Flags

Flag	Type	Description
port	int	Port of the Diameter Host. The default is 3868.
scheme	string	URI scheme of the Diameter Host. 'aas' is used for secure connectivity, and 'aaa' is used for unsecure connectivity. The default is 'aas'.
transport	string	Transport connectivity to Diameter Host. The default is 'tcp'.
initiateconnection	boolean	Specifies if the DLB must initiate connection to the configured Peer. The default is 'false'. When set to 'true', the DLB always considers the raisealarm flag as true for the configured peer.
raisealarm	boolean	Specifies if DLB must raise an alarm when a known peer is not connected. This flag is applicable only when the initiateconnection flag is set to 'false' for the configured Peer. The default is 'false'.

Example 15

```
beamctl diameter-interface add-diameter-peer cha1 peer1.com
```




```
beamctl diameter-interface add-diameter-peer cha1 10.1.1.1 --
port=3868 --scheme=aaa --transport=tcp --initiateconnection=true
--raisealarm=true
```

```
beamctl diameter-interface add-diameter-peer cha1
2001:db8:3333:4444:CCCC:DDDD:EEEE:FFFF --port=3868 --scheme=aaa
--transport=tcp --initiateconnection=true --raisealarm=true
```

3.2.1.2 Add Diameter Proxy

The `add-diameter-proxy` command adds a diameter proxy for a given realm for the specified AppGroup.

Syntax

```
beamctl diameter-interface add-diameter-proxy <appgroupname>
<diameterhost> <realm> [flags]
```

Arguments

Table 17 Mandatory Arguments

Argument	Description
appgroupname	Name of the app group.
diameterhost	FQDN, IPv4, or IPv6 of the Diameter Host.
realm	Realm served by the Diameter Proxy.

Flags

Table 18 Flags

Flag	Type	Description
port	int	Port of the Diameter Host. The default is 3868.
scheme	string	URI scheme of the Diameter Host. 'aaas' is used for secure connectivity, and 'aaa' is used for unsecure connectivity. The default is 'aaas'.
transport	string	Transport connectivity to Diameter Host. The default is 'tcp'.



Example 16

```
beamctl diameter-interface add-diameter-proxy chal
pgw2.ericsson.com "*"

beamctl diameter-interface add-diameter-proxy chal 10.1.1.1
"realm1" --port=3868 --scheme=aaa --transport=tcp

beamctl diameter-interface add-diameter-proxy chal
2001:db8:3333:4444:CCCC:DDDD:EEEE:FFFF "realm2" --port=3868 --
scheme=aaa --transport=tcp
```

3.2.1.3

Add Own Diameter Identity

The command `add-own-diameter-identity` adds or updates an own Diameter Identity for the specified DLB instance for the specified AppGroup.

Syntax

```
beamctl diameter-interface add-own-diameter-identity <appgroupname>
<dlbinstanceid> <owndiameteridentity>
```

Arguments

Table 19 Mandatory Arguments

Argument	Description
appgroupname	Name of the app group.
dlbinstanceid	ID of DLB instance. If DLB pod name is <i><eric-bss-cha-diameter-lb-0></i> , then <i><dlb-instance-id></i> is "0".
owndiameteridentity	The address to use as the FQDN of the Diameter Load Balancer.

Note: The addition or update of the *<OwnDiameterIdentity>*, restarts Diameter Stack in corresponding DLB instance due to which all peer connections are dropped and reconnected.

The change in DLB's Own FQDN must be updated in northbound and southbound peers to avoid connection errors.



Example 17

```
beamctl diameter-interface add-own-diameter-identity cha1 0
dlb0.appgroup1.com
```

3.2.1.4 Delete AppGroup

The command `delete-appgroup` removes diameter interface configuration for the specified AppGroup.

Syntax

```
beamctl diameter-interface delete-appgroup <appgroupname>
```

Table 20 Mandatory Arguments

Argument	Description
appgroupname	Name of the App Group.

Example 18

```
beamctl diameter-interface delete-appgroup cha1
```

3.2.1.5 Get All Config

The `get-all-config` command retrieves all diameter interface configuration.

Syntax

```
beamctl diameter-interface get-all-config
```

Arguments

The command has no arguments.

Example 19

```
beamctl diameter-interface get-all-config
```

Response:

```
"appGroups": [
  {
    "additionalServedRealms": [],
    "diameterPeerList": [
```



```
    "aas://peer1.com:3868;transport=tcp;initiateconnection:false;raisealarm:f  
  else"  
    ],  
    "diameterProxyList": [  
      "aas://pgw2.ericsson.com:3868;transport=tcp**"  
    ],  
    "name": "cha1",  
    "ownDiameterIdentity": {},  
    "ownRealm": null  
  },  
  {  
    "additionalServedRealms": [],  
    "diameterPeerList": [],  
    "diameterProxyList": [],  
    "name": "cha1",  
    "ownDiameterIdentity": {},  
    "ownRealm": "dlu0.appgroup1.com"  
  }  
],  
"globalConfig": {  
  "messageTimeout": 6000,  
  "ownRealm": null,  
  "restrictToPeerList": false,  
  "sendReplyToUnknownPeers": true,  
  "watchdogTimeout": 5000  
}
```

3.2.1.6

Get AppGroup

The `get-appgroup` command retrieves diameter interface configuration for the specified AppGroup.

Syntax

```
beamctl diameter-interface get-appgroup <appgroupname>
```

Arguments

Table 21 Mandatory Arguments

Argument	Description
appgroupname	Name of the AppGroup.

Example 20

```
beamctl diameter-interface get-appgroup cha1
```

Sample Response:

```
{  
  "additionalServedRealms": [],  
  "diameterPeerList": [  
    "aas://peer1.com:3868;transport=tcp#initiateconnection:false#raisealarm:false"  
  ],  
  "diameterProxyList": [  
    "aas://pgw2.ericsson.com:3868;transport=tcp**"  
  ],  
  "name": "cha1",
```



```
"ownDiameterIdentity": {
  "0": "d1b0.appgroup1.com"
},
"ownRealm": null
}
```

3.2.1.7 Get AppGroups

The `get-appgroups` command retrieves diameter interface configuration for all the AppGroups.

Syntax

```
beamctl diameter-interface get-appgroups
```

Arguments

The command has no arguments.

Example 21

```
beamctl diameter-interface get-appgroups
```

Response:

```
[
  {
    "additionalServedRealms": [],
    "diameterPeerList": [
      "aaas://peer1.com:3868;transport=tcp#initiateconnection:false#raisealarm:f →
    ],
    "diameterProxyList": [
      "aaas://pgw2.ericsson.com:3868;transport=tcp#"
    ],
    "name": "cha1",
    "ownDiameterIdentity": {
      "0": "d1b0.appgroup1.com"
    },
    "ownRealm": null
  },
  {
    "additionalServedRealms": [],
    "diameterPeerList": [],
    "diameterProxyList": [],
    "name": "CHA1",
    "ownDiameterIdentity": {},
    "ownRealm": null
  }
]
```

3.2.1.8 Get Diameter Peers

The `get-diameter-peers` command retrieves list of known peers for the specified AppGroup.



Syntax

```
beamctl diameter-interface get-diameter-peers <appgroupname>
```

Arguments

Table 22 Mandatory Arguments

Argument	Description
appgroupname	Name of the AppGroup.

Example 22

```
beamctl diameter-interface get-diameter-peers cha1
```

Sample Response:

```
[ "aaas://peer1.com:3868;transport=tcp#initiateconnection:false#raisealarm:false"
```

3.2.1.9

Get Diameter Proxies

The `get-diameter-proxies` command retrieves the list of Diameter Proxies for the specified AppGroup.

Syntax

```
beamctl diameter-interface get-diameter-proxies <appgroupname>
```

Arguments

Table 23 Mandatory Arguments

Argument	Description
appgroupname	Name of the AppGroup.

Example 23

```
beamctl diameter-interface get-diameter-proxies cha1
```

Sample Response:



```
[  
  "aas://pgw2.ericsson.com;3868;transport;tcp#*"
```

3.2.1.10 Get Global Config

The `get-global-config` command retrieves global level Diameter Interface configuration.

Syntax

```
beamctl diameter-interface get-global-config
```

Arguments

The command has no arguments.

Example 24

```
beamctl diameter-interface get-global-config
```

Sample Response:

```
{  
  "messageTimeout": 6000,  
  "ownRealm": null,  
  "restrictToPeerList": false,  
  "sendReplyToUnknownPeers": true,  
  "watchdogTimeout": 5000  
}
```

3.2.1.11 Get Own Diameter Identities

The command `get-own-diameter-identities` retrieves the list of own Diameter Identities for the specific AppGroup.

Syntax

```
beamctl diameter-interface get-own-diameter-identities  
<appgroupname>
```



Arguments

Table 24 Mandatory Arguments

Argument	Description
appgroupname	Name of the AppGroup.

Example 25

```
beamctl diameter-interface get-own-diameter-identities cha1
```

Sample Response

```
{
  "0": dlb0.appgroup1.com"
}
```

3.2.1.12

Get Own Diameter Identity

The command `get-own-diameter-identity` retrieves own Diameter Identity for the specified DLB instance for specified AppGroup.

Syntax

```
beamctl diameter-interface get-own-diameter-identity <appgroupname>
<dlbinstanceid>
```

Arguments

Table 25 Mandatory Arguments

Argument	Description
appgroupname	Name of the AppGroup.
dlbinstanceid	Id of DLB instance. If DLB pod name is <code>eric-bss-cha-diameter-lb-0</code> , then <code>dlb-instance-id</code> is <code>0</code> .

Example 26

```
beamctl diameter-interface get-own-diameter-identity cha1 0
```

Sample Response:



```
{
  "0": dlb0.appgroup1.com"
}
```

3.2.1.13 Remove Diameter Peer

The `remove-diameter-peer` command removes Diameter Peer from the list of known peers for the specified AppGroup.

Syntax

```
beamctl diameter-interface remove-diameter-peer <appgroupname>
<diameterhost>
```

Arguments

Table 26 Mandatory Arguments

Argument	Description
appgroupname	Name of the AppGroup.
diameterhost	FQDN, IPv4, or IPv6 of the Diameter Host.

Example 27

```
beamctl diameter-interface remove-diameter-peer cha1 peer1.com
beamctl diameter-interface remove-diameter-peer cha1 10.1.1.1
beamctl diameter-interface remove-diameter-peer cha1
2001:db8:3333:4444:CCCC:DDDD:EEEE:FFFF
```

3.2.1.14 Remove Diameter Proxy

The `remove-diameter-proxy` command, removes a Diameter Proxy-Realm from the list of proxies for the specified AppGroup.

Syntax

```
beamctl diameter-interface remove-diameter-proxy appgroupname
diameterhost realm
```



Arguments

Table 27 Mandatory Arguments

Argument	Description
appgroupname	Name of the AppGroup.
diameterhost	FQDN, IPv4, or IPv6 of the Diameter Host.
realm	Realm served by the Diameter Proxy.

Example 28

```
beamctl diameter-interface remove-diameter-proxy chal
pgw2.ericsson.com "*"

beamctl diameter-interface remove-diameter-proxy chal 10.1.1.1
"realm1"

beamctl diameter-interface remove-diameter-proxy chal
2001:db8:3333:4444:CCCC:DDDD:EEEE:FFFF "realm2"
```

3.2.1.15

Remove Own Diameter Identity

The command `remove-own-diameter-identity` removes own Diameter Identity for the specified DLB instance for the specified AppGroup.

Syntax

```
beamctl diameter-interface remove-own-diameter-identity CHA1 0
```

Arguments

Table 28 Mandatory Arguments

Argument	Description
appgroupname	Name of the AppGroup.
dlbinstanceid	ID of DLB instance. If DLB pod name is <code>eric-bss-cha-diameter-lb-0</code> , then <code>dlb-instance-id</code> is <code>'0'</code> .

Note: If `OwnDiameterIdentity` is removed for a DLB instance, then corresponding DLB service External IP is used as Own FQDN.



Example 29

```
beamctl diameter-interface remove-own-diameter-identity cha1 0
```

3.2.1.16

Set Message Timeout

The `set-message-timeout` command sets the value of Diameter MessageTimeout in milliseconds.

Syntax

```
beamctl diameter-interface set-message-timeout <messagetimeout>
```

Arguments

Table 29 Mandatory Arguments

Argument	Description
messagetimeout	This timer specifies how long to wait in milliseconds for the answer to a traffic request before timing out.

Example 30

```
beamctl diameter-interface set-message-timeout 6000
```

3.2.1.17

Set Restrict to Peer List

The `set-restrict-to-peer-list` command restricts connection to known peers. This command accepts a boolean flag *true* or *false*.

Syntax

```
beamctl diameter-interface set-restrict-to-peer-list  
<restricttopeerlist>
```

Arguments

Table 30 Mandatory Arguments

Argument	Description
restrictToPeerList	Specifies whether the DLB accepts requests from unknown peers or not.



Example 31

```
beamctl diameter-interface set-restrict-to-peer-list false
```

3.2.1.18

Set Send Reply to Unknown Peers

The `set-send-reply-to-unknown-peers` command sets sending connection denial response to unknown peers. This command accepts a boolean flag *true* or *false*. This applies only when `restrictToPeerList` is set to *true*.

Syntax

```
beamctl diameter-interface set-send-reply-to-unknown-peers  
<sendreplytounknownpeers>
```

Arguments

Table 31 Mandatory Arguments

Argument	Description
<code>sendReplyToUnknownPeers</code>	Specifies whether the DLB replies to unknown peers or not. Valid values are <i>true</i> and <i>false</i> . Any values other than true or false sets the <code>sendReplyToUnknownPeers</code> to <i>false</i> .

Example 32

```
beamctl diameter-interface set-send-reply-to-unknown-peers false
```

3.2.1.19

Set Watchdog Timeout

The `set-watchdog-timeout` command sets the value of Diameter WatchdogTimeout in milliseconds.

Syntax

```
beamctl diameter-interface set-watchdog-timeout <watchdogTimeout>
```



Arguments

Table 32 Mandatory Arguments

Argument	Description
watchdogTimeout	This timer specifies how long to wait in milliseconds for the answer to a watchdog request before timing out.

Example 33

```
beamctl diameter-interface set-watchdog-timeout 5000
```

3.2.1.20

Set Own Realm

Syntax

To set an own realm at the Global level, the below BEAM CLI command is used:

```
beamctl diameter-interface set-own-realm ownrealm
```

To set an own realm at the AppGroup, the below BEAM CLI command is used:

```
beamctl diameter-interface set-own-realm ownrealm --appgroup-name=<appgroupname>
```

- Note:**
- The AppGrp level value takes precedence over the value that is configured on the global level.
 - The Own Diameter Realm configured through BEAM CLI commands takes effect only after restarting the CHA Access and CHA DLB.

Arguments

Table 33 Mandatory Arguments

Argument	Description
ownRealm	The own realm name.

Table 34 Optional Arguments

Flag	Description
appGroupName	Application Group name.



Example 34

```
beamctl diameter-interface set-own-realm ericsson.com

beamctl diameter-interface set-own-realm ericsson.appgrp1.com --
appgroup-name=CHA1
```

3.2.1.21

Get Own Realm

Syntax

To retrieve a configured own diameter realm at global level, the below BEAM CLI command is used.

The Own Realm set at AppGroup level is retrieved if specified in the request, otherwise, the globally configured realm is retrieved.

```
beamctl diameter-interface get-own-realm
```

To retrieve a configured own diameter realm at the AppGroup level, the below BEAM CLI command is used:

```
beamctl diameter-interface get-own-realm --appgroup-
name=<appgroupname>
```

Arguments

Table 35 Optional Arguments

Argument	Description
appGroupName	Specifies the Application Group name.

Example 35

```
beamctl diameter-interface get-own-realm

beamctl diameter-interface get-own-realm --appgroup-name=CHA1
```

Sample Response:

```
{
  "ownRealm": "ericsson.appgrp1.com"
}
```



3.2.1.22 Remove Own Realm

To remove own diameter realm for the specified AppGroup.

Syntax

```
beamctl diameter-interface remove-own-realm --<appgroup-name>
```

Arguments

Table 36 Mandatory Arguments

Argument	Description
appGroupName	Name of the application group.

Example 36

```
beamctl diameter-interface remove-own-realm CHA1
```

3.2.1.23 Add Served Realm

To add an additional served diameter realm for the specified application group.

Syntax

```
beamctl diameter-interface add-additional-realm <appgroupname>  
<additionalRealm>
```

Arguments

Table 37 Mandatory Arguments

Argument	Description
ownRealm	Additional Diameter realm served by the system.
appGroupName	Name of the AppGroup.

Example 37

```
beamctl diameter-interface add-additional-realm --CHA1  
ericsson.appgrp1.com
```



3.2.1.24

Get Served Realm

This command is used to retrieve the additional served diameter realms for the specified AppGroup. If no AppGroup is specified, all available AppGroups are retrieved.

Syntax

```
beamctl diameter-interface get-additional-realms
```

```
beamctl diameter-interface get-additional-realms --appgroup-name=<appgroupname>
```

Arguments

Table 38 Mandatory Arguments

Argument	Description
appGroupName	Name of the AppGroup.

Example 38

```
beamctl diameter-interface get-additional-realms --app group-name=CHA1
```

Response:

```
{  
  "ericsson.appgrp1.com"  
}
```

3.2.1.25

Remove Served Realm

This command is used to remove the specified additional served diameter realm for the specified Application Group.

Note: If the realm to be removed is not specified, all the available additional served realms, for the specified Application Group, are removed.

Syntax

```
beamctl diameter-interface remove-additional-realm --appgroup-name
```

```
beamctl diameter-interface remove-additional-realm --appgroup-name --realm=<additionalRealm>
```




Arguments

Table 39 Mandatory Arguments

Argument	Description
ownRealm	Additional Diameter realm served by the system.
appGroupName	Name of the AppGroup.

Example 39

```
beamctl diameter-interface remove-additional-realm CHA1  
  
beamctl diameter-interface remove-additional-realm CHA1 --  
realm=ericsson.appgrp1.com
```

3.2.2 External Rating Configuration

The external rating commands provide the functionality to perform different operations towards the external rating interface. The interface manages the Diameter realms and peers used by CHA for performing rating in external systems.

The following command gives the specification for managing the external rating configuration.

```
beamctl external-rating -h
```

The available commands are described in the following subsections.

3.2.2.1 List Realm

The `list-realms` command shows all the currently configured realms for external rating.

Syntax

```
beamctl external-rating list-realms
```

Arguments

The command has no arguments.



Example 40

```
beamctl external-rating list-realms
```

3.2.2.2

Update Realms

The `update-realms` command replaces the existing external rating realms with the newly provided list. The command requires the list of external rating realms as a JSON from `stdin`.

Syntax

```
beamctl external-rating update-realms < <Input JSON file>
```

Arguments

The command has no arguments.

Parameters

Table 40 Update Realms JSON Parameters

The parameters used in `realms.json` are described in the below table:

Parameter	Parameter Type	Description
realm	Mandatory	Valid <i>External Rating Peer Realm</i> address. The value can also contain a <i>wildcard</i> (*) character at the beginning of the realm name. The wildcard entry indicates that any set of characters is allowed. Example: *.com indicates that this entry is valid for any realm ending with .com. (*) indicates that any realm is valid for the realm entry. Note: Only asterisk (*) is allowed as a wildcard entry, and only at the start of the realm.
appGzp	Optional	This field indicates the CHA Application Group the configuration is applicable for. The default value is <i>global</i>. If the value is <i>global</i>, then it is applicable for all CHA Application Groups across sites. It can also be made specific to a particular site by indicating the CHA Application Group name associated with it. Note: If a particular configuration entry is configured both as <i>global</i> and for a specific CHA Application Group, then the configuration with the CHA Application Group takes precedence.
sdp_id	Optional	Part of <code>optionalData</code>: Comma-separated list of SDP Virtual IPs (External Peer Identifiers). This is expressed as a String. This identifier is used by the application to resolve the Destination-Realm of the External Rating Peer. Identifiers expected to be configured in this list are: 1. The external rating peer identifier (SDPID) returned from Id-Translation. Note: The ID-Translation refers to the CPM ID-Translation look-up for a subscriber.



Parameter	Parameter Type	Description
		2. The Virtual-IP of the external Peer sent in the Redirect-Host AVP by SDP during redirections, owing to subscriber move in the External Rating Peer. Example: <pre>"sdp_id": ["sdp01.cs.", "10.216.230.37"]</pre>
applications	Mandatory	Comma-separated list of Diameter applications supported by the <i>External Rating Peer</i> of a realm. Example: — SBI-Sy(Nchf-SyPolicy) - 16777359 — SBI-Esy(Nchf-EsyPolicy) - 16777361 — Sy- 16777302 — Esy - 16777304 — CIP-IP - 16777232
initiateConnection	Optional	This field is deprecated. Use <code>initiateConnection</code> in <i>External-rating update-peer</i> command, described in chapter Update Peers on page 41.
strategy	Mandatory	Distribute the input traffic to multiple peers under a realm. The options are: round-robin, failover, or failover-failback. The default is round-robin.
addresses	Optional	Group of peer addresses for external rating. — <code>index</code>: Internally used to determine peers priority. Lowest value gets the highest priority. — <code>peerAddresses</code>: List of peer addresses for the group. Example: — Single Peer: <pre>[{"index":1,"peerAddresses":["aaa://10.63.22.21:3868;transport=tcp"]}]</pre> — Grouped Peer: <pre>[{"index":1,"peerAddresses":["aaa://10.63.22.21:3868;transport=tcp","aaa://10.63.22.22:3868;transport=tcp"]}, {"index":2,"peerAddresses":["aaa://10.63.22.23:3868;transport=tcp","aaa://10.63.22.24:3868;transport=tcp"]}, {"index":3,"peerAddresses":["aaa://[2001:db8:1::ab9:C0A8:102]:3868;transport=tcp","aaa://3.2.3.4:3868;transport=tcp"]}]</pre> Note: The parameters <code>index</code> and <code>peerAddresses</code> are mandatory. <code>peerAddress</code>: This is External rating peer Diameter URI address. The format of External Peer Diameter URI address: <code>aaa(s)://<fqdn/ipv4/[ipv6]>:<port>;transport=<tcp/sctp></code> IPv6 address should be enclosed within <code>[]</code> brackets.



Parameter	Parameter Type	Description
		Note: — The parameter <code>peer</code> is mandatory. — Over TCP, when the System is used as an External Rate and Charge application it supports both secured and unsecured communication. However over SCTP, when the System is used as an External Rate and Charge application it supports only unsecured communication.

Example 41

The following example includes two indices (SDP GeoRed configuration). The lowest index number is prioritized higher. *Index 1* is the primary SDP cluster, *Index 2* is the GeoRed backup SDP cluster. Without SDP GeoRed, *Index 2* can be omitted.

beamctl external-rating update-realms < realm.json

JSON for Update Realms:

```
[
  {
    "realm": "sdp01.sdp.example.com",
    "appGrp": "chal",
    "sdp_id": [
      "sdp01.cs."
    ],
    "applications": [
      "16777232",
      "16777359",
      "16777302",
      "16777304",
      "16777361"
    ],
    "initiateConnection": true,
    "strategy": "round-robin",
    "addresses": [
      {
        "index": 1,
        "peerAddresses": [
          "aaa://ka-cs-sdp01a.example.com:3868;transport=sctp",
          "aaa://ka-cs-sdp01b.example.com:3868;transport=sctp",
          "aaa://10.1.1.1:3868;transport=sctp"
        ]
      },
      {
        "index": 2,
        "peerAddresses": [
          "aaa://li-cs-geo-red-sdp02a.example.com:3868;transport=sctp",
          "aaa://li-cs-geo-red-sdp02b.example.com:3868;transport=sctp",
          "aaa://[2001:0db8:0001:0000:0000:0ab9:C0A8:0102]:3868;transport=sctp"
        ]
      }
    ]
  }
]
```

3.2.2.3

List Peers

The `list-peers` command lists the configured external rating diameter peers.



Syntax

```
beamctl external-rating list-peers
```

Arguments

The command has no arguments.

Example 42

```
beamctl external-rating list-peers
```

Sample Response:

```
{
  "address": [],
  "addresses": [
    {
      "index": 1,
      "peerAddresses": [
        "aaa://ka-cs-sdp01a.example.com:3868;transport=sctp",
        "aaa://ka-cs-sdp01b.example.com:3868;transport=sctp",
        "aaa://10.1.1.1:3868;transport=sctp"
      ]
    },
    {
      "index": 2,
      "peerAddresses": [
        "aaa://li-cs-geo-red-sdp02a.example.com:3868;transport=sctp",
        "aaa://li-cs-geo-red-sdp02b.example.com:3868;transport=sctp",
        "aaa://[2001:0db8:0001:0000:0000:0ab9:C0A8:0102]:3868;transport=sctp"
      ]
    }
  ],
  "appGrp": "cha1",
  "applications": [
    "16777359",
    "16777304",
    "16777302",
    "16777361",
    "16777232"
  ],
  "initiateConnection": true,
  "realm": "sdp01.sdp.example.com",
  "sdp_id": "sdp01.cs.",
  "strategy": "round-robin"
}
```

3.2.2.4 Update Peers

The `update-peers` command replaces the existing external rating Diameter peers with the newly provided list. The command requires the list of external peers as a JSON from `stdin`.

Syntax

```
beamctl external-rating update-peers < <Input JSON file>
```



Arguments

The command has no arguments.

Table 41 Update Peers JSON Parameters

The parameters used in peers.json are listed in the below table.

Parameter	Parameter Type	Description
Peer	Mandatory	Valid External Rating Diameter Peer URI (Diameter URI address). Format of External Diameter Peer address: <code>aaa(s)://<fqdn/ipv4/[ipv6]>:<port>;transport=<tcp/sctp></code> IPv6 address must be enclosed within [] brackets. Examples: — <code>"peer":aaa://10.24.3.41:3868;transport=tcp"</code> — <code>"peer":aaa://[2001:db8:1::ab9:C0A8:102]:3868;transport=tcp"</code> — <code>"peer":aaa://peer1.com:3868;transport=tcp"</code>
appGrp	Optional	This field indicates the CHA Application Group to which this configuration applies. The default value is <i>global</i>. If the value is <i>global</i>, then it is applicable for all CHA Application Groups across sites. It can also be made specific to a particular site by indicating the CHA Application Group name associated with it.



Parameter	Parameter Type	Description
		Note: If a particular configuration entry is configured both as <i>global</i> and for a specific CHA Application group, then the configuration with the CHA Application group takes precedence.
initiateConnection	Optional	Specifies if DLB must initiate connection to the configured Peer. The default is <i>false</i> . When set to <i>true</i> , the DLB always considers the <i>raiseAlarm</i> flag as <i>true</i> for the configured peer.
raiseAlarm	Optional	Specifies if DLB must raise an alarm when a known peer is not connected. This flag is applicable only when <i>initiateConnection</i> flag is set to <i>false</i> for the configured peer. Default is <i>false</i> .
connectAddresses	Optional	List of additional IP Addresses or Hostnames used to connect peer.

Example 43 JSON for Update Peers

beamctl external-rating update-peers < peers.json

```
[
  {
    "peer": "aaa://ka-cs-sdp01b.example.com:3868;transport=sctp",
    "appGrp": "cha1",
    "initiateConnection": true,
    "raiseAlarm": true,
    "connectAddresses": ["10.61.44.127", "10.61.44.128", "fc00:1000::f38b"]
  },
  {
    "peer": "aaa://ka-cs-sdp01a.example.com:3868;transport=sctp",
    "appGrp": "cha1",
    "initiateConnection": true,
    "raiseAlarm": true,
    "connectAddresses": ["10.61.44.137", "10.61.44.138", "fc00:1000::f38b"]
  },
  {
    "peer": "aaa://li-cs-geo-red-sdp02a.example.com:3868;transport=sctp",
    "appGrp": "cha1",
    "initiateConnection": false,
    "raiseAlarm": true,
    "connectAddresses": ["10.216.230.24", "10.216.230.25", "2001:db8:1::ab9:C0A8
:102"]
  },
  {
    "peer": "aaa://li-cs-geo-red-sdp02b.example.com:3868;transport=sctp",
    "appGrp": "cha1",
    "initiateConnection": false,
    "raiseAlarm": false,
    "connectAddresses": ["10.216.230.27", "10.216.230.28", "2001:db8:3333:4444:C
CCC:DDDD:EEEE:FFFF"]
  }
]
```



```
    "peer": "aaa://[fc00:1000::f38b]:3868;transport=sctp",
    "appGrp": "chal",
    "initiateConnection": true,
    "raiseAlarm": true,
    "connectAddresses": ["10.61.44.127", "10.61.44.128", "2001:db8:3333:4444:CCC →
C:DDDD:EEEE:FFFF"]
  },
  {
    "peer": "aaa://10.61.44.127:3868;transport=sctp",
    "appGrp": "chal",
    "initiateConnection": false,
    "raiseAlarm": false,
    "connectAddresses": ["10.216.230.27", "10.216.230.28", "2001:db8:3333:4444:C →
CCC:DDDD:EEEE:FFFF"]
  }
]
```

3.2.3 SCP Configuration

CHA SCP Configuration REST Interface and CHA SCP App Configuration REST Interface manages the configuration used by CHA for Service Communication Proxy (SCP) connections.

3.2.3.1 SCP Server Configuration

The following sections describe the configuration possible for SCP servers.

Add New SCP Server Configuration

The add-scp-server command adds a new SCP server configuration. The command requires a JSON object from standard input or shorthand format.

Syntax

```
beamctl scp-config add-scp-server < <SCP Config JSON file>
```

Arguments

This command has no arguments

Table 42 Attributes of SCP Config JSON file

Attribute Name	Attribute Type	Attribute Variable Name	Description
address	Mandatory	String	The URI or SCP server address. Example: https://<scp server fqdn> or <IPv4> or <IPv6>:<port>. IPv6 address should be



Attribute Name	Attribute Type	Attribute Variable Name	Description
			enclosed within [] brackets.
secured	Optional	Boolean	Deprecated. Indicates whether it is http or https communication. Possible Value: True or False.
priority	Optional	Integer	Servers with lower priority values will be used first.
appGroup	Optional	String	Name of the application group.
backoffWhenDownInSeconds	Optional	Integer	The time (in seconds) to wait after a connection failure before attempting to connect to the server again. A value of 0 means no delay. The default value is 0.

Example JSON for adding a SCP Server Configuration:

Example 1:

```
{
  "address": "https://ericsson.scp.host:8080",
  "priority": 1,
  "appGroup": "CHA1",
  "backoffWhenDownInSeconds": 60
}
```

Example 2:

```
{
  "address": "https://10.63.22.21:8080/scp",
  "priority": 1,
  "appGroup": "CHA1",
  "backoffWhenDownInSeconds": 60
}
```

Example 3:

```
{
  "address": "https://[1050:0:0:0:5:600:300c:326b]:8080/scp",
  "priority": 1,
  "appGroup": "CHA1",
  "backoffWhenDownInSeconds": 60
}
```

Example:

```
beamctl scp-config add-scp-server < scp_config.json
```



Response: SCP Server config added successfully with Id: 1

List All SCP Servers Configuration

The `list-scp-servers` command list the configuration for all SCP servers.

Syntax

```
beamctl scp-config list-scp-servers [--appgroup=<appgroup>]
```

Arguments

Table 43 Arguments

Argument Name	Argument Type	Argument Variable Name	Description
appGroup	Optional	String	List only the configuration for SCP servers with the specified app group.

Example

```
beamctl scp-config list-scp-servers --appgroup=CHA1
```

Example 1:

```
{
  "address": "https://ericsson.scp.host:8080",
  "priority": 1,
  "appGroup": "CHA1",
  "backoffWhenDownInSeconds": 60
}
```

Example 2:

```
{
  "address": "https://10.63.22.21:8080/scp",
  "priority": 1,
  "appGroup": "CHA1",
  "backoffWhenDownInSeconds": 60
}
```

Example 3:

```
{
  "address": "https://[1050:0:0:0:5:600:300c:326b]:8080/scp",
  "priority": 1,
  "appGroup": "CHA1",
  "backoffWhenDownInSeconds": 60
}
```

Note: The `appGroup` and `priority` are optional parameters. If these parameters are used in input.json file, always set to "null".

Get SCP Server Configuration

The `get-scp-server` command shows the configuration for a given SCP server.

Syntax



```
beamctl scp-config get-scp-server <id>
```

Arguments

Table 44 Argument

Argument Name	Argument Type	Argument Variable Name	Description
id	Mandatory	String	Server's unique identifier

Example

```
beamctl scp-config get-scp-server 1
```

Sample Output

```
{
  "address": "https://ericsson.scp.host",
  "appGroup": "CHA1",
  "id": "1"
}
```

Note: The `appGroup` and `priority` are optional parameters. If these parameters are used in `input.json` file, always set to "null".

Delete SCP Server Configuration

The `delete-scp-server` command removes the configuration for a given SCP server.

Syntax

```
beamctl scp-config delete-scp-server <id>
```

Arguments

Table 45 Argument

Argument Name	Argument Type	Argument Variable Name	Description
id	Mandatory	String	Server's unique identifier

Example

```
beamctl scp-config delete-scp-server 1
```



- Note:**
- For successful delete of SCP server, no response is given from BEAM CLI.
 - For delete command on non existing SCP server, response 404 is given from BEAM CLI.

Example:

Delete command on non existing SCP server:

```
./beamctl scp-config delete-scp-server 2
```

Error calling operation

HTTP 404: Resource 2 not found

3.2.3.2 SCP Application Configuration

The following sections describe the SCP application configurations:

List SCP Application Configuration

The `list-scp-app-config` command list all the SCP application configuration.

Syntax

```
beamctl scp-app-config list-scp-app-config
```

Arguments

The command has no arguments.

Example

```
beamctl scp-app-config list-scp-app-config
```

Add New SCP Application Configuration

The `add-scp-app-config` command adds a new SCP application configuration.

Syntax

```
beamctl scp-app-config add-scp-app-config < <SCP App Config JSON file>
```

Mandatory Arguments

This command has no arguments



- Note:**
- `appGroup` is always mandatory argument. The `scpTimeout` or `httpFailureCodes` has to go along with the `appGroup`.
 - Ensure the SCP timeout configuration is aligned with the configured SCP's request timeout value and configured transaction timeout.

Table 46 Attributes of SCP App Config JSON

Attribute Name	Attribute Type	Attribute Variable Name	Description
<code>appGroup</code>	Mandatory	String	Name of the application group
<code>scpTimeout</code>	Optional	String	Timeout in ms The <code>scpTimeout</code> or <code>httpFailureCodes</code> has to go along with the <code>appGroup</code> .
<code>httpFailureCodes</code>	Optional	Array of String	A list of http error codes that trigger reselection. The <code>scpTimeout</code> or <code>httpFailureCodes</code> has to go along with the <code>appGroup</code> .

Example

```
beamctl scp-app-config add-scp-app-config < scp_app_config.json
```

Sample `scp_app_config.json`

```
{
  "appGroup": "CHA1",
  "scpTimeout": 1000,
  "httpFailureCodes": [401, 402, 404, 500]
}
```

Get SCP Application Configuration

The `get-scp-app-config` command shows the application configuration for a given application group.

Syntax

```
beamctl scp-app-config get-scp-app-config <appGroup>
```



Arguments

Table 47 Argument

Argument Name	Argument Type	Argument Variable Name	Description
appGroup	Mandatory	String	Name of the application group

Example

```
beamctl scp-app-config get-scp-app-config CHA1
```

Delete SCP Application Configuration

The `delete-scp-app-config` command removes the remove the SCP application configuration for a given application group.

Syntax

```
beamctl scp-app-config delete-scp-app-config <appGroup>
```

Arguments

Table 48 Argument

Argument Name	Argument Type	Argument Variable Name	Description
appGroup	Mandatory	String	Name of the application group

Example

```
beamctl scp-app-config delete-scp-app-config CHA1
```

3.2.4

NRF Configuration

The Network Repository Function (NRF) configuration commands provide the functionality to perform different operations towards the NRF configuration interface. The interface manages the configuration used by CHA for NRF connections. The following command gives the specification for managing the NRF configuration.

```
beamctl nrf-config -h
```

The available commands are described in the following subsections.



3.2.4.1 Add NRF OAuth Server

The `add-nrf-oauth-server` command adds an NRF OAuth2 server to be used for network function authentication.

The command requires a JSON object from `stdin`.

Syntax

```
beamctl nrf-config add-nrf-oauth-server < oauth-server.json
```

Arguments

The command has no arguments.

Parameters

The parameters used in `oauth-server.json` are described below:

Table 49 Add NRF OAuth Server JSON Parameters

Parameter	Description
<code>address</code>	The NRF-OAuth endpoint address. Format: <code>http://<nrf server fqdn/ipv4/[ipv6]>:<port></code> Note: IPv6 address must be enclosed within <code>[]</code> brackets.
<code>failureCodes</code>	It represents the list of HTTP result Code for which OAuth server failover happens (switch to the next configured OAuth server). It can be any valid failure <i>HTTP status code</i>. For example: <code>[404,500]</code>.
<code>secured</code>	Indicates if TLS-based authentication is enabled or not.
<code>appGrp</code>	It indicates the CHA Application Group that the configuration is applicable for. The default value is <i>global</i>. If the value is <i>global</i>, then it is applicable for all CHA Application Groups across sites. It can also be made specific to a particular site by indicating the CHA Application Group name associated with it.



Parameter	Description
	Note: If a particular configuration entry is configured both as <i>global</i> and for a specific CHA Application Group, then the configuration with the CHA Application Group takes precedence.

Example 44

```
beamctl nrf-config add-nrf-oauth-server < oauth-server.json
```

Example 45 JSON for Add NRF OAuth Server

The following are JSON examples to add NRF OAuth Server:

```
{
  "address": "https://nrf.ericsson.com:3002",
  "failureCodes": [
    404,500
  ],
  "secured": true,
  "appGrp": "chal"
}
```

```
{
  "address": "https://10.63.22.21:3002",
  "failureCodes": [
    404,500
  ],
  "secured": true,
  "appGrp": "chal"
}
```

```
{
  "address": "https://[1050:0:0:0:5:600:300c:326b]:3002",
  "failureCodes": [
    404,500
  ],
  "secured": true,
  "appGrp": "chal"
}
```

Sample Response message for Add NRF OAuth Server:

```
Type: String
NRF Oauth Server added successfully.
```

3.2.4.2

Add NRF Server

The `add-nrf-server` command adds an NRF server to be used for network function registration. The command requires a JSON object from stdin.



Syntax

```
beamctl nrf-config add-nrf-server < <NRF Input JSON file>
```

Arguments

The command has no arguments.

Parameters

Table 50 Parameters of Add NRF Server JSON

The parameters used in nrf-server.json are described in the below table.

Parameter	Description
address	The NRF endpoint address. Format: <pre>http://<nrf server fqdn/ipv4/ [ipv6]>:<port></pre> Note: IPv6 address must be enclosed within [] brackets.
failureCodes	It represents the list of HTTP result Code for which NRF failover happens (switch to next configured NRF). It can be any valid failure <i>HTTP status code</i>. For example: [404,500].
secured	Indicates if TLS-based authentication is enabled or not.
compression	Indicates if the requests to the NRF are going to be compressed.
nfProfileValidation	Enable support for NFProfile validation after registration.
nfServiceType	Indicates type of NFService where configurable values are nfservices, nfservicelist, or both.
appGrp	It indicates the CHA Application Group that the configuration is applicable for. The default value is <i>global</i> . If the value is <i>global</i> , then it is applicable for all CHA Application Groups across sites. It can also be made specific to a particular site by indicating the CHA Application Group name associated with it.



Parameter	Description
	Note: If a particular configuration entry is configured both as <i>global</i> and for a specific CHA Application Group, then the configuration with the CHA Application Group takes precedence.

Sample Add NRF Server JSON

Example 46 Add NRF Server JSON

The following are JSON examples to add NRF Server:

```
{
  "address": "https://nrf.ericsson.com:3002",
  "failureCodes": [
    404,500
  ],
  "secured": true,
  "compression": true,
  "nfProfileValidation": true,
  "nfServiceType": "nfservices",
  "appGrp": "chal"
}
```

```
{
  "address": "https://10.63.22.21:3002",
  "failureCodes": [
    404,500
  ],
  "secured": true,
  "compression": true,
  "nfProfileValidation": true,
  "nfServiceType": "nfservices",
  "appGrp": "chal"
}
```

```
{
  "address": "https://[1050:0:0:0:5:600:300c:326b]:3002",
  "failureCodes": [
    404,500
  ],
  "secured": true,
  "compression": true,
  "nfProfileValidation": true,
  "nfServiceType": "nfservices",
  "appGrp": "chal"
}
```

Sample Response message for Add NRF Server:

```
Type: String
NRF Server added successfully.
```



3.2.4.3 Delete NRF Oauth Server

The `delete-nrf-oauth-server` command removes the configuration for a given NRF OAuth2 server. The command has one mandatory argument, `serverId`.

Syntax

```
beamctl nrf-config delete-nrf-oauth-server <serverId>
```

Arguments

Table 51 Mandatory Arguments

Argument	Description
<code>serverId</code>	NRF OAuth2 server identity.

Example 47

```
beamctl nrf-config delete-nrf-oauth-server 0
```

3.2.4.4 Delete NRF Server

The `delete-nrf-server` command removes the configuration for a given NRF server. The command has one mandatory argument, `serverId`.

Syntax

```
beamctl nrf-config delete-nrf-server <serverId>
```

Arguments

Table 52 Mandatory Arguments

Argument	Description
<code>serverId</code>	NRF server identity.

Example 48

```
beamctl nrf-config delete-nrf-server 0
```

3.2.4.5 Get NRF Oauth Server

The `get-nrf-oauth-server` command shows the configuration for a given NRF OAuth2 server. The command has one mandatory argument, `serverId`.

Syntax

```
beamctl nrf-config get-nrf-oauth-server <serverId>
```

Arguments



Table 53 Mandatory Arguments

Argument	Description
serverId	NRF OAuth server identity.

Example 49

```
beamctl nrf-config get-nrf-oauth-server 0
```

3.2.4.6

List NRF Oauth Servers

The `list-nrf-oauth-servers` command shows the NRF OAuth2 servers configured for NF authentication.

Syntax

```
beamctl nrf-config list-nrf-oauth-servers
```

Arguments

The command has no arguments.

Example 50

```
beamctl nrf-config list-nrf-oauth-servers
```

3.2.4.7

Get NRF Server

The `get-nrf-server` command shows the configuration for a given NRF server. The command has one mandatory argument, `serverId`.

Syntax

```
beamctl nrf-config get-nrf-server <serverId>
```

Arguments

Table 54 Mandatory Arguments

Argument	Description
serverId	NRF server identity

Example 51

```
beamctl nrf-config get-nrf-server 0
```



3.2.4.8 List NRF Servers

The `list-nrf-servers` command shows the NRF servers configured for NF registration.

Syntax

```
beamctl nrf-config list-nrf-servers
```

Arguments

The command has no arguments.

Example 52

```
beamctl nrf-config list-nrf-servers
```

3.2.4.9 List Registration Properties

The `list-registration-properties` command lists the current properties for NRF registration.

Syntax

```
beamctl nrf-config list-registration-properties
```

Arguments

The command has no arguments.

Example 53

```
beamctl nrf-config list-registration-properties
```

3.2.4.10 Update Registration Properties

The `update-registration-properties` command updates properties for NRF registration. The command requires a JSON object from stdin, or shorthand.

Update Registration Properties Using Stdin

The `update-registration-properties` command using stdin requires a JSON file.

Syntax

```
beamctl nrf-config update-registration-properties < config.json
```

Arguments

The command has no arguments.



Table 55 Update-Registration-Properties JSON Parameters

The parameters used in config.json are described in the below table.

Parameters	Description
nfRegistrationScope: <string>	Name of the NF registration scope.
retriesForNrfConnection: <integer>	Number of NRF connection retries, if connection fails.
retryIntervalForNrfConnection: <integer>	Retrial interval, in seconds, for NRF connection, if connection fails.
targetNfType: <string>	Name of the target NF type.
responseTimeout <integer>	Specifies the time to wait for a response from the NRF Client activity. Used to set the ReadTimeout properties for HTTP2 Client. Response time-out: In milliseconds granularity with range 0–1000000 ms. Default value is 1000 ms.
connectionTimeout <integer>	Specifies the maximum time allowed to wait for a NRF client to establish a connection with the NRF Server. Used to set the connectionTimeout properties for HTTP2 Client. Connection Timeout: In seconds granularity with range 0-1000 seconds. Default value is 10 seconds, if a connection timeout value is not configured.

Example: JSON for Update-Registration-Properties

```
{
  "nfRegistrationScope": "nnrf-nfm",
  "retriesForNrfConnection": 3,
  "retryIntervalForNrfConnection": 30,
  "targetNfType": "NRF",
  "responseTimeout": 1000
  "connectionTimeout": 10
}
```

Update Registration Properties Using Shorthand Commands

The update-registration-properties using shorthand commands require input arguments.

Syntax



```
beamctl nrf-config update-registration-properties <argument>
```

Command Arguments

See previous section about [Update-Registration-Properties JSON parameters](#)
[The parameters used in config.json are described in the below table.](#)

.

Example: Shorthand Commands for Update Registration Properties

```
beamctl nrf-config update-registration-properties  
nrfRegistrationScope: nnrf-nfm
```

```
beamctl nrf-config update-registration-properties  
retriesForNrfConnection: 3
```

```
beamctl nrf-config update-registration-properties  
retryIntervalForNrfConnection: 30
```

```
beamctl nrf-config update-registration-properties targetNfType:  
NRF
```

```
beamctl nrf-config update-registration-properties  
responseTimeout: 2000
```

```
beamctl nrf-config update-registration-properties  
connectionTimeout: 10
```

3.2.5

CHA Business Configuration

The `cha-businessconfiguration` command provides the capability to export Charging business configuration specifications in a zip format. The command uses the CHA Business Configuration Entity Management REST Interface to retrieve the Charging Business Configurations from CIL and package the entities in the zip format.

Table 56 CHA-BUSINESSCONFIGURATION Commands

Command Name	Description
<code>export-all</code>	Export all the charging business configuration specifications.
<code>export</code>	Exports the charging business configuration specifications specified in payload.



3.2.5.1 export-all

The `export-all` command exports all the CHA Business Configuration Specifications.

Syntax

```
beamctl cha-businessconfiguration export-all
```

The charging specifications are packaged in the generated zip file.

Arguments

The command has no arguments.

Example:

```
beamctl cha-businessconfiguration export-all
```

Sample Request

```
GET /cha/businessConfiguration/v1/export HTTPS/1.1
Host: charging.ericsson.com:13443
Accept: application/octet-stream
```

Sample Response:

```
HTTP/1.1 200 OK
Content-Disposition: attachment; filename="cha-business-config-1715165716.zip"
Content-Type: application/zip

Octet Stream containing all the Service Context Configurations, Protocol Definitions, and Other Ch →
arging Configuration Entities.
```

3.2.5.2 export

The command exports the CHA Business Configuration Specifications specified in JSON payload.

Syntax

```
beamctl cha-businessconfiguration export < <JSON file>
```

Arguments

The command has no arguments.

Example:

```
beamctl cha-businessconfiguration export < input.json
```




The input.json file contains the charging business configuration entities to be exported. The JSON input contains an array of entities. Each entity contains an entityType and an id field. The generated zip file contains all the exported specifications.

The export operations for AccountAdjustment Service Context and NormalizeRAT Dynamic Function are as follows:

Sample Request:

```
{
  "entities": [{
    "id": ["AccountAdjustment::1594126605012"],
    "entityType": "BalanceAdjustmentServiceContext"
  }, {
    "id": ["NormalizeRAT::1594126605012"],
    "entityType": "DynamicFunctions"
  }]
}
```

Table 57 Attributes for Input Json

Name		Type	Mandatory/Optional	Description
entities		Array of Entity	Optional	List of Entity ids and their associated Entity type to be exported
Entity		Object		
	id	Array of string	Optional	List of entity identifiers of the form <entity name>::<entity version> belonging to an Entity Type.
	entityType	string	Mandatory	Type of Charging Business Configuration Entity

Sample Response:

```
HTTP/1.1 200 OK
Content-Type:application/octet-stream
```

Octet Stream containing AccountAdjustment Service Context along with Active version of its depende →



nt entities such as DynamicFunctions, Protocol specification, and related NN rule and DynamicFunction "NormalizeRAT", are the responses to the REST Client. →

Note: The entity is an Optional parameter. When entities are null, all the business configurations are retrieved from CIL database.

The supported entityType corresponding to their respective external services are as follows:

— Service Context Configurations

- entityType: BalanceAdjustmentServiceContext , externalservice: balanceadjustment
- entityType: BalanceManagementServiceContext, externalservice : balancemanagement
- entityType: ConvergedRateAndChargeSBIServiceContext, externalservice : convergedrateandcharge
- entityType: OfflineRateAndChargeServiceContext, externalservice : offlinerateandchargebx
- entityType: OfflineRateAndChargeJSONFileServiceContext, externalservice : offlinerateandchargejsonfile
- entityType: OnlineRateAndChargeCipipServiceContext, externalservice : onlinerateandchargecipip
- entityType: OnlineRateAndChargeDCCAServiceContext, externalservice : onlinerateandchargero
- entityType: OnlineRateAndChargeRestServiceContext, externalservice : onlinerateandchargerest
- entityType: PolicyControlESyServiceContext, externalservice : policycontrolesy
- entityType: PolicyControlSyServiceContext, externalservice : policycontrolsy
- entityType: ProductAdminRateAndChargeServiceContext, externalservice : productadminrateandcharge
- entityType: PurchaseChargeServiceContext, externalservice : purchasecharge
- entityType: RecurrenceManagementServiceContext, externalservice : recurrencemanagement
- entityType: BillingCostCalculationServiceContext, externalservice : billingcostcalculation
- entityType: RefillServiceContext, externalservice : refill



- entityType: SpendingLimitControlSBIServiceContext, external service : spendinglimitcontrolsbi

— Protocol Definitions

- entityType: ConvergedRateAndChargeSBIProtocolDefinitions, externalservice : convergedrateandcharge/protocoldefinitions
- entityType: OfflineRateAndChargeJSONFileProtocolDefinitions, externalservice : offlineandchargejsonfile/protocoldefinitions
- entityType: OnlineRateAndChargeCipipProtocolDefinitions, externalservice : onlineandchargecipip/protocoldefinitions
- entityType: OnlineRateAndChargeDCCAProtocolDefinitions, externalservice : onlineandchargero/protocoldefinitions
- entityType: OnlineRateAndChargeRestProtocolDefinitions, externalservice : onlineandchargerest/protocoldefinitions
- entityType: PolicyControlESyProtocolDefinitions, externalservice : policycontrolesy/protocoldefinitions
- entityType: PolicyControlSyProtocolDefinitions, externalservice : policycontrolsy/protocoldefinitions
- entityType: SpendingLimitControlSBIProtocolDefinitions, externalservice : spendinglimitcontrolsbi/protocoldefinitions
- entityType: SBIReferenceResources, CAF service: sbireference/protocoldefinitions

— Other Charging Configuration Entities

- DynamicFunctions (dynamicfunctions)
- NumberNormalization (numbernormalization)
- NumberNormalizationPrefix (nnprefix)

3.2.6 CHA Event Topic Configuration

This interface manages the event topic configurations that is used for supporting message bus solution.

Note: The cha-event-topic-config command is to be run against each site.

Syntax

cha-event-topic-config <sub-command>



Available Sub Commands

Table 58 Available Sub Commands

Command	Description
list-topics	The command retrieves all configured event topic configurations for both MSG_KAFKA and CIL message bus types, and returns topic names, retention periods (in minutes), and partition information from the CHA business configuration.
get-topic-config	The command retrieves configuration for a specific event topic by name across both MSG_KAFKA and CIL platforms, and returns topic settings including retention period and partition details.

3.2.6.1

List Topics

The `list-topics` command retrieves a list of configured event topics.

Syntax

```
beamctl cha-event-topic-config list-topics
```

Arguments

The command has no arguments.

Parameters

Table 59 JSON Parameters

Parameter	Parameter Type	Description
retentionInMins	Integer	Retention period for the topic in minutes.
topicName	String	Name of the event topic.
topicPartition	Array	List of partition configurations for the topic.
partitionInfo	Integer	Partition information identifier.



Parameter	Parameter Type	Description
topicPartitionCount	Integer	Number of partitions for the topic.

Example 54

beamctl cha-event-topic-config list-topics

Sample Response

```
./beamctl cha-event-topic-config list-topics
{
  "CIL": [
    {
      "retentionInMins": 6,
      "topicName": "cha_cbc_event_customer",
      "topicPartition": [
        {
          "partitionInfo": 1,
          "topicPartitionCount": 20
        }
      ]
    },
    {
      "retentionInMins": 6,
      "topicName": "balanceSnapshot",
      "topicPartition": [
        {
          "partitionInfo": 1,
          "topicPartitionCount": 18
        }
      ]
    }
  ],
  "MSG_KAFKA": [
    {
      "retentionInMins": 6,
      "topicName": "cha_cbc_event_customer",
      "topicPartition": [
        {
          "partitionInfo": 1,
          "topicPartitionCount": 18
        }
      ]
    },
    {
      "retentionInMins": 6,
      "topicName": "balanceSnapshot",
      "topicPartition": [
        {
          "partitionInfo": 1,
          "topicPartitionCount": 18
        }
      ]
    }
  ]
}
```

3.2.6.2 Get Topic Configuration

The `get-topic-config` command retrieves configuration of the specified topic.



Note: This command operates on the `eric-bss-cha-business-config-event-messaging` configmap only. This configmap is used when topic config is centrally managed and defined during installation. The configmap cannot be updated, and the topic configuration is site specific. To see the local config, the command needs to be sent to all the sites.

Syntax

```
beamctl cha-event-topic-config get-topic-config <topicname>
```

Arguments

Table 60 Mandatory Arguments

Argument	Description
topicname	Name of the event topic configuration.

Parameters

Table 61 JSON Parameters

Parameter	Parameter Type	Description
retentionInMins	Integer	Retention period for the topic in minutes.
topicName	String	Name of the event topic.
topicPartition	Array	List of partition configurations for the topic.
partitionInfo	Integer	Partition information identifier.
topicPartitionCount	Integer	Number of partitions for the topic.

Example 55

```
beamctl cha-event-topic-config get-topic-config <topicname>
```

Sample Response

```
./beamctl cha-event-topic-config get-topic-config cha_cbc_event_customer
{
  "CIL": [
    {
      "retentionInMins": 6,
      "topicName": "cha_cbc_event_customer",
      "topicPartition": [
        {
          "partitionInfo": 1,
```



```

        "topicPartitionCount": 20
      }
    ]
  },
  "MSG_KAFKA": [
    {
      "retentionInMins": 6,
      "topicName": "cha_cbc_event_customer",
      "topicPartition": [
        {
          "partitionInfo": 1,
          "topicPartitionCount": 18
        }
      ]
    }
  ]
}

```

3.2.7 CHA Network Function Profile Configuration

The Network Function profile commands provide the functionality to perform different operations towards the NF profile interface. The interface manages the nfProfile that is used for registering the Charging Function (CHF) in the NRF.

The following command gives the specification for managing the NF profile configuration.

beamctl nf-profile-config -h

The available commands are described in the following subsections.

3.2.7.1 List NF Profile Config

The `list-nf-profile-config` command lists the configurable properties of the nfProfile used for NF registration along app group name.

Syntax

beamctl nf-profile-config list-nf-profile-config

Arguments

The command has no arguments.

Return the Map which has the key as appgroupname and value as Nfprofile.

Example 56

beamctl nf-profile-config list-nf-profile-config

Sample Response

```

"cha1": {
  "nfType": "CHF",
  "allowedNfTypes": [
    "SMF",

```



```
        "SMSF",
        "PCF"
    ],
    "heartBeatTimer": 120,
    "ipv4Addresses": [],
    "locality": "Karlskrona",
    "nfServicePersistence": false,
    "nfServices": [
        {
            "allowedNfTypes": [
                "PCF"
            ],
            "allowedOperationsPerNfInstance": →
            "allowedOperationsPerNfType": null →
            "allowedSnps": null,
            "ipEndPoints": [],
            "perPlmnSnssaiList": null,
            "sNssais": null,
            "scheme": null,
            "serviceName": "nchf-spendinglimit →
            "vendorId": null,
            "versions": [
                {
                    "apiFu" →
                    "apiVe" →
                }
            ]
        },
        {
            "allowedNfTypes": [
                "SMF",
                "SMSF"
            ],
            "allowedOperationsPerNfInstance": →
            "allowedOperationsPerNfType": null →
            "allowedSnps": null,
            "ipEndPoints": [],
            "perPlmnSnssaiList": null,
            "sNssais": null,
            "scheme": null,
            "serviceName": "nchf-convergedchar →
            "vendorId": null,
            "versions": [
                {
                    "apiFu" →
                    "apiVe" →
                }
            ]
        }
    ],
    "priority": 0,
    "capacity": null,
    "plmnList": [],
    "sNssais": null,
    "nsiList": [],
    "allowedNfDomains": [],
    "allowedNssais": [],
    "allowedPlmns": [],
    "chfInfo": null,
    "customInfo": null,
    "snpnList": [],
    "allowedSnps": [],
    "chfInfoList": null,
    "servingScope": [],
    "perPlmnSnssaiList": [],
    "scpDomains": [],
    "nfSetIdList": []
}
```




```
}
}
```

3.2.7.2

Add NF Profile Config

`add-nf-profile-config` command adds the configurable properties of the `nfProfile` used for NF registration. The command requires a JSON object from stdin, or shorthand.

Syntax

```
beamctl nf-profile-config add-nf-profile-config <appGroupName> <
Input Payload JSON file>
```

Arguments

Table 62 Mandatory Arguments

Argument	Description
<code>appGroupName</code>	Indicates the application group name.

Example

```
beamctl nf-profile-config add-nf-profile-config CHA1
<payload.json
```

Sample payload.json

```
{
  "nfType": "CHF",
  "allowedNfTypes": [
    "SMF",
    "SMSF",
    "PCF"
  ],
  "heartBeatTimer": 120,
  "ipv4Addresses": [],
  "locality": "Karlskrona",
  "nfServicePersistence": false,
  "nfServices": [
    {
      "allowedNfTypes": [
        "PCF"
      ],
      "allowedOperationsPerNfInstance": null,
      "allowedOperationsPerNfType": null,
      "allowedSnps": null,
      "ipEndpoints": [],
      "perPlmnSnssailist": null,
      "sNssais": null,
      "scheme": null,
      "serviceName": "nchf-spendinglimitcontro
    },
    {
      "vendorId": null,
      "versions": [
        {
          "apiFullVers
          "apiVersionI
        }
      ]
    }
  ]
}
```



```
    },
    {
        "allowedNfTypes": [
            "SMF",
            "SMSF"
        ],
        "allowedOperationsPerNfInstance": null,
        "allowedOperationsPerNfType": null,
        "allowedSnpsns": null,
        "ipEndPoints": [],
        "perPlmnSnssaiList": null,
        "sNssais": null,
        "scheme": null,
        "serviceName": "nchf-convergedcharging",
        "vendorId": null,
        "versions": [
            {
                "apiFullVers" →
                "apiVersionI" →
            }
        ]
    },
    {
        "priority": 0,
        "capacity": null,
        "plmnList": [],
        "sNssais": null,
        "nsiList": [],
        "allowedNfDomains": [],
        "allowedNssais": [],
        "allowedPlmns": [],
        "chfInfo": null,
        "customInfo": null,
        "snpnList": [],
        "allowedSnpsns": [],
        "chfInfoList": null,
        "servingScope": [],
        "perPlmnSnssaiList": [],
        "scpDomains": [],
        "nfSetIdList": []
    }
}

ion": "2.0.0",
nUri": "v2"
```

Table 63 Parameters for Payload JSON

Parameter	Description
nfType	Charging function type or Custom Type of network function, see 3GPP specification TS 29.510.
allowedNfTypes	Types of allowed Network Functions to access Charging Function. For more details, see 3GPP specification TS 29.510.
heartBeatTimer	Time in seconds expected between two consecutive heartbeat messages to the NRF.
capacity	Static capacity information in the range of 0-65535, expressed as a weight relative to other CHF instances. For more details, see 3GPP TS 29.510.
priority	Priority in the range of 0-65535, to be used for NF selection; lower values



Parameter	Description
	indicate a higher priority. For more details, refer 3GPP TS 29.510.
plmnList	PLMNs of the Charging Function. For example: [{"mcc":"001","mnc":"234"}]. Deprecated representation: mcc-mnc, pattern for mcc: $\wedge\{3\}$ \$ and mnc: $\wedge\{3\}$ \$. For example: [123-456,567-789]. For more details, see 3GPP TS 29.510.
snssais	S-NSSAI of the Charging Function. For example: [{"sst":255,"sd":"000001","sdRanges":[{"start":"abc001","end":"def002"}],"wildcardSd":true}]. Deprecated representation: sst-sd, pattern for sst: range of 0-255 and sd: $\wedge[A-Fa-f0-9]\{6\}$ \$. For example: [1-ABCD01, 2-cdef02]. For more details, see 3GPP TS 29.510.
nsiList	NSI identities of the Charging Function. For more details, see 3GPP TS 29.510.
allowedNfDomains	NF domain names allowed to access the Charging Function instance. For more details, see 3GPP TS 29.510.
allowedNssais	List of allowed S-NSSAI to access the Charging Function instance. For example: [{"sst":255,"sd":"000001","sdRanges":[{"start":"abc001","end":"def002"}],"wildcardSd":true}]. Deprecated representation: sst-sd, pattern for sst: range of 0-255 and sd: $\wedge[A-Fa-f0-9]\{6\}$ \$. For example: [1-ABCD01, 2-cdef02]. For more details, see 3GPP TS 29.510.
allowedPlmns	List of PLMN allowed to access the Charging Function instance. For example: [{"mcc":"001","mnc":"234"}]. Deprecated representation: mcc-mnc, pattern for mcc: $\wedge\{3\}$ \$ and mnc: $\wedge\{3\}$ \$. For example: [123-456,567-789]. For more details, see 3GPP TS 29.510.
nfServicePersistence	The capability to persist resource state in shared storage. For more details, see 3GPP TS 29.510.



Parameter	Description
nfServices	NfProfileConvergedChargingService or NfProfileSpendingLimitControlService parameters
chfInfo	Specific data for the Charging Function. For more details, see 3GPP TS 29.510.
customInfo	Specific data for custom Network Functions. For more details, see 3GPP TS 29.510.
retriesForNrfConnection	Count of retrying the NRF connection
retryIntervalForNrfConnection	Time out interval for retrying the NRF connection
locality	Location of the CHF instance, see 3GPP specification TS 29.510.
targetNfType	This specifies producer NF Type of OAuth2 server. For more details, see 3GPP specification TS 33.501.
nfRegistrationScope	This scope belongs to OAuth2 server for which the access token can be requested by NF Consumer. For more details, see 3GPP specification TS 29.510.
snpnList	Standalone Non-Public Network (SNPN) (s) of the Network Function. This IE can be configured if the CHF instance (represented by this app group) pertains to one or more SNPNs. For example: [{"mcc":"001","mnc":"123","nid":"11223344556"}], For more details, see 3GPP TS 29.510.
allowedSnpons	Standalone Non-Public Network (SNPN) (s) allowed to access this CHF Instance. For example: [{"mcc":"001","mnc":"123","nid":"11223344556"}], For more details, see 3GPP TS 29.510.
allowedSnpons	Standalone Non-Public Network (SNPN) (s) allowed to access this CHF Instance. For example: [{"mcc":"001","mnc":"123","nid":"11223344556"}], For more details, see 3GPP TS 29.510.



Parameter	Description
chfInfoList	Multiple entries of ChfInfo. This attribute provides additional information to the chfInfo. chfInfoList can be present even if the chfInfo is absent. The key of the map shall be a (unique) valid JSON string per clause 7 of IETF RFC 8259 [22], with a maximum of 32 characters.
servingScope	One or more served areas of the CHF instance. The absence of this attribute does not imply that the CHF instance can serve every area in the PLMN.
perPlmnSnssaiList	S-NSSAI of the CHF Service per PLMN. This can be a subset of the S-NSSAI supported per PLMN by the CHF instance. This IE can be included when the list of S-NSSAI supported by the CHF Service for each PLMN it is supporting is different. When present, this IE shall include the S-NSSAI supported by the CHF Service for each PLMN and it shall prevail over the list of S-NSSAI supported per PLMN by the CHF instance. When present, this IE shall override the sNssais IE. For example, :[{"plmnId": {"mcc":"string","mnc":"string"},"sNssaiList": [{"sst":255,"sd":"000001","sdRanges": [{"start":"abc001","end":"def002"}],"wildcardSd":true}],"nid":"11223344556"}]. For more details, see 3GPP TS 29.510.
scpDomains	SCP domains of CHF or SCP instances whose status is requested to be supervised.
nfSetIdList	NF Set ID defined in clause 28.12 of 3GPP TS 23.003. At most one NF Set ID shall be indicated per PLMN of the NF.

3.2.7.3

Get NF Profile Config

The `get-nf-profile-config` command gets the configurable properties of the `nfProfile` used for NF registration.

Syntax



```
beamctl nf-profile-config get-nf-profile-config <appGroupName>
```

Arguments

Table 64 Mandatory Arguments

Argument	Description
appGroupName	Indicates the application group name.

Example 57

```
beamctl nf-profile-config get-nf-profile-config CHA1
```

3.2.7.4 Update NF Profile Config

The update-nf-profile-config command updates the configurable properties of the nfProfile used for NF registration.

This command requires a JSON object from stdin, or shorthand.

Syntax

```
beamctl nf-profile-config update-nf-profile-config <appGroupName>  
< payload.json
```

Arguments

Table 65 Mandatory Arguments

Argument	Description
appGroupName	Indicates the application group name.

Example

```
beamctl nf-profile-config update-nf-profile-config CHA1  
<payload.json
```

payload JSON:

```
{  
  "nfType": "CHF",  
  "allowedNfTypes": [  
    "SMF",  
    "SMSF",  
  ]  
}
```



```
    "PCF"  
  ],  
  "heartBeatTimer": 120,  
  "capacity": null,  
  "priority": 0,  
  "plmnList": [],  
  "sNssais": null,  
  "nsiList": [],  
  "allowedNfDomains": [],  
  "allowedNssais": [],  
  "allowedPlmns": [],  
  "nfServicePersistence": false,  
  "chfInfo": null,  
  "customInfo": null,  
  "locality": "Karlskrona",  
  "snpnList": [],  
  "allowedSnpons": [],  
  "chfInfoList": null,  
  "servingScope": [],  
  "perPlmnSnssaiList": [],  
  "scpDomains": [],  
  "nfSetIdList": [],  
  "overloadReporting": {  
    "downwardsNormalLoadThreshold": 10,  
    "upwardsOverloadThreshold": 20  
  }  
}
```



```
}
```

Table 66 Update NF Profile Config - Payload JSON Parameters

Parameter	Description
nfType	Charging function type or Custom Type of network function, see 3GPP specification TS 29.510.
allowedNfTypes	Types of allowed Network Functions to access Charging Function. For more details, see 3GPP specification TS 29.510.
heartBeatTimer	Time in seconds expected between two consecutive heartbeat messages to the NRF.
capacity	Static capacity information in the range of 0-65535, expressed as a weight relative to other CHF instances. For more details, see 3GPP TS 29.510.
priority	Priority in the range of 0-65535, to be used for NF selection; lower values indicate a higher priority. For more details, refer 3GPP TS 29.510.
plmnList	PLMNs of the Charging Function. For example: [{"mcc":"001","mnc":"234"}]. Deprecated representation: mcc-mnc, pattern for mcc: ^\d{3}\$ and mnc: ^\d{3}\$. For example: [123-456,567-789]. For more details, see 3GPP TS 29.510.
snssais	S-NSSAI of the Charging Function. For example: [{"sst":255,"sd":"000001","sdRanges":[{"start":"abc001","end":"def002"}],"wildcardSd":true}]. Deprecated representation: sst-sd, pattern for sst: range of 0-255 and sd: ^[A-Za-f0-9]{6}\$. For example: [1-ABCD01, 2-cdef02]. For more details, see 3GPP TS 29.510.
nsiList	NSI identities of the Charging Function. For more details, see 3GPP TS 29.510.



Parameter	Description
allowedNfDomains	NF domain names allowed to access the Charging Function instance. For more details, see 3GPP TS 29.510.
allowedNssais	List of allowed S-NSSAI to access the Charging Function instance. For example: [{"sst":255,"sd":"000001","sdRanges":[{"start":"abc001","end":"def002"}],"wildcardSd":true}]. Deprecated representation: sst-sd, pattern for sst: range of 0-255 and sd: ^[A-Za-f0-9]{6}\$. For example: [1-ABCD01, 2-cdef02]. For more details, see 3GPP TS 29.510.
allowedPlmns	List of PLMN allowed to access the Charging Function instance. For example: [{"mcc":"001","mnc":"234"}]. Deprecated representation: mcc-mnc, pattern for mcc: ^\d{3}\$ and mnc: ^\d{3}\$. For example: [123-456,567-789]. For more details, see 3GPP TS 29.510.
nfServicePersistence	The capability to persist resource state in shared storage. For more details, see 3GPP TS 29.510.
nfServices	NfProfileConvergedChargingService or NfProfileSpendingLimitControlService parameters
chfInfo	Specific data for the Charging Function. For more details, see 3GPP TS 29.510.
customInfo	Specific data for custom Network Functions. For more details, see 3GPP TS 29.510.
retriesForNrfConnection	Count of retrying the NRF connection
retryIntervalForNrfConnection	Time out interval for retrying the NRF connection
locality	Location of the CHF instance, see 3GPP specification TS 29.510.
targetNfType	This specifies producer NF Type of OAuth2 server. For more details, see 3GPP specification TS 33.501.
nfRegistrationScope	This scope belongs to OAuth2 server for which the access token can be requested by NF Consumer. For more



Parameter	Description
	details, see 3GPP specification TS 29.510.
snpnList	Standalone Non-Public Network (SNPN) (s) of the Network Function. This IE can be configured if the CHF instance (represented by this app group) pertains to one or more SNPNs. For example: [{"mcc":"001","mnc":"123","nid":"11223344556"}], For more details, see 3GPP TS 29.510.
allowedSnpons	Standalone Non-Public Network (SNPN) (s) allowed to access this CHF Instance. For example: [{"mcc":"001","mnc":"123","nid":"11223344556"}], For more details, see 3GPP TS 29.510.
chfInfoList	Multiple entries of ChfInfo. This attribute provides additional information to the chfInfo. chfInfoList can be present even if the chfInfo is absent. The key of the map shall be a (unique) valid JSON string per clause 7 of IETF RFC 8259 [22], with a maximum of 32 characters.
servingScope	One or more served areas of the CHF instance. The absence of this attribute does not imply that the CHF instance can serve every area in the PLMN.
perPlmnSnssaiList	S-NSSAI of the CHF Service per PLMN. This can be a subset of the S-NSSAI supported per PLMN by the CHF instance. This IE can be included when the list of S-NSSAI supported by the CHF Service for each PLMN it is supporting is different. When present, this IE shall include the S-NSSAI supported by the CHF Service for each PLMN and it shall prevail over the list of S-NSSAI supported per PLMN by the CHF instance. When present, this IE shall override the sNssais IE. For example, :[{"plmnId": {"mcc":"string","mnc":"string"},"sNssaiList": [{"sst":255,"sd":"000001","sdRanges": [{"start":"abc001","end":"def002"}],"wildcardSd":true}], "nid":"11223344556"}]



Parameter	Description
	]. For more details, see 3GPP TS 29.510.
scpDomains	SCP domains of CHF or SCP instances whose status is requested to be supervised.
nfSetIdList	NF Set ID defined in clause 28.12 of 3GPP TS 23.003. At most one NF Set ID shall be indicated per PLMN of the NF.
upwardsOverloadThreshold	Threshold to indicate an overload situation for overload reporting. Must be between 1 and 99 inclusive. Must be higher or equal to downloadsNormalLoadThreshold.
downwardsNormalLoadThreshold	Threshold to indicate normal load in overload reporting. Must be between 1 and 99 inclusive. Must be lower or equal to upwardsOverloadThreshold.

When there is a partial update of the NFProfile for an existing CHF/CHF's Network Function Instance, the CHF sent an HTTP PATCH request with NFUpdate. This allows the addition, removal, or replacement of one or more parameters within the CHF/CHF's of Network Function Instance.

The NFProfile Partial Update aligns with RFC 6902, which specifies the JSON Patch format for expressing partial updates to JSON documents. However, it is important to note that the network function Profile Partial Update specifically supports the replacement of a list of objects as a whole, rather than modifying an individual entries within the list. Only exception to it is "nfServices" parameter.

Example: Patch Update nfProfile

Existing NFProfile

```
{
  "allowedNfDomains": [],
  "allowedNfTypes": [
    "SMF",
    "SMSF",
    "PCF"
  ],
  "allowedNssais": [],
  "allowedPlmns": [],
  "allowedSnpsns": [],
  "capacity": null,
  "chfInfo": null,
```



```
"chfInfoList": null,  
"customInfo": null,  
"heartBeatTimer": 120,  
"locality": null,  
"nfServicePersistence": false,  
"nfSetIdList": ["2322","3435"],  
"nfType": "CHF",  
"nsiList": [],  
"perPlmnSnssaiList": [],  
"plmnList": [],  
"priority": 0,  
"sNssais": null,  
"scpDomains": [],  
"servingScope": [],  
"snpnList": []  
}
```

PatchUpload Payload

Syntax

```
beamctl nf-profile-config update-nf-profile-config CHA1 <  
patchpayload.json
```

patchpayload JSON:

```
{  
  "allowedNfDomains": [".net",".com"], ↓-- Added  
  "allowedNfTypes": [  
    "SMF",  
    "SMSF",  
    "PCF",  
    "AMF"                ↓---- Updated  
  ],  
  "allowedNssais": [],  
  "allowedPlmns": [],  
  "allowedSnpns": [],  
  "capacity": null,  
  "chfInfo": null,  
  "chfInfoList": null,  
  "customInfo": null,  
  "heartBeatTimer": 140,  ↓--- Updated  
  "locality": null,  
  "nfServicePersistence": false,  
  "nfSetIdList": [],      ↓-- removed  
  "nfType": "CHF",  
  "nsiList": [],  
  "perPlmnSnssaiList": [],  
  "plmnList": [],  
  "priority": 0,  
}
```



```
"sNssais": null,
"scpDomains": [],
"servingScope": [],
"snpnList": []
}
```

Patch message request payload sent to NRF:

```
PATCH .../nf-instances/d28c5425-e2c1-4223-b84d-493bc38bf11f
Content-Length: 326
Content-Type: application/json-patch+json
[
  {
    "op": "add",
    "path": "/allowedNfDomains",
    "value": [
      ".net",
      ".com"
    ]
  },
  {
    "op": "remove",
    "path": "/nfSetIdList"
  },
  {
    "op": "replace",
    "path": "/allowedNfTypes",
    "value": [
      "SMF",
      "SMSF",
      "PCF",
      "AMF"
    ]
  },
  {
    "op": "replace",
    "path": "/heartBeatTimer",
    "value": 140
  }
]
```

3.2.7.5 Delete NF Profile Config

The `delete-nf-profile-config` command deletes the configurable properties of the `nfProfile` used for NF registration.

Syntax

```
beamctl nf-profile-config delete-nf-profile-config <appGroupName>
```



Arguments

Table 67 Mandatory Arguments

Argument	Description
appGroupName	Indicates the application group name.

Example 58

```
beamctl nf-profile-config delete-nf-profile-config CHA1
```

3.2.7.6

Get Converged Charging

The `get-converged-charging` command shows the configurable properties for the Converged Charging service in the appropriate appGroup based nfProfile.

Syntax

```
beamctl nf-profile-config get-converged-charging <appGroupName>
```

Arguments

Table 68 Mandatory Arguments

Argument	Description
appGroupName	Indicates the application group name.

Example 59

```
beamctl nf-profile-config get-converged-charging CHA1
```

3.2.7.7

Update Converged Charging

The `update-converged-charging` command updates the properties for the Converged Charging service in the appropriate appGroup based nfProfile.

The command requires a JSON object from stdin, or shorthand.

Syntax

```
beamctl nf-profile-config update-converged-charging  
<appGroupName> < nfServicePayload.json
```

Arguments

Table 69 Mandatory Arguments

Argument	Description
appGroupName	Indicates the application group name.



Table 70 Update Converged Charging - JSON Parameters

The parameters used in nfServicePayload.json are described in the below table.

Parameter	Description
allowedNfTypes	Types of allowed Network Functions to access Charging Function. For more details, refer 3GPP specification TS 29.510.
allowedSnpons	Standalone Non-Public Network (SNPN) (s) allowed to access this CHF Instance.
snssais	S-NSSAI of the CHF Service. This can be a subset of the S-NSSAI supported by the NF. When present, this IE shall represent the list of S-NSSAI supported by the CHF Service in all the PLMNs listed in the plmnList IE and it shall prevail over the list of S-NSSAI supported by the NF instance.
perPlmnSnssaiList	S-NSSAI of the CHF Service per PLMN. This can be a subset of the S-NSSAI supported per PLMN by the CHF instance. This IE can be included when the list of S-NSSAI supported by the CHF Service for each PLMN it is supporting is different. When present, this IE shall include the S-NSSAI supported by the CHF Service for each PLMN and it shall prevail over the list of S-NSSAI supported per PLMN by the CHF instance. When present, this IE shall override the sNssais IE.
allowedOperationsPerNfType	Map of allowed operations on resources for each type of CHF; the key of the map is the NF Type, and the value is an array of scopes. The scopes shall be any of those defined in the API that defines the current service (identified by the "serviceName" attribute).
allowedOperationsPerNfInstance	Map of allowed operations on resources for a given CHF Instance; the key of the map is the NF Instance Id, and the value is an array of scopes. The scopes shall be any of those defined in the API that defines the current service (identified by the "serviceName" attribute).



Example

```
beamctl nf-profile-config update-converged-charging CHA1 <
nfServicePayload.json
```

JSON for Update Converged Charging:

```
{
  "allowedSnpsns": null,
  "snssais": null,
  "vendorId": null,
  "perPlmnSnssaiList": null,
  "allowedOperationsPerNfType": null,
  "allowedOperationsPerNfInstance": null
}
```

3.2.7.8 Get Spending Limit Control

The `get-spending-limit-control` command shows the configurable properties for the Spending Limit Control service in the appropriate appGroup based nfProfile.

Syntax

```
beamctl nf-profile-config get-spending-limit-control
<appGroupName>
```

Arguments

Table 71 Mandatory Arguments

Argument	Description
appGroupName	Indicates the application group name.

Example 60

```
beamctl nf-profile-config get-spending-limit-control CHA1
```

3.2.7.9 Update Spending Limit Control

The `update-spending-limit-control` command updates the properties for the Spending Limit Control service in the appropriate appGroup based nfProfile.

The command requires a JSON object from stdin, or shorthand.

Syntax

```
beamctl nf-profile-config update-spending-limit-control
<appGroupName> < payload.json
```




Arguments

Table 72 Mandatory Arguments

Argument	Description
appGroupName	Indicates the application group name.

Table 73 Update NF Profile Config - Payload JSON Parameters

The parameters used in payload.json are described in the below table.

Parameter	Description
allowedNfTypes	Types of allowed Network Functions to access Charging Function. For more details, see 3GPP specification TS 29.510.
allowedSnpons	Standalone Non-Public Network (SNPN) (s) allowed to access this CHF Instance.
snssais	S-NSSAI of the CHF Service. This can be a subset of the S-NSSAI supported by the NF. When present, this IE shall represent the list of S-NSSAI supported by the CHF Service in all the PLMNs listed in the plmnList IE and it shall prevail over the list of S-NSSAI supported by the NF instance.
vendorId	Vendor ID of the CHF Service instance, according to the IANA-assigned "SMI Network Management Private Enterprise Codes"
perPlmnSnssaiList	S-NSSAI of the CHF Service per PLMN. This can be a subset of the S-NSSAI supported per PLMN by the CHF instance. This IE can be included when the list of S-NSSAI supported by the CHF Service for each PLMN it is supporting is different. When present, this IE shall include the S-NSSAI supported by the CHF Service for each PLMN and it shall prevail over the list of S-NSSAI supported per PLMN by the CHF instance. When present, this IE shall override the sNssais IE.
allowedOperationsPerNfType	Map of allowed operations on resources for each type of CHF; the key of the map is the NF Type, and the value is an array of scopes. The scopes shall be any of those



Parameter	Description
	defined in the API that defines the current service (identified by the "serviceName" attribute).
allowedOperationsPerNfInstance	Map of allowed operations on resources for a given CHF Instance; the key of the map is the NF Instance Id, and the value is an array of scopes. The scopes shall be any of those defined in the API that defines the current service (identified by the "serviceName" attribute).

Example

```
beamctl nf-profile-config update-spending-limit-control CHA1 < payload.json
```

JSON for Update spending limit control:

```
{
  "allowedSnps": null,
  "snssais": null,
  "vendorId": null,
  "perPlmnSnssaiList": null,
  "allowedOperationsPerNfType": null,
  "allowedOperationsPerNfInstance": null
}
```

3.2.7.10

Get Overload Reporting Configuration

The `get-overload-reporting-config` command gets the configurable properties of the `overloadReporting` in `nfProfile`.

Syntax

```
beamctl nf-profile-config get-overload-reporting-config appGroupName
```

Arguments

Table 74 Mandatory Arguments

Argument	Description
appGroupName	Indicates the application group name.



Example 61

```
beamctl nf-profile-config get-overload-reporting-config CHA1
```

3.2.7.11 Update Overload Reporting Configuration

The `update-overload-reporting-config` command updates the configurable properties of the `overloadReporting` in `nfProfile`.

Syntax

```
beamctl nf-profile-config update-overload-reporting-config
appGroupName < overloadReporting.json
```

Arguments

Table 75 Mandatory Arguments

Argument	Description
appGroupName	Indicates the application group name.

Example 62

- `beamctl nf-profile-config update-overload-reporting-config CHA1 < overloadReporting.json`
- `overloadReporting` JSON:

```
```json
{
 "upwardsOverloadThreshold": 20,
 "downwardsNormalLoadThreshold": 10
}
```
```

3.2.7.12 Delete Overload Reporting Configuration

The `delete-overload-reporting-config` command deletes the configurable properties of the `overloadReporting` in `nfProfile` and reset values to default.

Syntax

```
beamctl nf-profile-config delete-overload-reporting-config
appGroupName
```

Arguments



Table 76 Mandatory Arguments

| Argument | Description |
|--------------|---------------------------------------|
| appGroupName | Indicates the application group name. |

Example 63

```
beamctl nf-profile-config delete-overload-reporting-config CHA1
```

3.2.8 CHF Load Balancer Configurations

Charging communicates with CHF load balancer over Nchf service-based interface protocol. The connection towards the CHF load balancer is initiated by CHA for SNR/RAR traffic when feature `OperatorSpecificHandlingOfNotifyUriForCSA` is enabled. The external CHF load balancer address is configured in *eric-bss-cha-access-chf-load-balancer* configmap.

The following sections describe on how to update the CHF load balancer:

- List CHF Load Balancer Proxies
- Add CHF Load Balancer Proxy
- Get CHF Load Balancer Proxy
- Delete CHF Load Balancer Proxy

3.2.8.1 List CHF Load Balancer Proxies

The `list-chf-loadbalancer` command lists the CHF load balancer proxy configuration.

Syntax

```
beamctl chf-loadbalancer-config list-chf-load-balancer-proxies
```

Arguments

The command has no arguments.

Example 64

```
beamctl chf-loadbalancer-config list-chf-load-balancer-proxies
```

Sample Response:



```
{
  "1": {
    "address": "https://chf.example.com:8008/csa/notifyUri",
    "appGroup": "cha3",
    "customHeader": "customeHeader2"
  },
  "2": {
    "address": "https://10.63.22.21:8088/csa/notifyUri",
    "appGroup": "cha4",
    "customHeader": "customeHeader1"
  },
  "3": {
    "address": "https://[1050:0:0:0:5:600:300c:326b]:8088/csa/notifyUri",
    "appGroup": "cha1",
    "customHeader": "customeHeader3"
  }
}
```

3.2.8.2 Add CHF Load Balancer Proxy

The `add-chf-loadbalancer` command adds a CHF load balancer proxy configuration. The command requires a JSON object from stdin or shorthand.

Syntax

```
beamctl chf-loadbalancer-config add-chf-load-balancer-proxy < <CHF
Load Balancer Proxy JSON file>
```

Arguments

This command has no arguments

Attributes

Table 77 Attributes for CHF Load Balancer Proxy JSON

| Argument | Description |
|--------------|--|
| address | The CHF load balancer endpoint address.
For example: <code>chf.example.com</code> |
| customHeader | It customs request header for notification to an external client. |
| appGroup | Application Group name.
The default value is <i>global</i> . |



Example 65

```
beamctl chf-loadbalancer-config add-chf-load-balancer-proxy <
chf_load_balancer_proxy.json
```

Example 66 chf_load_balancer_proxy.json

Following, there are examples of a chf_load_balancer_proxy JSON:

```
— {
  "address": "https://chf.example.com/csa/notifyUri",
  "customHeader": "x-notify-Uri",
  "appGroup": "cha"
}
```

```
— {
  "address": "https://10.63.22.21:8008/csa/notifyUri",
  "customHeader": "x-notify-Uri",
  "appGroup": "cha"
}
```

```
{
  "address": "https://[1050:0:0:0:5:600:300c:326b]:8088/csa/notifyUri",
  "customHeader": "x-notify-Uri",
  "appGroup": "cha"
}
```

Sample Response:

CHF load balancer proxy added successfully.

3.2.8.3 Get CHF Load Balancer Proxy

The get-chf-loadbalancer command gets the CHF load balancer proxy configuration from listed array.

Syntax

```
beamctl chf-loadbalancer-config get-chf-load-balancer-proxy
<chfLoadbalancerProxyId>
```

Arguments

Table 78 Mandatory Arguments

| Argument | Description |
|------------------------|---|
| chfLoadbalancerProxyId | It is the CHF load balancer proxy identifier. |



Example 67

```
beamctl chf-loadbalancer-config get-chf-load-balancer-proxy 1
```

3.2.8.4 Delete CHF Load Balancer Proxy

The `delete-chf-loadbalancer` command deletes the CHF load balancer proxy configuration from listed array.

Syntax

```
beamctl chf-loadbalancer-config delete-chf-load-balancer-proxy  
<chfLoadbalancerProxyId>
```

Arguments

Table 79 Mandatory Arguments

| Argument | Description |
|------------------------|---|
| chfLoadbalancerProxyId | It is the CHF load balancer proxy identifier. |

Example 68

```
beamctl chf-loadbalancer-config delete-chf-load-balancer-proxy 1
```

3.2.9 Offline Mediation Server Configuration

Offline Mediation Server Configurations are used by CHA for performing the Offline rating.

The following sections describe on how to list, update, get, and add offline mediation server configurations:

- Get all configurations.
- Get the configuration of a specific attribute.
- Update configurations.
- Add a mediation server.



3.2.9.1 Get All Offline Configurations

The `get-all-config` command retrieves all the offline configurations.

Syntax

```
beamctl offline-rating-config get-all-config
```

Arguments

The command has no arguments.

Note: Sensitive information (hostaddress, username and password) is not responded in GET operation and can only be updated through POST/PUT.

Table 80 Offline Configuration Parameters

| Parameter | Mandatory/Optional | Description | Value/Default value | Data Type |
|--------------------------|--------------------|--|---------------------|----------------|
| checkDuplicateFiles | Mandatory | Indicates whether to enable or disable the duplicate file check. | true | Boolean |
| cdrFileExtension | Optional | Indicates the CDR extensions that are processed offline. | cdr | String |
| jsonFileExtension | Optional | Indicates the JSON extensions that are processed offline. | jsonx | String |
| thresholdValue | Optional | Indicates the maximum number of CDR files that can be processed at a time. | 20 | Integer |
| duplicateDetectionPeriod | Mandatory | Indicates the time for duplicate detection in hours. | 10 | Integer |
| mediationServers | Optional | Indicates the list of domain names created for each mediation instances. This is a hidden attribute. | [null] | List of String |



| Parameter | | Mandatory/Optional | Description | Value/Default value | Data Type |
|-----------|--------------------------|--------------------|--|-------------------------------|-----------|
| | activationStatus | Mandatory | Check whether this MM server is activated for CDR processing. The valid values are true or false only. | false | Boolean |
| | cdrAssignedDirectoryName | Mandatory | Indicates the assigned directory in which the CDRs are processed. CHA Access assumes that parent directory path already exists. If assigned directory does not exist, CHA creates the assigned directory. This requires SFTP umask must be configured to allow read and write permission. If assigned directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | /var/
offline/
assigned | String |
| | cdrInboxDirectoryName | Mandatory | Indicates the remote directory in which the input CDRs are located. CHA Access assumes that parent directory path already exists. If inbox directory does not exist, CHA creates the inbox directory. This | /var/
offline/
inbox | String |



| Parameter | | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|-----------|-----------------------|------------------------|---|--------------------------------|--------------|
| | | | requires SFTP umask must be configured to allow read and write permission. If inbox directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | | |
| | cdrProcessedDirectory | Mandatory | Indicates the processed directory in which the processed CDRs are located. CHA Access assumes that parent directory path already exists. If done directory does not exist, CHA creates the done directory. This requires SFTP umask must be configured to allow read and write permission. If done directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | /var/
offline/
done | String |
| | cdrResultDirName | Mandatory | Indicates the result directory in which the CDR result logs are located. CHA Access assumes | /var/
offline/
resultdir | String |



| Parameter | | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|-----------|---------------------------------|------------------------|---|----------------------------------|--------------|
| | | | that parent directory path already exists. If result directory does not exist, CHA creates the result directory. This requires SFTP umask must be configured to allow read and write permission. If result directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | | |
| | cdrUnprocessedDirectoryNa
me | Mandatory | Indicates the unprocessed directory in which duplicate and invalid CDRs are located. CHA Access assumes that parent directory path already exists. If unprocessed directory does not exist, CHA creates the unprocessed directory. This requires SFTP umask must be configured to allow read and write permission. If unprocessed directory already exists, then read and write permission must be granted to | /var/
offline/
unprocessed | String |



| Parameter | | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|-----------|-----------------------|------------------------|--|----------------------------|--------------|
| | | | mediationHostUsername (SFTP User). | | |
| | mediationHostAddress | Optional | Indicates the host name or IPv4/IPv6 address of the Mediation server containing CDRs. This attribute is mandatory for Mediationserver DN activation. | n/a | String |
| | mediationHostPassword | Optional | Indicates the password for the corresponding username. This is a hidden attribute. This attribute is mandatory for Mediationserver DN activation. | n/a | String |
| | mediationHostPort | Mandatory | Mediation server SSH port to communicate. | 22 | Integer |
| | mediationHostUsername | Optional | Indicates the username to login to Mediation server. This attribute is mandatory for Mediationserver DN activation. | n/a | String |
| | mediationServerIndex | Mandatory | Indicates the index to select from the array of mediation servers. | n/a | Integer |
| | retriesForConnection | Mandatory | Target latency for priority traffic in milliseconds. The value should not exceed 2000, after which there is a risk of timeouts. | 3 | Integer |
| | | | | | |



| Parameter | | Mandato
ry/
Optional | Description | Value/
Default
value | Data
Type |
|-----------|----------------------------|----------------------------|---|----------------------------|--------------|
| | retryIntervalForConnection | Mandato
ry | Indicates the number of retry interval in seconds for connection to the Mediation server. | 3 | Integer |

Example 69

beamctl offline-rating-config get-all-config

```
{
  "cdrFileExtension": "cdrFileExtensionData",
  "checkDuplicateFiles": true,
  "duplicateDetectionPeriod": 10,
  "jsonFileExtension": "jsonFileExtensionData",
  "mediationServers": [
    {
      "activationStatus": true,
      "cdrAssignedDirectoryName": "/var/offline/assigned",
      "cdrInboxDirectoryName": "/var/offline/inbox",
      "cdrProcessedDirectoryName": "/var/offline/done",
      "cdrResultDirName": "/var/offline/resultdir",
      "cdrUnprocessedDirectoryName": "/var/offline/unprocessed",
      "mediationHostAddress": null,
      "mediationHostPassword": null,
      "mediationHostPort": "22",
      "mediationHostUsername": null,
      "mediationServerIndex": "0",
      "retriesForConnection": 3,
      "retryIntervalForConnection": 3
    }
  ],
  "thresholdValue": 20
}
```

3.2.9.2

Get the Configuration of a Specific Attribute

The `get-config` command shows the configurations of the specified attribute.

Syntax

beamctl offline-rating-config get-config <attributeName>

Arguments



Table 81 Mandatory Arguments

| Argument | Description |
|--|---|
| attributeName <i><string></i> | The offline mediation server attribute name in string format. |

Example 70

```
beamctl offline-rating-config get-config checkDuplicateFiles
```

Sample Response

```
{
  "checkDuplicateFiles": True
}
```

3.2.9.3 Update Configurations

The `update-config` command updates the configurations for the specified offline server attribute. Given the attribute name and value pair, this command updates the particular key value pair alone. Given the JSON file, the command updates all the field given in the JSON.

Syntax

```
beamctl offline-rating-config update-config < update-config.json
```

```
beamctl offline-rating-config update-config <attributeName>: <value>
```

Arguments

This command has no arguments

Table 82 Offline Configuration Parameters

| Parameter | Mandatory/Optional | Description | Value/Default value | Data Type |
|---------------------|--------------------|--|---------------------|-----------|
| checkDuplicateFiles | Mandatory | Indicates whether to enable or disable the duplicate file check. | true | Boolean |
| cdrFileExtension | Optional | Indicates the CDR extensions that are processed offline. | cdr | String |
| jsonFileExtension | Optional | Indicates the JSON extensions | jsonx | String |



| Parameter | | Mandatory/Optional | Description | Value/Default value | Data Type |
|--------------------------|--------------------------|--------------------|---|-------------------------------|----------------|
| | | | that are processed offline. | | |
| thresholdValue | | Optional | Indicates the maximum number of CDR files that can be processed at a time. | 20 | Integer |
| duplicateDetectionPeriod | | Mandatory | Indicates the time for duplicate detection in hours. | 10 | Integer |
| mediationServers | | Optional | Indicates the list of domain names created for each mediation instances. This is a hidden attribute. | [null] | List of String |
| | activationStatus | Mandatory | Check whether this MM server is activated for CDR processing. The valid values are true or false only. | false | Boolean |
| | cdrAssignedDirectoryName | Mandatory | Indicates the assigned directory in which the CDRs are processed. CHA Access assumes that parent directory path already exists. If assigned directory does not exist, CHA creates the assigned directory. This requires SFTP umask must be configured to allow read and write permission. If assigned directory already exists, then read and write permission must | /var/
offline/
assigned | String |



| Parameter | | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|-----------|---------------------------|------------------------|---|----------------------------|--------------|
| | | | be granted to mediationHostUsername (SFTP User). | | |
| | cdrInboxDirectoryName | Mandatory | Indicates the remote directory in which the input CDRs are located. CHA Access assumes that parent directory path already exists. If inbox directory does not exist, CHA creates the inbox directory. This requires SFTP umask must be configured to allow read and write permission. If inbox directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | /var/
offline/
inbox | String |
| | cdrProcessedDirectoryName | Mandatory | Indicates the processed directory in which the processed CDRs are located. CHA Access assumes that parent directory path already exists. If done directory does not exist, CHA creates the done directory. This requires SFTP umask must be configured to | /var/
offline/
done | String |



| Parameter | | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|-----------|-----------------------------|------------------------|---|----------------------------------|--------------|
| | | | allow read and write permission. If done directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | | |
| | cdrResultDirName | Mandatory | Indicates the result directory in which the CDR result logs are located. CHA Access assumes that parent directory path already exists. If result directory does not exist, CHA creates the result directory. This requires SFTP umask must be configured to allow read and write permission. If result directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | /var/
offline/
resultdir | String |
| | cdrUnprocessedDirectoryName | Mandatory | Indicates the unprocessed directory in which duplicate and invalid CDRs are located. CHA Access assumes that parent directory path already exists. If | /var/
offline/
unprocessed | String |



| Parameter | | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|-----------|------------------------|------------------------|--|----------------------------|--------------|
| | | | unprocessed directory does not exist, CHA creates the unprocessed directory. This requires SFTP umask must be configured to allow read and write permission. If unprocessed directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | | |
| | mediationHost Address | Optional | Indicates the host name or IPv4/IPv6 address of the Mediation server containing CDRs. This attribute is mandatory for Mediationserver DN activation. | n/a | String |
| | mediationHost Password | Optional | Indicates the password for the corresponding username. This is a hidden attribute. This attribute is mandatory for Mediationserver DN activation. | n/a | String |
| | mediationHost Port | Mandatory | Mediation server SSH port to communicate. | 22 | Integer |
| | mediationHost Username | Optional | Indicates the username to login to Mediation server. This attribute is | n/a | String |



| Parameter | | Mandatory/Optional | Description | Value/Default value | Data Type |
|-----------|----------------------------|--------------------|---|---------------------|-----------|
| | | | mandatory for Mediationserver DN activation. | | |
| | mediationServerIndex | Mandatory | Indicates the index to select from the array of mediation servers. | n/a | Integer |
| | retriesForConnection | Mandatory | Target latency for priority traffic in milliseconds. The value should not exceed 2000, after which there is a risk of timeouts. | 3 | Integer |
| | retryIntervalForConnection | Mandatory | Indicates the number of retry interval in seconds for connection to the Mediation server. | 3 | Integer |

Example JSON for updating the entire configuration:

```
{
  "cdrFileExtension": "cdrFileExtensionData",
  "checkDuplicateFiles": true,
  "duplicateDetectionPeriod": 10,
  "jsonFileExtension": "jsonFileExtensionData",
  "thresholdValue": 20,
  "mediationServers": [
    {
      "mediationServerIndex": "0",
      "mediationHostAddress": "23.2.3.5",
      "mediationHostUsername": "admin",
      "mediationHostPassword": "testpwd",
      "mediationHostPort": 22,
      "cdrInboxDirectoryName": "/var/offline/inbox",
      "cdrAssignedDirectoryName": "/var/offline/assigned",
      "cdrUnprocessedDirectoryName": "/var/offline/unprocessed",
      "cdrProcessedDirectoryName": "/var/offline/done",
      "retriesForConnection": 3,
      "retryIntervalForConnection": 3,
      "cdrResultDirName": "/var/offline/resultdir",
      "activationStatus": true
    }
  ]
}
```

```
{
  "cdrFileExtension": "cdrFileExtensionData",
  "checkDuplicateFiles": true,
  "duplicateDetectionPeriod": 10,
  "jsonFileExtension": "jsonFileExtensionData",
  "thresholdValue": 20,
  "mediationServers": [
    {
```



```
    "mediationServerIndex": "0",
    "mediationHostAddress": "1050::600:300c:326b",
    "mediationHostUsername": "admin",
    "mediationHostPassword": "testpwd",
    "mediationHostPort": 22,
    "cdrInboxDirectoryName": "/var/offline/inbox",
    "cdrAssignedDirectoryName": "/var/offline/assigned",
    "cdrUnprocessedDirectoryName": "/var/offline/unprocessed",
    "cdrProcessedDirectoryName": "/var/offline/done",
    "retriesForConnection": 3,
    "retryIntervalForConnection": 3,
    "cdrResultDirName": "/var/offline/resultdir",
    "activationStatus": true
  }
}
```

Table 83 Optional Arguments

| Argument | Description |
|--|--|
| attributeName <i><string></i> | The offline mediation server attribute name in string format. |
| value <i><string/integer/boolean></i> | The offline mediation server attribute value in string/integer/boolean format. |

Example 1

```
beamctl offline-rating-config update-config < update-config.json
```

Response: Offline Configuration updated successfully

Example 2

```
beamctl offline-rating-config update-config thresholdValue:20
```

Response: Offline Configuration updated successfully

3.2.9.4

Add a Mediation Server

The `add-mediation-server` command adds a mediation server for offline configuration.

Syntax

```
beamctl offline-rating-config add-mediation-server < <mediation-server.json>
```

Arguments

This command has no arguments



Table 84 Offline Configuration Parameters

| Parameter | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|-----------------------|------------------------|--|----------------------------|--------------|
| mediationServerIndex | Mandatory | Indicates the index to select from the array of mediation servers. | n/a | Integer |
| mediationHostAddress | Optional | Indicates the host name or IPv4/IPv6 address of the Mediation server containing CDRs. This attribute is mandatory for Mediationserver DN activation. | n/a | String |
| mediationHostUsername | Optional | Indicates the username to login to Mediation server. This attribute is mandatory for Mediationserver DN activation. | n/a | String |
| mediationHostPassword | Optional | Indicates the password for the corresponding username. This is a hidden attribute. This attribute is mandatory for Mediationserver DN activation. | n/a | String |
| mediationHostPort | Mandatory | Mediation server SSH port to communicate. | 22 | Integer |
| cdrInboxDirectory | Mandatory | Indicates the remote directory in which the input CDRs are located. CHA Access assumes that parent directory path already exists. If inbox directory does not exist, CHA creates the inbox directory. This requires SFTP umask must be configured to allow read and write permission. If inbox directory already exists, then read and | /var/
offline/
inbox | String |



| Parameter | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|-----------------------------|------------------------|--|----------------------------------|--------------|
| | | write permission must be granted to mediationHostUsername (SFTP User). | | |
| cdrProcessedDirectoryName | Mandatory | Indicates the processed directory in which the processed CDRs are located. CHA Access assumes that parent directory path already exists. If done directory does not exist, CHA creates the done directory. This requires SFTP umask must be configured to allow read and write permission. If done directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | /var/
offline/
done | String |
| cdrUnprocessedDirectoryName | Mandatory | Indicates the unprocessed directory in which duplicate and invalid CDRs are located. CHA Access assumes that parent directory path already exists. If unprocessed directory does not exist, CHA creates the unprocessed directory. This requires SFTP umask must be configured to allow read and write permission. If unprocessed directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | /var/
offline/
unprocessed | String |



| Parameter | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|----------------------------|------------------------|--|--------------------------------|--------------|
| cdrAssignedDirectory | Mandatory | Indicates the assigned directory in which the CDRs are processed. CHA Access assumes that parent directory path already exists. If assigned directory does not exist, CHA creates the assigned directory. This requires SFTP umask must be configured to allow read and write permission. If assigned directory already exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | /var/
offline/
assigned | String |
| retriesForConnection | Mandatory | Target latency for priority traffic in milliseconds. The value should not exceed 2000, after which there is a risk of timeouts. | 3 | Integer |
| retryIntervalForConnection | Mandatory | Indicates the number of retry interval in seconds for connection to the Mediation server. | 3 | Integer |
| cdrResultDirectory | Mandatory | Indicates the result directory in which the CDR result logs are located. CHA Access assumes that parent directory path already exists. If result directory does not exist, CHA creates the result directory. This requires SFTP umask must be configured to allow read and write permission. If result directory already | /var/
offline/
resultdir | String |



| Parameter | Mandatory/
Optional | Description | Value/
Default
value | Data
Type |
|------------------|------------------------|--|----------------------------|--------------|
| | | exists, then read and write permission must be granted to mediationHostUsername (SFTP User). | | |
| activationStatus | Mandatory | Check whether this MM server is activated for CDR processing. The valid values are true or false only. | false | Boolean |

Example 71

```
beamctl offline-rating-config add-mediation-server < mediation-server.json
```

Sample JSON for adding a mediation server:

```
{
  "mediationServerIndex": "0",
  "mediationHostAddress": "23.2.3.5",
  "mediationHostUsername": "admin",
  "mediationHostPassword": "testpwd",
  "mediationHostPort": 22,
  "cdrInboxDirectoryName": "/var/offline/inbox",
  "cdrAssignedDirectoryName": "/var/offline/assigned",
  "cdrUnprocessedDirectoryName": "/var/offline/unprocessed",
  "cdrProcessedDirectoryName": "/var/offline/done",
  "retriesForConnection": 3,
  "retryIntervalForConnection": 3,
  "cdrResultDirName": "/var/offline/resultdir",
  "activationStatus": false
}
```

```
{
  "mediationServerIndex": "0",
  "mediationHostAddress": "1050::600:300c:326b",
  "mediationHostUsername": "admin",
  "mediationHostPassword": "testpwd",
  "mediationHostPort": 22,
  "cdrInboxDirectoryName": "/var/offline/inbox",
  "cdrAssignedDirectoryName": "/var/offline/assigned",
  "cdrUnprocessedDirectoryName": "/var/offline/unprocessed",
  "cdrProcessedDirectoryName": "/var/offline/done",
  "retriesForConnection": 3,
  "retryIntervalForConnection": 3,
  "cdrResultDirName": "/var/offline/resultdir",
  "activationStatus": false
}
```

Response: Mediation Server added successfully.

3.2.10

Voucher Server Configuration

CHA Voucher Server Configuration is used for supporting Voucher Based Refill.



The following sections describe on how to list, delete, get, and add voucher server configurations:

- List voucher server configurations.
- Add a voucher server.
- Get the configuration of a specific voucher attribute.
- Delete a voucher server.

3.2.10.1 Add a Voucher Server

The `add-voucher-server` command adds a voucher server. The command requires a JSON object from standard input or shorthand format.

Syntax

```
beamctl external-voucher-server-config add-voucher-server <
<Input JSON file>
```

Arguments

Table 85 Mandatory Arguments

| Argument | Description |
|--------------|---|
| appgroupname | Application Group name default value as global |
| address | The voucher endpoint address.
Example: <code>https://<voucher server fqdn> or <IPv4> or <IPv6>:<port></code> .
IPv6 address should be enclosed within <code>[]</code> brackets.. |
| secured | Indicates whether it is http or https communication.
Possible Value: True or False. |

Example 72 JSON for adding a voucher server:

```
Example 1:
{
  "appGroup": "cha",
  "address": "https://ericsson.ngvs.host:8080/VoucherUsage",
  "secured": true
}

Example 2:
{
  "appGroup": "cha",
  "address": "https://10.63.22.21:8080/VoucherUsage",
  "secured": true
}

Example 3:
{
  "appGroup": "cha",
  "address": "https://[1050:0:0:0:5:600:300c:326b]:8080/VoucherUsage",
```



```
} "secured": true
```

Example:

```
beamctl external-voucher-server-config add-voucher-server <
voucher-server.json
```

Response: Voucher Server created successfully with the Server Id: 1

3.2.10.2**List Voucher Server Configurations**

The `list-voucher-servers` command retrieves all the voucher servers available.

Syntax

```
beamctl external-voucher-server-config list-voucher-servers
```

Arguments

The command has no arguments.

Example 73

```
beamctl external-voucher-server-config list-voucher-servers
```

```
Example 1:
{
  "appGroup": "cha",
  "address": "https://ericsson.ngvs.host:8080/VoucherUsage",
  "secured": true
}

Example 2:
{
  "appGroup": "cha",
  "address": "https://10.63.22.21:8080/VoucherUsage",
  "secured": true
}

Example 3:
{
  "appGroup": "cha",
  "address": "https://[1050:0:0:0:5:600:300c:326b]:8080/VoucherUsage",
  "secured": true
}
```

3.2.10.3**Get the Configuration of a Specific Voucher Attribute**

The `get-voucher-server` command retrieves the configurations of the specified voucher server.

Syntax



```
beamctl external-voucher-server-config get-voucher-server
<serverid>
```

Arguments

Table 86 Mandatory Arguments

| Argument | Description |
|--|--|
| serverid <i><integer></i> | The voucher server id in integer format. |

Example

Example 74

```
beamctl external-voucher-server-config get-voucher-server 1
```

```
{
  "address": "https://example.ngvs.host:8080/VoucherUsage",
  "appGroup": "cha",
  "secured": true,
  "serverId": 1
}
```

3.2.10.4 Delete a Voucher Server

The `delete-voucher-server` command removes the configuration of the specified voucher server.

Syntax

```
beamctl external-voucher-server-config delete-voucher-server
<serverid>
```

Arguments

Table 87 Mandatory Arguments

| Argument | Description |
|--|--|
| serverid <i><integer></i> | The voucher server id in integer format. |

Example 75

```
beamctl external-voucher-server-config delete-voucher-server 1
```

3.2.11 NPDB Server Configuration

CHA Number Portability Database Server Configuration is used for Number Portability Lookup using DNS over HTTP.



The following sections describe on how to list, delete, get, and add NPDB server configurations:

- List Number Portability Database Servers.
- Add a Number Portability Database Server.
- Get the configuration of a specific Number Portability Database Server attribute.
- Delete a Number Portability Database Server.

3.2.11.1 Add a Number Portability Database Server

The `add-npdb-server` command adds a Number Probability database server. The command requires a JSON object from standard input or shorthand format.

Syntax

beamctl external-npdb-server-config add-npdb-server < <Input NPDB Server JSON file>

Arguments

Table 88 Mandatory Arguments

| Argument | Description |
|--------------|--|
| appgroupname | Application Group name default value as global |
| address | The NPDB endpoint address.
Example: <code>https://<npdb server fqdn> or <IPv4> or <IPv6>:<port></code> .
IPv6 address should be enclosed within <code>[]</code> brackets. |
| secured | Indicates whether it is http or https communication.
Possible Value: True or False. |
| serviceName | It indicates the service name of the Number Probability database server. |

Example 76 JSON for adding a voucher Number Probability database server

```
Example 1:
{
  "appGroup": "cha",
  "address": "https://ericsson.dns.host:8080",
  "secured": true,
  "serviceName": "EXTERNAL.DnsServer.NumberPortabilityDatabase"
}

Example 2:
{
  "appGroup": "cha",
  "address": "https://10.63.22.21:8080/npdb",
  "secured": true,
  "serviceName": "EXTERNAL.DnsServer.NumberPortabilityDatabase"
```



```

}
Example 3:
{
  "appGroup": "cha",
  "address": "https://[1050:0:0:0:5:600:300c:326b]:8088/npdb",
  "secured": true,
  "serviceName": "EXTERNAL.DnsServer.NumberPortabilityDatabase"
}

```

Example

beamctl external-npdb-server-config add-npdb-server < npdb-server.json

Response: Number Portability Server created successfully with the Server Id: 1

3.2.11.2

List Number Portability Database Servers

The `list-npdb-servers` command retrieves all the Number Probability database server configuration available.

Syntax

beamctl external-npdb-server-config list-npdb-servers

Arguments

The command has no arguments.

Example 77

beamctl external-npdb-server-config list-npdb-servers

Response

```

Example 1:
{
  "appGroup": "cha",
  "address": "https://ericsson.dns.host:8080",
  "secured": true,
  "serviceName": "EXTERNAL.DnsServer.NumberPortabilityDatabase"
}

Example 2:
{
  "appGroup": "cha",
  "address": "https://10.63.22.21:8008/npdb",
  "secured": true,
  "serviceName": "EXTERNAL.DnsServer.NumberPortabilityDatabase"
}

Example 3:
{
  "appGroup": "cha",
  "address": "https://[1050:0:0:0:5:600:300c:326b]:8088/npdb",
  "secured": true,
  "serviceName": "EXTERNAL.DnsServer.NumberPortabilityDatabase"
}

```



3.2.11.3 Get the Configuration of a Specific Number Portability Database Server Attribute

The `get-npdb-server` command shows the properties of the specified Number Probability database server.

Syntax

```
beamctl external-npdb-server-config get-npdb-server <serverid>
```

Arguments

Table 89 Mandatory Arguments

| Argument | Description |
|---------------------------------------|---------------------------------------|
| <code>serverid <integer></code> | The NPDB server id in integer format. |

Example

Example 78

```
beamctl external-npdb-server-config get-npdb-server 1
```

Sample Response

```
{
  "address": "https://example.dns.host",
  "appGroup": "cha",
  "secured": true,
  "serverId": 1
}
```

3.2.11.4 Delete a Number Portability Database Server

The `delete-npdb-server` command removes the configuration of the specified Number Probability database server.

Syntax

```
beamctl external-npdb-server-config delete-npdb-server <serverid>
```

Arguments

Table 90 Mandatory Arguments

| Argument | Description |
|---------------------------------------|---------------------------------------|
| <code>serverid <integer></code> | The NPDB server id in integer format. |

Example



Example 79

```
beamctl external-npdb-server-config delete-npdb-server 1
```

3.3 CPM Configuration Commands

Note: Several application components are listed but only the application components applicable for deployment in your environment are accessible. Those application components can list APIs or CLI commands that are not applicable to your deployment.

3.3.1 Partitions Configuration

The `partitions` command provides the information about the partitions configured in the system.

Syntax

```
beamctl partitions <sub-command>
```

Available Sub Commands

Table 91 Available Sub Commands

| Command | Description |
|------------------------|-------------------------------|
| <code>list-cust</code> | List all customer partitions. |

3.3.1.1 list-cust

The `list-cust` command lists all the customer partitions.

Syntax

```
beamctl partitions list-cust
```

Arguments

The command has no arguments.



Parameters

Table 92 JSON Parameters

| Parameter | Parameter Type | Description |
|------------------|----------------|--|
| activePartitions | Mandatory | List of active partitions.
Max length: 32 |

Example 80

beamctl partitions list-cust

Sample Response

```
{
  "resources": [
    {
      "activePartitions": [
        "1"
      ]
    }
  ]
}
```

3.3.2 Subscriber Account Location Management

The `sub-acct-loc-mgmt` command provides CRUD operations to manage the subscriber account location data. That is, the SDP name mapping with the SDP IP and its partition configuration.

Syntax

beamctl sub-acct-loc-mgmt <sub-command>

Commands

Table 93 Available Sub Commands

| Command | Description |
|-------------|---|
| create-conf | Creates SubscriberAccountLocation mapping. |
| delete-conf | Deletes SubscriberAccountLocation mapping for specific sdpName. |
| edit-conf | Updates SubscriberAccountLocation mapping. |



| Command | Description |
|-----------------------------|--|
| <code>list-all-confs</code> | Lists all SubscriberAccountLocation mapping. |
| <code>list-conf</code> | Lists SubscriberAccountLocation mapping for specific sdName. |

3.3.2.1

create-conf

The `create-conf` command can be used to create the Subscriber Account Location mapping. For example: SDP name, SDP IP, and CIL customer partition ID.

Syntax

```
beamctl sub-acct-loc-mgmt create-conf <request payload>
```

Arguments

Table 94 Mandatory Arguments

| Argument | Description |
|-----------------|---|
| Request payload | JSON payload for Subscriber Account Location. |

Flags

Table 95 Flags

| Flag | Type | Description |
|--------------------------|---------|---|
| <code>ip</code> | String | Indicates the IP address of the location. |
| <code>partitionId</code> | Integer | Indicates the CIL Customer partitionId. |
| <code>name</code> | String | Indicates the configuration name. |

Parameters

Table 96 JSON Parameters

| Parameter | Parameter Type | Description |
|---------------------------------|----------------|--|
| <code>ip</code> | Mandatory | Indicates the IP address of the location. |
| <code>lastUpdateDateTime</code> | Mandatory | Indicates the last Updated Date Time of the configuration. |



| Parameter | Parameter Type | Description |
|-------------|----------------|---|
| partitionId | Mandatory | Indicates the CIL Customer partitionId. |
| name | Mandatory | Indicates the configuration name. |

Example 81

```
beamctl sub-acct-loc-mgmt create-conf  
resource.configurations[0].ip:12.23.34.45, .partitionId:~1, .name  
:sdp01.cs
```

Or

```
beamctl sub-acct-loc-mgmt create-conf < request.json
```

Sample Request:

```
{  
  "resource": {  
    "configurations": [  
      {  
        "name": "sdp01.cs",  
        "ip": "12.23.34.45",  
        "partitionId": "1"  
      }  
    ]  
  }  
}
```

Sample Response:

```
{  
  "resources": [  
    {  
      "configurations": [  
        {  
          "ip": "12.23.34.45",  
          "lastUpdatedDateTime": "2021-05-07T04:49:10.133Z",  
          "name": "sdp01.cs",  
          "partitionId": "1"  
        }  
      ]  
    }  
  ]  
}
```

3.3.2.2

delete-conf

The delete-conf command can be used to delete an existing Subscriber Account Location mapping.

Syntax

```
beamctl sub-acct-loc-mgmt delete-conf <SDP Name>
```



Arguments

Table 97 Mandatory Arguments

| Argument | Description |
|----------|---|
| SDP Name | SDP name for which mapping needs to be deleted. |

Example 82

```
beamctl sub-acct-loc-mgmt delete-conf sdp01.cs
```

Sample Response:

```
{}
```

3.3.2.3

edit-conf

The `edit-conf` command can be used to update an existing Subscriber Account Location mapping.

Syntax

```
beamctl sub-acct-loc-mgmt edit-conf <SDP Name> <Request Payload>
```

Arguments

Table 98 Mandatory Arguments

| Argument | Description |
|-----------------|---|
| SDP Name | SDP name for which mapping needs to be updated. |
| Request payload | JSON payload for Subscriber Account Location. |

Flags

Table 99 Flags

| Flag | Type | Description |
|-------------|---------|---|
| ip | String | Indicates the IP address of the location. |
| partitionId | Integer | Indicates the CIL Customer partitionId. |



Parameters

Table 100 JSON Parameters

| Parameter | Parameter Type | Description |
|--------------------|----------------|--|
| ip | Mandatory | Indicates the IP address of the location. |
| lastUpdateDateTime | Mandatory | Indicates the last Updated Date Time of the configuration. |
| partitionId | Mandatory | Indicates the CIL Customer partitionId. |
| name | Mandatory | Indicates the configuration name. |

Example 83

```
beamctl sub-acct-loc-mgmt edit-conf sdp01.cs  
resource.ip:12.23.34.45, resource.partitionId:~1
```

Or

```
beamctl sub-acct-loc-mgmt edit-conf sdp01.cs < /home/request.json
```

Sample Request:

```
{  
  "resource": {  
    "ip": "12.23.34.45",  
    "partitionId": "1"  
  }  
}
```

Sample Response:

```
{  
  "resources": [  
    {  
      "ip": "12.23.34.45",  
      "lastUpdatedDateTime": "2021-05-07T04:57:36.221Z",  
      "name": "sdp01.cs",  
      "partitionId": "1"  
    }  
  ]  
}
```

3.3.2.4

list-all-confs

The `list-all-confs` command lists all the configured Subscriber Account Location mappings.

Syntax

```
beamctl sub-acct-loc-mgmt list-all-confs
```



Arguments

The command has no arguments.

Parameters

Table 101 JSON Parameters

| Parameter | Parameter Type | Description |
|--------------------|----------------|--|
| ip | Mandatory | Indicates the IP address of the location. |
| lastUpdateDateTime | Mandatory | Indicates the last Updated Date Time of the configuration. |
| partitionId | Mandatory | Indicates the CIL Customer partitionId. |
| name | Mandatory | Indicates the configuration name. |

Example 84

beamctl sub-acct-loc-mgmt list-all-confs

Sample Response:

```
{
  "resources": [
    {
      "ip": "12.23.34.45",
      "lastUpdatedDateTime": "2021-05-04T11:06:19.412Z",
      "name": "sdp01.cs",
      "partitionId": "1"
    },
    {
      "ip": "12.23.34.46",
      "lastUpdatedDateTime": "2021-05-07T04:49:10.133Z",
      "name": "sdp02.cs",
      "partitionId": "1"
    }
  ]
}
```

3.3.2.5

list-conf

The **list-conf** command lists the Subscriber Account Location mapping of a specific SDP.

Syntax

beamctl sub-acct-loc-mgmt list-conf *<SDP Name>*



Arguments

Table 102 Mandatory Arguments

| Argument | Description |
|----------|--|
| SDP Name | SDP name for which mapping needs to be read. |

Flags

The command has no flags.

Parameters

Table 103 JSON Parameters

| Parameter | Parameter Type | Description |
|--------------------|----------------|--|
| ip | Mandatory | Indicates the IP address of the location. |
| lastUpdateDateTime | Mandatory | Indicates the last Updated Date Time of the configuration. |
| partitionId | Mandatory | Indicates the CIL Customer partitionId. |
| name | Mandatory | Indicates the configuration name. |

Example 85

```
beamctl sub-acct-loc-mgmt list-conf sdp01.cs
```

Sample Response:

```
{
  "resources": [
    {
      "ip": "12.23.34.45",
      "lastUpdatedDateTime": "2021-05-03T10:26:12.883Z",
      "name": "sdp01.cs",
      "partitionId": "1"
    }
  ]
}
```

3.3.3 Subscriber Account Management

The sub-acct-mgmt command-line option can be used to manage the Subscriber Account application configurations.

Syntax

```
beamctl sub-acct-mgmt <sub-command>
```



Sub Commands

Table 104 Available Sub Commands

| Command | Description |
|--|---|
| import on page 123 | Provisions the subscriber account data (exported from AF) into CPM. |
| list-import-conf on page 128 | Provisions the information for subscriber account import configuration. |
| set-import-conf on page 128 | Creates or updates subscriber account import configuration. |
| list-all-confs on page 134 | Lists all the Subscriber Account application configurations. |
| list-all-keys on page 130 | Lists all the Subscriber Account keys configured in the system. |
| list-dns-conf on page 130 | Lists secure DNS communication is enabled or disabled. |
| list-dnssec-key on page 131 | Lists the key for secure DNS communication. |
| set-dns-conf on page 132 | Enables or disable secure DNS communication. |
| set-dnssec-key on page 133 | Sets the key for secure DNS communication. |

3.3.3.1

import

The `import` command provisions the subscriber account data (exported from AF) into CPM.

Syntax

```
beamctl sub-acct-mgmt import < <file path> [flags]
```

Arguments

The command has no arguments.

Flags

Table 105 Flags

| Flag | Type | Description |
|-------------------|--------|---|
| <code>type</code> | string | Type of communication ID and its valid values are MSISDN, IMSI. |



| Flag | Type | Description |
|--------------|---------|--|
| | | This is a mandatory parameter. |
| sub-acct-loc | string | Location information where the account of the subscriber exists, typically SDP ID. |
| upsert | string | Flag to determine whether to insert a new subscriber account lookup information, or update the existing account of the subscriber lookup information. |
| opType | integer | <p>opType determines the operation to be performed in the import API. For details, see Table 106.</p> <p>If upsert flag and opType is present in the query parameter, opType gets precedence and upsert is ignored.</p> <p>Operation Type performs the following operations:</p> <ul style="list-style-type: none">— insertOnly (1)— upsert (2)— insertIfNotPresent (3) <p>Note: This is also not valid for --type=IMSI.</p> <ul style="list-style-type: none">— validateandRemove (4)— remove (5) |

Table 106 Import Operation Types

| UI Operation | Description | Case |
|-----------------------|---|--------------|
| insertOnly (opType=1) | <p>This option inserts or overrides the record in database.</p> <p>Note: This option is recommended only if there is a greenfield deployment. Example: when the database is empty to begin with, or the given set of records have not been provisioned yet. Using this option on a non-empty database might result in data-integrity issues.</p> | Provisioning |
| upsert (opType=2) | This option inserts or updates the record in database. | Provisioning |



| UI Operation | Description | Case |
|-------------------------------|---|------------------------------------|
| | Note: This option is recommended when the database is non-empty. Using this option would maintain data integrity. However, it would incur slight performance overhead for managing integrity constraints as compared to opType=1. | |
| InsertIfNotPresent (opType=3) | This option inserts entities which do not exist in the database. This option is valid for MSISDN and is guarded by the FC CPM_FEATURE_MANAGE_NON_EXISTING_SUBSCRIBER. | Provisioning |
| validateandRemove (opType=4) | This option validates the corresponding mapping from the request within the database. <ul style="list-style-type: none"> For MSISDN, record if the SDP mapping from the request matches with the existing SDP from the database. For IMSI, remove the record if the MSISDN mapping from the request matches with the existing MSISDN from database. | De-provisioning with validation |
| remove (opType=5) | This option deletes data without performing any validation. | De-provisioning without validation |

Parameters

Table 107 JSON Parameters

| Parameter | Parameter Type | Description |
|--------------|----------------|--|
| type | Mandatory | Type of communication ID and its valid values are remove the MSISDN, IMSI. |
| successCount | Mandatory | Indicates success count of imports. |
| failureCount | Mandatory | Indicates failure count of imports. |

Example 86 Importing Account Data, i.e., MSISDN

File /home/msisdndata contains the following data:

```
919990934345
919990932333
```

```
beamctl sub-acct-mgmt import < /home/msisdndata.msisdn --
type=msisdn --sub-acct-loc=sdpo1.cs --upsert=false
```

Sample Response:



```
{
  "resources": [
    {
      "type": "MSISDN",
      "successCount": "2",
      "failureCount": "0"
    }
  ]
}
```

Example 87 Importing Subscriber Data, i.e., IMSI Data

File /home/imsidata contains the following data:

```
31346000000231 919990934345
31346000000232 919990932333
```

```
beamctl sub-acct-mgmt import < /home/imsidata.txt --type=imsi --
upsert=false
```

Sample Response:

```
{
  "resources": [
    {
      "type": "IMSI",
      "successCount": "2",
      "failureCount": "0"
    }
  ]
}
```

Example 88 Importing Account Data, i.e., MSISDN

File /home/msisdndata contains the following data:

```
919990934345
919990932333
```

```
beamctl sub-acct-mgmt import < /home/msisdndata msisdn --
type=msisdn --sub-acct-loc=sdp01.cs --opType=2
```

Sample Response:

```
{
  "resources": [
    {
      "type": "MSISDN",
      "successCount": "2",
      "failureCount": "0"
    }
  ]
}
```



Example 89 Importing Subscriber Data, i.e., IMSI Data

File /home/imsidata contains the following data:

```
31346000000231 919990934345
31346000000232 919990932333
```

```
beamctl sub-acct-mgmt import < /home/imsidata.txt --type=imsi --opType=2
```

Sample Response:

```
{
  "resources": [
    {
      "type": "IMSI",
      "successCount": "2",
      "failureCount": "0"
    }
  ]
}
```

Example 90 Deprovisioning Account Data, i.e., MSISDN

File /home/msisdndata contains the following data:

```
919990934345
919990932333
```

```
beamctl sub-acct-mgmt import < /home/msisdndata msisdn --type=msisdn --sub-acct-loc=sdp01.cs --opType=4
```

Sample Response:

```
{
  "resources": [
    {
      "type": "MSISDN",
      "successCount": "2",
      "failureCount": "0"
    }
  ]
}
```

Example 91 Deprovisioning Subscriber Data, i.e., IMSI Data

File /home/imsidata contains the following data:

```
31346000000231 919990934345
31346000000232 919990932333
```

```
beamctl sub-acct-mgmt import < /home/imsidata.txt --type=imsi --opType=4
```

Sample Response:



```
{
  "resources": [
    {
      "type": "IMSI",
      "successCount": "2",
      "failureCount": "0"
    }
  ]
}
```

3.3.3.2 list-import-conf

The `list-import-conf` command provisions the information for subscriber account import configuration.

Syntax

```
beamctl sub-acct-mgmt list-import-conf
```

Arguments

The command has no arguments.

Parameters

Table 108 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------|----------------|---|
| threadCount | Mandatory | This parameter specifies the number of threads to execute the tasks. |
| batchSize | Mandatory | This parameter specifies the number of tasks in a batch in one CIL transaction. |

Example 92

```
beamctl sub-acct-mgmt list-import-conf
```

Sample Response:

```
{
  "batchSize": "5000",
  "threadCount": "10"
}
```

3.3.3.3 set-import-conf

The `set-import-conf` command creates or updates the subscriber account import configuration.



Syntax

```
beamctl sub-acct-mgmt set-import-conf < config.json
```

Arguments

The command has no arguments.

Mandatory

Table 109 Mandatory Input

| Input | Description |
|-------------|--|
| config.json | The configuration in JSON format from stdin. |

Parameters

Table 110 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------|----------------|---|
| threadCount | Mandatory | This parameter specifies the number of threads to execute the tasks. |
| batchSize | Mandatory | This parameter specifies the number of tasks in a batch in one CIL transaction. |

Example 93

```
beamctl sub-acct-mgmt set-import-conf < /home/config.json
```

Sample Request:

```
{
  "batchSize": 5000,
  "threadCount": 10
}
```

Sample Response:

```
{
  "batchSize": "5000",
  "threadCount": "10"
}
```



3.3.3.4 list-all-keys

The `list-all-keys` command lists all the Subscriber Account keys configured in the system.

Syntax

```
beamctl sub-acct-mgmt list-all-keys
```

Arguments

The command has no arguments.

Parameters

Table 111 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------------|----------------|-----------------------------------|
| Kupdate_key.key | Mandatory | Secret key for DNS authentication |
| name | Mandatory | Name of the secret |

Example 94

```
beamctl sub-acct-mgmt list-all-keys
```

Sample Response:

```
{
  "secrets": [
    {
      "data": {
        "Kupdate_key.key": "dXBkYXRlX2tleS4gSU4gS0VZIDUxMiAzIDE2MyBvOUdCMlRyakIx →
ZTNuU0VnUjZ0eUNraCt5RlYrUEIrdW82 T3YxM2JWU3pRPQ=="
      },
      "name": "eric-bss-cpm-app-dnssecurity"
    }
  ]
}
```

3.3.3.5 list-dns-conf

The `list-dns-conf` command lists if secure DNS communication is enabled or disabled.

Syntax

```
beamctl sub-acct-mgmt list-dns-conf
```



Arguments

The command has no arguments.

Parameters

Table 112 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------|----------------|---|
| enableDnsSecurity | Mandatory | Enable or disable DNS security for provisioning flows of MSISDN and IMSI. |

Example 95

```
beamctl sub-acct-mgmt list-dns-conf
```

Sample Response:

```
{
  "enableDnsSecurity": "true"
}
```

3.3.3.6

list-dnssec-key

The `list-dnssec-key` command lists the key for secure DNS communication.

Syntax

```
beamctl sub-acct-mgmt list-dnssec-key
```

Arguments

The command has no arguments.

Parameters

Table 113 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------------|----------------|------------------------------------|
| Kupdate_key.key | Mandatory | Secret key for DNS authentication. |

Example 96

```
beamctl sub-acct-mgmt list-dnssec-key
```



Sample Response:

```
{
  "Kupdate_key.key": "dXBkYXR1X2tleS4gSU4gS0VZIDUxMiAzIDE2MyBvOUdCMlRyakIxZTNuU0
  VnUjZ0eUNraCt5R1YrUEIrdW82T3YxM2JWU3pRPQ=="
}
```

3.3.3.7

set-dns-conf

The `set-dns-conf` command can be used to enable or disable secure DNS communication.

Syntax

```
beamctl sub-acct-mgmt set-dns-conf < <Request Payload>
```

Arguments

Table 114 Mandatory Arguments

| Argument | Description |
|-----------------|---|
| Request Payload | JSON payload for DNS configuration to enable or disable DNS security. |

Flags

Table 115 Flags

| Flag | Type | Description |
|--------------------------------|---------|---|
| <code>enableDnsSecurity</code> | boolean | Enable or disable DNS security for provisioning flows of MSISDN and IMSI. |

Parameters

Table 116 JSON Parameters

| Parameter | Parameter Type | Description |
|--------------------------------|----------------|---|
| <code>enableDnsSecurity</code> | Mandatory | Enable or disable DNS security for provisioning flows of MSISDN and IMSI. |

Example 97

```
— beamctl sub-acct-mgmt set-dns-conf enableDnsSecurity:true
— beamctl sub-acct-mgmt set-dns-conf < /home/request.json
```

Sample Request:



```
{
  "enableDnsSecurity": true
}
```

Sample Response:

```
{
  "enableDnsSecurity": true
}
```

3.3.3.8

set-dnssec-key

The `set-dnssec-key` command can be used to set the key for secure DNS communication.

Syntax

```
beamctl sub-acct-mgmt set-dnssec-key < <Request Payload>
```

Arguments

Table 117 Mandatory Arguments

| Argument | Description |
|-----------------|------------------------------|
| Request Payload | JSON payload for DNS secret. |

Parameters

Table 118 JSON Parameters

| Parameter | Parameter Type | Description |
|------------------------------|----------------|------------------------------------|
| <code>Kupdate_key.key</code> | Mandatory | Secret key for DNS authentication. |

Example 98

```
beamctl sub-acct-mgmt set-dnssec-key < /home/request.json
```

Sample Request:

```
{
  "Kupdate_key.key": "update_key. IN KEY 512 3 163 o9GB2TrjB1e3nSEgR6NyCkh+yFV+P →
B+uo60v13bVSzQ="
}
```

Sample Response:



```
{
  "Kupdate_key.key": "update_key. IN KEY 512 3 163 o9GB2TrjB1e3nSEgR6NyCkh+yFV+P →
B+uo60v13bVSzQ="
}
```

3.3.3.9 list-all-confs

The `list-all-confs` command lists all the Subscriber Account application configurations.

Syntax

```
beamctl sub-acct-mgmt list-all-confs
```

Arguments

The command has no arguments.

Parameters

Table 119 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------------------------|----------------|---|
| <code>data.enableDnsSecurity</code> | Mandatory | Enable or disable DNS security for provisioning flows of MSISDN and IMSI. |
| <code>name</code> | Mandatory | Name of ConfigMap. |

Example 99

```
beamctl sub-acct-mgmt list-all-confs
```

Sample Response:

```
{
  "configmaps": [
    {
      "data": {
        "enableDnsSecurity": "true"
      },
      "name": "eric-bss-cpm-app-dnsconfiguration"
    }
  ]
}
```

3.3.4 CPM APP Application Configuration

The `cpm-app-config` command-line option can be used to manage the CPM APP-related application configurations.



Syntax

```
beamctl cpm-app-config
```

Available Sub Commands

Table 120 Available Sub Commands

| Command | Description |
|---------|--|
| get | Gets the app configuration parameter. |
| get-all | Gets all the attributes and their values of all application configurations or of a particular application configuration. |
| set | Sets the app configuration. |

3.3.4.1

get

The get command gets an attribute with its value of application configuration.

Syntax

```
beamctl cpm-app-config get <context> <attribute>
```

Arguments

Table 121 Mandatory Arguments

| Argument | Description |
|-----------|-------------------------------|
| context | The context of the ConfigMap. |
| attribute | Key in ConfigMap. |

Parameters

Table 122 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------|----------------|---------------------------------------|
| ttlInterval | Mandatory | The Time to Live interval in seconds. |

Example 100

```
beamctl cpm-app-config get ttlconfig ttlInterval
```

Sample Response



```
{
  "ttlInterval": 86400
}
```

Example 101

beamctl cpm-app-config get currentTopicConfig topicConfiguration

Sample Response

```
{
  "topicConfiguration": {
    "topicNames": [
      {
        "retentionInMins": 240,
        "topicName": "cpm_event_customer",
        "topicPartition": [
          {
            "partitionInfo": 1,
            "topicPartitionCount": 18
          },
          {
            "partitionInfo": 2,
            "topicPartitionCount": 18
          },
          {
            "partitionInfo": 3,
            "topicPartitionCount": 18
          }
        ]
      },
      {
        "topicName": "cpm_contract_update_notification_event",
        "topicPartition": [
          {
            "partitionInfo": 1,
            "topicPartitionCount": 18
          },
          {
            "partitionInfo": 2,
            "topicPartitionCount": 18
          },
          {
            "partitionInfo": 3,
            "topicPartitionCount": 18
          }
        ]
      },
      {
        "topicName": "cpm_characteristic_value_update_notification_acknowledge_event",
        "topicPartition": [
          {
            "partitionInfo": 1,
            "topicPartitionCount": 18
          },
          {
            "partitionInfo": 2,
            "topicPartitionCount": 18
          },
          {
            "partitionInfo": 3,
            "topicPartitionCount": 18
          }
        ]
      },
      {
        "topicName": "cpm_customer_data_update_event",
        "topicPartition": [
          {
            "partitionInfo": 1,
            "topicPartitionCount": 18
          },
          {
            "partitionInfo": 2,
            "topicPartitionCount": 18
          },
          {
            "partitionInfo": 3,
            "topicPartitionCount": 18
          }
        ]
      }
    ]
  }
}
```



```

        "partitionInfo": 2,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 3,
        "topicPartitionCount": 18
      }
    ]
  },
  {
    "topicName": "cpm_notification_event",
    "topicPartition": [
      {
        "partitionInfo": 1,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 2,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 3,
        "topicPartitionCount": 18
      }
    ]
  },
  {
    "topicName": "cpm_product_update_notification_acknowledge_event",
    "topicPartition": [
      {
        "partitionInfo": 1,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 2,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 3,
        "topicPartitionCount": 18
      }
    ]
  },
  {
    "topicName": "cpm_event_queue",
    "topicPartition": [
      {
        "partitionInfo": 1,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 2,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 3,
        "topicPartitionCount": 18
      }
    ]
  },
  {
    "topicName": "cpm_lifecycle_action",
    "topicPartition": [
      {
        "partitionInfo": 1,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 2,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 3,
        "topicPartitionCount": 18
      }
    ]
  },
  {
    "retentionInMins": 244,
    "topicName": "cpm_lifecycle_event",

```



```
    "topicPartition": [
      {
        "partitionInfo": 1,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 2,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 3,
        "topicPartitionCount": 18
      }
    ]
  },
  {
    "topicName": "cpm_product_update_notification_event",
    "topicPartition": [
      {
        "partitionInfo": 1,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 2,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 3,
        "topicPartitionCount": 18
      }
    ]
  },
  {
    "topicName": "cpm_02_lifecycle_event",
    "topicPartition": [
      {
        "partitionInfo": 1,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 2,
        "topicPartitionCount": 18
      },
      {
        "partitionInfo": 3,
        "topicPartitionCount": 18
      }
    ]
  }
]
}
```

3.3.4.2 get-all

The `get-all` command lists all the application configuration along with the attributes and their values.

Syntax

```
beamctl cpm-app-config get-all
```

Arguments

The command has no arguments.



Parameters

Table 123 JSON Parameters

| Parameter | Parameter Type | Description |
|--|----------------|--|
| name | Mandatory | Name of the Configuration. |
| ttlInterval | Mandatory | The Time to Live interval in seconds. |
| maxAllowedBusinessRestConcurrentRequests | Mandatory | The parameter defines the number of concurrent requests CPM can handle. Any number of concurrent requests above this is rejected with Overload Exception. The default value is the optimum value which is set based on peak load traffic case. It is recommended not to increase it further. This value is common for Business Logic REST Interface therefore that increasing its value impacts latency. |
| consumerEntrySize | Mandatory | The parameter to specify the size of consumer entry. |
| batchSize | Mandatory | This parameter specifies the number of tasks in a batch in one CIL transaction. |
| threadCount | Mandatory | This parameter specifies the number of threads to execute the tasks. |
| enableDnsSecurity | Mandatory | Enable or disable DNS security for provisioning flows of MSISDN and IMSI. |
| failureRateThreshold | Mandatory | The parameter specifies the number of failures that must occur to open the circuit breaker. |
| isCircuitBreakerEnabled | Mandatory | The parameter enables and disables the circuit breaker capability in CPM. |
| maxSearchCount | Mandatory | The maximum number of records returned by the search operation. |
| cleanupGracePeriod | Mandatory | It is used to set cleanup grace period in seconds in case the cleanup request is for the current date (system). |
| maxNumOfHops | Optional | The maximum number of hops (remote reads) allowed per request. |
| topicConfiguration | Optional | This parameter shows the topic configuration currently used by CPM containing retentionInMins per topic and topicPartitionCount per active partition per topics. |
| minAvailable | Mandatory | The minimum capacity that must be always available. If the available capacity is below this point, failover is triggered. The definition can be in percentage, or absolute number of pods. |
| monitoringDelay | Mandatory | SS delays the next monitoring by this period. It is defined in seconds. |
| monitoringFailureCount | Mandatory | This parameter shows the number of successive monitoring failures |



| Parameter | Parameter Type | Description |
|-------------------|----------------|--|
| | | that qualifies the appgroup for failover. |
| monitoringStatus | Mandatory | This parameter indicates whether the pod health monitoring process is currently enabled or disabled. |
| overloadThreshold | Mandatory | It is the overload percentage. Equal and above this percentage is considered an overloaded pod. |

Example 102

```
beamctl cpm-app-config get-all
```

Sample Response

```
{
  "app-configs": [
    {
      "data": {
        "supervisor": {
          "schemaVersion": "1.0.0",
          "supervisor": {
            "appMonitoringConfig": [
              {
                "podHealthChecks": {
                  "minAvailable": "25%",
                  "monitoringDelay": 2,
                  "monitoringFailureCount": 10,
                  "monitoringStatus": "enabled",
                  "overloadThreshold": 95
                }
              }
            ]
          }
        },
        "version": "1.0.0"
      },
      "name": "supervisorconfig"
    },
    {
      "data": {
        "consumerEntrySize": 10
      },
      "name": "consumerentryconfig"
    },
    {
      "data": {
        "cpm_02_lifecycle_event_topicPartitionCount": 16,
        "cpm_characteristic_value_update_notification_acknowledge_event_topicPartitionCount": 16,
        "cpm_contract_update_notification_event_topicPartitionCount": 16,
        "cpm_customer_data_update_event_topicPartitionCount": 16,
        "cpm_event_customer_topicPartitionCount": 16,
        "cpm_event_queue_topicPartitionCount": 16,
        "cpm_lifecycle_action_topicPartitionCount": 16,
        "cpm_lifecycle_event_topicPartitionCount": 16,
        "cpm_notification_event_topicPartitionCount": 16,
        "cpm_product_update_notification_acknowledge_event_topicPartitionCount": 16,
        "cpm_product_update_notification_event_topicPartitionCount": 16
      },
      "name": "topicconfig"
    },
    {
      "data": {
        "failureRateThreshold": 3,
        "isCircuitBreakerEnabled": true
      },
      "name": "circuitbreakerconfig"
    },
    {
      "data": {
```




```

        "enableMetrics": false,
        "frequency": 15
      },
      {
        "name": "metricsgeneratorconfig"
      },
      {
        "data": {
          "maxSearchCount": 500
        },
        "name": "commonconfig"
      },
      {
        "data": {
          "ttlInterval": 86400
        },
        "name": "ttlconfig"
      },
      {
        "data": {
          "maxAllowedBusinessRestConcurrentRequests": 50
        },
        "name": "trafficmonitorconfig"
      },
      {
        "data": {
          "leaseTime": 300,
          "leaseTimeForGurId": 900,
          "waitTimeEvent": 30,
          "waitTimeRest": 2,
          "waitTimeScheduler": 30
        },
        "name": "lockmanagerconfig"
      },
      {
        "data": {
          "cpmSensitiveDataEncryptionEnabled": "",
          "cpmTranslationDataEncryptionEnabled": "",
          "cpmTransientDataEncryptionEnabled": ""
        },
        "name": "sensitivedataconfig"
      }
    ]
  }

```

```

{
  "app-configs": [
    {
      "data": {
        "supervisor": {
          "schemaVersion": "1.0.0",
          "supervisor": {
            "appMonitoringConfig": [
              {
                "podHealthChecks": {
                  "minAvailable": "25%",
                  "monitoringDelay": 2,
                  "monitoringFailureCount": 10,
                  "monitoringStatus": "enabled",
                  "overloadThreshold": 95
                }
              }
            ]
          }
        },
        "version": "1.0.0"
      },
      "name": "supervisorconfig"
    },
    {
      "data": {
        "consumerEntrySize": 10
      },
      "name": "consumerentryconfig"
    },
    {
      "data": {
        "cpm_02_lifecycle_event_topicPartitionCount": 16,
        "cpm_characteristic_value_update_notification_acknowledge_event_topicPar →

```



```

titionCount": 16,
  "cpm_contract_update_notification_event_topicPartitionCount": 16,
  "cpm_customer_data_update_event_topicPartitionCount": 16,
  "cpm_event_customer_topicPartitionCount": 16,
  "cpm_event_queue_topicPartitionCount": 16,
  "cpm_lifecycle_action_topicPartitionCount": 16,
  "cpm_lifecycle_event_topicPartitionCount": 16,
  "cpm_notification_event_topicPartitionCount": 16,
  "cpm_product_update_notification_acknowledge_event_topicPartitionCount"
: 16,
  "cpm_product_update_notification_event_topicPartitionCount": 16
},
"name": "topicconfig"
},
{
  "data": {
    "failureRateThreshold": 3,
    "isCircuitBreakerEnabled": true
  },
  "name": "circuitbreakerconfig"
},
{
  "data": {
    "enableMetrics": false,
    "frequency": 15
  },
  "name": "metricsgeneratorconfig"
},
{
  "data": {
    "maxSearchCount": 500
  },
  "name": "commonconfig"
},
{
  "data": {
    "ttlInterval": 86400
  },
  "name": "ttlconfig"
},
{
  "data": {
    "maxAllowedBusinessRestConcurrentRequests": 50
  },
  "name": "trafficmonitorconfig"
},
{
  "data": {
    "leaseTime": 300,
    "leaseTimeForGuxId": 900,
    "waitTimeEvent": 30,
    "waitTimeRest": 2,
    "waitTimeScheduler": 30
  },
  "name": "lockmanagerconfig"
},
{
  "data": {
    "cpmSensitiveDataEncryptionEnabled": "",
    "cpmTranslationDataEncryptionEnabled": ""
  },
  "name": "sensitivedataconfig"
}
]
}

```

3.3.4.2.1 get-all --context

The get-all command with context gets an application configuration with its attributes and values.



Syntax

```
beamctl cpm-app-config get-all --context <context>
```

Arguments

Table 124 Mandatory Arguments

| Argument | Description |
|----------|-------------------------------|
| context | The context of the ConfigMap. |

Parameters

Table 125 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------|----------------|---------------------------------------|
| ttlInterval | Mandatory | The Time to Live interval in seconds. |
| name | Mandatory | Name of the Configuration. |

Example 103

```
beamctl cpm-app-config get-all --context ttlconfig
```

Sample Response

```
{
  "app-configs": [
    {
      "data": {
        "ttlInterval": 86400
      },
      "name": "ttlconfig"
    }
  ]
}
```

3.3.4.3

set

The set command updates single or multiple attributes of an application configuration.

Syntax

```
beamctl cpm-app-config set <context> [<|<<|<<<] <RedirectedInput>
```

```
beamctl cpm-app-config set <context> <attribute>:<value>
[, <anotherattribute>:<anothervalue>]
```



Arguments

Table 126 Mandatory Arguments

| Arguments | Description |
|-----------------------|---|
| context | The context of the ConfigMap. |
| attribute | Key in ConfigMap. |
| value | Value of the key. |
| < <RedirectedInput> | Input from a JSON file. |
| << <RedirectedInput> | Here document, which tells the CLI to continue to read input until a specified delimiter is found, usually EOF. |
| <<< <RedirectedInput> | Here string, which is like here document, but it only takes one word as input without a delimiter. |

Flags

Table 127 Flags

| Flag | Type | Description |
|-------------|---------|---------------------------------------|
| ttlInterval | boolean | The Time to Live interval in seconds. |

Parameters

Table 128 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|--|
| message | Mandatory | The message that is displayed after the command is executed. |
| status | Mandatory | Status = 200, command executed Successfully. |

Note: BEAM CLI set command is not allowed for the `currenttopicconfig`. Upon setting the `currenttopicconfig`, the following response is returned:

```
HTTP 405: {"status":405,"code":"InvalidUpdateOperation","message":"Update operation is not allowed.", "details":"Update operation for eric-bss-cpm -app-currenttopicconfig is not allowed"} →
```

Example 104

```
beamctl cpm-app-config set ttlconfig < config.json
```

```
beamctl cpm-app-config set ttlconfig << EOF
{
```



```
"ttlInterval": 86400
}
EOF
```

```
beamctl cpm-app-config set ttlconfig <<< '{"ttlInterval": 86400}'
```

```
beamctl cpm-app-config set ttlconfig ttlInterval:86400
```

Example of config.json

```
{
  "ttlInterval":86400
}
```

Sample Response

```
{
  "message": "Service config 'ttlInterval' updated successfully",
  "status": 200
}
```

3.3.5 CPM BSR Application Configuration

The **cpm-bsr-config** command-line option can be used to manage the CPM BSR-related application configurations.

Syntax

```
beamctl cpm-bsr-config
```

Available Sub Commands

Table 129 Available Sub Commands

| Command | Description |
|---------|---|
| get | Get the app configuration parameter. |
| get-all | Get all the attributes and their values of all application configurations or of a particular application configuration. |
| set | Set app configuration. |

3.3.5.1 get

The **get** command gets an attribute with its value of application configuration.



Syntax

```
beamctl cpm-bsr-config get <context> <attribute>
```

Arguments

Table 130 Mandatory Arguments

| Argument | Description |
|-----------|-------------------------------|
| context | The context of the ConfigMap. |
| attribute | Key in ConfigMap. |

Parameters

Table 131 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------|----------------|---------------------------------------|
| ttlInterval | Mandatory | The Time to Live interval in seconds. |

3.3.5.2

get-all

The `get-all` command lists all the application configuration along with the attributes and their values.

Syntax

```
beamctl cpm-bsr-config get-all
```

Arguments

The command has no arguments.

Parameters

Table 132 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------------------|----------------|---|
| name | Mandatory | Name of the Configuration. |
| ttlInterval | Mandatory | The Time to Live interval in seconds. |
| genericSchedulerRetryCount | Mandatory | The parameter to specify the generic scheduler retry counter. |
| genericSchedulerRetryInterval | Mandatory | Generic scheduler retry interval. |
| connectionTimeout | Mandatory | The parameter to specify the REST client connection time-out. |
| fallingBehindLimit | Mandatory | The parameter fallingBehindLimit defines |



| Parameter | Parameter Type | Description |
|--|----------------|---|
| | | how far behind <i>now</i> the <code>follower_start_minute</code> can fall before raising an alarm. |
| <code>followerStartMinuteFailedSiteBufferTime</code> | Mandatory | The number of minutes back in time from failed site follower start min or failover site follower start min whichever earliest. Maximum one hour back in time. |
| <code>followerStartMinuteInPast</code> | Mandatory | The parameter <code>followerStartMinuteInPast</code> defines how far in past the scheduler looks for tasks at start up. |
| <code>forcedRescanInterval</code> | Mandatory | The parameter <code>forcedRescanInterval</code> defines the time period in minutes between follower rescans from <code>follower_start_minute</code> . Doing a rescan makes sure that there are no missed chunks. |
| <code>initialDelay</code> | Mandatory | The parameter <code>initialDelay</code> specifies a time delay in minutes before the follower starts to execute tasks at start up. The purpose with this is to stabilize the system before starting to process tasks. |
| <code>leaderExecutionInterval</code> | Mandatory | The parameter <code>leaderExecutionInterval</code> defines the time period in minutes between leader executions to move the <code>follower_start_minute</code> forward to the oldest chunk with unprocessed tasks |
| <code>maxExecTimeForChunk</code> | Mandatory | The parameter <code>maxExecTimeForChunk</code> defines for how long processing of a chunk can go on before considering it to be a failure. After this period the chunk is deleted and an alarm is raised, in minutes. |
| <code>noOfConnectionPerRoute</code> | Mandatory | The number of connections allowed for CPM. |
| <code>socketTimeout</code> | Mandatory | The parameter to specify the REST client socket time-out. |
| <code>minAvailable</code> | Mandatory | The minimum capacity that must be always available. If the available capacity is below this point, failover is triggered. The definition can be in percentage, or absolute number of pods. |
| <code>monitoringDelay</code> | Mandatory | SS delays the next monitoring by this period. It is defined in seconds. |
| <code>monitoringFailureCount</code> | Mandatory | This parameter shows the number of successive monitoring failures that qualifies the appgroup for failover. |
| <code>monitoringStatus</code> | Mandatory | This parameter indicates whether the pod health monitoring process is currently enabled or disabled. |
| <code>overloadThreshold</code> | Mandatory | It is the overload percentage. Equal and above this percentage is considered an overloaded pod. |



Example 105

beamctl cpm-bsr-config get-all

Sample Response

```
{
  "app-configs": [
    {
      "data": {
        "ttlInterval": 86400
      },
      "name": "ttlconfig"
    },
    {
      "data": {
        "genericSchedulerRetryInterval": 60000,
        "getGenericSchedulerRetryCount": 3
      },
      "name": "genericschedulerconfig"
    },
    {
      "data": {
        "supervisor": {
          "schemaVersion": "1.0.0",
          "supervisor": {
            "appMonitoringConfig": [
              {
                "podHealthChecks": {
                  "minAvailable": "25%",
                  "monitoringDelay": 2,
                  "monitoringFailureCount": 10,
                  "monitoringStatus": "enabled",
                  "overloadThreshold": 95
                }
              }
            ]
          }
        },
        "version": "1.0.0"
      },
      "name": "supervisorconfig"
    },
    {
      "data": {
        "connectionTimeout": 500,
        "fallingBehindLimit": 15,
        "followerStartMinuteFailedSiteBufferTime": 0,
        "followerStartMinuteInPast": 2880,
        "forcedRescanInterval": 12,
        "initialDelay": 1,
        "leaderExecutionInterval": 5,
        "maxExecTimeForChunk": 300,
        "noOfConnectionPerRoute": 15,
        "socketTimeout": 60000
      },
      "name": "schedulerconfig"
    }
  ]
}
```

3.3.5.2.1 get-all --context

The `get-all` command with context gets an application configuration with its attributes and values.

Syntax

beamctl cpm-bsr-config get-all --context <context>



Arguments

Table 133 Mandatory Arguments

| Argument | Description |
|----------|-------------------------------|
| context | The context of the ConfigMap. |

Parameters

Table 134 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------|----------------|---------------------------------------|
| ttlInterval | Mandatory | The Time to Live interval in seconds. |
| name | Mandatory | Name of the Configuration. |

Example 106

```
beamctl cpm-bsr-config get-all --context ttlconfig
```

Sample Response

```
{
  "app-configs": [
    {
      "data": {
        "ttlInterval": 86400
      },
      "name": "ttlconfig"
    }
  ]
}
```

3.3.5.3

set

The set command updates single or multiple attributes of an application configuration.

Syntax

```
beamctl cpm-bsr-config set <context> [<|<<|<<<] <RedirectedInput>
```

```
beamctl cpm-bsr-config set <context>:<attribute> <value>
[, <anotherattribute>:<anothervalue>]
```



Arguments

Table 135 Mandatory Arguments

| Argument | Description |
|-----------------------|---|
| context | The context of the ConfigMap. |
| attribute | Key in ConfigMap. |
| value | Value of the key. |
| < <RedirectedInput> | Input from a JSON file. |
| << <RedirectedInput> | Here document, which tells the CLI to continue to read input until a specified delimiter is found, usually EOF. |
| <<< <RedirectedInput> | Here string, which is like the here document, but it only takes one word as input without a delimiter. |

Flags

Table 136 Flags

| Flag | Type | Description |
|-------------|---------|---------------------------------------|
| ttlInterval | boolean | The Time to Live interval in seconds. |

Parameters

Table 137 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|--|
| message | Mandatory | The message that is displayed after the command is executed. |
| status | Mandatory | Status = 200, command executed Successfully. |

Example 107

```
beamctl cpm-bsr-config set ttlconfig < config.json
```

```
beamctl cpm-bsr-config set ttlconfig << EOF
{
  "ttlInterval": 86400
}
EOF
```

```
beamctl cpm-bsr-config set ttlconfig <<< '{"ttlInterval": 86400}'
```

```
beamctl cpm-bsr-config set ttlconfig ttlInterval:86400
```



Sample Response

```
{
  "message": "Service config 'ttlInterval' updated successfully",
  "status": 200
}
```

Example of config.json

```
{
  "ttlInterval": 86400
}
```

3.3.6 CPM BATCH Application Configuration

The `cpm-batch-config` command-line option is used to manage the CPM-Batch related application configurations.

Syntax

beamctl cpm-batch-config

Available Sub Commands

Table 138 Available Sub Commands

| Command | Description |
|---------|--|
| get | Gets the CPM-BATCH app configuration parameter. |
| get-all | Gets all the attributes and their values of all application configurations or of a particular application configuration. |
| set | Sets the CPM-BATCH app configuration. |

3.3.6.1 get

The `get` command gets an attribute with its value of application configuration.

Syntax

beamctl cpm-batch-config get <context> <attribute>



Arguments

Table 139 Mandatory Arguments

| Argument | Description |
|-----------|-------------------------------|
| context | The context of the ConfigMap. |
| attribute | Key in ConfigMap. |

Parameters

Table 140 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|--|
| Consumers | Mandatory | A customer who has a contract which acts as a consumer in a sharing setup. |

Example 108

```
beamctl cpm-batch-config get serviceconfig consumers
```

Sample Response

```
{
  "consumers": 10
}
```

3.3.6.2

get-all

The `get-all` command lists all the application configuration along with the attributes and their values.

Syntax

```
beamctl cpm-batch-config get-all
```

Arguments

The command has no arguments.

Parameters

Table 141 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|----------------------------|
| name | Mandatory | Name of the Configuration. |



| Parameter | Parameter Type | Description |
|---------------------------|----------------|---|
| consumers | Mandatory | Number of BatchConsumers to use. |
| maxConcurrentJobs | Mandatory | The maximum number of concurrently running jobs. |
| onlyFailedRecordsInResult | Mandatory | If it is set to <i>true</i> , only failed records are reported in the result of a batch. |
| recordQueueSize | Mandatory | The number of records to hold in the queue between BatchProducer and BatchConsumer. Configurations related to hidden parameters in SFTPFileServe. |

Example 109

```
beamctl cpm-batch-config get-all
```

Sample Response

```
{
  "app-configs": [
    {
      "data": {
        "consumers": 10,
        "maxConcurrentJobs": 1,
        "onlyFailedRecordsInResult": false,
        "recordQueueSize": 2000
      },
      "name": "serviceconfig"
    }
  ]
}
```

3.3.6.2.1 get-all --context

The `get-all` command with context gets an application configuration with its attributes and values.

Syntax

```
beamctl cpm-batch-config get-all --context <context>
```

Arguments

The command has no arguments.

Parameters

Table 142 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|----------------------------|
| name | Mandatory | Name of the Configuration. |



| Parameter | Parameter Type | Description |
|---------------------------|----------------|---|
| consumers | Mandatory | Number of BatchConsumers to use. |
| maxConcurrentJobs | Mandatory | The maximum number of concurrently running jobs. |
| onlyFailedRecordsInResult | Mandatory | If it is set to <i>true</i> , only failed records are reported in the result of a batch. |
| recordQueueSize | Mandatory | The number of records to hold in the queue between BatchProducer and BatchConsumer. Configurations related to hidden parameters in SFTPFileServe. |

Example 110

```
beamctl cpm-batch-config get-all --context serviceconfig
```

Sample Response

```
{
  "app-configs": [
    {
      "data": {
        "consumers": 10,
        "maxConcurrentJobs": 1,
        "onlyFailedRecordsInResult": false,
        "recordQueueSize": 2000
      },
      "name": "serviceconfig"
    }
  ]
}
```

3.3.6.3 set

The set command updates single or multiple attributes of an application configuration.

Syntax

```
beamctl cpm-batch-config set <context> [<|<<|<<<] <RedirectedInput>
```

```
beamctl cpm-batch-config set <context> <attribute>:<value>
[, <anotherattribute>:<anothervalue>]
```

Arguments

Table 143 Mandatory Arguments

| Argument | Description |
|-----------|-------------------------------|
| context | The context of the ConfigMap. |
| attribute | Key in ConfigMap. |



| Argument | Description |
|-----------------------|---|
| value | Value of the key. |
| < <RedirectedInput> | Input from a JSON file. |
| << <RedirectedInput> | Here document, which tells the CLI to continue to read input until a specified delimiter is found, usually EOF. |
| <<< <RedirectedInput> | Here string, which is like here document, but it only takes one word as input without a delimiter. |

Flags

Table 144 Flags

| Flag | Type | Description |
|-----------|---------|---------------------------------------|
| consumers | boolean | The Time to Live interval in seconds. |

Parameters

Table 145 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|--|
| message | Mandatory | The message that is displayed after the command is executed. |
| status | Mandatory | Status = 200, command executed Successfully. |

Example 111

```
beamctl cpm-batch-config set serviceconfig < config.json
```

```
beamctl cpm-batch-config set serviceconfig << EOF
```

```
{
  "consumers": 2
}
EOF
```

```
beamctl cpm-batch-config set serviceconfig <<< '{"consumers": 2}'
```

```
beamctl cpm-batch-config set serviceconfig consumers: 2
```

Sample Response

```
{
  "message": "Service config 'consumers' updated successfully",
  "status": 200
}
```



Example of config.json

```
{
  "consumers": 2
}
```

3.4 CUS Configuration Commands

3.4.1 CUS APP Application Configuration

CUS Application Configuration REST Interface provides the `cus-appconfig` command, which allows users to manage (Read and Update) a resource to direct the order of import export sequence.

Syntax

beamctl cus-appconfig

Commands

Table 146 Available Commands

| Command | Description |
|---------|---|
| get-all | It retrieves all the keys available in the resource. |
| get | It retrieves the value for a key passed as an argument. |
| set | It takes a key and sets the value of a key. |

3.4.1.1 get-all

The `get-all` command retrieves the all the keys present in the ConfigMap **eric-bss-cus-app-appconfig**.

Syntax

beamctl cus-appconfig get-all --server https://<FQDN>:<PORT>

Arguments

The command has no arguments.

Example 112

beamctl cus-appconfig get-all --server https://<FQDN>:<PORT>



Sample Response:

```
{
  "keys": [
    "desc",
    "importexportsequence",
    "name"
  ]
}
```

3.4.1.2

get

The get command retrieves the value for the provided key present in the ConfigMap **eric-bss-cus-app-appconfig**.

Syntax

```
beamctl cus-appconfig get <existingKey> --server https://
<FQDN>: <PORT>
```

Arguments

Table 147 Mandatory Arguments

| Argument | Description |
|-------------|---|
| existingKey | The value for this must be a String and it must exist in the ConfigMap resource.

If an incorrect or nonexistence key is provided, the response is <i>404 Not Found</i> . |

Example 113

```
beamctl cus-appconfig get importexportsequence --server https://
<FQDN>: <PORT>
```

Sample Response:

```
{
  "key": "importexportsequence",
  "value": "NTF,CHA,RMCA,CPM,TAX,INV,FIN"
}
```

3.4.1.3

set

The set command sets the value for a key passed in the payload with the help of three available options.



Syntax

```
beamctl cus-appconfig set <existingKey> <available input options> --  
server https://<FQDN>:<PORT>
```

Arguments

Table 148 Mandatory Arguments

| Argument | Description |
|-------------|---|
| existingKey | The key must exist in the ConfigMap resource.
After the input is correctly validated against the schema, the value is successfully set.
If an incorrect or nonexistence key is provided, the response is <i>404 Not Found</i> . |

Parameters

Table 149 JSON Parameters

| Parameter | Parameter Type | Description |
|----------------------|----------------|--|
| importexportsequence | Mandatory | This sequence denotes the order in which export and import of business configuration entities occur. |

Example 114

Example passing the value for the key through the JSON file:

```
beamctl cus-appconfig set importexportsequence --server https://  
<FQDN>:<PORT> < option1.json
```

Sample Response:

Service Config importexportsequence updated successfully.

Example 115

Example passing the value for the key in the command line as a formatted argument:

```
beamctl cus-appconfig set importexportsequence --server https://  
<FQDN>:<PORT> << EOF
```

```
{  
  "key": "importexportsequence",  
  "value": "value"  
}  
EOF
```

Sample Response:



Service Config importexportsequence updated successfully.

Example 116

Example passing the value for the key in the command line as an argument in a line:

```
beamctl cus-appconfig set importexportsequence --
server https://<FQDN>:<PORT> <<<' { "key":"importexportsequence",
"value":"value" } '
```

Sample Response:

Service Config importexportsequence updated successfully.

3.5 INV Configuration Commands

3.5.1 INV APP Application Configuration

INV Configuration Management REST Interface provides the `inv-app-config` command, which provides the options to manage (Read and Update) a resource.

Note: When failover of an AG is done, the O&M service from the failed AG, it gives a response as *Service Unavailable* with error code 503.

Syntax

```
beamctl inv-app-config
```

Commands

Table 150 INV App Config Commands

| Command | Description |
|---------|---|
| get-all | It retrieves all the keys available in the resource. |
| get | It retrieves the value for a key passed as an argument. |
| set | It sets the value of a key, available in three different ways of input. |



Valid Context

Table 151 INV Valid Context

| Context | Parameter | Description |
|-------------------------------|-----------------------------------|--|
| controllerconsumertopicconfig | consumerConfig | List of topics polled on controller service.
Each topic contains the parameters: topicName, noOfConsumers, pollingIntervalInMs, recoveryIntervalInMinutes, and isReadOnly. |
| controllereventcontrolconfig | billRunStatisticsEvent | Controls generation of BillRunStatisticsEvent.
Valid values are <i>enabled</i> or <i>disabled</i> .
The default value is <i>disabled</i> . |
| | settlementRunStatisticsEvent | Controls generation of SettlementRunStatisticsEvent.
Valid values are <i>enabled</i> or <i>disabled</i> .
The default value is <i>disabled</i> . |
| cpmgrconfig | centralPartition | Central Site boolean Variable. |
| dataconfig | invSensitiveDataEncryptionEnabled | Flag to enable or disable sensitive data encryption in INV. Supported values are Y, N, and EMPTY_STRING. |
| | invTransientDataEncryptionEnabled | Flag to enable or disable transient data encryption in INV. Supported values are Y, N, and EMPTY_STRING. |
| docstoreconfig | host | Hostname or IP address of the CMIS server. |
| | isCmisConnectionDisabled | This flag denotes that CMIS is disabled or not. |
| | port | Port number of the CMIS server. |
| | protocol | Protocol of the document store. |
| | url | URL of the document store. |
| docstoresecret | username | User name of the document store.

Note:
This is sensitive information and must be stored securely. Only the set command is supported. The get and get-all commands are not supported and will return an error message stating Context Attribute not valid. |
| | password | Password of the document store.

Note:
This is sensitive information and must be stored securely. Only the set command is supported. The get and get-all commands are not supported and will return an error message stating Context Attribute not valid. |
| docupdateconfig | noOfDocUpdateEventPartitions | Number of partitions created for the Document Update event.

Note: |



| Context | Parameter | Description |
|------------------------------|--|--|
| | | The value of the parameter must not be reduced. |
| edmprodconfig | noOfAggregateMappingPartitions | Number of Partition Aggregate Mapping. |
| | timeToLiveInSecondsAggregateMapping | Time to Live Aggregate Mapping. |
| | noOfPartitonEdmPosting | Number of Queue Partition. |
| finqueueconfig | timeToLiveInSeconds | Time to Live for topic partitions to use for Invoicing customer events. |
| | noOfPartitions | Partitions to use for Invoicing customer events. |
| formatterconfig | formattingAdapterName | <p>Name of the Formatter adapter used for the XML formatting.</p> <p>With FeatureAdvancedExternalFormatterIntegration feature switched on, the valid values of <code><formattingAdapterName></code> are <code>sftp</code>, <code>rest</code>, and <code>internal</code> only.</p> <p>Note:</p> <ul style="list-style-type: none"> — In case of <code>sftp</code>, INV Raw Invoice file is formatted by an external Formatter through SFTP server. — In case of <code>rest</code>, INV Raw Invoice file is formatted by an external Formatter through REST. — In case of <code>internal</code>, INV Raw Invoice file is formatted by the INV Formatter. |
| | endPoint | End point used for sending INV Raw Invoice File for formatting. An empty value represents the end point is retrieved from the service registry. |
| | formattedDocumentReturned | It represents whether the external Formatter returns the INV Formatted Invoice File to INV or not. |
| formatterconsumertopicconfig | consumerConfig | <p>List of topics polled on formatter service.</p> <p>Each topic contains the parameters: <code>topicName</code>, <code>noOfConsumers</code>, <code>pollingIntervalInMs</code>, <code>recoveryIntervalInMinutes</code>, and <code>isReadOnly</code>.</p> |
| hkconfig | purgeIndividualCustomerUnbilledDataAfterDaysOfInvoiceGen | Number of days starting from invoice generation, after which, the unbilled data can be deleted for an individual customer. |



| Context | Parameter | Description |
|---------------|--|--|
| | purgePaymentResponsibleUnbilledDataAfterDaysOfInvoiceGen | Number of days starting from invoice generation, after which, the unbilled data can be deleted for payment responsible. |
| | purgeIndividualCustomerInvoiceDataAfterDays | The number of days starting from invoice generation, after which, the invoice can be deleted for an individual customer. Must be greater than <code>purgeBillRunStatisticsAfterDays</code> and <code>purgeBillingRequestDataAfterDays</code> . |
| | purgePaymentResponsibleInvoiceDataAfterDays | Number of days starting from invoice generation, after which, the invoice can be deleted for payment responsible. Must be greater than <code>purgeBillRunStatisticsAfterDays</code> and <code>purgeBillingRequestDataAfterDays</code> . |
| | purgeBillRunStatisticsAfterDays | Number of days from bill run request is received, post which, the bill run data and statistics can be deleted.
Note: No invoices are deleted. |
| | purgeBillingRequestDataAfterDays | Number of days from the day the request is received, so the request specific data can be deleted.
Note: No invoices are deleted. |
| | purgePartnerUnsettledDataAfterDaysOfBillGen | Number of days starting from the settlement note generation, after which, the unsettled data can be deleted for partners. |
| | purgePartnerBillDataAfterDays | The number of days starting from bill generation, after which, the bill can be deleted for partners. Must be greater than <code>purgeBillRunStatisticsAfterDays</code> and <code>purgeBillingRequestDataAfterDays</code> . |
| metricsconfig | trafficStatisticsIntervalSeconds | Time interval (in seconds) to generate statistics for SA, SCA, SPCA, CA, RRA, and Zookeeper. |
| | nonTrafficStatisticsIntervalSeconds | Time interval (in seconds) to generate statistics for Activity and Custom MBeans. |
| | cilcircuitBreakerStatisticsIntervalSeconds | Time interval (in seconds) to generate statistics for CilcircuitBreaker. |
| | activityStatisticsIntervalSeconds | Maximum file size in MBs for Activity and Custom MBeans. |
| | activityStatisticsMaxLogFileSizeMB | Maximum file size in MBs for CTPA, CWPA, PCA, and PA. |
| | trafficStatisticsMaxLogFileSizeMB | Maximum file size in MBs for SA, SCA, SPCA, CA, RRA, and Zookeeper. |
| | nonTrafficStatisticsMaxLogFileSizeMB | Maximum file size in MBs for CilcircuitBreaker. |



| Context | Parameter | Description |
|------------------------------|---|--|
| | cilcircuitBreakerStatisticsMaxLogFileSizeMB | Time interval (in days) for deletion of archived log files. |
| | maxArchiveAgeDays | Time interval (in seconds) to generate statistics for SA, SCA, SPCA, CA, RRA, and Zookeeper. |
| notifconfig | periodicBillRunTemplate | On cycle template. |
| | rebillTemplate | Rebill template. |
| | onDemandTemplate | Off-cycle template. |
| | docGenTemplate | Credit Debit Note template. |
| | invoiceGenerationCategory | Invoice generation category. |
| | invoiceRebillCategory | Invoice Rebill category. |
| | invoiceSuppressionTemplate | Invoice suppression template. |
| | invoiceSuppressionCategory | Invoice suppression category. |
| | documentGenerationCategory | Document Generation category. |
| | finalInvoiceTemplate | Final Invoice template. |
| | finalInvoiceCategory | Final Invoice Category. |
| | settlementNoteGenerationTemplate | Template for settlement note generation. |
| | settlementNoteGenerationCategory | Category for settlement note generation. |
| numconfig | strategyStartIndex | StartWith Index Name with their values (e.g. S1=5;S2=10) delimited with (;) semicolon, its case sensitive. |
| | periodicStrategy | Periodic Run Index Name. |
| | simulationStrategy | Simulation Index Name. |
| | onDemandStrategy | On-Demand Index Name. |
| | periodicPrefix | Periodic Bill Run Invoice Number Prefix. |
| | onDemandPrefix | On-Demand Bill Run Invoice Number Prefix. |
| postaggregatorconsumerconfig | simulationPrefix | Simulation Bill Run Invoice Number Prefix. |
| | consumerConfig | List of topics polled on postaggregator service. Each topic contains the parameters: topicName, noOfConsumers, pollingIntervalInMs, recoveryIntervalInMinutes, and isReadOnly. |
| processoreventcontrolconfig | invoiceGenerationEvent | Controls generation of InvoiceGenerationEvent. Valid values – enabled/disabled Valid values are <i>enabled</i> or <i>disabled</i> . The default value is <i>enabled</i> . |
| | settlementNoteGenerationEvent | Controls generation of SettlementNoteGenerationEvent. Valid values – enabled/disabled Valid values are <i>enabled</i> or <i>disabled</i> . The default value is <i>enabled</i> . |
| | externalCalculatedChargeEvent | Controls generation of ExternalCalculatedChargeEvent. |



| Context | Parameter | Description |
|---------------------|--|---|
| | | Valid values are <i>enabled</i> or <i>disabled</i> .
The default value is <i>enabled</i> . |
| producertopicconfig | producerTopicConfigList | List of topics on what data is produced from controller, processor or formatter service. Each topic contains the parameters: topicName, numberOfTopicPartition, and isReadOnly. |
| rbaconfig | vbaCountPerScalingPartition | Number of VBAs to be stored in database per scaling partition. |
| recomputeconfig | maximumNumberOfAutomaticRecompute | Maximum number of automatic recomputations for bill run and settlement run. |
| restformconfig | cmisIntegrationSupported | Represents whether the external Formatter returns the formatted document to INV or not. |
| | formatterServerList | List of external formatter server endpoints. |
| rtconfig | timeToLiveInMilliseconds | Time to Live for topic partitions to use for invoicing customer events. |
| | isNRTAPriorityMoreThanBillRun | Indicates whether NRTA must take higher processing priority over bill running task. |
| shardconfig | noOfShardAndTopicPartition | Number of shards and topic partitions for bill run and on-demand bill cycles. |
| periodicStrategy | The strategy name to be used for periodic (on-cycle) bill runs. Must match one of the defined strategyName values. | |
| simulationStrategy | | |
| strategies | | |
| strategyType | | |
| sftpconfig | retriesForConnection | Number of retries to be made for SFTP connection/operation. |
| | connectionRetryIntervalInSeconds | Time interval between the retries. |
| | pollSize | Number of files to be picked in one SFTP poll. |
| | sftpStrictHostKeyCheck | Represents whether known hosts need to be verified or not. |
| timeoutconfig | timeoutPerContractForFirstBillForAccountInSec | Time-out (in seconds) per contract for first bill. |
| | factorForTimeoutOfBillGenForAccount | Time-out (in seconds) factor for duration of bill generation. |



| Context | Parameter | Description |
|---------|-------------------------|--|
| | noOfLastBillsToConsider | Number of previous bills to consider for computing historic duration per contract. |

For external application configuration details, see INV Application Configuration Function Guide.

3.5.1.1

get-all

The `get-all` command provides attributes with their values for all the application configurations.

Syntax

```
beamctl inv-app-config get-all [ --context <context> ]
```

Note: Use `--context` to get all configs of that particular context.

Arguments

Table 152 Mandatory Arguments

| Argument | Description |
|----------|-------------------------------|
| context | The context of the ConfigMap. |

Parameters

Table 153 JSON Parameters

| Parameter | Parameter Type | Description |
|---|----------------|---|
| activityStatisticsIntervalSeconds | Mandatory | Maximum file size in MBs for Activity and Custom MBeans. |
| activityStatisticsMaxLogFileSizeMB | Mandatory | Maximum file size in MBs for CTPA, CWPA, PCA, and PA. |
| centralPartition | Mandatory | Central Site boolean variable. |
| centralPartition | Mandatory | Central Site boolean variable. |
| cilcircuitBreakerStatisticsIntervalSeconds | Mandatory | Time interval (in seconds) to generate statistics for CilcircuitBreaker. |
| cilcircuitBreakerStatisticsMaxLogFileSizeMB | Mandatory | Time interval (in days) for deletion of archived log files. |
| cmisIntegrationSupported | Mandatory | Represents whether the external Formatter returns the formatted document to INV or not. |
| connectionRetryIntervalInSeconds | Mandatory | Time interval between the retries. |
| consumerConfig | Mandatory | List of topics polled on a service. Each topic contains the parameters: <code>topicName</code> , <code>noOfConsumers</code> , |



| Parameter | Parameter Type | Description |
|-------------------------------------|----------------|---|
| | | pollingIntervalInMs, recoveryIntervalInMinutes, and isReadOnly. |
| docGenTemplate | Mandatory | Credit Debit Note template. |
| documentGenerationCategory | Mandatory | Document Generation category. |
| endPoint | Mandatory | End point used for sending INV Raw Invoice File for formatting. An empty value represents the end point is retrieved from the service registry.

Note:
This becomes irrelevant if the FeatureAdvancedExternalFormatterIntegration feature is switched on. |
| factorForTimeoutOfBillGenForAccount | Mandatory | Time-out (in seconds) factor for duration of bill generation. |
| finalInvoiceCategory | Mandatory | Final Invoice Category. |
| finalInvoiceTemplate | Mandatory | Final Invoice template. |
| formattedDocumentReturned | Mandatory | It represents whether the external Formatter returns the INV Formatted Invoice File to INV or not. |
| formatterServerList | Mandatory | List of external formatter server endpoints. |
| formattingAdapterName | Mandatory | Name of the Formatter adapter used for the XML formatting.
With FeatureAdvancedExternalFormatterIntegration feature switched on, the valid values of <i><formattingAdapterName></i> are sftp, rest, and internal only.

Note: <ul style="list-style-type: none">— In case of sftp, INV Raw Invoice file is formatted by an external Formatter through SFTP server.— In case of rest, INV Raw Invoice file is formatted by an external Formatter through REST.— In case of internal, INV Raw Invoice file is formatted by the INV Formatter. |
| host | Mandatory | Hostname of the external system or server to connect to. |
| invoiceGenerationCategory | Mandatory | Invoice generation category. |
| invoiceRebillCategory | Mandatory | Invoice Rebill category. |



| Parameter | Parameter Type | Description |
|--------------------------------------|----------------|--|
| invoiceSuppressionCategory | Mandatory | Invoice suppression category. |
| invoiceSuppressionTemplate | Mandatory | Invoice suppression template. |
| isCmisConnectionDisabled | Mandatory | Indicates whether the CMIS connection is disabled or enabled. |
| isNRTAPriorityMoreThanBillRun | Mandatory | Indicates whether NRTA must take higher processing priority over bill running task. |
| isReadOnly | Mandatory | This flag represents whether the noOfConsumers field can be updated or not. |
| maxArchiveAgeDays | Mandatory | Time interval (in seconds) to generate statistics for SA, SCA, SPCA, CA, RRA, and Zookeeper. |
| maximumNumberOfAutomaticRecompute | Mandatory | Maximum number of automatic recomputations for bill run and settlement run. |
| newInvoiceNumberGenerationForRebill | Mandatory | Determines whether a new invoice number must be generated when performing a rebill operation. |
| nonTrafficStatisticsIntervalSeconds | Mandatory | Time interval (in seconds) to generate statistics for Activity and Custom MBeans. |
| nonTrafficStatisticsMaxLogFileSizeMB | Mandatory | Maximum file size in MBs for CircuitBreaker. |
| noOfAggregateMappingPartitions | Mandatory | Number of Partition Aggregate Mapping. |
| noOfConsumers | Mandatory | It represents the number of consumers for each customer partition. It must not be decreased from the existing value. |
| noOfDocUpdateEventPartitions | Mandatory | Number of partitions created for the Document Update event.

Note:
The value of the parameter must not be reduced. |
| noOfLastBillsToConsider | Mandatory | Number of previous bills to consider for computing historic duration per contract. |
| noOfPartitions | Mandatory | Partitions to use for Invoicing customer events. |
| noOfPartitonEdmPosting | Mandatory | Number of Queue Partition. |
| noOfShardAndTopicPartition | Mandatory | Number of shards and topic partitions for bill run and on-demand bill cycles. |
| numberOfTopicPartition | Mandatory | Total number of partitions to be used for the topic. |
| onDemandPrefix | Mandatory | On-Demand Bill Run Invoice Number Prefix. |
| onDemandStrategy | Mandatory | On-Demand Index Name. |
| onDemandTemplate | Mandatory | Off-cycle template. |
| periodicBillRunTemplate | Mandatory | On cycle template. |
| periodicPrefix | Mandatory | Periodic Bill Run Invoice Number Prefix. |
| periodicStrategy | Mandatory | Periodic Run Index Name. |
| pollingIntervalInMs | Mandatory | It represents the poll interval time between two polls, in milliseconds. |



| Parameter | Parameter Type | Description |
|--|----------------|--|
| | | If no data is present in the topic partition, it pauses configured Ms. |
| pollSize | Mandatory | Number of files to be picked in one SFTP poll. |
| port | Mandatory | The network port number used to connect to the host. |
| producerTopicConfigList | Mandatory | List of topics on what data is produced from a controller, processor, or formatter service. Each topic contains the parameters: topicName, numberOfTopicPartition, and isReadOnly. |
| protocol | Mandatory | Communication protocol used. |
| purgeBillingRequestDataAfterDays | Mandatory | Number of days from the day the request is received, so the request-specific data can be deleted.
Note:
No invoices are deleted. |
| purgeBillRunStatisticsAfterDays | Mandatory | Number of days from bill run request is received, post which, the bill run data and statistics can be deleted.
Note:
No invoices are deleted. |
| purgeIndividualCustomerInvoiceDataAfterDays | Mandatory | The number of days starting from invoice generation, after which, the invoice can be deleted for an individual customer. Must be greater than <code>purgeBillRunStatisticsAfterDays</code> and <code>purgeBillingRequestDataAfterDays</code> . |
| purgeIndividualCustomerUnbilledDataAfterDaysOfInvoiceGen | Mandatory | Number of days starting from invoice generation, after which, the unbilled data can be deleted for an individual customer. |
| purgePartnerBillDataAfterDays | Mandatory | The number of days starting from bill generation, after which, the bill can be deleted for partners. Must be greater than <code>purgeBillRunStatisticsAfterDays</code> and <code>purgeBillingRequestDataAfterDays</code> . |
| purgePartnerUnsettledDataAfterDaysOfBillGen | Mandatory | Number of days starting from the settlement note generation, after which, the unsettled data can be deleted for partners. |
| purgePaymentResponsibleInvoiceDataAfterDays | Mandatory | Number of days starting from invoice generation, after which, the invoice can be deleted for payment responsible. Must be greater than <code>purgeBillRunStatisticsAfterDays</code> and <code>purgeBillingRequestDataAfterDays</code> . |
| purgePaymentResponsibleUnbilledDataAfterDaysOfInvoiceGen | Mandatory | Number of days starting from invoice generation, after which, the unbilled data can be deleted for payment responsible. |
| rebillTemplate | Mandatory | rebillTemplate. |



| Parameter | Parameter Type | Description |
|---|----------------|--|
| recoveryIntervalInMinutes | Mandatory | It represents a recovery interval time in minutes for recovery. |
| retriesForConnection | Mandatory | Number of retries to be made for SFTP connection/operation. |
| settlementNoteGenerationCategory | Mandatory | Category for settlement note generation. |
| settlementNoteGenerationTemplate | Mandatory | Template for settlement note generation. |
| sftpStrictHostKeyCheck | Mandatory | Represents whether known hosts need to be verified or not. |
| simulationPrefix | Mandatory | Simulation Bill Run Invoice Number Prefix. |
| simulationStrategy | Mandatory | Simulation Index Name. |
| strategyStartIndex | Mandatory | StartWith Index Name with their values (for example: S1=5;S2=10) delimited with (;) semicolon, its case sensitive. |
| timeoutPerContractForFirstBillForAccountInSeconds | Mandatory | Number of previous bills to consider for computing historic duration per contract. |
| timeToLiveInMilliseconds | Mandatory | Time to Live for topic partitions to use for invoicing customer events. |
| timeToLiveInSeconds | Mandatory | Time to Live for topic partitions to use for Invoicing customer events. |
| timeToLiveInSecondsAggregateMapping | Mandatory | Time to Live Aggregate Mapping. |
| topicName | Mandatory | Name of the topic. |
| trafficStatisticsIntervalSeconds | Mandatory | Time interval (in seconds) to generate statistics for SA, SCA, SPCA, CA, RRA, and Zookeeper. |
| trafficStatisticsMaxLogFileSizeMB | Mandatory | Maximum file size in MBs for SA, SCA, SPCA, CA, RRA, and Zookeeper. |
| url | Mandatory | Full endpoint URL for accessing the service or resource. |
| vbaCountPerScalingPartition | Mandatory | Number of VBAs to be stored in the database per scaling partition. |

Example 117

beamctl inv-app-config get-all --context cpmgrconfig

Sample Response:

```
{
  "configs": [
    {
      "data": {
        "centralPartition": 1
      },
      "name": "cpmgrconfig"
    }
  ]
}
```



Example 118

beamctl inv-app-config get-all

Sample Response:

```
{
  "configs": [
    {
      "name": "postaggregatorconsumerconfig",
      "data": {
        "consumerConfig": "[ { topicName:inv_charge_redirection_event, no
OfConsumers:1, pollingIntervalInMs:10000, recoveryIntervalInMinutes:5,
isReadOnly:true }]"
      },
    },
    {
      "name": "producerconfig",
      "data": {
        "producerTopicConfigList": "[ { topicName: inv_notification_event
, numberOfTopicPartition: 3, isReadOnly:false }, { topicName: inv_c
ustomer_invoice_event, numberOfTopicPartition: 1, isReadOnly:false },
{ topicName: inv_doc_format_request, numberOfTopicPartition: 1, isReadO
nly:false }, { topicName: inv_doc_generate_request, numberOfTopicPartiti
on: 1, isReadOnly:false }, { topicName: inv_doc_format_event, number
OfTopicPartition: 4, isReadOnly:false }, { topicName: inv_doc_update_re
quest, numberOfTopicPartition: 4, isReadOnly:false }, { topicName: i
nv_aggregate_mapping_event, numberOfTopicPartition: 1, isReadOnly:false }
, { topicName: inv_stats_event, numberOfTopicPartition: 1, isReadOnly:
true }, { topicName: inv_external_doc_update_event, numberOfTopicPartiti
on: 3, isReadOnly:false }, { topicName: inv_housekeeping_event, numb
erOfTopicPartition: 5, isReadOnly:false }, { topicName: inv_internal_fai
l_event, numberOfTopicPartition: 1, isReadOnly:false }, { topicName: i
nv_provider_event, numberOfTopicPartition: 4, isReadOnly:false },
{ topicName: inv_partyrole_report_event, numberOfTopicPartition: 1, isR
eadOnly:false }, { topicName: inv_settlement_formatting_event, numberOfT
opicPartition: 10, isReadOnly:false }, { topicName: inv_stats_combine_ev
ent, numberOfTopicPartition: 1, isReadOnly:true }, { topicName: inv_u
nBilled_Adjustment_request, numberOfTopicPartition: 8, isReadOnly:false }
, { topicName: inv_batch_partyrole_formatting_event, numberOfTopicPartiti
on: 1, isReadOnly:false }, { topicName: inv_orghierarchy_view, number
OfTopicPartition: 1, isReadOnly:true }]"
      },
    },
    {
      "name": "metricsconfig",
      "data": {
        "activityStatisticsIntervalSeconds": 300,
        "trafficStatisticsIntervalSeconds": 300,
        "nonTrafficStatisticsIntervalSeconds": 300,
        "circuitBreakerStatisticsIntervalSeconds": 5,
        "activityStatisticsMaxLogFileSizeMB": 100,
        "trafficStatisticsMaxLogFileSizeMB": 100,
        "nonTrafficStatisticsMaxLogFileSizeMB": 100,
        "circuitBreakerStatisticsMaxLogFileSizeMB": 100,
        "maxArchiveAgeDays": 5
      },
    },
    {
      "name": "edmprodconfig",
      "data": {
        "noOfAggregateMappingPartitions": 10,
        "timeToLiveInSecondsAggregateMapping": 2419200,
        "noOfPartitonEdmPosting": 10
      },
    },
    {
      "name": "timeoutconfig",
      "data": {
        "timeoutPerContractForFirstBillForAccountInSec": 60000,
        "factorForTimeoutOfBillGenForAccount": 500,
        "noOfLastBillsToConsider": 6
      },
    },
    {
      "name": "formatterconsumerconfig",
```



```

      "data": {
        "consumerConfig": "[ { topicName:inv_doc_generate_request, noOfC →
onsumers:4, pollingIntervalInMs:1000, isReadOnly:false }, { topicNam →
e:inv_external_doc_update_event, noOfConsumers:4, pollingIntervalInMs:5000 →
, isReadOnly:false }]"
      },
      {
        "name": "recomputeconfig",
        "data": {
          "maximumNumberOfAutomaticRecompute": 0
        }
      },
      {
        "name": "notifconfig",
        "data": {
          "periodicBillRunTemplate": "InvoiceGenerationNotification",
          "rebillTemplate": "InvoiceRebillNotification",
          "onDemandTemplate": "InvoiceGenerationNotification",
          "docGenTemplate": "CDNGeneration",
          "invoiceGenerationCategory": "LifeCycle",
          "invoiceRebillCategory": "LifeCycle",
          "invoiceSuppressionTemplate": "InvoiceSuppressionNotification",
          "invoiceSuppressionCategory": "LifeCycle",
          "documentGenerationCategory": "DocumentGeneration",
          "finalInvoiceTemplate": "FinalInvoice",
          "finalInvoiceCategory": "LifeCycle",
          "settlementNoteGenerationTemplate": "SettlementNoteGenerationInformation →
ForPartyRole"
          "settlementNoteGenerationCategory": "LifeCycle"
        }
      },
      {
        "name": "controllereventcontrolconfig",
        "data": {
          "billRunStatisticsEvent": "disabled",
          "settlementRunStatisticsEvent": "disabled"
        }
      },
      {
        "name": "processoreventcontrolconfig",
        "data": {
          "invoiceGenerationEvent": "enabled",
          "settlementNoteGenerationEvent": "enabled",
          "externalCalculatedChargeEvent": "enabled"
        }
      },
      {
        "name": "rbaconfig",
        "data": {
          "vbaCountPerScalingPartition": 2000
        }
      },
      {
        "name": "rtaconfig",
        "data": {
          "timeToLiveInMilliseconds": 259200,
          "isNRTAPriorityMoreThanBillRun": true
        }
      },
      {
        "name": "formatterconfig",
        "data": {
          "endPoint": "dummy",
          "formattedDocumentReturned": true
        }
      },
      {
        "name": "cpmgrconfig",
        "data": {
          "centralPartition": 1
        }
      },
      {
        "name": "docstoreconfig",
        "data": {
          "isCmisConnectionDisabled": true,
          "host": "localhost",
          "port": 8443,
          "url": "chemistry-opencmis-server-fileshare-1.1.0/services11/",

```



```
    "protocol": "https"
  },
  {
    "name": "finqueueconfig",
    "data": {
      "timeToLiveInSeconds": 604800,
      "noOfPartitions": 10,
      "onDemandRetryAfter": 60,
      "onCycleRetryAfter": 60
    }
  },
  {
    "name": "restformconfig",
    "data": {
      "cmisIntegrationSupported": false,
      "formatterServerList": "null"
    }
  },
  {
    "name": "controllerconsumertopicconfig",
    "data": {
      "consumerConfig": "[ { topicName:inv_customer_batch, noOfConsumers:10, pollingIntervalInMs:5000, isReadOnly:true }, { topicName:inv_async_event_request, noOfConsumers:10, pollingIntervalInMs:1000, isReadOnly:true }, { topicName:fin_customer_invoice_event, noOfConsumers:1, pollingIntervalInMs:1000, isReadOnly:false }, { topicName:cha_cbc_event_customer, noOfConsumers:5, pollingIntervalInMs:1000, isReadOnly:false }, { topicName:inv_batch_partyrole_formatting_event, noOfConsumers:4, pollingIntervalInMs:1000, recoveryIntervalInMinutes:10, isReadOnly:false }, { topicName:inv_doc_update_request, noOfConsumers:1, pollingIntervalInMs:5000, isReadOnly:false }, { topicName:inv_doc_format_request, noOfConsumers:1, pollingIntervalInMs:1000, recoveryIntervalInMinutes:120, isReadOnly:false }, { topicName:inv_unBilled_Adjustment_request, noOfConsumers:8, pollingIntervalInMs:1000, recoveryIntervalInMinutes:10, isReadOnly:false }, { topicName:inv_housekeeping_event, noOfConsumers:5, pollingIntervalInMs:30000, isReadOnly:false }, { topicName:inv_settlement_formatting_event, noOfConsumers:10, pollingIntervalInMs:1000, isReadOnly:false }, { topicName:inv_stats_combine_event, noOfConsumers:1, pollingIntervalInMs:5000, isReadOnly:true }, { topicName:inv_stats_event, noOfConsumers:1, pollingIntervalInMs:5000, isReadOnly:true }, { topicName:inv_internal_fail_fin_event, noOfConsumers:1, pollingIntervalInMs:30000, isReadOnly:true }, { topicName:inv_orghierarchy_view, noOfConsumers:1, pollingIntervalInMs:5000, recoveryIntervalInMinutes:1440, isReadOnly:true }, { topicName:inv_provider_event, noOfConsumers:1, pollingIntervalInMs:1000, isReadOnly:false }, { topicName:inv_charge_redirection_event, noOfConsumers:1, pollingIntervalInMs:10000, recoveryIntervalInMinutes:5, isReadOnly:true }, { topicName:recovery_cha_cbc_event_customer, noOfConsumers:5, pollingIntervalInMs:30000, isReadOnly:false } ]"
    }
  },
  {
    "name": "numconfig",
    "data": {
      "strategyStartIndex": "S1=5;S2=10",
      "periodicStrategy": "S1",
      "simulationStrategy": "S2",
      "onDemandStrategy": "S1",
      "periodicPrefix": "OnCycle",
      "onDemandPrefix": "OffCycle",
      "simulationPrefix": "Simulation",
      "newInvoiceNumberGenerationForRebill": false
    }
  },
  {
    "name": "hkconfig",
    "data": {
      "purgeIndividualCustomerUnbilledDataAfterDaysOfInvoiceGen": 0,
      "purgePaymentResponsibleUnbilledDataAfterDaysOfInvoiceGen": 0,
      "purgeIndividualCustomerInvoiceDataAfterDays": 184,
      "purgePaymentResponsibleInvoiceDataAfterDays": 366,
      "purgeBillRunStatisticsAfterDays": 184,
      "purgeBillingRequestDataAfterDays": 7,
      "purgePartnerUnsettledDataAfterDaysOfBillGen": 0,
      "purgePartnerBillDataAfterDays": 366
    }
  },
  {
    "name": "sftpconfig",
```




```

      "data": {
        "retriesForConnection": 3,
        "connectionRetryIntervalInSeconds": 3,
        "pollSize": 20,
        "sftpStrictHostKeyCheck": "yes",
        "formatterServerList": "[ { hostServerAddress=localhost, hostUser →
name=root, hostPort=22, rawDocumentDirectory=/var/raw/input, rawDocumen →
tMetadataDirectory=/var/raw/metadata, formattedDocumentMetadataDirectory=/var →
/formatted/metadata, formattedDocumentMetadataOngoingDirectory=/var/formatted →
/metadata/ongoing, formattedDocumentMetadataDoneDirectory=/var/formatted/meta →
data/done, activationStatus=true } ]"
      }
    }
    {
      "data": {
        "invSensitiveDataEncryptionEnabled": "",
        "invTransientDataEncryptionEnabled": ""
      },
      "name": "dataconfig"
    }
  ]
}

```

3.5.1.2

get

The get command provides the attribute with its value of an application configuration. This command has two arguments.

Syntax

```
beamctl inv-app-config get <context> <attribute>
```

Arguments

Table 154 Mandatory Arguments

| Arguments | Description |
|-----------|-------------------------------|
| context | The context of the ConfigMap. |
| attribute | Key in ConfigMap. |

Parameters

Table 155 JSON Parameters

| Parameter | Parameter Type | Description |
|------------------|----------------|--------------------------------|
| centralPartition | Mandatory | Central Site boolean variable. |

Example 119

Sample Request:

```
beamctl inv-app-config get cpmgrconfig centralPartition
```



Sample Response:

```
{
  "centralPartition": 1
}
```

3.5.1.3

set

The set command updates a single or multiple attributes of an application configuration, including both ConfigMap and Secret.

Syntax

```
beamctl inv-app-config set <context> [<|<<] <RedirectedInput>
```

```
beamctl inv-app-config set <context> <attribute>:<value>
[, <anotherattribute>:<anothervalue> ]
```

Note: Use the comma to separate the key: value pairs if multiple attributes need to be changed.

Arguments

Table 156 Mandatory Arguments

| Argument | Description |
|-------------------|--|
| <context> | The context of the ConfigMap or secret. |
| <attribute> | Key in ConfigMap or secret. |
| <value> | Value of the key. |
| <RedirectedInput> | Input from a JSON file. |
| <RedirectedInput> | Here document which tells the CLI to continue to read input until a specified delimiter is found, usually EOF. |

Parameters

Table 157 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|---|
| message | Mandatory | Message is displayed after the command is executed. |
| status | Mandatory | Status = 200, command executed successfully. |



Example 120

Sample Request:

```
beamctl inv-app-config set cpmgrconfig < config.json
```

Sample content of config.json:

```
{
  "centralPartition": 2
}
```

Sample Response:

```
{
  "message": "Configuration updated successfully",
  "status": 200
}
```

Example 121

Sample Request:

```
beamctl inv-app-config set cpmgrconfig << EOF
```

```
{
  "centralPartition": 2
}
EOF
```

Sample Response:

```
{
  "message": "Configuration updated successfully",
  "status": 200
}
```

Example 122

Sample Request:

```
beamctl inv-app-config set cpmgrconfig centralPartition: 2
```

Sample Response:

```
{
  "message": "Configuration updated successfully",
  "status": 200
}
```



```
} "status": 200
```

The following are examples of updating the **docstoresecret**:

Example 123

Sample Request:

```
beamctl inv-app-config set docstoresecret username:test
```

```
beamctl inv-app-config set docstoresecret < config.json
```

Sample content of config.json:

```
{
  "password": ""test"
}
```

Note: It is not advisable to update sensitive information, for example, passwords, through BEAM CLI console due to security reasons. Instead, this information must be provided as a JSON file to the BEAM CLI command. Also, the GET operation for the secret does not return such sensitive information in the response.

Sample Response:

```
{
  "message": "Configuration updated successfully",
  "status": 200
}
```

Example 124

Sample Request:

```
beamctl inv-app-config set docstoreconfig < config.json
```

Sample content of config.json:

```
{
  "host": "localhost",
  "isCmisConnectionDisabled": false,
  "port": 8443,
  "protocol": "https",
  "url": "chemistry-opencmis-server-inmemory-1.1.0/services11/"
}
```

Sample Response:



```
{
  "message": "Configuration updated successfully",
  "status": 200
}
```

3.6 EDM Configuration Commands

Note: When failover of an AG is done, the O&M services from the failed AG will give response as “Service Unavailable” with error code 503.

3.6.1 Publishing Transform Configuration

3.6.1.1 list

The `list` command lists all the transformation metadata present or the transformation metadata of the specified schema. Use the `schema-name` parameter to list the metadata of the specified schema.

Syntax

Command Deprecated in API v1.1: `beamctl pub-transform-config list`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 list`

Command Deprecated in API v1.1: `beamctl pub-transform-config list --schema-name <schema-name>`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 list --schema-name <schema-name>`

Arguments

The command has no arguments.

Flags

Table 158 Flags

| Name | Description |
|-------------|--------------------------|
| schema-name | To list the schema name. |

Example 125

Command Deprecated in API v1.1: `beamctl pub-transform-config list`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 list`



Command Deprecated in API v1.1: `beamctl pub-transform-config list --schema-name ChargingNetworkFunctionEvent`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 list --schema-name ChargingNetworkFunctionEvent`

3.6.1.2 get-active-metadatas

The `get-active-metadatas` command list all the active versions of transformation metadata present.

Syntax

Command Deprecated in API v1.1: `beamctl pub-transform-config get-active-metadatas`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 get-active-metadatas`

Arguments

The command has no arguments.

Example 126

Command Deprecated in API v1.1: `beamctl pub-transform-config get-active-metadatas`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 get-active-metadatas`

3.6.1.3 delete-metadata

The `delete-metadata` command deletes the transformation metadata present. Ensure to mention the source-name, source-version, target-name, target-version, and source-api-version.

Syntax

Command Deprecated in API v1.1: `beamctl pub-transform-config delete-metadata --source-name <source-name> --source-version <source-version> --target-name <target-name> --target-version <target-version> --source-api-version <source-api-version>`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 delete-metadata --source-name <source-name> --source-version <source-version> --target-name <target-name> --target-version <target-version> --source-api-version <source-api-version>`

Arguments



The command has no arguments.

Flags

Table 159 Flags

| Name | Description |
|--------------------|--------------------------------------|
| source-name | To delete the source schema name. |
| source-version | To delete the source schema version. |
| target-name | To delete the target schema name. |
| target-version | To delete the target schema version. |
| source-api-version | To delete the source API version. |

Example 127

Command Deprecated in API v1.1: `beamctl pub-transform-config delete-metadata --source-name ChargingNetworkFunctionEvent --source-version 1 --target-name BssfChfCdrAsn1 --target-version 3 --source-api-version v2`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 delete-metadata --source-name ChargingNetworkFunctionEvent --source-version 1 --target-name BssfChfCdrAsn1 --target-version 3 --source-api-version v2`

3.6.1.4

activate-metadata

The `activate-metadata` command activates the metadata. Ensure to mention the source-name, source-version, target-name, target-version, source-api-version, and tracking-version.

Syntax

Command Deprecated in API v1.1: `beamctl pub-transform-config activate-metadata --source-name <source-name> --source-version <source-version> --target-name <target-name> --target-version <target-version> --source-api-version <source-api-version>`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 activate-metadata --source-name <source-name> --source-version <source-version> --target-name <target-name> --target-version <target-version> --source-api-version <source-api-version> --tracking-version <tracking-version>`

Arguments

The command has no arguments.

Flags



Table 160 Flags

| Name | Description |
|--------------------|--|
| source-name | To activate the source schema name. |
| source-version | To activate the source schema version. |
| target-name | To activate the target schema name. |
| target-version | To activate the target schema version. |
| source-api-version | To activate the source API version. |
| tracking-version | To activate the tracking version. |

Example 128

Command Deprecated in API v1.1: `beamctl pub-transform-config activate-metadata --source-name ChargingNetworkFunctionEvent --source-version 1 --target-name BssfChfCdrAsn1 --target-version 3 --source-api-version v2`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 activate-metadata --source-name ChargingNetworkFunctionEvent --source-version 1 --target-name BssfChfCdrAsn1 --target-version 3 --source-api-version v2 --tracking-version t1`

3.6.1.5

add-metadata

The `add-metadata` command imports the required transformation metadata. Ensure to mention the imported metadata in the metadata path.

Syntax

Command Deprecated in API v1.1: `beamctl pub-transform-config add-metadata < metadata-path`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 add-metadata < metadata-path`

Arguments

The command has no arguments.

Flags

Table 161 Flags

| Name | Description |
|---------------|--------------------------------|
| metadata-path | Path of the metadata to import |



Example 129

Command Deprecated in API v1.1:

```
beamctl pub-transform-config add-metadata < /var/tmp/ASN1/
ChargingNetworkFunctionEvent_2_BssfChfCdrAsn1Rel16_1_v3.json
```

Command Introduced in API v2.0: beamctl

```
pub-transform-config-v2 add-metadata < /var/tmp/ASN1/
ChargingNetworkFunctionEvent_2_BssfChfCdrAsn1Rel16_1_v3.json
```

Example 130 add-metadata

```
ChargingNetworkFunctionEvent_2_BssfChfCdrAsn1Rel16_1_v3.json
{
  "sourceSchemaDetails": {
    "schemaNameSpace": "com.ericsson.bss.cha.event.schema",
    "schemaName": "ChargingNetworkFunctionEvent",
    "schemaVersion": "2.0.0",
    "apiVersionInUri": "v3"
  },
  "targetSchemaDetails": {
    "schemaNameSpace": "com.ericsson.bss.sys.event.schema",
    "schemaName": "BssfChfCdrAsn1Rel16",
    "schemaVersion": "1.0.1"
  },
  "consolidation": {
    "defaultGroupingOperation": "first"
  },
  "definitions": {
    "ratingIndicatorMapping": {
      "ONLINE_CHARGING": true,
      "OFFLINE_CHARGING": false
    },
    "quotaManagementIndicatorMapping": {
      "ONLINE_CHARGING": 0,
      "OFFLINE_CHARGING": 1
    },
    "3gppPSDataOffStatusMapping": {
      "ACTIVE": 0,
      "INACTIVE": 1
    },
    "partialRecordMethodEnumMapping": {
      "DEFAULT": 0,
      "INDIVIDUAL": 1
    },
    "triggerCategoryEnumMapping": {
      "IMMEDIATE_REPORT": 0,
      "DEFERRED_REPORT": 1
    },
    "triggerTypeMapping": {
      "QHT": 410,
      "FINAL": 500,
```



```
"FINAL_UPDATE": 500,  
"FINAL_RELEASE": 503,  
"VALIDITY_TIME": 406,  
"OTHER_QUOTA_TYPE": 409,  
"FORCED_REAUTHORISATION": 407,  
"UNUSED_QUOTA_TIMER": 502,  
"ABNORMAL_RELEASE": 506,  
"QOS_CHANGE": 100,  
"VOLUME_LIMIT": 201,  
"TIME_LIMIT": 200,  
"PLMN_CHANGE": 107,  
"USER_LOCATION_CHANGE": 101,  
"RAT_CHANGE": 108,  
"UE_TIMEZONE_CHANGE": 106,  
"TARIFF_TIME_CHANGE": 105,  
"MAX_NUMBER_OF_CHANGES_IN_CHARGING_CONDITIONS": 203,  
"MANAGEMENT_INTERVENTION": 501,  
"CHANGE_OF_UE_PRESENCE_IN_PRESENCE_REPORTING_AREA": 103,  
"CHANGE_OF_3GPP_PS_DATA_OFF_STATUS": 104,  
"SERVING_NODE_CHANGE": 102,  
"REMOVAL_OF_UPF": 111,  
"ADDITION_OF_UPF": 110,  
"START_OF_SERVICE_DATA_FLOW": 2,  
"START_OF_SERVICE_DATA_FLOW_INITIAL": 2,  
"START_OF_SERVICE_DATA_FLOW_UPDATE": 408,  
"SESSION_AMBR_CHANGE": 109,  
"GFRB_GUARANTEED_STATUS_CHANGE": 115,  
"UNIT_COUNT_INACTIVITY_TIMER": 502,  
"HANDOVER_START": 703,  
"HANDOVER_CANCEL": 702,  
"HANDOVER_COMPLETE": 704,  
"INSERTION_OF_ISMF": 112,  
"REMOVAL_OF_ISMF": 113,  
"CHANGE_OF_ISMF": 114,  
"START_OF_SDF_ADDITIONAL_ACCESS": 411,  
"ECGI_CHANGE": 700,  
"TAI_CHANGE": 701,  
"ADDITION_OF_ACCESS": 116,  
"REMOVAL_OF_ACCESS": 117  
},  
"preemptCapMapping": {  
  "NOT_PREEMPT": 0,  
  "MAY_PREEMPT": 1  
},  
"preemptVulnMapping": {  
  "NOT_PREEMPTABLE": 0,  
  "PREEMPTABLE": 1  
},  
"presenceStateMapping": {  
  "IN_AREA": 0,  
  "OUT_OF_AREA": 1,  
  "INACTIVE": 2,
```



```

        "UNKNOWN": 3
    },
    "raTypeMapping": {
        "NR": 51,
        "EUTRA": 6,
        "WLAN": 3,
        "VIRTUAL": 7,
        "NB-IOT": 8,
        "WIREFLINE": 55,
        "WIREFLINE_CABLE": 56,
        "WIREFLINE_BBF": 57,
        "LTE-M": 9,
        "NR_U": 51,
        "EUTRA_U": 6,
        "TRUSTED_N3GA": 65,
        "TRUSTED_WLAN": 3,
        "UTRA": 1,
        "GERA": 2
    },
    "networkFunctionalityMapping": {
        "SMF": 1,
        "AMF": 2,
        "SMSF": 3,
        "SMS": 3,
        "PGW_C_SMF": 9,
        "NEFF": 8,
        "NEF": 8,
        "SGW": 4,
        "I_SMF": 5,
        "ePDG": 6,
        "CEF": 7
    },
    "UserRoamerInOutMapping": {
        "IN_BOUND": 0,
        "OUT_BOUND": 1
    },
    "pduTypeMapping": {
        "IPV4V6": 0,
        "IPV4": 1,
        "IPV6": 2,
        "UNSTRUCTURED": 3,
        "ETHERNET": 4
    },
    "chChSelectionModeMapping": {
        "HOME_DEFAULT": 3,
        "ROAMING_DEFAULT": 4,
        "VISITING_DEFAULT": 5
    },
    "sSCModeMapping": {
        "SSC_MODE_1": 1,
        "SSC_MODE_2": 2,
        "SSC_MODE_3": 3
    }

```



```
},
"servedPEIMapping": {
  "imeisv": 0,
  "mAC": 1,
  "eUI64": 2,
  "imei": 0,
  "mac": 1,
  "eui": 2
},
"sMAddressTypeMapping": {
  "EMAIL_ADDRESS": 0,
  "MSISDN": 1,
  "IPV4_ADDRESS": 2,
  "IPV6_ADDRESS": 3,
  "NUMERIC_SHORTCODE": 4,
  "ALPHANUMERIC_SHORTCODE": 5,
  "OTHER": 6,
  "IMSI": 7
},
"deliveryReportRequestedMapping": {
  "YES": 0,
  "NO": 1
},
"classIdentifierMapping": {
  "PERSONAL": 0,
  "ADVERTISEMENT": 1,
  "INFORMATIONAL": 2,
  "AUTO": 3
},
"priorityTypeMapping": {
  "LOW": 0,
  "NORMAL": 1,
  "HIGH": 2
},
"sMServiceTypeMapping": {
  "VAS4SMS_SHORT_MESSAGE_CONTENT_PROCESSING": 0,
  "VAS4SMS_SHORT_MESSAGE_FORWARDING": 1,
  "VAS4SMS_SHORT_MESSAGE_FORWARDING_MULTIPLE_SUBSCRIPTIONS": 2,
  "VAS4SMS_SHORT_MESSAGE_FILTERING": 3,
  "VAS4SMS_SHORT_MESSAGE_RECEIPT": 4,
  "VAS4SMS_SHORT_MESSAGE_NETWORK_STORAGE": 5,
  "VAS4SMS_SHORT_MESSAGE_TO_MULTIPLE_DESTINATIONS": 6,
  "VAS4SMS_SHORT_MESSAGE_VIRTUAL_PRIVATE_NETWORK(VPN)": 7,
  "VAS4SMS_SHORT_MESSAGE_AUTO_REPLY": 8,
  "VAS4SMS_SHORT_MESSAGE_PERSONAL_SIGNATURE": 9,
  "VAS4SMS_SHORT_MESSAGE_DEFERRED_DELIVERY": 10
},
"sMReplyPathRequestedMapping": {
  "NO_REPLY_PATH_SET": 0,
  "REPLY_PATH_SET": 1
},
"sMMessageTypeMapping": {
```



```

        "SUBMISSION": 0,
        "DELIVERY_REPORT": 1,
        "SM_SERVICE_REQUEST": 2
    },
    "SMInterfaceTypeMapping": {
        "UNKNOWN": 0,
        "MOBILE_ORIGINATING": 1,
        "MOBILE_TERMINATING": 2,
        "APPLICATION_ORIGINATING": 3,
        "APPLICATION_TERMINATING": 4
    },
    "causeForRecClosingMapping": {
        "0": 1,
        "1": 1,
        "2": 0
    },
    "SubscriptionIDTypeMapping": {
        "imsi": 1,
        "nai": 3,
        "gci": 3,
        "gli": 3
    },
    "dNNSelectionModeMapping": {
        "VERIFIED": 0,
        "UE_DNN_NOT_VERIFIED": 1,
        "NW_DNN_NOT_VERIFIED": 2
    },
    "mAPDUSteeringFunctionalityMapping": {
        "MPTCP": 0,
        "ATSSS_LL": 1
    },
    "activeMapping": {
        "3GPP_ACCESS": 0,
        "NON_3GPP_ACCESS": 1
    },
    "steerModeValueMapping": {
        "ACTIVE_STANDBY": 0,
        "LOAD_BALANCING": 1,
        "SMALLEST_DELAY": 2,
        "PRIORITY_BASED": 3
    },
    "mAPDUSessionIndicatorMapping": {
        "MA_PDU_REQUEST": 0,
        "MA_PDU_NETWORK_UPGRADE_ALLOWED": 1
    }
},
"fields": [
    {
        "sourcePath": "requestInformation.sessionId",
        "targetPath": "CHFRecord.chargingFunctionRecord.chargingSessionId",
        "operation": "toHexString($-SV)"
    }
]

```



```

    },
    {
      "targetPath": "CHFRecord.chargingFunctionRecord.recordType",
      "operation": "toInt(200)"
    },
    {
      "sourcePath": "chargingData.chargingId",
      "targetPath": "CHFRecord.chargingFunctionRecord.chargingID"
    },
    {
      "sourcePath": "chargingData.subscriberIdentifier",
      "targetPath": "CHFRecord.chargingFunctionRecord.subscriberIdentifie →
r.subscriptionIDType",
      "operation": "ignoreUnknownEnumValue(stringToEnum(splitStringWithRe →
gex($-SV,-,0),SubscriptionIDTypeMapping))"
    },
    {
      "sourcePath": "chargingData.subscriberIdentifier",
      "targetPath": "CHFRecord.chargingFunctionRecord.subscriberIdentifie →
r.subscriptionIDData",
      "operation": "splitStringWithRegex($-SV,-,1)"
    },
    {
      "targetPath": "CHFRecord.chargingFunctionRecord.recordingNetworkFun →
ctionID",
      "operation": "generateUUID()"
    },
    {
      "sourcePath": "eventHeader.eventCreationTime.timestamp",
      "targetPath": "CHFRecord.chargingFunctionRecord.recordOpeningTime",
      "operation": "toHexString(getTimeStampOctetArray($-SV))"
    },
    {
      "sourcePath": "chargingData.invocationSequenceNumber",
      "targetPath": "CHFRecord.chargingFunctionRecord.recordSequenceNumbe →
r"
    },
    {
      "targetPath": "CHFRecord.chargingFunctionRecord.localRecordSequence →
Number",
      "operation": "getRecordIdNo()"
    },
    {
      "sourcePath": "chargingData.triggers[]",
      "targetPath": "CHFRecord.chargingFunctionRecord.triggers[].Trigger" →
,
      "operation": "arrayToSequenceOf",
      "fields": [
        {
          "sourcePath": ".",
          "conditonValueSource": "triggerType",
          "operation": "conditionBasedHandling(=)",

```



```

        "conditionValueMapping": [
          {
            "conditionValue": "FINAL",
            "fields": [
              {
                "sourcePath": "triggerType",
                "targetPath": "sMFTrigger",
                "operation": "stringToEnum(getTriggerTypeBy →
SessionType($-SV),triggerTypeMapping)"
              }
            ]
          },
          {
            "conditionValue": "*",
            "fields": [
              {
                "sourcePath": "triggerType",
                "targetPath": "sMFTrigger",
                "operation": "ignoreUnknownEnumValue(string →
ToEnum($-SV,triggerTypeMapping))"
              }
            ]
          }
        ]
      },
      {
        "sourcePath": "chargingData.multipleUnitUsage[]",
        "targetPath": "CHFRecord.chargingFunctionRecord.listOfMultipleUnitU →
sage[].MultipleUnitUsage",
        "operation": "arrayToSequenceOf",
        "groupingOperation": "merge",
        "fields": [
          {
            "sourcePath": "ratingGroup",
            "targetPath": "ratingGroup"
          },
          {
            "sourcePath": "multihomedPDUAddress.ipv4dynamicAddressFlag" →
,
            "targetPath": "multihomedPDUAddress.iPV4dynamicAddressFlag"
          },
          {
            "sourcePath": "multihomedPDUAddress.ipv6dynamicPrefixFlag",
            "targetPath": "multihomedPDUAddress.iPV6dynamicPrefixFlag"
          },
          {
            "sourcePath": "multihomedPDUAddress.pduIPv4Address",
            "targetPath": "multihomedPDUAddress.pDUIPv4Address.iPTextRe →
presentedAddress.iPTextV4Address"
          }
        ]
      }
    ],
    {
      "sourcePath": "chargingData.multipleUnitUsage[]",
      "targetPath": "CHFRecord.chargingFunctionRecord.listOfMultipleUnitU →
sage[].MultipleUnitUsage",
      "operation": "arrayToSequenceOf",
      "groupingOperation": "merge",
      "fields": [
        {
          "sourcePath": "ratingGroup",
          "targetPath": "ratingGroup"
        },
        {
          "sourcePath": "multihomedPDUAddress.ipv4dynamicAddressFlag" →
,
          "targetPath": "multihomedPDUAddress.iPV4dynamicAddressFlag"
        },
        {
          "sourcePath": "multihomedPDUAddress.ipv6dynamicPrefixFlag",
          "targetPath": "multihomedPDUAddress.iPV6dynamicPrefixFlag"
        },
        {
          "sourcePath": "multihomedPDUAddress.pduIPv4Address",
          "targetPath": "multihomedPDUAddress.pDUIPv4Address.iPTextRe →
presentedAddress.iPTextV4Address"
        }
      ]
    }
  ],
  {
    "sourcePath": "chargingData.multipleUnitUsage[]",
    "targetPath": "CHFRecord.chargingFunctionRecord.listOfMultipleUnitU →
sage[].MultipleUnitUsage",
    "operation": "arrayToSequenceOf",
    "groupingOperation": "merge",
    "fields": [
      {
        "sourcePath": "ratingGroup",
        "targetPath": "ratingGroup"
      },
      {
        "sourcePath": "multihomedPDUAddress.ipv4dynamicAddressFlag" →
,
        "targetPath": "multihomedPDUAddress.iPV4dynamicAddressFlag"
      },
      {
        "sourcePath": "multihomedPDUAddress.ipv6dynamicPrefixFlag",
        "targetPath": "multihomedPDUAddress.iPV6dynamicPrefixFlag"
      },
      {
        "sourcePath": "multihomedPDUAddress.pduIPv4Address",
        "targetPath": "multihomedPDUAddress.pDUIPv4Address.iPTextRe →
presentedAddress.iPTextV4Address"
      }
    ]
  }
]

```



```

        {
            "sourcePath": "multihomedPDUAddress.pduIPv6AddresswithPrefi →
x",
            "targetPath": "multihomedPDUAddress.pDUIPv6AddresswithPrefi →
x.iPTextRepresentedAddress.iPTextV6Address"
        },
        {
            "sourcePath": "uPFID",
            "targetPath": "uPFID"
        },
        {
            "sourcePath": "usedUnitContainer[]",
            "alternateSourcePath": "UsedUnitContainer[]",
            "targetPath": "usedUnitContainers[].UsedUnitContainer",
            "operation": "arrayToSequenceOf",
            "fields": [
                {
                    "sourcePath": "localSequenceNumber",
                    "targetPath": "localSequenceNumber"
                },
                {
                    "sourcePath": "serviceId",
                    "targetPath": "serviceIdentifier"
                },
                {
                    "sourcePath": "quotaManagementIndicator",
                    "targetPath": "quotaManagementIndicatorExt",
                    "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,quotaManagementIndicatorMapping))"
                },
                {
                    "sourcePath": ".",
                    "operation": "conditionBasedHandling(=)",
                    "conditionValueSource": "quotaManagementIndicator",
                    "conditionValueMapping": [
                        {
                            "conditionValue": "ONLINE_CHARGING",
                            "fields": [
                                {
                                    "targetPath": "quotaManagementIndic →
ator",
                                    "operation": "toBoolean(true)"
                                }
                            ]
                        },
                        {
                            "conditionValue": "OFFLINE_CHARGING",
                            "fields": [
                                {
                                    "targetPath": "quotaManagementIndic →
ator",
                                    "operation": "toBoolean(false)"
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
}

```




```

    }
  ]
}
],
{
  "sourcePath": "triggerTimestamp",
  "targetPath": "triggerTimeStamp",
  "operation": "toHexString(getTimeStampOctetArray($- →
SV))"
},
{
  "sourcePath": "totalVolume",
  "targetPath": "dataTotalVolume"
},
{
  "sourcePath": "uplinkVolume",
  "targetPath": "dataVolumeUplink"
},
{
  "sourcePath": "downlinkVolume",
  "targetPath": "dataVolumeDownlink"
},
{
  "sourcePath": "serviceSpecificUnits",
  "targetPath": "serviceSpecificUnits"
},
{
  "sourcePath": "eventTimeStamps[]",
  "targetPath": "eventTimeStamp",
  "operation": "toHexString(getTimeStampOctetArray($- →
SV))"
},
{
  "sourcePath": "time",
  "targetPath": "time"
},
{
  "sourcePath": "triggers[]",
  "targetPath": "triggers[].Trigger",
  "operation": "arrayToSequenceOf",
  "fields": [
    {
      "sourcePath": ".",
      "conditionValueSource": "triggerType",
      "operation": "conditionBasedHandling(=)",
      "conditionValueMapping": [
        {
          "conditionValue": "QUOTA_EXHAUSTED" →
,
          "fields": [
            {

```

```

        "sourcePath": ".",
        "operation": "conditionBase",
        "conditionValueMapping": [
            {
                "conditionValueSource": ".chargingData.multipleUnitUsage[].usedUnitContainer[].totalVolume",
                "conditionValueAlternateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].totalVolume",
                "conditionValue": true,
                "fields": [
                    {
                        "targetPath": "sMFTrigger",
                        "operation": "toInt(404)"
                    }
                ]
            },
            {
                "conditionValueSource": ".chargingData.multipleUnitUsage[].usedUnitContainer[].uplinkVolume",
                "conditionValueAlternateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].uplinkVolume",
                "conditionValue": true,
                "fields": [
                    {
                        "targetPath": "sMFTrigger",
                        "operation": "toInt(404)"
                    }
                ]
            },
            {
                "conditionValueSource": ".chargingData.multipleUnitUsage[].usedUnitContainer[].downlinkVolume",
                "conditionValueAlternateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].downlinkVolume",
                "conditionValue": true,
                "fields": [
                    {
                        "targetPath": "sMFTrigger",
                        "operation": "toInt(404)"
                    }
                ]
            }
        ]
    },
    "sourcePath": ".",
    "operation": "conditionBase",
    "conditionValueMapping": [
        {
            "conditionValueSource": ".chargingData.multipleUnitUsage[].usedUnitContainer[].totalVolume",
            "conditionValueAlternateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].totalVolume",
            "conditionValue": true,
            "fields": [
                {
                    "targetPath": "sMFTrigger",
                    "operation": "toInt(404)"
                }
            ]
        },
        {
            "conditionValueSource": ".chargingData.multipleUnitUsage[].usedUnitContainer[].uplinkVolume",
            "conditionValueAlternateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].uplinkVolume",
            "conditionValue": true,
            "fields": [
                {
                    "targetPath": "sMFTrigger",
                    "operation": "toInt(404)"
                }
            ]
        },
        {
            "conditionValueSource": ".chargingData.multipleUnitUsage[].usedUnitContainer[].downlinkVolume",
            "conditionValueAlternateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].downlinkVolume",
            "conditionValue": true,
            "fields": [
                {
                    "targetPath": "sMFTrigger",
                    "operation": "toInt(404)"
                }
            ]
        }
    ]
}

```



```

    }
  ],
  {
    "conditionValueSource": ".chargingData.multipleUnitUsage[].usedUnitContainer[].time",
    "conditionValueAlternateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].time",
    "conditionValue": true,
    "fields": [
      {
        "targetPath": "sMFTrigger",
        "operation": "toInt(403)"
      }
    ]
  },
  {
    "conditionValueSource": ".chargingData.multipleUnitUsage[].usedUnitContainer[].serviceSpecificUnits",
    "conditionValueAlternateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].serviceSpecificUnits",
    "conditionValue": true,
    "fields": [
      {
        "targetPath": "sMFTrigger",
        "operation": "toInt(405)"
      }
    ]
  }
],
{
  "conditionValue": "QUOTA_THRESHOLD",
  "fields": [
    {
      "sourcePath": ".",
      "operation": "conditionBaseHandling(!null)",
      "conditionValueMapping": [
        {
          "conditionValueSource":

```



```

e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].totalVolume",
                                "conditionValueAlte →
rateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].totalVolum →
e",
                                "conditionValue": t →
rue,
                                "fields": [
                                    {
                                        "targetPath →
": "sMFTrigger",
                                        "operation" →
: "toInt(401)"
                                    }
                                ],
                                },
                                {
                                    "conditonValueSourc →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].uplinkVolume",
                                    "conditionValueAlte →
rateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].uplinkVolu →
me",
                                    "conditionValue": t →
rue,
                                    "fields": [
                                        {
                                            "targetPath →
": "sMFTrigger",
                                            "operation" →
: "toInt(401)"
                                        }
                                    ],
                                    },
                                    {
                                        "conditonValueSourc →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].downlinkVolume",
                                        "conditionValueAlte →
rateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].downlinkVo →
lume",
                                        "conditionValue": t →
rue,
                                        "fields": [
                                            {
                                                "targetPath →
": "sMFTrigger",
                                                "operation" →
: "toInt(401)"
                                            }
                                        ],
                                        },
                                        {
                                            "conditonValueSourc →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].time",

```



```

"conditionValueAlte →
rateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].time",
"conditionValue": t →
rue,
"fields": [
{
"targetPath →
"operation" →
}
],
},
{
"conditonValueSourc →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].serviceSpecificUnits →
",
"conditionValueAlte →
rateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].serviceSpe →
cificUnits",
"conditionValue": t →
rue,
"fields": [
{
"targetPath →
"operation" →
}
]
}
]
}
]
},
{
"conditionValue": "*",
"fields": [
{
"sourcePath": "triggerType" →
,
"targetPath": "sMFTrigger",
"operation": "defaultValue( →
stringToEnum(getTriggerTypeBySessionType($-SV),triggerTypeMapping),507)"
}
]
}
]
}
],
},
},

```



```

{
  "sourcePath": ".",
  "operation": "conditionBasedHandling(=)",
  "conditionValueSource": "quotaManagementIndicator",
  "conditionValueMapping": [
    {
      "conditionValue": "ONLINE_CHARGING",
      "fields": [
        {
          "targetPath": "ratingIndicator",
          "operation": "toBoolean(true)"
        }
      ]
    }
  ]
},
{
  "sourcePath": ".",
  "operation": "conditionBasedHandling(!=)",
  "conditionValueSource": "quotaManagementIndicator",
  "conditionValueMapping": [
    {
      "conditionValue": "ONLINE_CHARGING",
      "fields": [
        {
          "sourcePath": ".",
          "conditionValueSource": "chargingData.multipleUnitUs →
age[].ratingGroup",
          "operation": "conditionBasedHandling(isPresent, sess →
ionInformation.offlineRatingIndication[])",
          "conditionValueMapping": [
            {
              "conditionValue": true,
              "fields": [
                {
                  "targetPath": "ratingIndicator",
                  "operation": "toBoolean(true)"
                }
              ]
            }
          ],
          {
            "conditionValue": false,
            "fields": [
              {
                "targetPath": "ratingIndicator",
                "operation": "toBoolean(false)"
              }
            ]
          }
        ]
      ]
    }
  ]
}
]

```



```

    }
  ]
},
{
  "sourcePath": ".",
  "operation": "conditionBasedHandling(!null)",
  "conditionValueSource": "quotaManagementIndicator",
  "conditionValueMapping": [
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age[].ratingGroup",
          "operation": "conditionBasedHandling(isPresent, sess →
ionInformation.offlineRatingIndication[])",
          "conditionValueMapping": [
            {
              "conditionValue": true,
              "fields": [
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                  "targetPath": "ratingIndicator",
                  "operation": "toBoolean(true)"
                }
              ]
            }
          ],
          "conditionValue": false,
          "fields": [
            {
              "targetPath": "ratingIndicator",
              "operation": "toBoolean(false)"
            }
          ]
        }
      ]
    }
  ],
  "sourcePath": "pDUContainerInformation.chargingRule →
BaseName",
  "targetPath": "pDUContainerInformation.chargingRule →
BaseName"
},
{
  "sourcePath": "pDUContainerInformation.afChargingId →
entifier",
  "targetPath": "pDUContainerInformation.afChargingId →

```



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entifier"
    },
    {
      "sourcePath": "pDUContainerInformation.rATType",
      "targetPath": "pDUContainerInformation.rATType",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,raTypeMapping))"
    },
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      "sourcePath": "pDUContainerInformation.sponsorIdent
ity",
      "targetPath": "pDUContainerInformation.sponsorIdent
ity",
      "operation": "toHexString($-SV)"
    },
    {
      "sourcePath": "pDUContainerInformation.applications
erviceProviderIdentity",
      "targetPath": "pDUContainerInformation.applicationS
erviceProviderIdentity",
      "operation": "toHexString($-SV)"
    },
    {
      "sourcePath": "pDUContainerInformation.userLocation
Information",
      "targetPath": "pDUContainerInformation.userLocation
Information",
      "operation": "toHexString(getBytes($-SV))"
    },
    {
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      "targetPath": "pDUContainerInformation.ueTimeZone",
      "operation": "convertUETimezoneToTwoOctets($-SV)"
    },
    {
      "sourcePath": "pDUContainerInformation.servingNodeI
D[]",
      "targetPath": "pDUContainerInformation.servingNetwo
rkFunctionID[].ServingNetworkFunctionID",
      "operation": "arrayToSequenceOf",
      "fields": [
        {
          "sourcePath": "aMFId",
          "targetPath": "aMFIdentifier"
        },
        {
          "sourcePath": "servingNetworkFunctionInform
ation.nodeFunctionality",
          "targetPath": "servingNetworkFunctionInform
ation.networkFunctionality",
          "operation": "stringToEnum($-SV,networkFunc
tionalityMapping)"
        }
      ]
    }
  ]
}
```




```

        },
        {
            "sourcePath": "servingNetworkFunctionInform →
ation.nFIPv4Address",
            "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
        },
        {
            "sourcePath": "servingNetworkFunctionInform →
ation.nFIPv6Address",
            "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionIPv6Address.iPTextRepresentedAddress.iPTextV6Address"
        },
        {
            "sourcePath": "servingNetworkFunctionInform →
ation.nFPLMNID",
            "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionPLMNIdentifier",
            "operation": "toHexString(generatePLMN_ID($ →
-SV))"
        },
        {
            "sourcePath": "servingNetworkFunctionInform →
ation.nFFqdn",
            "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionFQDN.domainName"
        },
        {
            "sourcePath": "servingNetworkFunctionInform →
ation.nFName",
            "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionName"
        }
    ]
},
{
    "sourcePath": "pDUContainerInformation.timeofFirstU →
sage",
    "targetPath": "pDUContainerInformation.timeOfFirstU →
sage",
    "operation": "toHexString(getTimeStampOctetArray($- →
SV))"
},
{
    "sourcePath": "pDUContainerInformation.timeofLastUs →
age",
    "targetPath": "pDUContainerInformation.timeOfLastUs →
age",
    "operation": "toHexString(getTimeStampOctetArray($- →
SV))"
},
{

```



```
on.5qi",
on.fiveQi"
    },
    {
on.qnc",
on.qoSNotificationControl"
    },
    {
on.reflectiveQos",
on.reflectiveQos"
    },
    {
on.maxbrUL",
on.maxbitrateUL",
    },
    {
on.maxbrDL",
on.maxbitrateDL",
    },
    {
on.gbrUL",
on.guaranteedbitrateUL",
    },
    {
on.gbrDL",
on.guaranteedbitrateDL",
    },
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on.priorityLevel",
on.priorityLevel"
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    "sourcePath": "pDUContainerInformation.qoSInformati →
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    "sourcePath": "pDUContainerInformation.qoSInformati →
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    "operation": "toHexString($-SV)"
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    "targetPath": "pDUContainerInformation.qoSInformati →
    "operation": "toHexString($-SV)"
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    "targetPath": "pDUContainerInformation.qoSInformati →
    "operation": "toHexString($-SV)"
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    "targetPath": "pDUContainerInformation.qoSInformati →
    "operation": "toHexString($-SV)"
    "sourcePath": "pDUContainerInformation.qoSInformati →
    "targetPath": "pDUContainerInformation.qoSInformati →
```



```

        "sourcePath": "pDUContainerInformation.qoSInformation",
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    },
    {
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        "targetPath": "pDUContainerInformation.qoSInformation",
    },
    {
        "sourcePath": "pDUContainerInformation.qoSInformation",
        "targetPath": "pDUContainerInformation.qoSInformation",
    },
    {
        "sourcePath": "pDUContainerInformation.qoSInformation",
        "targetPath": "pDUContainerInformation.qoSInformation",
    },
    {
        "sourcePath": "pDUContainerInformation.presenceReportingAreaInformation",
        "targetPath": "pDUContainerInformation.presenceReportingAreaInfo",
        "operation": "mapToSequence",
        "fields": [
            {
                "sourcePath": "praId",
                "targetPath": "presenceReportingAreaIdentifier",
                "operation": "toHexString($-SV)"
            },
            {
                "sourcePath": "presenceState",
                "targetPath": "presenceReportingAreaStatus",
                "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,presenceStateMapping))"
            }
        ]
    },
    {
        "sourcePath": "pDUContainerInformation.presenceReportingAreaInformation",
        "targetPath": "pDUContainerInformation.presenceReportingAreaInfo.presenceReportingAreaElementsList",
        "operation": "toHexString(getBytes(toJSONString(getElementsFromMap($-SV,ncqiList,trackingAreaList,globalRanNodeIdList,ecqiList,globalRanNodeList)))"
    }
]

```



```

baleNbIdList))))"
    },
    {
      "sourcePath": "pDUContainerInformation.presenceRepo
rtingAreaInformation",
      "targetPath": "pDUContainerInformation.presenceRepo
rtingAreaInfo.presenceReportingAreaNode",
      "operation": "toString(10)"
    },
    {
      "sourcePath": "pDUContainerInformation.qoSInformati
on.arp.priorityLevel",
      "targetPath": "pDUContainerInformation.qoSInformati
on.aRP.priorityLevel"
    },
    {
      "sourcePath": "pDUContainerInformation.qoSInformati
on.arp.preemptCap",
      "targetPath": "pDUContainerInformation.qoSInformati
on.aRP.preemptionCapability",
      "operation": "stringToEnum($-SV,preemptCapMapping)"
    },
    {
      "sourcePath": "pDUContainerInformation.qoSInformati
on.arp.preemptVuln",
      "targetPath": "pDUContainerInformation.qoSInformati
on.aRP.preemptionVulnerability",
      "operation": "stringToEnum($-SV,preemptVulnMapping)"
    },
    {
      "sourcePath": "pDUContainerInformation.3gppPSDataOf
fStatus",
      "targetPath": "pDUContainerInformation.threeGPPPSDa
taOffStatus",
      "operation": "ignoreUnknownEnumValue(stringToEnum($
-SV,3gppPSDataOffStatusMapping))"
    },
    {
      "sourcePath": "pDUContainerInformation.qoSCharacter
istics",
      "targetPath": "pDUContainerInformation.qoSCharacter
istics",
      "operation": "toHexString(getBytes($-SV))"
    },
    {
      "sourcePath": "pDUContainerInformation.mAPDUSteerin
gFunctionality",
      "targetPath": "pDUContainerInformation.mAPDUSteerin
gFunctionality",
      "operation": "ignoreUnknownEnumValue(stringToEnum($
-SV,mAPDUSteeringFunctionalityMapping))"
    }
  ]
}

```



```

    },
    {
      "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.3gLoad",
      "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.threegLoad"
    },
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      "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.active",
      "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.active",
      "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,activeMapping))"
    },
    {
      "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.prioAcc",
      "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.prioAcc",
      "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,activeMapping))"
    },
    {
      "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.standby",
      "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.standby",
      "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,activeMapping))"
    },
    {
      "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.steerModeValue",
      "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.steerModeValue",
      "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,steerModeValueMapping))"
    },
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      "sourcePath": "pDUContainerInformation.afChargingId →
String",
      "targetPath": "pDUContainerInformation.afChargingId →
String"
    }
  ]
}
],
{
  "sourcePath": "chargingData.roamingQBCInformation.uPFID",
  "targetPath": "CHFRecord.chargingFunctionRecord.roamingQBCInformati →

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```
on.uPFID"
  },
  {
    "sourcePath": "chargingData.roamingQBCInformation.roamingChargingPr →
ofile.partialRecordMethod",
    "targetPath": "CHFRecord.chargingFunctionRecord.roamingQBCInformati →
on.roamingChargingProfile.partialRecordMethod",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,partialRecor →
dMethodEnumMapping))"
  },
  {
    "sourcePath": "chargingData.roamingQBCInformation.roamingChargingPr →
ofile.triggers[]",
    "targetPath": "CHFRecord.chargingFunctionRecord.roamingQBCInformati →
on.roamingChargingProfile.roamingTriggers[].RoamingTrigger",
    "operation": "arrayToSequenceOf",
    "fields": [
      {
        "sourcePath": ".",
        "conditonValueSource": "triggerType",
        "operation": "conditionBasedHandling(=)",
        "conditionValueMapping": [
          {
            "conditionValue": "FINAL",
            "fields": [
              {
                "sourcePath": "triggerType",
                "targetPath": "trigger",
                "operation": "stringToEnum(getTriggerTypeBy →
SessionType($-SV),triggerTypeMapping)"
              }
            ]
          }
        ],
        "conditionValue": "*",
        "fields": [
          {
            "sourcePath": "triggerType",
            "targetPath": "trigger",
            "operation": "ignoreUnknownEnumValue(string →
ToEnum($-SV,triggerTypeMapping))"
          }
        ]
      }
    ]
  },
  {
    "sourcePath": "triggerCategory",
    "targetPath": "triggerCategory",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,trig →
gerCategoryEnumMapping))"
  },

```



```

        {
            "sourcePath": "timeLimit",
            "targetPath": "timeLimit"
        },
        {
            "sourcePath": "volumeLimit64",
            "targetPath": "volumeLimit"
        },
        {
            "sourcePath": "maxNumberOfccc",
            "targetPath": "maxNbChargingConditions"
        }
    ]
},
{
    "sourcePath": "chargingData.roamingQBCInformation.multipleQFIcontai →
ner[]",
    "targetPath": "CHFRecord.chargingFunctionRecord.roamingQBCInformati →
on.multipleQFIcontainer[].MultipleQFIContainer",
    "operation": "arrayToSequenceOf",
    "groupingOperation": "merge",
    "fields": [
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            "sourcePath": "qFIContainerInformation.qFI",
            "targetPath": "qosFlowId"
        },
        {
            "sourcePath": "triggers[]",
            "targetPath": "triggers[].Trigger",
            "operation": "arrayToSequenceOf",
            "fields": [
                {
                    "sourcePath": ".",
                    "conditionValueSource": "triggerType",
                    "operation": "conditionBasedHandling(=)",
                    "conditionValueMapping": [
                        {
                            "conditionValue": "FINAL",
                            "fields": [
                                {
                                    "sourcePath": "triggerType",
                                    "targetPath": "sMFTrigger",
                                    "operation": "stringToEnum(getTrigg →
erTypeBySessionType($-SV),triggerTypeMapping)"
                                }
                            ]
                        }
                    ]
                },
                {
                    "conditionValue": "*",
                    "fields": [
                        {
                            "sourcePath": "triggerType",

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Enum($-SV,triggerTypeMapping),507)"
    }
  ]
}
]
},
{
  "sourcePath": "triggerTimestamp",
  "targetPath": "triggerTimeStamp",
  "operation": "toHexString(getTimeStampOctetArray($-SV))"
},
{
  "sourcePath": "totalVolume",
  "targetPath": "dataTotalVolume"
},
{
  "sourcePath": "uplinkVolume",
  "targetPath": "dataVolumeUplink"
},
{
  "sourcePath": "downlinkVolume",
  "targetPath": "dataVolumeDownlink"
},
{
  "sourcePath": "time",
  "targetPath": "time"
},
{
  "sourcePath": "localSequenceNumber",
  "targetPath": "localSequenceNumber"
},
{
  "sourcePath": "qFIContainerInformation.timeofFirstUsage",
  "targetPath": "timeOfFirstUsage",
  "operation": "toHexString(getTimeStampOctetArray($-SV))"
},
{
  "sourcePath": "qFIContainerInformation.reportTime",
  "targetPath": "reportTime",
  "operation": "toHexString(getTimeStampOctetArray($-SV))"
},
{
  "sourcePath": "qFIContainerInformation.timeofLastUsage",
  "targetPath": "timeOfLastUsage",
  "operation": "toHexString(getTimeStampOctetArray($-SV))"
},
{
  "sourcePath": "qFIContainerInformation.qoSInformation.5qi",
  "targetPath": "sMFTrigger",
  "operation": "defaultValue(stringTo

```




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    "targetPath": "qoSInformation.fiveQi"
  },
  {
    "sourcePath": "qFIContainerInformation.qoSInformation.arp.p →
priorityLevel",
    "targetPath": "qoSInformation.arp.priorityLevel"
  },
  {
    "sourcePath": "qFIContainerInformation.qoSInformation.arp.p →
reemptCap",
    "targetPath": "qoSInformation.arp.preemptionCapability",
    "operation": "stringToEnum($-SV,preemptCapMapping)"
  },
  {
    "sourcePath": "qFIContainerInformation.qoSInformation.arp.p →
reemptVuln",
    "targetPath": "qoSInformation.arp.preemptionVulnerability",
    "operation": "stringToEnum($-SV,preemptVulnMapping)"
  },
  {
    "sourcePath": "qFIContainerInformation.qoSInformation.qnc",
    "targetPath": "qoSInformation.qoSNotificationControl"
  },
  {
    "sourcePath": "qFIContainerInformation.qoSInformation.refle →
ctiveQos",
    "targetPath": "qoSInformation.reflectiveQos"
  },
  {
    "sourcePath": "qFIContainerInformation.qoSInformation.maxbr →
UL",
    "targetPath": "qoSInformation.maxbitrateUL",
    "operation": "toHexString($-SV)"
  },
  {
    "sourcePath": "qFIContainerInformation.qoSInformation.maxbr →
DL",
    "targetPath": "qoSInformation.maxbitrateDL",
    "operation": "toHexString($-SV)"
  },
  {
    "sourcePath": "qFIContainerInformation.qoSInformation.gbrUL →
",
    "targetPath": "qoSInformation.guaranteedbitrateUL",
    "operation": "toHexString($-SV)"
  },
  {
    "sourcePath": "qFIContainerInformation.qoSInformation.gbrDL →
",
    "targetPath": "qoSInformation.guaranteedbitrateDL",
    "operation": "toHexString($-SV)"
  },
  },

```



```

    {
      "sourcePath": "qFIContainerInformation.qoSInformation.prior →
ityLevel",
      "targetPath": "qoSInformation.priorityLevel"
    },
    {
      "sourcePath": "qFIContainerInformation.qoSInformation.averW →
indow",
      "targetPath": "qoSInformation.averWindow"
    },
    {
      "sourcePath": "qFIContainerInformation.qoSInformation.maxDa →
taBurstVol",
      "targetPath": "qoSInformation.maxDataBurstVol"
    },
    {
      "sourcePath": "qFIContainerInformation.qoSInformation.maxPa →
cketLossRateDl",
      "targetPath": "qoSInformation.maxPacketLossRateDL"
    },
    {
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cketLossRateUl",
      "targetPath": "qoSInformation.maxPacketLossRateUL"
    },
    {
      "sourcePath": "qFIContainerInformation.userLocationInformat →
ion",
      "targetPath": "userLocationInformation",
      "operation": "toHexString(getBytes($-SV))"
    },
    {
      "sourcePath": "qFIContainerInformation.uetimeZone",
      "targetPath": "uETimeZone",
      "operation": "convertUETimezoneToTwoOctets($-SV)"
    },
    {
      "sourcePath": "qFIContainerInformation.presenceReportingAre →
aInformation{}",
      "targetPath": "presenceReportingAreaInfo",
      "operation": "mapToSequence",
      "fields": [
        {
          "sourcePath": "praId",
          "targetPath": "presenceReportingAreaIdentifier",
          "operation": "toHexString($-SV)"
        },
        {
          "sourcePath": "presenceState",
          "targetPath": "presenceReportingAreaStatus",
          "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,presenceStateMapping))"
        }
      ]
    }
  ]
}

```



```

    }
  ],
  {
    "sourcePath": "qFIContainerInformation.presenceReportingAreaInformation",
    "targetPath": "presenceReportingAreaInfo.presenceReportingAreaElementsList",
    "operation": "toHexString(getBytes(toJSONString(getElementsFromMap($-SV,ncgiList,trackingAreaList,globalRanNodeIdList,ecgiList,globaleNbIdList))))"
  },
  {
    "sourcePath": "qFIContainerInformation.presenceReportingAreaInformation",
    "targetPath": "presenceReportingAreaInfo.presenceReportingAreaNode",
    "operation": "toString(10)"
  },
  {
    "sourcePath": "qFIContainerInformation.rATType",
    "targetPath": "rATType",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,rATTypeMapping))"
  },
  {
    "sourcePath": "qFIContainerInformation.servingNetworkFunctionID[]",
    "targetPath": "servingNetworkFunctionID[].ServingNetworkFunctionID",
    "operation": "arrayToSequenceOf",
    "fields": [
      {
        "sourcePath": "aMFId",
        "targetPath": "aMFIdentifier"
      },
      {
        "sourcePath": "servingNetworkFunctionInformation.networkFunctionality",
        "targetPath": "servingNetworkFunctionInformation.networkFunctionality",
        "operation": "stringToEnum($-SV,networkFunctionalityMapping)"
      }
    ],
    {
      "sourcePath": "servingNetworkFunctionInformation.networkFunctionIPv4Address",
      "targetPath": "servingNetworkFunctionInformation.networkFunctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
    },
    {
      "sourcePath": "servingNetworkFunctionInformation.networkFunctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
    }
  ]
}

```



```

IPv6Address",
    "targetPath": "servingNetworkFunctionInformation.networkFunctionIPv6Address.iPTextRepresentedAddress.iPTextV6Address"
  },
  {
    "sourcePath": "servingNetworkFunctionInformation.nF
PLMNID",
    "targetPath": "servingNetworkFunctionInformation.networkFunctionPLMNIdentifier",
    "operation": "toHexString(generatePLMN_ID($-SV))"
  },
  {
    "sourcePath": "servingNetworkFunctionInformation.nF
Fqdn",
    "targetPath": "servingNetworkFunctionInformation.networkFunctionFQDN.domainName"
  },
  {
    "sourcePath": "servingNetworkFunctionInformation.nF
Name",
    "targetPath": "servingNetworkFunctionInformation.networkFunctionName"
  }
]
},
{
  "sourcePath": "qFIContainerInformation.3gppPSDataOffStatus"
,
  "targetPath": "threeGPPPSDataOffStatus",
  "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,3gpp
PSDataOffStatusMapping))"
},
{
  "sourcePath": "qFIContainerInformation.3gppChargingId",
  "targetPath": "threeGPPChargingID"
},
{
  "sourcePath": "qFIContainerInformation.enhancedDiagnostics[
]",
  "targetPath": "extensionDiagnostics.rANNASCause[.RANNASCau
se",
  "operation": "simpleArrayToSequenceOf(toHexString($-SV))"
},
{
  "sourcePath": "qFIContainerInformation.qoSCharacteristics",
  "targetPath": "qoSCharacteristics",
  "operation": "toHexString(getBytes($-SV))"
}
]
},
{
  "sourcePath": "chargingData.nfConsumerIdentification.nodeFunctional

```



```

ity",
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionality",
    "operation": "stringToEnum($-SV,networkFunctionalityMapping)"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFIPv4Address" →
  },
  {
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFIPv6Address" →
  },
  {
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionIPv6Address.iPTextRepresentedAddress.iPTextV6Address"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFPLMNID",
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionPLMNIdentifier",
    "operation": "toHexString(generatePLMN_ID($-SV))"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFFqdn",
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionFQDN.domainName"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFName",
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionName"
  },
  {
    "sourcePath": "requestInformation.sessionOperationType",
    "targetPath": "CHFRecord.chargingFunctionRecord.causeForRecClosing" →
  },
  {
    "operation": "intToEnum($-SV,causeForRecClosingMapping)",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.chargingI →
d",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUSessionChargingID"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.homeProvi →
dedChargingId",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.homeProvidedChargingID",
    "groupingOperation": "last"
  }
}

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```
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.mAPDUSessionInformation.mAPDUSessionIndicator",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.mAPDUSessionInformation.mAPDUSessionIndicator",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,mAPDUSessionIndicatorMapping))",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.mAPDUNon3GPPUserLocationInfo",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.mAPDUNonThreeGPPUserLocationInfo",
      "operation": "toHexString(getBytes($-SV))",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.enhancedDiagnostics[]",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.enhancedDiagnostics.rANNASRelCause[].RANNASRelCause",
      "operation": "arrayToSequenceOf",
      "groupingOperation": "last",
      "fields": [
        {
          "sourcePath": "ngApCause.group",
          "targetPath": "ngApCause.group"
        },
        {
          "sourcePath": "ngApCause.value",
          "targetPath": "ngApCause.value"
        },
        {
          "sourcePath": "5gMmCause",
          "targetPath": "fivegMmCause"
        },
        {
          "sourcePath": "5gSmCause",
          "targetPath": "fivegSmCause"
        },
        {
          "sourcePath": "epsCause",
          "targetPath": "epsCause",
          "operation": "toHexString($-SV)"
        }
      ]
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.userInformation",

```



```

    "operation": "conditionBasedHandling(^)",
    "conditionValueSource": "servedGPSI",
    "conditionValueMapping": [
      {
        "conditionValue": "msisdn",
        "fields": [
          {
            "sourcePath": "servedGPSI",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDU →
SessionChargingInformation.userIdentifier.iSDN-E164",
            "operation": "splitStringWithRegex($-SV,-,1)"
          }
        ]
      },
      {
        "conditionValue": "extid",
        "fields": [
          {
            "sourcePath": "servedGPSI",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDU →
SessionChargingInformation.userIdentifier.externalId",
            "operation": "splitStringWithRegex($-SV,-,1)"
          }
        ]
      }
    ],
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.userInfor →
mation.servedPEI",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.userEquipmentInfo.subscriberEquipmentNumberType",
      "operation": "stringToEnum(splitStringWithRegex($-SV,-,0),servedPEI →
Mapping)"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.userInfor →
mation.servedPEI",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.userEquipmentInfo.subscriberEquipmentNumberData",
      "operation": "toHexString(splitStringWithRegex($-SV,-,1))"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.userInfor →
mation",
      "operation": "conditionBasedHandling(=)",
      "conditionValueSource": "unauthenticatedFlag",
      "conditionValueMapping": [
        {
          "conditionValue": true,
          "fields": [
            {

```



```
        "targetPath": "CHFRecord.chargingFunctionRecord.pDU →
SessionChargingInformation.sUPIunauthenticatedFlag",
        "operation": "setNULL()"
    }
    ]
}
],
{
    "sourcePath": "chargingData.pDUSessionChargingInformation.userInfor →
mation.roamerInOut",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.userRoamerInOut",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,UserRoamerIn →
OutMapping))"
},
{
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.pduSessionID",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pduSessionId"
},
{
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.networkSlicingInfo.sNSSAI.sst",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.networkSliceInstanceID.sST"
},
{
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.networkSlicingInfo.sNSSAI.sd",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.networkSliceInstanceID.sD",
    "operation": "toHexString(hexToByte($-SV,3))"
},
{
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.pduType",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUType",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,pduTypeMappi →
ng))"
},
{
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.sscMode",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.sSCMode",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sSCModeMappi →
ng))"
},
{
```




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        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.servingCNPlmnId",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.servingCNPLMNID",
        "operation": "toHexString(generatePLMN_ID($-SV))"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.hPlmnId",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.sUPIPLMNIdentifier",
        "operation": "toHexString(generatePLMN_ID($-SV))"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.mAPDUNon3GPPRAType",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.mAPDUNonThreeGPPRAType",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,raTypeMappin →
g))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.ratType",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.rATType",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,raTypeMappin →
g))"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.dnnId",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.dataNetworkNameIdentifier",
        "operation": "splitStringWithRegex($-SV,.mnc,0)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.dnnSelectionMode",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.dnnSelectionMode",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,dNNSelection →
ModeMapping))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.pduAddress.pduIPv4Address",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUAddress.pDUIPV4Address.iPTextRepresentedAddress.iPTextV4Address"
    },

```



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{
  "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.pduAddress.pduIPv6AddresswithPrefix",
  "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.pDUAddress.pDUIPv6AddresswithPrefix.iPTextRepresentedAddress.iPTextV6Address"
},
{
  "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.pduAddress.iPv4dynamicAddressFlag",
  "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.pDUAddress.iPv4dynamicAddressFlag"
},
{
  "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.pduAddress.iPv6dynamicPrefixFlag",
  "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.pDUAddress.iPv6dynamicPrefixFlag"
},
{
  "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.authorizedQoSInformation.5qi",
  "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.authorizedQoSInformation.fiveQi"
},
{
  "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.authorizedQoSInformation.priorityLevel",
  "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.authorizedQoSInformation.priorityLevel"
},
{
  "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.authorizedQoSInformation.averWindow",
  "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.authorizedQoSInformation.averWindow"
},
{
  "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.authorizedQoSInformation.maxDataBurstVol",
  "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.authorizedQoSInformation.maxDataBurstVol"
},
{
  "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.subscribedQoSInformation.5qi",
  "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.subscribedQoSInformation.fiveQi"
},
{
  "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.subscribedQoSInformation.priorityLevel",
  "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.subscribedQoSInformation.priorityLevel"
}
```



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        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.subscribedQoSInformation.priorityLevel"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedSessionAMBR.uplink",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.authorizedSessionAMBR.ambrUL",
        "operation": "toHexString(getBytes($-SV))"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedSessionAMBR.downlink",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.authorizedSessionAMBR.ambrDL",
        "operation": "toHexString(getBytes($-SV))"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.subscribedSessionAMBR.uplink",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.subscribedSessionAMBR.ambrUL",
        "operation": "toHexString(getBytes($-SV))"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.subscribedSessionAMBR.downlink",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.subscribedSessionAMBR.ambrDL",
        "operation": "toHexString(getBytes($-SV))"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.ueTimeZon →
e",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.ueTimeZone",
        "operation": "convertUETimezoneToTwoOctets($-SV)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.rANSecond →
aryRATUsageReport",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.rANSecondaryRATUsageReport[].NGRANSecondaryRATUsageReport",
        "operation": "elementToSequenceOf",
        "fields": [
            {
                "sourcePath": "rANSecondaryRATType",
                "targetPath": "nGRANSecondaryRATType",
                "operation": "toHexString($-SV)"
            },
            {
                "sourcePath": "qosFlowsUsageReports[]",

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        "targetPath": "qosFlowsUsageReports[].QosFlowsUsageReport",
        "operation": "arrayToSequenceOf",
        "fields": [
            {
                "sourcePath": "qFI",
                "targetPath": "qosFlowId"
            },
            {
                "sourcePath": "startTimestamp",
                "targetPath": "startTime",
                "operation": "toHexString(getTimeStampOctetArray($- →
SV))"
            },
            {
                "sourcePath": "endTimestamp",
                "targetPath": "endTime",
                "operation": "toHexString(getTimeStampOctetArray($- →
SV))"
            },
            {
                "sourcePath": "downlinkVolume",
                "targetPath": "dataVolumeDownlink"
            },
            {
                "sourcePath": "uplinkVolume",
                "targetPath": "dataVolumeUplink"
            }
        ]
    },
    {
        "targetPath": "CHFRecord.chargingFunctionRecord.duration",
        "operation": "toInt(0)",
        "groupingOperation": "difference"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.chargingCharacteristics",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.chargingCharacteristics",
        "operation": "paddingZerosForOctets($-SV,2)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.chargingCharacteristicsSelectionMode",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.chChSelectionMode",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,chChSelectio →
nModeMapping))"
    },
    {
```



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        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.arP.priorityLevel",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.authorizedQoSInformation.aRP.priorityLevel"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.arP.preemptCap",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.authorizedQoSInformation.aRP.preemptionCapability",
        "operation": "stringToEnum($-SV,preemptCapMapping)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.arP.preemptVuln",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.authorizedQoSInformation.aRP.preemptionVulnerability",
        "operation": "stringToEnum($-SV,preemptVulnMapping)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.subscribedQoSInformation.arP.priorityLevel",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.subscribedQoSInformation.aRP.priorityLevel"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.subscribedQoSInformation.arP.preemptCap",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.subscribedQoSInformation.aRP.preemptionCapability",
        "operation": "stringToEnum($-SV,preemptCapMapping)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.subscribedQoSInformation.arP.preemptVuln",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.subscribedQoSInformation.aRP.preemptionVulnerability",
        "operation": "stringToEnum($-SV,preemptVulnMapping)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.3gppPSDataOffStatus",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.threeGPPPSDataOffStatus",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,3gppPSDataOf →
fStatusMapping))"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.userLocat →
ioninfo",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.userLocationInformation",

```



```
        "operation": "toHexString(getBytes($-SV))"
      },
      {
        "sourcePath": "chargingData.pDUSessionChargingInformation.presenceR →
eportingAreaInformation{}",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.presenceReportingAreaInfo",
        "operation": "mapToSequence",
        "fields": [
          {
            "sourcePath": "praId",
            "targetPath": "presenceReportingAreaIdentifier",
            "operation": "toHexString($-SV)"
          },
          {
            "sourcePath": "presenceState",
            "targetPath": "presenceReportingAreaStatus",
            "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,pres →
enceStateMapping))"
          }
        ]
      },
      {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.startTime",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUSessionstartTime",
        "operation": "toHexString(getTimeStampOctetArray($-SV))"
      },
      {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.stopTime",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUSessionstopTime",
        "operation": "toHexString(getTimeStampOctetArray($-SV))"
      },
      {
        "sourcePath": "chargingData.pDUSessionChargingInformation.presenceR →
eportingAreaInformation",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.presenceReportingAreaInfo.presenceReportingAreaElementsList",
        "operation": "toHexString(getBytes(toJSONString(getElementsFromMap( →
$-SV,ncgiList,trackingAreaList,globalRanNodeIdList,ecgiList,globaleNbIdList)))"
      },
      {
        "sourcePath": "chargingData.pDUSessionChargingInformation.presenceR →
eportingAreaInformation",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.presenceReportingAreaInfo.presenceReportingAreaNode",
        "operation": "toString(10)"
      }
    ],
  },
}
```



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    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.servingNetworkFunctionID",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.servingNetworkFunctionID[].ServingNetworkFunctionID",
      "operation": "elementToSequenceOf",
      "fields": [
        {
          "sourcePath": "aMFId",
          "targetPath": "aMFIdentifier"
        },
        {
          "sourcePath": "servingNetworkFunctionInformation.nFIPv4Addr →
ess",
          "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
        },
        {
          "sourcePath": "servingNetworkFunctionInformation.nodeFunc →
tionality",
          "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionality",
          "operation": "stringToEnum($-SV,networkFunctionalityMapping →
)"
        },
        {
          "sourcePath": "servingNetworkFunctionInformation.nFIPv4Addr →
ess",
          "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
        },
        {
          "sourcePath": "servingNetworkFunctionInformation.nFIPv6Addr →
ess",
          "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionIPv6Address.iPTextRepresentedAddress.iPTextV6Address"
        },
        {
          "sourcePath": "servingNetworkFunctionInformation.nFPLMNID",
          "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionPLMNIdentifier",
          "operation": "toHexString(generatePLMN_ID($-SV))"
        },
        {
          "sourcePath": "servingNetworkFunctionInformation.nFFqdn",
          "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionFQDN.domainName"
        },
        {
          "sourcePath": "servingNetworkFunctionInformation.nFName",
          "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionName"
        }
      ]
    }

```



```
    }
  ],
  {
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorSUPI",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorIMSI",
    "operation": "toHexString(parseTBCD(splitStringWithRegex($-SV,-,1)) →
)",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorGPSI",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorMSISDN",
    "operation": "convertToAddressString(splitStringWithRegex($-SV,-,1) →
)",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorOtherAddress.sMAddressType",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorOtherAddress.sMAddressType",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMAddressTyp →
eMapping))",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorOtherAddress.sMAddressData",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorOtherAddress.sMAddressData",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorOtherAddress.sMAddressDomain.domainName",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorOtherAddress.sMAddressDomain.sMDomainName",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorOtherAddress.sMAddressDomain.3GPPIMSIMCCMNC",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorOtherAddress.sMAddressDomain.threeGPPIMSI-MCC-MNC" →
,
    "operation": "toHexString(generatePLMN_ID($-SV))",
    "groupingOperation": "last"
  }
],
{
  "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorOtherAddress.sMAddressDomain.threeGPPIMSI-MCC-MNC",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorOtherAddress.sMAddressDomain.threeGPPIMSI-MCC-MNC" →
,
  "operation": "toHexString(generatePLMN_ID($-SV))",
  "groupingOperation": "last"
}
```




```

    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorSCCPAddress",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorSCCPAddress",
      "operation": "convertToAddressString($-SV)",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorReceivedAddress.sMAddressType",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorReceivedAddress.sMAddressType",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMAddressTyp →
eMapping))"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorReceivedAddress.sMAddressData",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorReceivedAddress.sMAddressData",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorReceivedAddress.sMAddressDomain.domainName",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorReceivedAddress.sMAddressDomain.sMDomainName",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorReceivedAddress.sMAddressDomain.3GPPIMSIMCCMNC",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorReceivedAddress.sMAddressDomain.threeGPPIMSI-MCC-M →
NC",
      "operation": "toHexString(generatePLMN_ID($-SV))",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorInterface.interfaceId",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorInterface.interfaceId",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorInterface.interfacePort",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorInterface.interfacePort",

```



```
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorInterface.interfaceText",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorInterface.interfaceText",
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorInterface.interfaceType",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorInterface.interfaceType",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,SMInterfaceT →
ypeMapping))",
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorProtocolId",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorProtocolId",
        "operation": "toHexString($-SV)",
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.recipientInfo[]" →
,
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.recipientInfos[].RecipientInfo",
        "operation": "arrayToSequenceOf",
        "groupingOperation": "last",
        "fields": [
          {
            "sourcePath": "recipientSUPI",
            "targetPath": "recipientIMSI",
            "operation": "toHexString(parseTBCD(splitStringWithRegex($- →
SV,-,1)))"
          },
          {
            "sourcePath": "recipientGPSI",
            "targetPath": "recipientMSISDN",
            "operation": "convertToAddressString(splitStringWithRegex($ →
-SV,-,1))"
          },
          {
            "sourcePath": "recipientOtherAddress.sMAddressType",
            "targetPath": "recipientOtherAddress.sMAddressType",
            "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMAd →
dressTypeMapping))"
          }
        ]
      }
    ]
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.recipientInfo[]" →
,
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.recipientInfos[].RecipientInfo",
    "operation": "arrayToSequenceOf",
    "groupingOperation": "last",
    "fields": [
      {
        "sourcePath": "recipientSUPI",
        "targetPath": "recipientIMSI",
        "operation": "toHexString(parseTBCD(splitStringWithRegex($- →
SV,-,1)))"
      },
      {
        "sourcePath": "recipientGPSI",
        "targetPath": "recipientMSISDN",
        "operation": "convertToAddressString(splitStringWithRegex($ →
-SV,-,1))"
      },
      {
        "sourcePath": "recipientOtherAddress.sMAddressType",
        "targetPath": "recipientOtherAddress.sMAddressType",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMAd →
dressTypeMapping))"
      }
    ]
  }
]
```



```

        {
            "sourcePath": "recipientOtherAddress.sMaddressData",
            "targetPath": "recipientOtherAddress.sMaddressData"
        },
        {
            "sourcePath": "recipientOtherAddress.sMaddressDomain.domain" →
            "targetPath": "recipientOtherAddress.sMaddressDomain.sMDoma →
Name",
            "targetPath": "recipientOtherAddress.sMaddressDomain.sMDoma →
inName"
        },
        {
            "sourcePath": "recipientOtherAddress.sMaddressDomain.3GPPIM →
SIMCCMNC",
            "targetPath": "recipientOtherAddress.sMaddressDomain.threeG →
PPIMSI-MCC-MNC",
            "operation": "toHexString(generatePLMN_ID($-SV))"
        },
        {
            "sourcePath": "recipientSCCPAddress",
            "targetPath": "recipientSCCPAddress",
            "operation": "convertToAddressString($-SV)"
        },
        {
            "sourcePath": "recipientReceivedAddress.sMaddressType",
            "targetPath": "recipientReceivedAddress.sMaddressType",
            "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMad →
dressTypeMapping))"
        },
        {
            "sourcePath": "recipientReceivedAddress.sMaddressData",
            "targetPath": "recipientReceivedAddress.sMaddressData"
        },
        {
            "sourcePath": "recipientReceivedAddress.sMaddressDomain.dom →
ainName",
            "targetPath": "recipientReceivedAddress.sMaddressDomain.sMD →
omainName"
        },
        {
            "sourcePath": "recipientReceivedAddress.sMaddressDomain.3GP →
PIMSIMCCMNC",
            "targetPath": "recipientReceivedAddress.sMaddressDomain.thr →
eeGPPIMSI-MCC-MNC",
            "operation": "toHexString(generatePLMN_ID($-SV))"
        },
        {
            "sourcePath": "sMDestinationInterface.interfaceId",
            "targetPath": "sMDestinationInterface.interfaceId"
        },
        {
            "sourcePath": "sMDestinationInterface.interfacePort",
            "targetPath": "sMDestinationInterface.interfacePort"
        }
    ]
}

```



```

    },
    {
      "sourcePath": "sMDestinationInterface.interfaceText",
      "targetPath": "sMDestinationInterface.interfaceText"
    },
    {
      "sourcePath": "sMDestinationInterface.interfaceType",
      "targetPath": "sMDestinationInterface.interfaceType",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV, SMin
interfaceTypeMapping))"
    },
    {
      "sourcePath": "sMrecipientProtocolId",
      "targetPath": "sMRecipientProtocolID",
      "operation": "toHexString($-SV)"
    }
  ]
},
{
  "sourcePath": "chargingData.sMSChargingInformation.userEquipmentInf
o",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat
ion.userEquipmentInfo.subscriberEquipmentNumberType",
  "operation": "stringToEnum(splitStringWithRegex($-SV, -, 0), servedPEI
Mapping)",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.userEquipmentInf
o",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat
ion.userEquipmentInfo.subscriberEquipmentNumberData",
  "operation": "toHexString(splitStringWithRegex($-SV, -, 1))"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.userLocationinfo
",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat
ion.userLocationInformation",
  "operation": "toHexString(getBytes($-SV))",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.ueTimeZone",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat
ion.uETimeZone",
  "operation": "convertUETimezoneToTwoOctets($-SV)",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.rATType",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat

```



```

ion.raType",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,raTypeMapping))",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSCChargingInformation.sMSCAddress",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSCChargingInformation.sMSCAddress",
    "operation": "convertToAddressString($-SV)",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.invocationTimeStamp",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSCChargingInformation.eventTimeStamp",
    "operation": "toHexString(getTimeStampOctetArray($-SV))",
    "validateParentPath": "chargingData.sMSCChargingInformation",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSCChargingInformation.sMDataCodingScheme",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSCChargingInformation.sMDataCodingScheme",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSCChargingInformation.sMMessageType",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSCChargingInformation.sMMessageType",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMMessageTypeMapping))",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSCChargingInformation.sMReplyPathRequested",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSCChargingInformation.sMReplyPathRequested",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMReplyPathRequestedMapping))",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSCChargingInformation.sMUserDataHeader",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSCChargingInformation.sMUserDataHeader",
    "operation": "toHexString($-SV)",
    "groupingOperation": "last"
  },

```



```
{
  "sourcePath": "chargingData.sMSChargingInformation.sMDischargeTime" →
,
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMDischargeTime",
  "operation": "toHexString(getTimeStampOctetArray($-SV))",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.numberofMessages →
Sent",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMTotalsNumber",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.sMServiceType",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMServiceType",
  "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMServiceTyp →
eMapping))",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.sMSequenceNumber →
",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMSequenceNumber",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.submissionTime",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.submissionTime",
  "operation": "toHexString(getTimeStampOctetArray($-SV))",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.sMPriority",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMPriority",
  "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,priorityType →
Mapping))",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.messageReference →
",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.messageReference",
  "operation": "defaultValue(toHexString($-SV),0)",
  "groupingOperation": "last"
}
```



```

    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.messageSize",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.messageSize",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.messageClass.cla →
ssIdentifier",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.messageClass",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,classIdentif →
ierMapping))",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.deliveryReportRe →
quested",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMdeliveryReportRequested",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,deliveryRepo →
rtRequestedMapping))",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.serviceSpecificationInfo",
      "targetPath": "CHFRecord.chargingFunctionRecord.serviceSpecificatio →
nInformation",
      "operation": "toHexString($-SV)",
      "groupingOperation": "last"
    },
    {
      "targetPath": "CHFRecord.chargingFunctionRecord.incompleteCDRIndica →
tion.initialLost",
      "operation": "optionalTargetField",
      "groupingOperation": "substitute($-initialLost)"
    },
    {
      "targetPath": "CHFRecord.chargingFunctionRecord.incompleteCDRIndica →
tion.updateLost",
      "operation": "optionalTargetField",
      "groupingOperation": "substitute($-updateLost)"
    },
    {
      "targetPath": "CHFRecord.chargingFunctionRecord.recordExtensions[1] →
.ManagementExtension",
      "operation": "indexedArraytoSequence",
      "fields": [
        {
          "targetPath": "identifier",
          "operation": "toString(1 2 826 0 1249 0 5 7 0 0)"
        }
      ]
    }
  ]
}

```



```
    },
    {
      "targetPath": "significance",
      "operation": "toBoolean(false)"
    },
    {
      "targetPath": "information[1].ProtocolEnhancement",
      "operation": "indexedArraytoSequence",
      "fields": [
        {
          "targetPath": "ieKey",
          "operation": "toString(apiVersionInUri)"
        },
        {
          "sourcePath": "apiVersionInUri",
          "targetPath": "ieRaw",
          "operation": "toHexString($-SV)"
        }
      ]
    }
  ]
}
```

3.6.1.6 validate-metadata

The `validate-metadata` command validate the required transformation metadata. Ensure to mention the validated metadata in the metadata path.

Syntax

Command Deprecated in API v1.1: `beamctl pub-transform-config validate-metadata --metadata-path <metadata-path> --source-file-path <source-file-path> --target-file-path <target-file-path> <payload>`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 validate-metadata --metadata-path <metadata-path> --source-file-path <source-file-path> --target-file-path <target-file-path> <payload>`

Flags

Table 162 Flags

| Name | Description |
|------------------|---|
| metadata-path | To validate the path of the metadata |
| source-file-path | To validate the path of the source-file |



| Name | Description |
|------------------|---|
| target-file-path | To validate the path of the target-file |

Example 131

Command Deprecated in API v1.1: `beamctl pub-transform-config validate-metadata --metadata-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1_BssfChfCdrAsn1_3_v2.json < /var/tmp/ASN1/ASN1.zip`, where ASN1.zip should contain the metadata file.

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 validate-metadata --metadata-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1_BssfChfCdrAsn1_3_v2.json < /var/tmp/ASN1/ASN1.zip`, where ASN1.zip should contain the metadata file.

Command Deprecated in API v1.1: `beamctl pub-transform-config validate-metadata --metadata-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1_BssfChfCdrAsn1_3_v2.json --source-file-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1.json < /var/tmp/ASN1/ASN2.zip`, where ASN2.zip should the metadata file and source-schema files.

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 validate-metadata --metadata-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1_BssfChfCdrAsn1_3_v2.json --source-file-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1.json < /var/tmp/ASN1/ASN2.zip`, where ASN2.zip should the metadata file and source-schema files.

Command Deprecated in API v1.1: `beamctl pub-transform-config validate-metadata --metadata-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1_BssfChfCdrAsn1_3_v2.json --source-file-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1.json --target-file-path /var/tmp/ASN1/bssf_chf_cdr_asn1.asn < /var/tmp/ASN1/ASN3.zip`, where ASN3.zip should contain the metadata file, source-schema and target-schema files.

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 validate-metadata --metadata-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1_BssfChfCdrAsn1_3_v2.json --source-file-path /var/tmp/ASN1/ChargingNetworkFunctionEvent_1.json --target-file-path /var/tmp/ASN1/bssf_chf_cdr_asn1.asn < /var/tmp/ASN1/ASN3.zip`, where ASN3.zip should contain the metadata file, source-schema and target-schema files.

Note: Solution Integrator (SI) must prepare the zip file with the required ASN.1 schemas included. The same needs to be placed in the path which is given as the input for CLI.

Example 132 validate-metadata

```
ChargingNetworkFunctionEvent_2_BssfChfCdrAsn1Rel16_1_v3.json
{
  "sourceSchemaDetails": {
    "schemaNameSpace": "com.ericsson.bss.cha.event.schema",
    "schemaName": "ChargingNetworkFunctionEvent",
    "schemaVersion": "2.0.0",
    "apiVersionInUri": "v3"
  },
  "targetSchemaDetails": {
    "schemaNameSpace": "com.ericsson.bss.sys.event.schema",
    "schemaName": "BssfChfCdrAsn1Rel16",
    "schemaVersion": "1.0.1"
  },
  "consolidation": {
    "defaultGroupingOperation": "first"
  },
  "definitions": {
    "ratingIndicatorMapping": {
      "ONLINE_CHARGING": true,
      "OFFLINE_CHARGING": false
    },
    "quotaManagementIndicatorMapping": {
      "ONLINE_CHARGING": 0,
      "OFFLINE_CHARGING": 1
    },
    "3gppPSDataOffStatusMapping": {
      "ACTIVE": 0,
      "INACTIVE": 1
    },
    "partialRecordMethodEnumMapping": {
      "DEFAULT": 0,
      "INDIVIDUAL": 1
    },
    "triggerCategoryEnumMapping": {
      "IMMEDIATE_REPORT": 0,
      "DEFERRED_REPORT": 1
    },
    "triggerTypeMapping": {
      "QHT": 410,
      "FINAL": 500,
      "FINAL_UPDATE": 500,
      "FINAL_RELEASE": 503,
      "VALIDITY_TIME": 406,
      "OTHER_QUOTA_TYPE": 409,
      "FORCED_REAUTHORISATION": 407,
      "UNUSED_QUOTA_TIMER": 502,
      "ABNORMAL_RELEASE": 506,
      "QOS_CHANGE": 100,
      "VOLUME_LIMIT": 201,
      "TIME_LIMIT": 200,
    }
  }
}
```



```

    "PLMN_CHANGE": 107,
    "USER_LOCATION_CHANGE": 101,
    "RAT_CHANGE": 108,
    "UE_TIMEZONE_CHANGE": 106,
    "TARIFF_TIME_CHANGE": 105,
    "MAX_NUMBER_OF_CHANGES_IN_CHARGING_CONDITIONS": 203,
    "MANAGEMENT_INTERVENTION": 501,
    "CHANGE_OF_UE_PRESENCE_IN_PRESENCE_REPORTING_AREA": 103,
    "CHANGE_OF_3GPP_PS_DATA_OFF_STATUS": 104,
    "SERVING_NODE_CHANGE": 102,
    "REMOVAL_OF_UPF": 111,
    "ADDITION_OF_UPF": 110,
    "START_OF_SERVICE_DATA_FLOW": 2,
    "START_OF_SERVICE_DATA_FLOW_INITIAL": 2,
    "START_OF_SERVICE_DATA_FLOW_UPDATE": 408,
    "SESSION_AMBR_CHANGE": 109,
    "GFBF_GUARANTEED_STATUS_CHANGE": 115,
    "UNIT_COUNT_INACTIVITY_TIMER": 502,
    "HANDOVER_START": 703,
    "HANDOVER_CANCEL": 702,
    "HANDOVER_COMPLETE": 704,
    "INSERTION_OF_ISMF": 112,
    "REMOVAL_OF_ISMF": 113,
    "CHANGE_OF_ISMF": 114,
    "START_OF_SDF_ADDITIONAL_ACCESS": 411,
    "ECGI_CHANGE": 700,
    "TAI_CHANGE": 701,
    "ADDITION_OF_ACCESS": 116,
    "REMOVAL_OF_ACCESS": 117
  },
  "preemptCapMapping": {
    "NOT_PREEMPT": 0,
    "MAY_PREEMPT": 1
  },
  "preemptVulnMapping": {
    "NOT_PREEMPTABLE": 0,
    "PREEMPTABLE": 1
  },
  "presenceStateMapping": {
    "IN_AREA": 0,
    "OUT_OF_AREA": 1,
    "INACTIVE": 2,
    "UNKNOWN": 3
  },
  "raTypeMapping": {
    "NR": 51,
    "EUTRA": 6,
    "WLAN": 3,
    "VIRTUAL": 7,
    "NBIOT": 8,
    "WIRELIN": 55,
    "WIRELIN_CABLE": 56,

```



```
    "WIRELINE_BBF": 57,  
    "LTE-M": 9,  
    "NR_U": 51,  
    "EUTRA_U": 6,  
    "TRUSTED_N3GA": 65,  
    "TRUSTED_WLAN": 3,  
    "UTRA": 1,  
    "GERA": 2  
  },  
  "networkFunctionalityMapping": {  
    "SMF": 1,  
    "AMF": 2,  
    "SMSF": 3,  
    "SMS": 3,  
    "PGW_C_SMF": 9,  
    "NEFF": 8,  
    "NEF": 8,  
    "SGW": 4,  
    "I_SMF": 5,  
    "ePDG": 6,  
    "CEF": 7  
  },  
  "UserRoamerInOutMapping": {  
    "IN_BOUND": 0,  
    "OUT_BOUND": 1  
  },  
  "pduTypeMapping": {  
    "IPV4V6": 0,  
    "IPV4": 1,  
    "IPV6": 2,  
    "UNSTRUCTURED": 3,  
    "ETHERNET": 4  
  },  
  "chChSelectionModeMapping": {  
    "HOME_DEFAULT": 3,  
    "ROAMING_DEFAULT": 4,  
    "VISITING_DEFAULT": 5  
  },  
  "sSCModeMapping": {  
    "SSC_MODE_1": 1,  
    "SSC_MODE_2": 2,  
    "SSC_MODE_3": 3  
  },  
  "servedPEIMapping": {  
    "imeisv": 0,  
    "mAC": 1,  
    "eUI64": 2,  
    "imei": 0,  
    "mac": 1,  
    "eui": 2  
  },  
  "sMAddressTypeMapping": {
```



```

        "EMAIL_ADDRESS": 0,
        "MSISDN": 1,
        "IPV4_ADDRESS": 2,
        "IPV6_ADDRESS": 3,
        "NUMERIC_SHORTCODE": 4,
        "ALPHANUMERIC_SHORTCODE": 5,
        "OTHER": 6,
        "IMSI": 7
    },
    "deliveryReportRequestedMapping": {
        "YES": 0,
        "NO": 1
    },
    "classIdentifierMapping": {
        "PERSONAL": 0,
        "ADVERTISEMENT": 1,
        "INFORMATIONAL": 2,
        "AUTO": 3
    },
    "priorityTypeMapping": {
        "LOW": 0,
        "NORMAL": 1,
        "HIGH": 2
    },
    "sMServiceTypeMapping": {
        "VAS4SMS_SHORT_MESSAGE_CONTENT_PROCESSING": 0,
        "VAS4SMS_SHORT_MESSAGE_FORWARDING": 1,
        "VAS4SMS_SHORT_MESSAGE_FORWARDING_MULTIPLE_SUBSCRIPTIONS": 2,
        "VAS4SMS_SHORT_MESSAGE_FILTERING": 3,
        "VAS4SMS_SHORT_MESSAGE_RECEIPT": 4,
        "VAS4SMS_SHORT_MESSAGE_NETWORK_STORAGE": 5,
        "VAS4SMS_SHORT_MESSAGE_TO_MULTIPLE_DESTINATIONS": 6,
        "VAS4SMS_SHORT_MESSAGE_VIRTUAL_PRIVATE_NETWORK(VPN)": 7,
        "VAS4SMS_SHORT_MESSAGE_AUTO_REPLY": 8,
        "VAS4SMS_SHORT_MESSAGE_PERSONAL_SIGNATURE": 9,
        "VAS4SMS_SHORT_MESSAGE_DEFERRED_DELIVERY": 10
    },
    "sMReplyPathRequestedMapping": {
        "NO_REPLY_PATH_SET": 0,
        "REPLY_PATH_SET": 1
    },
    "sMMessageTypeMapping": {
        "SUBMISSION": 0,
        "DELIVERY_REPORT": 1,
        "SM_SERVICE_REQUEST": 2
    },
    "SMInterfaceTypeMapping": {
        "UNKNOWN": 0,
        "MOBILE_ORIGINATING": 1,
        "MOBILE_TERMINATING": 2,
        "APPLICATION_ORIGINATING": 3,
        "APPLICATION_TERMINATING": 4
    }

```



```
    },
    "causeForRecClosingMapping": {
      "0": 1,
      "1": 1,
      "2": 0
    },
    "SubscriptionIDTypeMapping": {
      "imsi": 1,
      "nai": 3,
      "gci": 3,
      "gli": 3
    },
    "dNNSelectionModeMapping": {
      "VERIFIED": 0,
      "UE_DNN_NOT_VERIFIED": 1,
      "NW_DNN_NOT_VERIFIED": 2
    },
    "mAPDUSteeringFunctionalityMapping": {
      "MPTCP": 0,
      "ATSSS_LL": 1
    },
    "activeMapping": {
      "3GPP_ACCESS": 0,
      "NON_3GPP_ACCESS": 1
    },
    "steerModeValueMapping": {
      "ACTIVE_STANDBY": 0,
      "LOAD_BALANCING": 1,
      "SMALLEST_DELAY": 2,
      "PRIORITY_BASED": 3
    },
    "mAPDUSessionIndicatorMapping": {
      "MA_PDU_REQUEST": 0,
      "MA_PDU_NETWORK_UPGRADE_ALLOWED": 1
    }
  },
  "fields": [
    {
      "sourcePath": "requestInformation.sessionId",
      "targetPath": "CHFRecord.chargingFunctionRecord.chargingSessionId",
      "operation": "toHexString($-SV)"
    },
    {
      "targetPath": "CHFRecord.chargingFunctionRecord.recordType",
      "operation": "toInt(200)"
    },
    {
      "sourcePath": "chargingData.chargingId",
      "targetPath": "CHFRecord.chargingFunctionRecord.chargingID"
    }
  ]
}
```



```

        "sourcePath": "chargingData.subscriberIdentifier",
        "targetPath": "CHFRecord.chargingFunctionRecord.subscriberIdentifie →
r.subscriberIDType",
        "operation": "ignoreUnknownEnumValue(stringToEnum(splitStringWithRe →
gex($-SV,-,0),SubscriptionIDTypeMapping))"
    },
    {
        "sourcePath": "chargingData.subscriberIdentifier",
        "targetPath": "CHFRecord.chargingFunctionRecord.subscriberIdentifie →
r.subscriberIDData",
        "operation": "splitStringWithRegex($-SV,-,1)"
    },
    {
        "targetPath": "CHFRecord.chargingFunctionRecord.recordingNetworkFun →
ctionID",
        "operation": "generateUUID()"
    },
    {
        "sourcePath": "eventHeader.eventCreationTime.timestamp",
        "targetPath": "CHFRecord.chargingFunctionRecord.recordOpeningTime",
        "operation": "toHexString(getTimeStampOctetArray($-SV))"
    },
    {
        "sourcePath": "chargingData.invocationSequenceNumber",
        "targetPath": "CHFRecord.chargingFunctionRecord.recordSequenceNumbe →
r"
    },
    {
        "targetPath": "CHFRecord.chargingFunctionRecord.localRecordSequence →
Number",
        "operation": "getRecordIdNo()"
    },
    {
        "sourcePath": "chargingData.triggers[]",
        "targetPath": "CHFRecord.chargingFunctionRecord.triggers[].Trigger" →
,
        "operation": "arrayToSequenceOf",
        "fields": [
            {
                "sourcePath": ".",
                "conditonValueSource": "triggerType",
                "operation": "conditionBasedHandling(=)",
                "conditionValueMapping": [
                    {
                        "conditionValue": "FINAL",
                        "fields": [
                            {
                                "sourcePath": "triggerType",
                                "targetPath": "sMFTrigger",
                                "operation": "stringToEnum(getTriggerTypeBy →
SessionType($-SV),triggerTypeMapping)"
                            }
                        ]
                    }
                ]
            }
        ]
    }

```



```
    ],
    {
      "conditionValue": "*",
      "fields": [
        {
          "sourcePath": "triggerType",
          "targetPath": "sMFTrigger",
          "operation": "ignoreUnknownEnumValue(string" →
ToEnum($-SV,triggerTypeMapping))"
        }
      ]
    }
  ]
},
{
  "sourcePath": "chargingData.multipleUnitUsage[]",
  "targetPath": "CHFRecord.chargingFunctionRecord.listOfMultipleUnitU →
sage[].MultipleUnitUsage",
  "operation": "arrayToSequenceOf",
  "groupingOperation": "merge",
  "fields": [
    {
      "sourcePath": "ratingGroup",
      "targetPath": "ratingGroup"
    },
    {
      "sourcePath": "multihomedPDUAddress.ipv4dynamicAddressFlag" →
,
      "targetPath": "multihomedPDUAddress.iPV4dynamicAddressFlag"
    },
    {
      "sourcePath": "multihomedPDUAddress.ipv6dynamicPrefixFlag",
      "targetPath": "multihomedPDUAddress.iPV6dynamicPrefixFlag"
    },
    {
      "sourcePath": "multihomedPDUAddress.pduIPv4Address",
      "targetPath": "multihomedPDUAddress.pDUIPv4Address.iPTextRe →
presentedAddress.iPTextV4Address"
    },
    {
      "sourcePath": "multihomedPDUAddress.pduIPv6AddresswithPrefi →
x",
      "targetPath": "multihomedPDUAddress.pDUIPv6AddresswithPrefi →
x.iPTextRepresentedAddress.iPTextV6Address"
    },
    {
      "sourcePath": "uPFID",
      "targetPath": "uPFID"
    }
  ],
}
```




```

{
  "sourcePath": "usedUnitContainer[]",
  "alternateSourcePath": "UsedUnitContainer[]",
  "targetPath": "usedUnitContainers[].UsedUnitContainer",
  "operation": "arrayToSequenceOf",
  "fields": [
    {
      "sourcePath": "localSequenceNumber",
      "targetPath": "localSequenceNumber"
    },
    {
      "sourcePath": "serviceId",
      "targetPath": "serviceIdentifier"
    },
    {
      "sourcePath": "quotaManagementIndicator",
      "targetPath": "quotaManagementIndicatorExt",
      "operation": "ignoreUnknownEnumValue(stringToEnum($
-SV,quotaManagementIndicatorMapping))"
    },
    {
      "sourcePath": ".",
      "operation": "conditionBasedHandling(=)",
      "conditionValueSource": "quotaManagementIndicator",
      "conditionValueMapping": [
        {
          "conditionValue": "ONLINE_CHARGING",
          "fields": [
            {
              "targetPath": "quotaManagementIndic
ator",
              "operation": "toBoolean(true)"
            }
          ]
        },
        {
          "conditionValue": "OFFLINE_CHARGING",
          "fields": [
            {
              "targetPath": "quotaManagementIndic
ator",
              "operation": "toBoolean(false)"
            }
          ]
        }
      ]
    }
  ],
  {
    "sourcePath": "triggerTimestamp",
    "targetPath": "triggerTimeStamp",
    "operation": "toHexString(getTimeStampOctetArray($-
SV))"
  }
}

```



```

    },
    {
      "sourcePath": "totalVolume",
      "targetPath": "dataTotalVolume"
    },
    {
      "sourcePath": "uplinkVolume",
      "targetPath": "dataVolumeUplink"
    },
    {
      "sourcePath": "downlinkVolume",
      "targetPath": "dataVolumeDownlink"
    },
    {
      "sourcePath": "serviceSpecificUnits",
      "targetPath": "serviceSpecificUnits"
    },
    {
      "sourcePath": "eventTimeStamps[]",
      "targetPath": "eventTimeStamp",
      "operation": "toHexString(getTimeStampOctetArray($- →
SV))"
    },
    {
      "sourcePath": "time",
      "targetPath": "time"
    },
    {
      "sourcePath": "triggers[]",
      "targetPath": "triggers[].Trigger",
      "operation": "arrayToSequenceOf",
      "fields": [
        {
          "sourcePath": ".",
          "conditionValueSource": "triggerType",
          "operation": "conditionBasedHandling(=)",
          "conditionValueMapping": [
            {
              "conditionValue": "QUOTA_EXHAUSTED" →
,
              "fields": [
                {
                  "sourcePath": ".",
                  "operation": "conditionBase →
dHandling(!null)",
                  "conditionValueMapping": [
                    {
                      "conditionValueSource" →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].totalVolume",
                      "conditionValueAlte →
rnateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].totalVolum
e",

```



```

"conditionValue": t →
rue,
    "fields": [
        {
            "targetPath" →
            "operation" →
        }
    ]
},
{
    "conditonValueSource" →
    "conditionValueAlte" →
    "rateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].uplinkVolu" →
    "me",
    "conditionValue": t →
    rue,
        "fields": [
            {
                "targetPath" →
                "operation" →
            }
        ]
    },
    {
        "conditonValueSource" →
        "conditionValueAlte" →
        "rateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].downlinkVo" →
        "lume",
        "conditionValue": t →
        rue,
            "fields": [
                {
                    "targetPath" →
                    "operation" →
                }
            ]
        },
        {
            "conditonValueSource" →
            "conditionValueAlte" →
            "rateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[]." →
            "time",
            "conditionValue": t →
            rue,

```



```

                                "fields": [
                                    {
                                        "targetPath" →
                                        "operation" →
                                    }
                                ]
                                },
                                {
                                    "conditonValueSourc →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].serviceSpecificUnits →
",
                                    "conditionValueAlte →
rnateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].serviceSpe →
cificUnits",
                                    "conditionValue": t →
rue,
                                    "fields": [
                                        {
                                            "targetPath" →
                                            "operation" →
                                        }
                                    ]
                                }
                            ]
                        },
                        {
                            "conditionValue": "QUOTA_THRESHOLD" →
                            "fields": [
                                {
                                    "sourcePath": ".",
                                    "operation": "conditionBase →
                                    "conditionValueMapping": [
                                        {
                                            "conditonValueSourc →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].totalVolume",
                                            "conditionValueAlte →
rnateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].totalVolum →
e",
                                            "conditionValue": t →
rue,
                                            "fields": [
                                                {
                                                    "targetPath" →

```



```

                                "operation" →
: "toInt(401)"
                                }
                                ]
                                },
                                {
                                    "conditonValueSource" →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].uplinkVolume",
                                    "conditionValueAlte" →
rinateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].uplinkVolu →
me",
                                    "conditionValue": t →
rue,
                                    "fields": [
                                        {
                                            "targetPath" →
": "sMFTrigger",
                                            "operation" →
: "toInt(401)"
                                        }
                                    ]
                                },
                                {
                                    "conditonValueSource" →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].downlinkVolume",
                                    "conditionValueAlte" →
rinateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].downlinkVo →
lume",
                                    "conditionValue": t →
rue,
                                    "fields": [
                                        {
                                            "targetPath" →
": "sMFTrigger",
                                            "operation" →
: "toInt(401)"
                                        }
                                    ]
                                },
                                {
                                    "conditonValueSource" →
e": ".chargingData.multipleUnitUsage[].usedUnitContainer[].time",
                                    "conditionValueAlte" →
rinateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].time",
                                    "conditionValue": t →
rue,
                                    "fields": [
                                        {
                                            "targetPath" →
": "sMFTrigger",
                                            "operation" →
: "toInt(400)"
                                        }
                                    ]
                                }
                            ],
                        },
                    ],
                },
            ],
        ],
    ],
}

```



```
    }
  ],
  },
  {
    "conditionValueSource": ".chargingData.multipleUnitUsage[].usedUnitContainer[].serviceSpecificUnits",
    "conditionValueAlternateSource": ".chargingData.multipleUnitUsage[].UsedUnitContainer[].serviceSpecificUnits",
    "conditionValue": true,
    "fields": [
      {
        "targetPath": "sMFTrigger",
        "operation": "toInt(402)"
      }
    ]
  }
],
},
{
  "conditionValue": "*",
  "fields": [
    {
      "sourcePath": "triggerType",
      "targetPath": "sMFTrigger",
      "operation": "defaultValue(stringToEnum(getTriggerTypeBySessionType($-SV),triggerTypeMapping),507)"
    }
  ]
}
],
},
{
  "sourcePath": "quotaManagementIndicator",
  "targetPath": "ratingIndicator",
  "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,ratingIndicatorMapping))"
},
{
  "sourcePath": "pDUContainerInformation.chargingRuleBaseName",
  "targetPath": "pDUContainerInformation.chargingRuleBaseName"
}
```



```

    },
    {
      "sourcePath": "pDUCContainerInformation.afChargingId →
entifier",
      "targetPath": "pDUCContainerInformation.afChargingId →
entifier"
    },
    {
      "sourcePath": "pDUCContainerInformation.rATType",
      "targetPath": "pDUCContainerInformation.rATType",
      "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,raTypeMapping))"
    },
    {
      "sourcePath": "pDUCContainerInformation.sponsorIdent →
ity",
      "targetPath": "pDUCContainerInformation.sponsorIdent →
ity",
      "operation": "toHexString($-SV)"
    },
    {
      "sourcePath": "pDUCContainerInformation.applications →
erviceProviderIdentity",
      "targetPath": "pDUCContainerInformation.applicationS →
erviceProviderIdentity",
      "operation": "toHexString($-SV)"
    },
    {
      "sourcePath": "pDUCContainerInformation.userLocation →
Information",
      "targetPath": "pDUCContainerInformation.userLocation →
Information",
      "operation": "toHexString(getBytes($-SV))"
    },
    {
      "sourcePath": "pDUCContainerInformation.ueTimeZone",
      "targetPath": "pDUCContainerInformation.uETimeZone",
      "operation": "convertUETimezoneToTwoOctets($-SV)"
    },
    {
      "sourcePath": "pDUCContainerInformation.servingNodeI →
D[]",
      "targetPath": "pDUCContainerInformation.servingNetwo →
rkFunctionID[].ServingNetworkFunctionID",
      "operation": "arrayToSequenceOf",
      "fields": [
        {
          "sourcePath": "aMFId",
          "targetPath": "aMFIdentifier"
        },
        {
          "sourcePath": "servingNetworkFunctionInform →

```



```
ation.nodeFunctionality",
                                "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionality",
                                "operation": "stringToEnum($-SV,networkFunc →
tionalityMapping)"
                                },
                                {
                                    "sourcePath": "servingNetworkFunctionInform →
ation.nFIPv4Address",
                                    "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
                                },
                                {
                                    "sourcePath": "servingNetworkFunctionInform →
ation.nFIPv6Address",
                                    "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionIPv6Address.iPTextRepresentedAddress.iPTextV6Address"
                                },
                                {
                                    "sourcePath": "servingNetworkFunctionInform →
ation.nFPLMNID",
                                    "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionPLMNIdentifier",
                                    "operation": "toHexString(generatePLMN_ID($ →
-SV))"
                                },
                                {
                                    "sourcePath": "servingNetworkFunctionInform →
ation.nFFqdn",
                                    "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionFQDN.domainName"
                                },
                                {
                                    "sourcePath": "servingNetworkFunctionInform →
ation.nFName",
                                    "targetPath": "servingNetworkFunctionInform →
ation.networkFunctionName"
                                }
                            ]
                        },
                        {
                            "sourcePath": "pDUContainerInformation.timeofFirstU →
sage",
                            "targetPath": "pDUContainerInformation.timeOfFirstU →
sage",
                            "operation": "toHexString(getTimeStampOctetArray($- →
SV))"
                        },
                        {
                            "sourcePath": "pDUContainerInformation.timeofLastUs →
age",
                            "targetPath": "pDUContainerInformation.timeOfLastUs →
```




```

age",
                                "operation": "toHexString(getTimeStampOctetArray($- →
SV))"
                                },
                                {
on.5qi",
                                "sourcePath": "pDUContainerInformation.qoSInformati →
on.fiveQi"
                                "targetPath": "pDUContainerInformation.qoSInformati →
                                },
                                {
on.qnc",
                                "sourcePath": "pDUContainerInformation.qoSInformati →
on.qoSNotificationControl"
                                "targetPath": "pDUContainerInformation.qoSInformati →
                                },
                                {
on.reflectiveQos",
                                "sourcePath": "pDUContainerInformation.qoSInformati →
on.reflectiveQos"
                                "targetPath": "pDUContainerInformation.qoSInformati →
                                },
                                {
on.maxbrUL",
                                "sourcePath": "pDUContainerInformation.qoSInformati →
on.maxbitrateUL",
                                "targetPath": "pDUContainerInformation.qoSInformati →
                                "operation": "toHexString($-SV) "
                                },
                                {
on.maxbrDL",
                                "sourcePath": "pDUContainerInformation.qoSInformati →
on.maxbitrateDL",
                                "targetPath": "pDUContainerInformation.qoSInformati →
                                "operation": "toHexString($-SV) "
                                },
                                {
on.gbrUL",
                                "sourcePath": "pDUContainerInformation.qoSInformati →
on.guaranteedbitrateUL",
                                "targetPath": "pDUContainerInformation.qoSInformati →
                                "operation": "toHexString($-SV) "
                                },
                                {
on.gbrDL",
                                "sourcePath": "pDUContainerInformation.qoSInformati →
on.guaranteedbitrateDL",
                                "targetPath": "pDUContainerInformation.qoSInformati →
                                "operation": "toHexString($-SV) "
                                },
                                {
                                "sourcePath": "pDUContainerInformation.qoSInformati →

```



```

on.priorityLevel",
on.priorityLevel"
    },
    {
        "sourcePath": "pDUContainerInformation.qoSInformati →
on.averWindow",
on.averWindow"
        "targetPath": "pDUContainerInformation.qoSInformati →
    },
    {
        "sourcePath": "pDUContainerInformation.qoSInformati →
on.maxDataBurstVol",
on.maxDataBurstVol"
        "targetPath": "pDUContainerInformation.qoSInformati →
    },
    {
        "sourcePath": "pDUContainerInformation.qoSInformati →
on.maxPacketLossRateDl",
on.maxPacketLossRateDL "
        "targetPath": "pDUContainerInformation.qoSInformati →
    },
    {
        "sourcePath": "pDUContainerInformation.qoSInformati →
on.maxPacketLossRateUl",
on.maxPacketLossRateUL "
        "targetPath": "pDUContainerInformation.qoSInformati →
    },
    {
        "sourcePath": "pDUContainerInformation.presenceRepo →
rtingAreaInformation{}",
rtingAreaInfo",
        "targetPath": "pDUContainerInformation.presenceRepo →
        "operation": "mapToSequence",
        "fields": [
            {
                "sourcePath": "praId",
                "targetPath": "presenceReportingAreaIdentif →
ier",
                "operation": "toHexString($-SV)"
            },
            {
                "sourcePath": "presenceState",
                "targetPath": "presenceReportingAreaStatus" →
            },
            {
                "operation": "ignoreUnknownEnumValue(string →
ToEnum($-SV,presenceStateMapping))"
            }
        ]
    },
    {
        "sourcePath": "pDUContainerInformation.presenceRepo →

```



```

    "presenceReportingAreaInformation",
      "targetPath": "pDUContainerInformation.presenceReportingAreaInfo.presenceReportingAreaElementsList",
      "operation": "toHexString(getBytes(toJSONString(getElementsFromMap($-SV,ncgiList,trackingAreaList,globalRanNodeIdList,ecgiList,globalNbIdList))))"
    },
    {
      "sourcePath": "pDUContainerInformation.presenceReportingAreaInformation",
      "targetPath": "pDUContainerInformation.presenceReportingAreaInfo.presenceReportingAreaNode",
      "operation": "toString(10)"
    },
    {
      "sourcePath": "pDUContainerInformation.qoSInformation.on.ar.p.priorityLevel",
      "targetPath": "pDUContainerInformation.qoSInformation.on.aRP.priorityLevel"
    },
    {
      "sourcePath": "pDUContainerInformation.qoSInformation.on.ar.p.preemptCap",
      "targetPath": "pDUContainerInformation.qoSInformation.on.aRP.preemptionCapability",
      "operation": "stringToEnum($-SV,preemptCapMapping)"
    },
    {
      "sourcePath": "pDUContainerInformation.qoSInformation.on.ar.p.preemptVuln",
      "targetPath": "pDUContainerInformation.qoSInformation.on.aRP.preemptionVulnerability",
      "operation": "stringToEnum($-SV,preemptVulnMapping)"
    },
    {
      "sourcePath": "pDUContainerInformation.3gppPSDataOffStatus",
      "targetPath": "pDUContainerInformation.threeGPPPSDataOffStatus",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,3gppPSDataOffStatusMapping))"
    },
    {
      "sourcePath": "pDUContainerInformation.qoSCharacteristics",
      "targetPath": "pDUContainerInformation.qoSCharacteristics",
      "operation": "toHexString(getBytes($-SV))"
    },
    {
      "sourcePath": "pDUContainerInformation.mAPDUSTeering"
    }
  ]
}

```



```
gFunctionality",
    "targetPath": "pDUContainerInformation.mAPDUSteerin →
gFunctionality",
    "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,mAPDUSteeringFunctionalityMapping))"
    },
    {
    "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.3gLoad",
    "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.threegLoad"
    },
    {
    "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.active",
    "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.active",
    "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,activeMapping))"
    },
    {
    "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.prioAcc",
    "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.prioAcc",
    "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,activeMapping))"
    },
    {
    "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.standby",
    "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.standby",
    "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,activeMapping))"
    },
    {
    "sourcePath": "pDUContainerInformation.mAPDUSteerin →
gMode.steerModeValue",
    "targetPath": "pDUContainerInformation.mAPDUSteerin →
gMode.steerModeValue",
    "operation": "ignoreUnknownEnumValue(stringToEnum($ →
-SV,steerModeValueMapping))"
    },
    {
    "sourcePath": "pDUContainerInformation.afChargingId →
String",
    "targetPath": "pDUContainerInformation.afChargingId →
String"
    }
  ]
}
```



```

    ],
    {
      "sourcePath": "chargingData.roamingQBCInformation.uPFID",
      "targetPath": "CHFRecord.chargingFunctionRecord.roamingQBCInformati →
on.uPFID"
    },
    {
      "sourcePath": "chargingData.roamingQBCInformation.roamingChargingPr →
ofile.partialRecordMethod",
      "targetPath": "CHFRecord.chargingFunctionRecord.roamingQBCInformati →
on.roamingChargingProfile.partialRecordMethod",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,partialRecor →
dMethodEnumMapping))"
    },
    {
      "sourcePath": "chargingData.roamingQBCInformation.roamingChargingPr →
ofile.triggers[]",
      "targetPath": "CHFRecord.chargingFunctionRecord.roamingQBCInformati →
on.roamingChargingProfile.roamingTriggers[].RoamingTrigger",
      "operation": "arrayToSequenceOf",
      "fields": [
        {
          "sourcePath": ".",
          "conditonValueSource": "triggerType",
          "operation": "conditionBasedHandling(=)",
          "conditionValueMapping": [
            {
              "conditionValue": "FINAL",
              "fields": [
                {
                  "sourcePath": "triggerType",
                  "targetPath": "trigger",
                  "operation": "stringToEnum(getTriggerTypeBy →
SessionType($-SV),triggerTypeMapping)"
                }
              ]
            }
          ],
          "conditionValue": "*",
          "fields": [
            {
              "sourcePath": "triggerType",
              "targetPath": "trigger",
              "operation": "ignoreUnknownEnumValue(string →
ToEnum($-SV,triggerTypeMapping))"
            }
          ]
        }
      ]
    }
  ],
  {
    {

```



```

        "sourcePath": "triggerCategory",
        "targetPath": "triggerCategory",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,triggerCategoryEnumMapping))"
      },
      {
        "sourcePath": "timeLimit",
        "targetPath": "timeLimit"
      },
      {
        "sourcePath": "volumeLimit64",
        "targetPath": "volumeLimit"
      },
      {
        "sourcePath": "maxNumberOfccc",
        "targetPath": "maxNbChargingConditions"
      }
    ]
  },
  {
    "sourcePath": "chargingData.roamingQBCInformation.multipleQFIcontainer[]",
    "targetPath": "CHFRecord.chargingFunctionRecord.roamingQBCInformation.multipleQFIcontainer[].MultipleQFIContainer",
    "operation": "arrayToSequenceOf",
    "groupingOperation": "merge",
    "fields": [
      {
        "sourcePath": "qFIContainerInformation.qFI",
        "targetPath": "qosFlowId"
      },
      {
        "sourcePath": "triggers[]",
        "targetPath": "triggers[].Trigger",
        "operation": "arrayToSequenceOf",
        "fields": [
          {
            "sourcePath": ".",
            "conditionValueSource": "triggerType",
            "operation": "conditionBasedHandling(=)",
            "conditionValueMapping": [
              {
                "conditionValue": "FINAL",
                "fields": [
                  {
                    "sourcePath": "triggerType",
                    "targetPath": "sMFTTrigger",
                    "operation": "stringToEnum(getTriggerTypeBySessionType($-SV),triggerTypeMapping)"
                  }
                ]
              }
            ]
          }
        ]
      }
    ]
  },

```



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        {
            "conditionValue": "*",
            "fields": [
                {
                    "sourcePath": "triggerType",
                    "targetPath": "sMFTrigger",
                    "operation": "defaultValue(stringTo →
Enum($-SV,triggerTypeMapping),507)"
                }
            ]
        }
    ],
    {
        {
            "sourcePath": "triggerTimestamp",
            "targetPath": "triggerTimeStamp",
            "operation": "toHexString(getTimeStampOctetArray($-SV))"
        },
        {
            "sourcePath": "totalVolume",
            "targetPath": "dataTotalVolume"
        },
        {
            "sourcePath": "uplinkVolume",
            "targetPath": "dataVolumeUplink"
        },
        {
            "sourcePath": "downlinkVolume",
            "targetPath": "dataVolumeDownlink"
        },
        {
            "sourcePath": "time",
            "targetPath": "time"
        },
        {
            "sourcePath": "localSequenceNumber",
            "targetPath": "localSequenceNumber"
        },
        {
            "sourcePath": "qFIContainerInformation.timeofFirstUsage",
            "targetPath": "timeOfFirstUsage",
            "operation": "toHexString(getTimeStampOctetArray($-SV))"
        },
        {
            "sourcePath": "qFIContainerInformation.reportTime",
            "targetPath": "reportTime",
            "operation": "toHexString(getTimeStampOctetArray($-SV))"
        },
        {
            "sourcePath": "qFIContainerInformation.timeofLastUsage",

```



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        "targetPath": "timeOfLastUsage",
        "operation": "toHexString(getTimeStampOctetArray($-SV))"
      },
      {
        "sourcePath": "qFIContainerInformation.qoSInformation.5qi",
        "targetPath": "qoSInformation.fiveQi"
      },
      {
        "sourcePath": "qFIContainerInformation.qoSInformation.arp.p →
priorityLevel",
        "targetPath": "qoSInformation.aRP.priorityLevel"
      },
      {
        "sourcePath": "qFIContainerInformation.qoSInformation.arp.p →
reemptCap",
        "targetPath": "qoSInformation.aRP.preemptionCapability",
        "operation": "stringToEnum($-SV,preemptCapMapping)"
      },
      {
        "sourcePath": "qFIContainerInformation.qoSInformation.arp.p →
reemptVuln",
        "targetPath": "qoSInformation.aRP.preemptionVulnerability",
        "operation": "stringToEnum($-SV,preemptVulnMapping)"
      },
      {
        "sourcePath": "qFIContainerInformation.qoSInformation.qnc",
        "targetPath": "qoSInformation.qoSNotificationControl"
      },
      {
        "sourcePath": "qFIContainerInformation.qoSInformation.refle →
ctiveQos",
        "targetPath": "qoSInformation.reflectiveQos"
      },
      {
        "sourcePath": "qFIContainerInformation.qoSInformation.maxbr →
UL",
        "targetPath": "qoSInformation.maxbitrateUL",
        "operation": "toHexString($-SV)"
      },
      {
        "sourcePath": "qFIContainerInformation.qoSInformation.maxbr →
DL",
        "targetPath": "qoSInformation.maxbitrateDL",
        "operation": "toHexString($-SV)"
      },
      {
        "sourcePath": "qFIContainerInformation.qoSInformation.gbrUL →
",
        "targetPath": "qoSInformation.guaranteedbitrateUL",
        "operation": "toHexString($-SV)"
      },
      {
```




```

",
    {
        "sourcePath": "qFIContainerInformation.qoSInformation.gbrDL",
        "targetPath": "qoSInformation.guaranteedbitrateDL",
        "operation": "toHexString($-SV)"
    },
    {
        "sourcePath": "qFIContainerInformation.qoSInformation.prior",
        "targetPath": "qoSInformation.priorityLevel"
    },
    {
        "sourcePath": "qFIContainerInformation.qoSInformation.averW",
        "targetPath": "qoSInformation.averWindow"
    },
    {
        "sourcePath": "qFIContainerInformation.qoSInformation.maxDa",
        "targetPath": "qoSInformation.maxDataBurstVol"
    },
    {
        "sourcePath": "qFIContainerInformation.qoSInformation.maxPa",
        "targetPath": "qoSInformation.maxPacketLossRateDL"
    },
    {
        "sourcePath": "qFIContainerInformation.qoSInformation.maxPa",
        "targetPath": "qoSInformation.maxPacketLossRateUL"
    },
    {
        "sourcePath": "qFIContainerInformation.userLocationInformat",
        "targetPath": "userLocationInformation",
        "operation": "toHexString(getBytes($-SV))"
    },
    {
        "sourcePath": "qFIContainerInformation.uetimeZone",
        "targetPath": "uETimeZone",
        "operation": "convertUETimezoneToTwoOctets($-SV)"
    },
    {
        "sourcePath": "qFIContainerInformation.presenceReportingAre",
        "targetPath": "presenceReportingAreaInfo",
        "operation": "mapToSequence",
        "fields": [
            {
                "sourcePath": "praId",
                "targetPath": "presenceReportingAreaIdentifier",
                "operation": "toHexString($-SV)"
            },
        ],
    },
    "aInformation{}"

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        {
            "sourcePath": "presenceState",
            "targetPath": "presenceReportingAreaStatus",
            "operation": "ignoreUnknownEnumValue(stringToEnum($
-SV,presenceStateMapping)))"
        }
    ]
},
{
    "sourcePath": "qFIContainerInformation.presenceReportingAre
aInformation",
    "targetPath": "presenceReportingAreaInfo.presenceReportingA
reaElementsList",
    "operation": "toHexString(getBytes(toJSONString(getElements
FromMap($-SV,ncgiList,trackingAreaList,globalRanNodeIdList,ecgiList,globaleNbId
List))))"
},
{
    "sourcePath": "qFIContainerInformation.presenceReportingAre
aInformation",
    "targetPath": "presenceReportingAreaInfo.presenceReportingA
reaNode",
    "operation": "toString(10)"
},
{
    "sourcePath": "qFIContainerInformation.rATType",
    "targetPath": "rATType",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,raTy
peMapping)))"
},
{
    "sourcePath": "qFIContainerInformation.servingNetworkFuncTi
onID[]",
    "targetPath": "servingNetworkFunctionID[].ServingNetworkFun
ctionID",
    "operation": "arrayToSequenceOf",
    "fields": [
        {
            "sourcePath": "aMFId",
            "targetPath": "aMFIdentifier"
        },
        {
            "sourcePath": "servingNetworkFunctionInformation.no
deFunctionality",
            "targetPath": "servingNetworkFunctionInformation.ne
tworkFunctionality",
            "operation": "stringToEnum($-SV,networkFunctionalit
yMapping)"
        },
        {
            "sourcePath": "servingNetworkFunctionInformation.nF
IPv4Address",

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        "targetPath": "servingNetworkFunctionInformation.nf
workFunctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
    },
    {
        "sourcePath": "servingNetworkFunctionInformation.nf
IPv6Address",
        "targetPath": "servingNetworkFunctionInformation.nf
workFunctionIPv6Address.iPTextRepresentedAddress.iPTextV6Address"
    },
    {
        "sourcePath": "servingNetworkFunctionInformation.nf
PLMNID",
        "targetPath": "servingNetworkFunctionInformation.nf
workFunctionPLMNIdentifier",
        "operation": "toHexString(generatePLMN_ID($-SV))"
    },
    {
        "sourcePath": "servingNetworkFunctionInformation.nf
Fqdn",
        "targetPath": "servingNetworkFunctionInformation.nf
workFunctionFQDN.domainName"
    },
    {
        "sourcePath": "servingNetworkFunctionInformation.nf
Name",
        "targetPath": "servingNetworkFunctionInformation.nf
workFunctionName"
    }
]
},
{
    "sourcePath": "qFIContainerInformation.3gppPSDataOffStatus"
,
    "targetPath": "threeGPPPSDataOffStatus",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,3gpp
PSDataOffStatusMapping))"
},
{
    "sourcePath": "qFIContainerInformation.3gppChargingId",
    "targetPath": "threeGPPChargingID"
},
{
    "sourcePath": "qFIContainerInformation.enhancedDiagnostics[
]",
    "targetPath": "extensionDiagnostics.rANNASCause[.RANNASCau
se",
    "operation": "simpleArrayToSequenceOf(toHexString($-SV))"
},
{
    "sourcePath": "qFIContainerInformation.qoSCharacteristics",
    "targetPath": "qoSCharacteristics",
    "operation": "toHexString(getBytes($-SV))"
}

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    }
  ],
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nodeFunctional
ity",
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn
formation.networkFunctionality",
    "operation": "stringToEnum($-SV,networkFunctionalityMapping)"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFIPv4Address" →
    ,
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFIPv6Address" →
    ,
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionIPv6Address.iPTextRepresentedAddress.iPTextV6Address"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFPLMNID",
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionPLMNIdentifier",
    "operation": "toHexString(generatePLMN_ID($-SV))"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFFqdn",
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionFQDN.domainName"
  },
  {
    "sourcePath": "chargingData.nfConsumerIdentification.nFName",
    "targetPath": "CHFRecord.chargingFunctionRecord.nFunctionConsumerIn →
formation.networkFunctionName"
  },
  {
    "sourcePath": "requestInformation.sessionOperationType",
    "targetPath": "CHFRecord.chargingFunctionRecord.causeForRecClosing" →
    ,
    "operation": "intToEnum($-SV,causeForRecClosingMapping)",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.chargingI →
d",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUSessionChargingID"
  },
  {
```



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        "sourcePath": "chargingData.pDUSessionChargingInformation.homeProvidedChargingId",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.homeProvidedChargingID",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.mAPDUSessionInformation.mAPDUSessionIndicator",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.mAPDUSessionInformation.mAPDUSessionIndicator",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,mAPDUSessionIndicatorMapping))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.mAPDUNon3GPPUserLocationInfo",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.mAPDUNonThreeGPPUserLocationInfo",
        "operation": "toHexString(getBytes($-SV))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.enhancedDiagnostics[]",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.enhancedDiagnostics.rANNASRelCause[].RANNASRelCause",
        "operation": "arrayToSequenceOf",
        "groupingOperation": "last",
        "fields": [
            {
                "sourcePath": "ngApCause.group",
                "targetPath": "ngApCause.group"
            },
            {
                "sourcePath": "ngApCause.value",
                "targetPath": "ngApCause.value"
            },
            {
                "sourcePath": "5gMmCause",
                "targetPath": "fivegMmCause"
            },
            {
                "sourcePath": "5gSmCause",
                "targetPath": "fivegSmCause"
            },
            {
                "sourcePath": "epsCause",
                "targetPath": "epsCause",
                "operation": "toHexString($-SV)"
            }
        ]
    }

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    ],
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.userInfor →
mation",
      "operation": "conditionBasedHandling(^)",
      "conditonValueSource": "servedGPSI",
      "conditionValueMapping": [
        {
          "conditionValue": "msisdn",
          "fields": [
            {
              "sourcePath": "servedGPSI",
              "targetPath": "CHFRecord.chargingFunctionRecord.pDU →
SessionChargingInformation.userIdentifier.iSDN-E164",
              "operation": "splitStringWithRegex($-SV,-,1)"
            }
          ]
        },
        {
          "conditionValue": "extid",
          "fields": [
            {
              "sourcePath": "servedGPSI",
              "targetPath": "CHFRecord.chargingFunctionRecord.pDU →
SessionChargingInformation.userIdentifier.externalId",
              "operation": "splitStringWithRegex($-SV,-,1)"
            }
          ]
        }
      ]
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.userInfor →
mation.servedPEI",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.userEquipmentInfo.subscriberEquipmentNumberType",
      "operation": "stringToEnum(splitStringWithRegex($-SV,-,0),servedPEI →
Mapping)"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.userInfor →
mation.servedPEI",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.userEquipmentInfo.subscriberEquipmentNumberData",
      "operation": "toHexString(splitStringWithRegex($-SV,-,1))"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.userInfor →
mation",
      "operation": "conditionBasedHandling(=)",
      "conditonValueSource": "unauthenticatedFlag",
```



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        "conditionValueMapping": [
            {
                "conditionValue": true,
                "fields": [
                    {
                        "targetPath": "CHFRecord.chargingFunctionRecord.pDU →
SessionChargingInformation.sUPIunauthenticatedFlag",
                        "operation": "setNULL()"
                    }
                ]
            }
        ],
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.userInfor →
mation.roamerInOut",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.userRoamerInOut",
            "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,UserRoamerIn →
OutMapping))"
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.pduSessionID",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pduSessionId"
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.networkSlicingInfo.sNSSAI.sst",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.networkSliceInstanceID.sST"
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.networkSlicingInfo.sNSSAI.sd",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.networkSliceInstanceID.sd",
            "operation": "toHexString(hexToByte($-SV,3))"
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.pduType",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pduType",
            "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,pduTypeMappi →
ng))"
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.sscMode",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →

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information.sSCMode",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sSCModeMapping))"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.servingCNPlmnId",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.servingCNPLMNID",
    "operation": "toHexString(generatePLMN_ID($-SV))"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.hPlmnId",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.sUPIPLMNIdentifier",
    "operation": "toHexString(generatePLMN_ID($-SV))"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.mAPDUNon3GPPRAType",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.mAPDUNonThreeGPPRAType",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,raTypeMapping))",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.ratType",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.rATType",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,raTypeMapping))"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.dnnId",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.dataNetworkNameIdentifier",
    "operation": "splitStringWithRegex($-SV,.mnc,0)"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.dnnSelectionMode",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.dnnSelectionMode",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,dnnSelectionModeMapping))",
    "groupingOperation": "last"
  },
  {
```




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        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.pduAddress.pduIPv4Address",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUAddress.pDUIPV4Address.iPTextRepresentedAddress.iPTextV4Address"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.pduAddress.pduIPv6AddresswithPrefix",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUAddress.pDUIPV6AddresswithPrefix.iPTextRepresentedAddress.iPTextV →
6Address"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.pduAddress.iPv4dynamicAddressFlag",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUAddress.iPV4dynamicAddressFlag"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.pduAddress.iPv6dynamicPrefixFlag",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUAddress.iPV6dynamicPrefixFlag"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.5qi",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.authorizedQoSInformation.fiveQi"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.priorityLevel",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.authorizedQoSInformation.priorityLevel"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.averWindow",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.authorizedQoSInformation.averWindow"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.maxDataBurstVol",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.authorizedQoSInformation.maxDataBurstVol"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.subscribedQoSInformation.5qi",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →

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information.subscribedQoSInformation.fiveQi"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.subscribedQoSInformation.priorityLevel",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.subscribedQoSInformation.priorityLevel"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.authorizedSessionAMBR.uplink",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.authorizedSessionAMBR.ambrUL",
      "operation": "toHexString(getBytes($-SV))"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.authorizedSessionAMBR.downlink",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.authorizedSessionAMBR.ambrDL",
      "operation": "toHexString(getBytes($-SV))"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.subscribedSessionAMBR.uplink",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.subscribedSessionAMBR.ambrUL",
      "operation": "toHexString(getBytes($-SV))"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.subscribedSessionAMBR.downlink",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.subscribedSessionAMBR.ambrDL",
      "operation": "toHexString(getBytes($-SV))"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.ueTimeZone",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.ueTimeZone",
      "operation": "convertUETimezoneToTwoOctets($-SV)"
    },
    {
      "sourcePath": "chargingData.pDUSessionChargingInformation.rANSecondaryRATUsageReport",
      "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.rANSecondaryRATUsageReport[].NGRANSecondaryRATUsageReport",
      "operation": "elementToSequenceOf",
      "fields": [
        {
          "sourcePath": "rANSecondaryRATType",
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        "targetPath": "nGRANSecondaryRATType",
        "operation": "toHexString($-SV)"
    },
    {
        "sourcePath": "qosFlowsUsageReports[]",
        "targetPath": "qosFlowsUsageReports[].QosFlowsUsageReport",
        "operation": "arrayToSequenceOf",
        "fields": [
            {
                "sourcePath": "qFI",
                "targetPath": "qosFlowId"
            },
            {
                "sourcePath": "startTimestamp",
                "targetPath": "startTime",
                "operation": "toHexString(getTimeStampOctetArray($-SV))"
            },
            {
                "sourcePath": "endTimestamp",
                "targetPath": "endTime",
                "operation": "toHexString(getTimeStampOctetArray($-SV))"
            },
            {
                "sourcePath": "downlinkVolume",
                "targetPath": "dataVolumeDownlink"
            },
            {
                "sourcePath": "uplinkVolume",
                "targetPath": "dataVolumeUplink"
            }
        ]
    }
],
{
    "targetPath": "CHFRecord.chargingFunctionRecord.duration",
    "operation": "toInt(0)",
    "groupingOperation": "difference"
},
{
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.chargingCharacteristics",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingInformation.chargingCharacteristics",
    "operation": "paddingZerosForOctets($-SV,2)"
},
{
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessionInformation.chargingCharacteristicsSelectionMode",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI

```



```
information.chChSelectionMode",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,chChSelectio →
nModeMapping))"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.arp.priorityLevel",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.authorizedQoSInformation.aRP.priorityLevel"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.arp.preemptCap",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.authorizedQoSInformation.aRP.preemptionCapability",
        "operation": "stringToEnum($-SV,preemptCapMapping)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.authorizedQoSInformation.arp.preemptVuln",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.authorizedQoSInformation.aRP.preemptionVulnerability",
        "operation": "stringToEnum($-SV,preemptVulnMapping)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.subscribedQoSInformation.arp.priorityLevel",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.subscribedQoSInformation.aRP.priorityLevel"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.subscribedQoSInformation.arp.preemptCap",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.subscribedQoSInformation.aRP.preemptionCapability",
        "operation": "stringToEnum($-SV,preemptCapMapping)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.subscribedQoSInformation.arp.preemptVuln",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.subscribedQoSInformation.aRP.preemptionVulnerability",
        "operation": "stringToEnum($-SV,preemptVulnMapping)"
    },
    {
        "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.3gppPSDataOffStatus",
        "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nInformation.threeGPPPSDataOffStatus",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,3gppPSDataOf →
fStatusMapping))"
    },
    },
```



```

        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.userLocat →
ioninfo",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.userLocationInformation",
            "operation": "toHexString(getBytes($-SV))"
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.presenceR →
eportingAreaInformation{}",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.presenceReportingAreaInfo",
            "operation": "mapToSequence",
            "fields": [
                {
                    "sourcePath": "praId",
                    "targetPath": "presenceReportingAreaIdentifier",
                    "operation": "toHexString($-SV)"
                },
                {
                    "sourcePath": "presenceState",
                    "targetPath": "presenceReportingAreaStatus",
                    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,pres →
enceStateMapping))"
                }
            ]
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.startTime",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUSessionstartTime",
            "operation": "toHexString(getTimeStampOctetArray($-SV))"
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.stopTime",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.pDUSessionstopTime",
            "operation": "toHexString(getTimeStampOctetArray($-SV))"
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.presenceR →
eportingAreaInformation",
            "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.presenceReportingAreaInfo.presenceReportingAreaElementsList",
            "operation": "toHexString(getBytes(toJSONString(getElementsFromMap( →
$-SV,ncgilist,trackingAreaList,globalRanNodeIdList,ecgilist,globaleNbIdList)))) →
"
        },
        {
            "sourcePath": "chargingData.pDUSessionChargingInformation.presenceR →

```



```
reportingAreaInformation",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.presenceReportingAreaInfo.presenceReportingAreaNode",
    "operation": "toString(10)"
  },
  {
    "sourcePath": "chargingData.pDUSessionChargingInformation.pduSessio →
nInformation.servingNetworkFunctionID",
    "targetPath": "CHFRecord.chargingFunctionRecord.pDUSessionChargingI →
nformation.servingNetworkFunctionID[].ServingNetworkFunctionID",
    "operation": "elementToSequenceOf",
    "fields": [
      {
        "sourcePath": "aMFid",
        "targetPath": "aMFIdentifier"
      },
      {
        "sourcePath": "servingNetworkFunctionInformation.nFIPv4Addr →
ess",
        "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
      },
      {
        "sourcePath": "servingNetworkFunctionInformation.nodeFunc →
tionality",
        "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionality",
        "operation": "stringToEnum($-SV,networkFunctionalityMapping →
)"
      },
      {
        "sourcePath": "servingNetworkFunctionInformation.nFIPv4Addr →
ess",
        "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionIPv4Address.iPTextRepresentedAddress.iPTextV4Address"
      },
      {
        "sourcePath": "servingNetworkFunctionInformation.nFIPv6Addr →
ess",
        "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionIPv6Address.iPTextRepresentedAddress.iPTextV6Address"
      },
      {
        "sourcePath": "servingNetworkFunctionInformation.nFPLMNID",
        "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionPLMNIdentifier",
        "operation": "toHexString(generatePLMN_ID($-SV))"
      },
      {
        "sourcePath": "servingNetworkFunctionInformation.nFFqdn",
        "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionFQDN.domainName"
```



```

        },
        {
            "sourcePath": "servingNetworkFunctionInformation.nFName",
            "targetPath": "servingNetworkFunctionInformation.networkFun →
ctionName"
        }
    ]
},
{
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorSUPI",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorIMSI",
    "operation": "toHexString(parseTBCD(splitStringWithRegex($-SV,-,1)) →
)",
    "groupingOperation": "last"
},
{
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorGPSI",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorMSISDN",
    "operation": "convertToAddressString(splitStringWithRegex($-SV,-,1) →
)",
    "groupingOperation": "last"
},
{
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorOtherAddress.sMaddressType",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorOtherAddress.sMaddressType",
    "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMaddressTyp →
eMapping))",
    "groupingOperation": "last"
},
{
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorOtherAddress.sMaddressData",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorOtherAddress.sMaddressData",
    "groupingOperation": "last"
},
{
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorOtherAddress.sMaddressDomain.domainName",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorOtherAddress.sMaddressDomain.sMDomainName",
    "groupingOperation": "last"
},
{
    "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorOtherAddress.sMaddressDomain.3GPPIMSIMCCMNC",

```



```
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorOtherAddress.sMAddressDomain.threeGPPIMSI-MCC-MNC" →
    ,
        "operation": "toHexString(generatePLMN_ID($-SV))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorSCCPAddress",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorSCCPAddress",
        "operation": "convertToAddressString($-SV)",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorReceivedAddress.sMAddressType",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorReceivedAddress.sMAddressType",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMAddressTyp →
eMapping))"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorReceivedAddress.sMAddressData",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorReceivedAddress.sMAddressData",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorReceivedAddress.sMAddressDomain.domainName",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorReceivedAddress.sMAddressDomain.sMDomainName",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.o →
riginatorReceivedAddress.sMAddressDomain.3GPPIMSIMCCMNC",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.originatorReceivedAddress.sMAddressDomain.threeGPPIMSI-MCC-M →
NC",
        "operation": "toHexString(generatePLMN_ID($-SV))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorInterface.interfaceId",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorInterface.interfaceId",
        "groupingOperation": "last"
    },
    },
```




```

    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorInterface.interfacePort",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorInterface.interfacePort",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorInterface.interfaceText",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorInterface.interfaceText",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorInterface.interfaceType",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorInterface.interfaceType",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,SMInterfaceT →
ypeMapping))",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.originatorInfo.s →
MOriginatorProtocolId",
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.originatorInfo.sMOriginatorProtocolID",
      "operation": "toHexString($-SV)",
      "groupingOperation": "last"
    },
    {
      "sourcePath": "chargingData.sMSChargingInformation.recipientInfo[]" →
,
      "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.recipientInfos[].RecipientInfo",
      "operation": "arrayToSequenceOf",
      "groupingOperation": "last",
      "fields": [
        {
          "sourcePath": "recipientSUPI",
          "targetPath": "recipientIMSI",
          "operation": "toHexString(parseTBCD(splitStringWithRegex($- →
SV,-,1)))"
        },
        {
          "sourcePath": "recipientGPSI",
          "targetPath": "recipientMSISDN",
          "operation": "convertToAddressString(splitStringWithRegex($ →
-SV,-,1))"
        }
      ]
    }
  ]
}

```



```
dressTypeMapping))"
    },
    {
      "sourcePath": "recipientOtherAddress.sMaddressType",
      "targetPath": "recipientOtherAddress.sMaddressType",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMAd →
dressTypeMapping))"
    },
    {
      "sourcePath": "recipientOtherAddress.sMaddressData",
      "targetPath": "recipientOtherAddress.sMaddressData"
    },
    {
      "sourcePath": "recipientOtherAddress.sMaddressDomain.domain →
Name",
      "targetPath": "recipientOtherAddress.sMaddressDomain.sMDoma →
inName"
    },
    {
      "sourcePath": "recipientOtherAddress.sMaddressDomain.3GPPIM →
SIMCCMNC",
      "targetPath": "recipientOtherAddress.sMaddressDomain.threeG →
PPIMSI-MCC-MNC",
      "operation": "toHexString(generatePLMN_ID($-SV))"
    },
    {
      "sourcePath": "recipientSCCPAddress",
      "targetPath": "recipientSCCPAddress",
      "operation": "convertToAddressString($-SV)"
    },
    {
      "sourcePath": "recipientReceivedAddress.sMaddressType",
      "targetPath": "recipientReceivedAddress.sMaddressType",
      "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMAd →
dressTypeMapping))"
    },
    {
      "sourcePath": "recipientReceivedAddress.sMaddressData",
      "targetPath": "recipientReceivedAddress.sMaddressData"
    },
    {
      "sourcePath": "recipientReceivedAddress.sMaddressDomain.dom →
ainName",
      "targetPath": "recipientReceivedAddress.sMaddressDomain.sMD →
omainName"
    },
    {
      "sourcePath": "recipientReceivedAddress.sMaddressDomain.3GP →
PIMSIMCCMNC",
      "targetPath": "recipientReceivedAddress.sMaddressDomain.thr →
eeGPPIMSI-MCC-MNC",
      "operation": "toHexString(generatePLMN_ID($-SV))"
    },
    {
      "sourcePath": "sMDestinationInterface.interfaceId",
```



```

        "targetPath": "sMDestinationInterface.interfaceId"
      },
      {
        "sourcePath": "sMDestinationInterface.interfacePort",
        "targetPath": "sMDestinationInterface.interfacePort"
      },
      {
        "sourcePath": "sMDestinationInterface.interfaceText",
        "targetPath": "sMDestinationInterface.interfaceText"
      },
      {
        "sourcePath": "sMDestinationInterface.interfaceType",
        "targetPath": "sMDestinationInterface.interfaceType",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,SMIn →
terfaceTypeMapping))"
      },
      {
        "sourcePath": "sMrecipientProtocolId",
        "targetPath": "sMRecipientProtocolID",
        "operation": "toHexString($-SV)"
      }
    ]
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.userEquipmentInf →
o",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.userEquipmentInfo.subscriberEquipmentNumberType",
    "operation": "stringToEnum(splitStringWithRegex($-SV,-,0),servedPEI →
Mapping)",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.userEquipmentInf →
o",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.userEquipmentInfo.subscriberEquipmentNumberData",
    "operation": "toHexString(splitStringWithRegex($-SV,-,1))"
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.userLocationinfo →
",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.userLocationInformation",
    "operation": "toHexString(getBytes($-SV))",
    "groupingOperation": "last"
  },
  {
    "sourcePath": "chargingData.sMSChargingInformation.uetimeZone",
    "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.uETimeZone",
    "operation": "convertUETimezoneToTwoOctets($-SV)",

```



```
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.rATType",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.rATType",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,raTypeMappin →
g))",
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.sMSCAddress",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMSCAddress",
        "operation": "convertToAddressString($-SV)",
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.invocationTimeStamp",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.eventtimestamp",
        "operation": "toHexString(getTimeStampOctetArray($-SV))",
        "validateParentPath": "chargingData.sMSChargingInformation",
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.sMDataCodingSche →
me",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMDataCodingScheme",
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.sMMessageType",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMMessageType",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMMessageTyp →
eMapping))",
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.sMReplyPathReque →
sted",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMReplyPathRequested",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMReplyPathR →
equestedMapping))",
        "groupingOperation": "last"
      },
      {
        "sourcePath": "chargingData.sMSChargingInformation.sMUserDataHeader →
",
```



```

        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMUserDataHeader",
        "operation": "toHexString($-SV)",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.sMDischargeTime" →
,
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMDischargeTime",
        "operation": "toHexString(getTimeStampOctetArray($-SV))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.numberofMessages →
Sent",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMTotalNumber",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.sMServiceType",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMServiceType",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,sMServiceTyp →
eMapping))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.sMSequenceNumber →
",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMSequenceNumber",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.submissionTime",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.submissionTime",
        "operation": "toHexString(getTimeStampOctetArray($-SV))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.sMPriority",
        "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMPriority",
        "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,priorityType →
Mapping))",
        "groupingOperation": "last"
    },
    {
        "sourcePath": "chargingData.sMSChargingInformation.messageReference →

```



```
"",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.messageReference",
  "operation": "defaultValue(toHexString($-SV),0)",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.messageSize",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.messageSize",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.messageClass.cla →
ssIdentifier",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.messageClass",
  "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,classIdentif →
ierMapping))",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.sMSChargingInformation.deliveryReportRe →
quested",
  "targetPath": "CHFRecord.chargingFunctionRecord.sMSChargingInformat →
ion.sMdeliveryReportRequested",
  "operation": "ignoreUnknownEnumValue(stringToEnum($-SV,deliveryRepo →
rtRequestedMapping))",
  "groupingOperation": "last"
},
{
  "sourcePath": "chargingData.serviceSpecificationInfo",
  "targetPath": "CHFRecord.chargingFunctionRecord.serviceSpecificatio →
nInformation",
  "operation": "toHexString($-SV)",
  "groupingOperation": "last"
},
{
  "targetPath": "CHFRecord.chargingFunctionRecord.incompleteCDRIndica →
tion.initialLost",
  "operation": "optionalTargetField",
  "groupingOperation": "substitute($-initialLost)"
},
{
  "targetPath": "CHFRecord.chargingFunctionRecord.incompleteCDRIndica →
tion.updateLost",
  "operation": "optionalTargetField",
  "groupingOperation": "substitute($-updateLost)"
},
{
  "targetPath": "CHFRecord.chargingFunctionRecord.recordExtensions[1] →
.ManagementExtension",
```



```

"operation": "indexedArraytoSequence",
"fields": [
  {
    "targetPath": "identifier",
    "operation": "toString(1 2 826 0 1249 0 5 7 0 0)"
  },
  {
    "targetPath": "significance",
    "operation": "toBoolean(false)"
  },
  {
    "targetPath": "information[1].ProtocolEnhancement",
    "operation": "indexedArraytoSequence",
    "fields": [
      {
        "targetPath": "ieKey",
        "operation": "toString(apiVersionInUri)"
      },
      {
        "sourcePath": "apiVersionInUri",
        "targetPath": "ieRaw",
        "operation": "toHexString($-SV)"
      }
    ]
  }
]
}
]
}
}

```

3.6.1.7

download-metadata

The download-metadata command exports the transformation metadata available of the specified schema in zip format. Ensure to mention the source-name, source-version, target-name, target-version, source-api-version, and config-type.

Syntax

Command Deprecated in API v1.1: `beamctl pub-transform-config download-metadata --source-name <source-name> --source-version <source-version> --target-name <target-name> --target-version <target-version> --source-api-version <source-api-version> --config-type <config-type>`

Command Deprecated in API v2.0: `beamctl pub-transform-config-v2 download-metadata --source-name <source-name> --source-version <source-version> --target-name <target-name> --target-version`



```
<target-version> --source-api-version <source-api-version> --config-type <config-type>
```

Arguments

The command has no arguments.

Flags

Table 163 Flags

| Name | Description |
|--------------------|--------------------------------------|
| source-name | To export the source schema name. |
| source-version | To export the source schema version. |
| target-name | To export the target schema name. |
| target-version | To export the target schema version. |
| source-api-version | To export the source API version. |
| config-type | To export the config type. |

Example 133

Command Deprecated in API v1.1: `beamctl pub-transform-config download-metadata --source-name ChargingNetworkFunctionEvent --source-version 1 --target-name BssfChfCdrAsn1 --target-version 3 --source-api-version v2 --config-type ReferenceVersion`

Command Introduced in API v2.0: `beamctl pub-transform-config-v2 download-metadata --source-name ChargingNetworkFunctionEvent --source-version 1 --target-name BssfChfCdrAsn1 --target-version 3 --source-api-version v2 --config-type ReferenceVersion`

3.6.2 Publishing Destination Configuration

3.6.2.1 list-partitions

The `list-partitions` command is to list all the publishing endpoint configuration. Ensure to mention the customer-partition-id to list the configuration of the required partition.

Syntax

Command Deprecated in API v1.1: `beamctl pub-dest-config list-partitions`

Command Introduced in API v2.0:



```
beamctl pub-dest-config-v2 list-partitions
```

```
beamctl pub-dest-config-v2 list-partitions --customer-partition
<customer-partition>
```

Flag

Table 164 Flags

| Name | Description |
|--------------------|-----------------------|
| customer-partition | Customer Partition ID |

Example

Command Deprecated in API v1.1: `beamctl pub-dest-config list-partitions`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 list-partitions`

```
beamctl pub-dest-config-v2 list-partitions --customer-partition 1
```

Note: customer-partition flag takes a single valid customer partition value. Comma-Separated Values for customer-partition is not supported in 'list-partitions' command. Use 'get-config' command to download publishing configurations for more than one partition.

3.6.2.2

add-config

The add-config command is used to upload the configuration based on the customer partition. Customer partition is an optional flag and when given, the configuration belonging to the respective partition is uploaded. If not given, all the configurations of the partitions provided in the payload is uploaded.

Syntax

Command Deprecated in API v1.1: `beamctl pub-dest-config add-config --customer-partition <customer-partition> < payload`

Command Introduced in API v2.0:

```
beamctl pub-dest-config-v2 add-config --customer-partition
<customer-partition> < payload
```

```
beamctl pub-dest-config-v2 add-config < payload
```

Flag

Table 165 Flags

| Name | Description |
|--------------------|-----------------------|
| customer-partition | Customer Partition ID |



Example

Command Deprecated in API v1.1: `beamctl pub-dest-config add-config --customer-partition 1 < /home/admin/config.json`

Command Introduced in API v2.0:

`beamctl pub-dest-config-v2 add-config --customer-partition 1 < /home/admin/config.json`

`beamctl pub-dest-config-v2 add-config < /home/admin/config.json`

Example 134 JSON Format

```
{
  "sftp": {
    "commonConfig": {
      "sftpRetryAttempts": "0",
      "sftpConnectionTimeout": "5000",
      "sftpRetryInterval": "0",
      "sftpBinaryMode": "false",
      "sftpStrictHostKeyCheck": "no",
      "maxFileSizeInMb": "30",
      "maxLogSizeInMb": "2",
      "maxNoOfRecordsPerLogFile": "1",
      "maxNoOfRecordsPerFile": "1"
    },
    "partitionConfigs": [
      {
        "partitionId": "0",
        "secondaryDestination": {
          "sftpPortNumber": "22",
          "sftpHostname": "vmx-edm016.ete.ka.sw.ericsson.se",
          "sftpPassword": "<password>",
          "sftpDestinationFolder": "../var/tmp/sftpTestDirGlobal_sec/",
          "sftpErrorDestinationFolder": "../var/tmp/sftpTestDirGlobal_Error_ →
sec/",
          "sftpUserName": "admin"
        },
        "primaryDestination": {
          "sftpPortNumber": "22",
          "sftpHostname": "vmx-edm016.ete.ka.sw.ericsson.se",
          "sftpPassword": "<password>",
          "sftpDestinationFolder": "../var/tmp/sftpTestDirGloba/",
          "sftpErrorDestinationFolder": "../var/tmp/sftpTestDirGlobal_Error/ →
",
          "sftpUserName": "admin"
        }
      },
      {
        "partitionId": "1",
        "primaryDestination": {
```



```

    "sftpHostname": "vmx-edm016.ete.ka.sw.ericsson.se",
    "sftpUserName": "admin",
    "sftpPassword": "<password>",
    "sftpPortNumber": "22",
    "sftpDestinationFolder": "../..home/admin/sftpTestDir/",
    "sftpErrorDestinationFolder": "../..home/admin/sftpTestDir_Error/",
    "sftpChecksumDestinationFolder": "../..home/admin/Checksum",
    "sftpLogConfig": {
      "sftpLogUsername": "admin",
      "sftpLogHostname": "vmx-edm016.ete.ka.sw.ericsson.se",
      "sftpLogPassword": "<password>",
      "sftpLogPortNumber": "",
      "sftpLogDestinationFolder": "../..home/admin/sftpLogTestDir/"
    }
  },
  "secondaryDestination": {
    "sftpHostname": "vmx-edm017.ete.ka.sw.ericsson.se",
    "sftpUserName": "admin",
    "sftpPassword": "<password>",
    "sftpPortNumber": "22",
    "sftpDestinationFolder": "../..home/admin/sftpTestDir/",
    "sftpErrorDestinationFolder": "../..home/admin/sftpTestDir_Error/",
    "sftpChecksumDestinationFolder": "../..home/admin/Checksum",
    "sftpLogConfig": {
      "sftpLogUsername": "admin",
      "sftpLogHostname": "vmx-edm017.ete.ka.sw.ericsson.se",
      "sftpLogPassword": "<password>",
      "sftpLogPortNumber": "",
      "sftpLogDestinationFolder": "../..home/admin/sftpLogTestDir/"
    }
  }
}
]
},
"kafka": {
  "commonConfig": {
    "kafkaRetries": "3",
    "kafkaSecurityEnabled": "ENABLE",
    "connectionTimeout": "6"
  },
  "partitionConfigs": [
    {
      "partitionId": "1",
      "primaryDestination": {
        "kafkaHostName": "vmx-edm-132.ete.ka.sw.ericsson.se",
        "kafkaPort": "55555"
      },
      "secondaryDestination": {
        "kafkaHostName": "vmx-edm-132.ete.ka.sw.ericsson.se",
        "kafkaPort": "55555"
      }
    }
  ]
}

```



```
}  
}  
]
```

fileType Validation Rules

The following are the list of validations performed while importing JSON with `fileType` attribute:

- Same file type configuration cannot not be repeated.
- SFTP Generic Configuration (partition configuration without file type attribute) contains duplicates. It must be configured once per partition. User must not import the JSON without generic configuration.
- CDR file control block configurations are applicable only for CHAD and generic configuration.
- The newly introduced attribute `fileType` must be a valid one. It must be one of the Event Type or CDR Type or ASN Indicators as specified in the interface document. It must be case-sensitive. For more information on Event Data Object, see EDM Event Publishing File Interface 4.25.

AliasConfig Validation Rules

The following are the list of validations performed while importing JSON with alias configurations in it:

- Alias name must start with a \$ sign and cannot not contain any other special characters.
- The defined Alias name under `aliasConfigs` section must be used at least once in `partitionConfigs` section of the JSON file.
- Alias names defined in partition configuration section must be present in `aliasConfigs` section.
- The `aliasConfigs` section specified in the input JavaScript Object Notation file must not contain duplicate Alias Configurations.

Example 135 JSON Format with Alias and Filetype

```
{  
  "sftp": {  
    "commonConfig": {  
      "sftpRetryAttempts": 0,  
      "sftpRetryInterval": 5000,  
      "maxFileSizeInMb": 30,  
      "maxNoOfRecordsPerFile": "1",  
      "sftpBinaryMode": "false",  
      "sftpStrictHostKeyCheck": "yes",  
    },  
  },  
}
```



```

    "maxNoOfRecordsPerLogFile": "1",
    "maxLogSizeInMb": 30,
    "sftpConnectionTimeout": "5000"
  },
  "aliasConfigs": [
    {
      "aliasName": "$primaryDestinationAliasCHAD",
      "sftpHostname": "seliicvm12152.bss-te.seli.gic.ericsson.se",
      "sftpUserName": "admin",
      "sftpPassword": "<password>",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "../..home/admin/sftpFileTypeTestDir/fileType →
CHADDir/",
      "sftpErrorDestinationFolder": "../..home/admin/sftpFileTypeErrorDir/fi →
leTypeCHADErrorDir/",
      "sftpChecksumDestinationFolder": "sftpGenericDestinationFolder/Checksum →
",
      "sftpLogConfig": {
        "sftpLogPassword": "<password>",
        "sftpLogPortNumber": "22",
        "sftpLogUsername": "admin",
        "sftpLogDestinationFolder": "../..home/admin/sftpFileTypeTestDir/fil →
eTypeCHADDir/logDir",
        "sftpLogHostname": "seliicvm12152.bss-te.seli.gic.ericsson.se"
      }
    },
    {
      "aliasName": "$primaryDestinationAliasCHF",
      "sftpHostname": "seliicvm12153.bss-te.seli.gic.ericsson.se",
      "sftpUserName": "admin",
      "sftpPassword": "<password>",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "../..home/admin/sftpFileTypeTestDir/fileType →
CHFDir/",
      "sftpErrorDestinationFolder": "../..home/admin/sftpFileTypeErrorDir/fi →
leTypeCHFErrorDir/"
    },
    {
      "aliasName": "$secondaryDestinationAlias",
      "sftpHostname": "seliicvm12153.bss-te.seli.gic.ericsson.se",
      "sftpUserName": "admin",
      "sftpPassword": "<password>",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "sftpFileTypeTestDir",
      "sftpErrorDestinationFolder": "sftpFileTypeErrorDir"
    }
  ],
  "partitionConfigs": [
    {
      "partitionId": "1",
      "fileType": "CHF",
      "primaryDestination": "$primaryDestinationAliasCHF",

```



```

    "secondaryDestination": "$secondaryDestinationAlias"
  },
  {
    "partitionId": "1",
    "fileType": "CHAD",
    "primaryDestination": "$primaryDestinationAliasCHAD",
    "secondaryDestination": "$secondaryDestinationAlias"
  },
  {
    "partitionId": "1",
    "primaryDestination": {
      "sftpHostname": "seliicvm12152.bss-te.seli.gic.ericsson.se",
      "sftpUserName": "admin",
      "sftpPassword": "<password>",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "sftpGenericDestinationFolder",
      "sftpErrorDestinationFolder": "sftpGenericErrorDestinationFolder",
      "sftpChecksumDestinationFolder": "sftpGenericDestinationFolder/Checks →
um",
      "sftpLogConfig": {
        "sftpLogPassword": "<password>",
        "sftpLogPortNumber": "22",
        "sftpLogUsername": "admin",
        "sftpLogDestinationFolder": "../home/admin/genericDestination/lo →
gtmp",
        "sftpLogHostname": "seliicvm12152.bss-te.seli.gic.ericsson.se"
      }
    },
    "secondaryDestination": {
      "sftpHostname": "seliicvm12153.bss-te.seli.gic.ericsson.se",
      "sftpUserName": "admin",
      "sftpPassword": "<password>",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "sftpGenericDestinationFolder",
      "sftpErrorDestinationFolder": "sftpGenericErrorDestinationFolder",
      "sftpChecksumDestinationFolder": "sftpGenericDestinationFolder/Checks →
um",
      "sftpLogConfig": {
        "sftpLogPassword": "<password>",
        "sftpLogPortNumber": "22",
        "sftpLogUsername": "admin",
        "sftpLogDestinationFolder": "../home/admin/genericDestination/lo →
gtmp",
        "sftpLogHostname": "seliicvm12153.bss-te.seli.gic.ericsson.se"
      }
    }
  }
]
},
"kafka": {
  "commonConfig": {
    "kafkaRetries": "3",

```



```

    "kafkaSecurityEnabled": "DISABLE",
    "connectionTimeout": "6"
  },
  "partitionConfigs": [
    {
      "partitionId": "1",
      "primaryDestination": {
        "kafkaHostName": "seliicvm12152.bss-te.seli.gic.ericsson.se",
        "kafkaPort": "55555"
      },
      "secondaryDestination": {
        "kafkaHostName": "seliicvm12153.bss-te.seli.gic.ericsson.se",
        "kafkaPort": "55555"
      }
    }
  ]
}

```

Note: All the mandatory parameters must be populated in the above JSON to configure any endpoint.

Example:

When configuring the SFTP endpoint, populate all the required parameters under the SFTP section with actual values and populate only the mandatory parameters under the KAFKA section with default values.

For more information on the SFTP and KAFKA configuration, see [EDM Application Configuration Function Guide](#).

3.6.2.3 add-type-config

The `add-type-config` command is used to upload the configuration based on the type value.

Type is a mandatory flag and uploads the configuration for the given external configuration type value.

Syntax

Command Deprecated in API v1.1: `beamctl pub-dest-config add-type-config --type <type> <payload>`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 add-type-config --type <type> <payload>`

Flag



Table 166 Flags

| Name | Description |
|------|-----------------------------|
| type | External Configuration type |

Example

Command Deprecated in API v1.1: `beamctl pub-dest-config add-type-config --type extConfig < /home/admin/config.json`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 add-type-config --type extConfig < /home/admin/config.json`

Example 136 Config JSON

```
[{
  "name": "CreditControlEvent",
  "namespace": "com.ericsson.bss.cha.event.schema",
  "eventTypeIndicator": "CR",
  "asnTypeIndicator": "occr",
  "checksum": "true"
}]
```

Example 137 Config JSON with subDirectory

```
[{
  "name": "CreditControlEvent",
  "namespace": "com.ericsson.bss.cha.event.schema",
  "eventTypeIndicator": "CR",
  "asnTypeIndicator": "CHAD",
  "subDirectory": "BssfChargingCdrAsn1",
  "checksum": "true"
}]
```

- Note:**
- "subDirectory" is an optional attribute and if configured, the CDRs is published to that directory in the destination server.
 - Base directory is used from publishing endpoint configuration.
 - For example, File for CCE events is generated in the destination server under `/<base directory>/BssfChargingCdrAsn1/EDMEventPublishing_ASN1BER_L1_eric-bss-edm-publishing-b7d4df9f5-r66n9_20230515-134528_2858-0-0_4_customer_1.CHAD`.

3.6.2.4

get-config

The `get-config` command is used to download all the publishing endpoint configuration and ensure to mention the customer-partition-id to get the configuration of the required partition.



Syntax

Command Deprecated in API v1.1: `beamctl pub-dest-config get-config --customer-partition <customer-partition>`

Command Introduced in API v2.0:

`beamctl pub-dest-config-v2 get-config --customer-partition <customer-partition>`

`beamctl pub-dest-config-v2 get-config`

Flag

Table 167 Flags

| Name | Description |
|--------------------|-----------------------|
| customer-partition | Customer Partition ID |

Note: If any alias configuration is present for the partition, then the configured aliases for the respective partition are listed.

Example

Command Deprecated in API v1.1: `beamctl pub-dest-config get-config --customer-partition 1`

Command Introduced in API v2.0:

`beamctl pub-dest-config-v2 get-config --customer-partition 1`

`beamctl pub-dest-config-v2 get-config`

3.6.2.5

get-type-config

The `get-type-config` command is used to download the configuration based on the type value. Type is a mandatory flag.

Syntax

Command Deprecated in API v1.1: `beamctl pub-dest-config get-type-config <type>`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 get-type-config <type>`

Flag

Table 168 Flags

| Name | Description |
|------|-----------------------------|
| type | External Configuration type |



Example

Command Deprecated in API v1.1: `beamctl pub-dest-config get-type-config extConfig`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 get-type-config extConfig`

3.6.2.6

delete-config

The `delete-config` command is used to delete the configuration from edmpub. Customer partition is a mandatory flag.

Syntax

Command Deprecated in API v1.1: `beamctl pub-dest-config delete-config <customer-partition>`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 delete-config <customer-partition> --file-type <file-type>`

Flag

Table 169 Flags

| Name | Description |
|--------------------|---|
| customer-partition | Customer Partition ID |
| file-type | Deletes the publishing configuration for the specified file type. |

Example

Command Deprecated in API v1.1: `beamctl pub-dest-config delete-config 1`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 delete-config 1 --file-type CHF`



- Note:**
- SI needs to give the customer-partition id to delete the configuration.
 - Filetype parameter input must be case-sensitive and it must be one of the event or asn indicators specified in the EDM Event Publishing interface. See section Event Data Object in EDM Event Publishing File Interface 4.23.
 - File type is an optional parameter and it is applicable only for API Version v2.0.
 - During deletion, if only the customer-partition parameter is provided and no file-type option is given, then the entire partition configuration along with all the filetype configurations in it gets deleted.

3.6.2.7

Retransmit

The put command is used to retransmit public keys. Customer partition id and Sftp destination type are mandatory flags.

Syntax

Command Deprecated in API v1.1: `beamctl pub-dest-config retransmit-key <customer-partition> <sftp-destination-type>`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 retransmit-key <customer-partition> <sftp-destination-type>`

Flag

Table 170 Flags

| Name | Description |
|-----------------------|------------------------------------|
| customer-partition | Customer Partition ID |
| sftp-destination-type | Provides the SFTP destination type |

Example

Command Deprecated in API v1.1: `beamctl pub-dest-config retransmit-key 1 primary`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 retransmit-key 1 primary`

Note: Input parameter <sftp-destination-type> has either primary/secondary/all.



3.6.2.8 retransmit-all

The `put-all` command is used to retransmit public keys. Customer partition id is a mandatory flag.

Syntax

Command Deprecated in API v1.1: `beamctl pub-dest-config retransmit-keys <customer-partition>`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 retransmit-keys <customer-partition>`

Flag

Table 171 Flags

| Name | Description |
|--------------------|-----------------------|
| customer-partition | Customer Partition ID |

Example

Command Deprecated in API v1.1: `beamctl pub-dest-config retransmit-keys 1`

Command Introduced in API v2.0: `beamctl pub-dest-config-v2 retransmit-keys 1`

3.6.3 Snapshot Publisher Destination Configuration

3.6.3.1 list-partitions

The `list-partitions` command lists all snapshot publisher endpoint configurations. User can optionally provide a customer-partition ID to list the configuration for a specific partition.

Syntax

`beamctl snapshot-pub-dest-config list-partitions`

`beamctl snapshot-pub-dest-config list-partitions --customer-partition <customer-partition>`

Flag

Table 172 Flags

| Name | Description |
|--------------------|-----------------------|
| customer-partition | Customer Partition ID |



Example

```
beamctl snapshot-pub-dest-config list-partitions
```

```
beamctl snapshot-pub-dest-config list-partitions --customer-partition 1
```

Note: The customer-partition parameter is optional and accepts a single valid value. Comma-separated values **are not supported**. Use the **get-config** command to download configurations for multiple partitions.

3.6.3.2

add-config

The add-config command uploads snapshot publisher configurations. The customer-partition flag is optional. If provided, only the configuration for that partition would be uploaded. If omitted, configurations for all partitions in the payload would be uploaded.

Syntax

```
beamctl snapshot-pub-dest-config add-config --customer-partition <customer-partition> < payload
```

```
beamctl snapshot-pub-dest-config add-config < payload
```

Flag

Table 173 Flags

| Name | Description |
|--------------------|-----------------------|
| customer-partition | Customer Partition ID |

Example

```
beamctl snapshot-pub-dest-config add-config --customer-partition 1 < /home/admin/config.json
```

```
beamctl snapshot-pub-dest-config add-config < /home/admin/config.json
```

Example JSON Format with Alias and Filetype

```
{
  "sftp": {
    "commonConfig": {
      "sftpRetryAttempts": "0",
      "snapshotMaxFileSizeInMb": "30",
      "maxSnapshotEventsToPublishPerFile": "2000",
      "sftpBinaryMode": "false",
      "sftpStrictHostKeyCheck": "yes",

```



```
    "sftpConnectionTimeout": "5000"
  },
  "aliasConfigs": [
    {
      "aliasName": "$primaryDestinationAliasESP",
      "sftpHostname": "seliicvm03352.bss-te.seli.gic.ericsson. →
se",
      "sftpUserName": "admin",
      "sftpPassword": "EvaiKi01",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "../..home/admin/sftpFileTypeT →
estDir/fileTypeESDir/",
      "sftpErrorDestinationFolder": "../..home/admin/sftpFile →
TypeErrorDir/fileTypeESPErrDir/"
    },
    {
      "aliasName": "$primaryDestinationAliasBSP",
      "sftpHostname": "seliicvm03352.bss-te.seli.gic.ericsson. →
se",
      "sftpUserName": "admin",
      "sftpPassword": "EvaiKi01",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "../..home/admin/sftpFileTypeT →
estDirBSP",
      "sftpErrorDestinationFolder": "../..home/admin/sftpFile →
TypeErrorDir/fileTypeBSPErrDir/"
    },
    {
      "aliasName": "$secondaryDestinationAlias",
      "sftpHostname": "seliicvm03352.bss-te.seli.gic.ericsson. →
se",
      "sftpUserName": "admin",
      "sftpPassword": "EvaiKi01",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "../..home/admin/sftpFileTypeT →
estDir/new",
      "sftpErrorDestinationFolder": "../..home/admin/sftpFile →
TypeErrorDir/new"
    },
    {
      "aliasName": "$secondaryDestinationAlias2",
      "sftpHostname": "seliicvm03352.bss-te.seli.gic.ericsson. →
se",
      "sftpUserName": "admin",
      "sftpPassword": "EvaiKi01",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "../..home/admin/sftpFileTypeT →
estDir2/new",
      "sftpErrorDestinationFolder": "../..home/admin/sftpFile →
TypeErrorDir2/new"
    }
  ]
}
```



```

],
"partitionConfigs": [
  {
    "partitionId": "1",
    "fileType": "BSP",
    "primaryDestination": "$primaryDestinationAliasBSP",
    "secondaryDestination": "$secondaryDestinationAlias"
  },
  {
    "partitionId": "2",
    "fileType": "BSP",
    "primaryDestination": "$primaryDestinationAliasBSP",
    "secondaryDestination": "$secondaryDestinationAlias2"
  },
  {
    "partitionId": "2",
    "fileType": "ESP",
    "primaryDestination": "$primaryDestinationAliasESP",
    "secondaryDestination": "$secondaryDestinationAlias"
  },
  {
    "partitionId": "1",
    "primaryDestination": {
      "sftpHostname": "seliicvm03352.bss-te.seli.gic.ericsson.se",
      "sftpUserName": "admin",
      "sftpPassword": "EvaiKi01",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "../home/admin/sftpGenericDestinationFolder",
      "sftpErrorDestinationFolder": "../home/admin/sftpGenericErrorDestinationFolder"
    },
    "secondaryDestination": {
      "sftpHostname": "seliicvm03352.bss-te.seli.gic.ericsson.se",
      "sftpUserName": "admin",
      "sftpPassword": "EvaiKi01",
      "sftpPortNumber": "22",
      "sftpDestinationFolder": "../home/admin/sftpGenericDestinationFolder_sec",
      "sftpErrorDestinationFolder": "../home/admin/sftpGenericErrorDestinationFolder_sec"
    }
  }
]
}

```



fileType Validation Rules

The following validations apply when importing JSON with **fileType** attribute:

- The same file type cannot be configured more than once.
- SFTP generic configuration (that is, configuration without a **fileType** attribute) must appear only once per partition. It must not be omitted.
- The **fileType** value must be either ESP (Entity Snapshot) or BSP (Balance Snapshot), and it is **case-sensitive**.

AliasConfig Validation Rules

The following validations apply when importing JSON with alias configurations:

- Alias names must start with a \$ symbol and cannot contain other special characters.
- Every alias defined under **aliasConfigs** must be referenced at least once in **partitionConfigs**.
- All alias names used in **partitionConfigs** must be defined in **aliasConfigs**.
- Duplicate alias definitions are not allowed.

Note: All required fields must be present in the JSON to configure an endpoint.

Example

When configuring an SFTP endpoint, fill in all the required parameters under the SFTP section.

For more details, refer to the section **EDM Snapshot Publisher** in EDM Application Configuration Function Guide.

3.6.3.3

get-config

The get-config command downloads snapshot publisher configurations. User may optionally specify a customer-partition to retrieve a specific configuration.

Syntax

```
beamctl snapshot-pub-dest-config get-config --customer-partition  
<customer-partition>
```

```
beamctl snapshot-pub-dest-config get-config
```

Flag



Table 174 Flags

| Name | Description |
|--------------------|-----------------------|
| customer-partition | Customer Partition ID |

Note: If alias configurations exist for a partition, they would be included in the output.

Example

```
beamctl snapshot-pub-dest-config get-config --customer-partition 1
```

```
beamctl snapshot-pub-dest-config get-config
```

3.6.3.4

delete-config

The delete-config command deletes a snapshot publisher configuration from **edmsnapshotpub**. The customer-partition flag is mandatory.

Syntax

```
beamctl snapshot-pub-dest-config delete-config <customer-partition> --file-type <file-type>
```

Flag

Table 175 Flags

| Name | Description |
|--------------------|--|
| customer-partition | Customer Partition ID |
| file-type | Deletes the snapshot publisher configuration for the specified file type (BSP or ESP). |

Example

```
beamctl snapshot-pub-dest-config delete-config 1 --file-type BSP
```

- Note:**
- Customer-partition is required to delete any configuration.
 - File-type is an optional, case-sensitive parameter and must be either BSP or ESP.
 - If file-type is omitted, the entire configuration for the specified partition including all file types would be deleted.



3.6.3.5 Retransmit

The `retransmit` command retransmits public keys. Both `customer-partition` and `sftp-destination-type` are required.

Syntax

```
beamctl snapshot-pub-dest-config retransmit-key <customer-partition> <sftp-destination-type>
```

Flag

Table 176 Flags

| Name | Description |
|-----------------------|--|
| customer-partition | Customer Partition ID |
| sftp-destination-type | Provides the SFTP destination type (primary, secondary or all) |

Example

```
beamctl snapshot-pub-dest-config retransmit-key 1 primary
```

3.6.3.6 retransmit-all

The `retransmit-all` command retransmits public keys for a specified customer partition. Here, Customer partition id is a mandatory flag.

Syntax

```
beamctl snapshot-pub-dest-config retransmit-keys <customer-partition>
```

Flag

Table 177 Flags

| Name | Description |
|--------------------|-----------------------|
| customer-partition | Customer Partition ID |

Example

```
beamctl snapshot-pub-dest-config retransmit-keys 1
```



3.6.4 Publishing ASN Schema

3.6.4.1 list

The `list` command is to list all the ASN schemas present in the system.

Syntax

Command Deprecated in API v1.1: `beamctl pub-asn-schema list`

Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 list`

Example

Command Deprecated in API v1.1: `beamctl pub-asn-schema list`

Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 list`

3.6.4.2 delete-schema

The `delete-schema` command is used to delete the specified ASN1 schema present in the system. Schema name is a mandatory flag.

Syntax

Command Deprecated in API v1.1: `beamctl pub-asn-schema delete-schema --schema-name <asn1 schema name>`

Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 delete-schema --schema-name <asn1 schema name>`

Flag

Table 178 Flags

| Name | Description |
|-------------|--------------------------------|
| schema-name | To delete the ASN schema name. |

Example

Command Deprecated in API v1.1:

```
beamctl pub-asn-schema delete-schema --schema-name  
com.ericsson.bss.sys.event.schema.BssfChfCdrAsn1_1.0.0
```

Command Introduced in API v2.0:

```
beamctl pub-asn-schema-v2 delete-schema --schema-name  
com.ericsson.bss.sys.event.schema.BssfChfCdrAsn1_1.0.0
```

3.6.4.3 add-schema

The `add-schema` command is used to add asn1 schemas to the system.



Syntax

Command Deprecated in API v1.1: `beamctl pub-asn-schema add-schema --overwrite-status <overwrite-status> < filename`

Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 add-schema --overwrite-status <overwrite-status> < filename`

Flag

Table 179 Flags

| Name | Description | Possible Values |
|------------------|--|-----------------|
| overwrite-status | Flag to determine if overwrite is required or not. | true/false |

Example

Command Deprecated in API v1.1: `beamctl pub-asn-schema add-schema --overwrite-status true < /var/tmp/asn.zip`

Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 add-schema --overwrite-status true < /var/tmp/asn.zip`

Note: SI must prepare the zip file with the required ASN.1 schemas included. The same needs to be placed in the path which is given as the input for CLI.

Example 138 addSchema

```
-- ASN.1 Formal Description

-- schemaNameSpace : "com.ericsson.bss.sys.event.schema"
-- schemaName : "BssfChargingCdrOccAsn1"
-- schemaVersion : "5.0.0"
--
-- Definition of schemaVersion format:
-- Major field = Incompatible changes (e.g. to
-- introduce a new Mandatory parameter)
-- Minor field = Backwards compatible changes (e.g. to
-- introduce a new Optional parameter)
-- Patch field = Backwards compatible bug fixes
-- (i.e. related to existing parameter use)
--
-- Note: Code changes below this line (e.g. assignment of new
-- field Ids) must be coordinated with other documents/files that
-- are making use of the given record type
--
```



```

CHARGINGCDR {iso(1) member-body(2) bsi(826) disc(0)
ericsson(1249) mobileDomain (0) charging (5) cCHARGINGCDR (1)
asn1Module (0)}

DEFINITIONS IMPLICIT TAGS ::=

BEGIN

EXPORTS ChargingDataOutputRecord;
ChargingDataOutputRecord ::= CHOICE {
  -- Reserved, not used [0]-[5]
  onlineCreditControlRecord [6] OnlineCreditControlRecord
  -- Above line: Comma (,) is only used with ellipsis notations
  -- Reserved, not used [7]
  -- Ellipsis notations (...) aren't supported by BSSF
}

OnlineCreditControlRecord ::= SEQUENCE {
  resultCode [0] Unsigned32,
  resultCodeExtension [1] Unsigned32 OPTIONAL,
  triggerTime [2] TimeStamp,
  nodeName [3] UTF8String (SIZE(1..255)),
  serviceContextID [4] UTF8String,
  chargingContextID [5] ChargingContext,
  serviceSessionID [6] UTF8String,
  recordIdentificationNumber [7] RecordIdentificationNumber,
  partialSequenceNumber [8] INTEGER OPTIONAL,
  lastPartialOutput [9] BOOLEAN OPTIONAL,
  servedSubscriptionID [10] SEQUENCE OF SubscriptionID
  OPTIONAL,
  correlationID [11] EXPLICIT CorrelationID OPTIONAL,
  servingElement [12] EXPLICIT ServingElement OPTIONAL,
  creditControlRecords [13] SEQUENCE OF SDPCreditControlRecord
  OPTIONAL,
  usedUnchargedServiceUnits [14] UsedServiceUnits OPTIONAL,
  -- Reserved, not used [15]-[24]
  iMSChargingIdentifier [25] IMSChargingIdentifier OPTIONAL,
  outgoingSessionId [26] OutgoingSessionId OPTIONAL,
  userSessionID [27] UserSessionID OPTIONAL,
  causeCode [28] CauseCode OPTIONAL,
  creditControlFailureHandling [29] CreditControlFailureHandling OPTIONAL,
  -- Reserved, not used [30]-[32]
  relatedICIDs [33] SEQUENCE OF RelatedICID OPTIONAL,
  -- Reserved, not used [34]-[44]
  supplementaryService [45] SEQUENCE OF SupplementaryServices OPTIONAL →
L,
  -- Reserved, not used [46]-[51]
  nodeFunctionality [52] NodeFunctionality OPTIONAL,
  reasonHeader [53] SEQUENCE OF ReasonHeader OPTIONAL,
  accessNetworkInformation [54] SEQUENCE OF AccessNetworkInformation OPTI →
ONAL,
  reportingReason [55] ReportingReason OPTIONAL,

```



```
-- Reserved, not used          [56]-[64]
sessionStartTime               [65] TimeStamp OPTIONAL,
sessionStopTime               [66] TimeStamp OPTIONAL,
fUIRedirectAddress            [67] UTF8String OPTIONAL,
retargetAddress               [68] UTF8String OPTIONAL,
failOpenRecord                [69] BOOLEAN OPTIONAL,
failOpenChargingContextSpecific [70] SEQUENCE OF ContextParameter
                                OPTIONAL
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

-----
--
--  ONLINE CREDIT CONTROL RECORD DATA TYPES
--
-----

AccessNetworkInformation ::= OCTET STRING (SIZE (1..255))
AcctMultiSessionID ::= OCTET STRING (SIZE (1..255))
AddressString ::= OCTET STRING (SIZE (1..20))
AssociatedPartyAddress ::= UTF8String (SIZE (1..255))
CauseCode ::= Integer32
ChangeTime ::= TimeStamp
ChargingContext ::= UTF8String (SIZE (1..255))
ContextParameter ::= SEQUENCE
{
  parameterID      [0] Integer64,
  parameterValue   [1] ContextParameterValueType
}
ContextParameterValueType ::= CHOICE
{
  boolean          [0] BOOLEAN,
  integer32        [1] Integer32,
  -- Reserved, not used [2]
  unsigned32       [3] Unsigned32,
  unsigned64       [4] Unsigned64,
  -- Reserved, not used [5]
  octetString      [6] OCTET STRING,
  string           [7] UTF8String,
  time             [8] Time,
  partyInformation [9] PartyInformation,
  group            [10] Group,
  -- Reserved, not used [11]
```



```

strings          [12] Strings,
groups           [13] Groups,
-- Reserved, not used [14]-[15]
unsigned32s      [16] Unsigned32s
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

CorrelationID ::= CHOICE {
    -- Reserved, not used [0]-[2]
    dccCorrelationId      [3] DccCorrelationID
}

CreditControlFailureHandling ::= ENUMERATED {
    terminate (0),
    continue  (1)
}

CreditControlRecord ::= SEQUENCE
{
    -- Reserved, used [1]-[28] ASN.1 BER encoded blob data,
    -- no decoding required
    -- Ellipsis notations (...) aren't supported by BSSF
}

CUGInformation ::= OCTET STRING (SIZE (1..255))

DccCorrelationID ::= SEQUENCE {
    acctMultiSessionID [0] AcctMultiSessionID OPTIONAL,
    ccCorrelationId     [1] OCTET STRING OPTIONAL,
    sessionId           [2] UTF8String OPTIONAL,
    ccRequestNumber     [3] Unsigned32 OPTIONAL
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

Group ::= SEQUENCE OF ContextParameter

Groups ::= SEQUENCE OF Group

GSNAddress ::= IPBinaryAddress

IMSChargingIdentifier ::= UTF8String (SIZE (1..255))

Integer8 ::= INTEGER (-128..127)

Integer32 ::= INTEGER (-2147483648..2147483647)

Integer64 ::= INTEGER
            (-9223372036854775808..9223372036854775807)

IPBinaryAddress ::= CHOICE {

```



```
    iPBInV4Address      [0] OCTET STRING (SIZE(4)),
    iPBInV6Address      [1] OCTET STRING (SIZE(16))
}

MonetaryUnits ::= SEQUENCE
{
    amount      [0] Integer64,
    decimals    [1] Integer32,
    currency    [2] Unsigned32
}

NodeFunctionality ::= ENUMERATED {
sCSCF          (0),
pCSCF          (1),
iCSCF          (2),
mRFC           (3),
mGCF           (4),
bGCF           (5),
aS             (6),
iBCF           (7),
sGW            (8),
pGW            (9),
hSGW           (10),
eCSCF          (11),
mME            (12),
tRF            (13),
tF             (14),
aTCF           (15),
proxyFunction  (16),
ePDG           (17),
tDF            (18),
tWAG           (19)
}

NumberOfDiversions ::= Unsigned32

NumberOfParticipants ::= Unsigned32

OriginHost ::= OCTET STRING (SIZE (1..255))

OriginInfo ::= SEQUENCE {
    originRealm      [0] OriginRealm OPTIONAL,
    originHost       [1] OriginHost OPTIONAL
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

OriginRealm ::= OCTET STRING (SIZE (1..255))

OutgoingSessionId ::= UTF8String (SIZE (1..255))

ParticipantActionType ::= ENUMERATED {
```




```

createConf          (0),
joinConf            (1),
inviteIntoConf      (2),
quitConf            (3)
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

PartyInformation ::= CHOICE
{
  msisdn            [0] AddressString,
  imsi              [1] AddressString,
  sipUri            [2] UTF8String,
  nai               [3] UTF8String
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

ReasonHeader ::= UTF8String (SIZE (1..255))

RecordIdentificationNumber ::= OCTET STRING (SIZE(1..4))

RelatedICID ::= UTF8String (SIZE (1..255))

ReportingReason ::= ENUMERATED {
  threshold          (0),
  qht                (1),
  final              (2),
  quotaExhausted     (3),
  validityTime       (4),
  otherQuotaType      (5),
  ratingConditionChanged (6),
  forcedReAuthorisation (7),
  poolExhausted      (8)
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

SDPCreditControlRecord ::= CHOICE
{
  creditControlRecord [0] CreditControlRecord
  -- Above line: Comma (,) is only used with ellipsis notations
  -- Ellipsis notations (...) aren't supported by BSSF
}

ServiceID ::= UTF8String (SIZE (1..255))

ServiceMode ::= Unsigned32

ServiceType ::= Unsigned32

ServingElement ::= CHOICE {

```



```
originInfo          [0] OriginInfo,
mSCAddress          [1] AddressString,
ggsnAddress         [2] GSNAddress,
sgsnAddress         [3] GSNAddress
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

Strings ::= SEQUENCE OF UTF8String

SubscriptionID ::= SEQUENCE {
    subscriptionIDType    [0] SubscriptionIDType,
    subscriptionIDValue   [1] SubscriptionIDData
}

SubscriptionIDData ::= UTF8String (SIZE (1..255))

SubscriptionIDType ::= ENUMERATED {
    endUserE164          (0),
    endUserIMSI          (1),
    -- Reserved, not used (2),
    endUserNAI           (3),
    -- Reserved, not used (4)
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

SupplementaryServices ::= SEQUENCE {
    serviceType          [0] ServiceType OPTIONAL,
    serviceMode          [1] ServiceMode OPTIONAL,
    numberOfDiversions   [2] NumberOfDiversions OPTIONAL,
    associatedPartyAddress [3] AssociatedPartyAddress OPTIONAL,
    serviceID            [4] ServiceID OPTIONAL,
    changeTime           [5] ChangeTime OPTIONAL,
    numberOfParticipants [6] NumberOfParticipants OPTIONAL,
    participantActionType [7] ParticipantActionType OPTIONAL,
    cUGInformation        [8] CUGInformation OPTIONAL
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

TariffChangeUsage ::= ENUMERATED
{
    before(0),
    after(1),
    indeterminate(2)
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

Time ::= SEQUENCE
```



```

{
  sec      [0] Integer32,
  tz       [1] Integer8
}

TimeStamp ::= OCTET STRING (SIZE(9))

Unsigned32 ::= INTEGER (0..4294967295)

Unsigned32s ::= SEQUENCE OF Unsigned32

Unsigned64 ::= INTEGER (0..18446744073709551615)

UsedServiceUnit ::= SEQUENCE
{
  tariffChangeUsage      [0] TariffChangeUsage OPTIONAL,
  timeUnit               [1] Unsigned32 OPTIONAL,
  moneyUnit              [2] MonetaryUnits OPTIONAL,
  totalOctetsUnit        [3] Unsigned64 OPTIONAL,
  uplinkOctetsUnit       [4] Unsigned64 OPTIONAL,
  downlinkOctetsUnit     [5] Unsigned64 OPTIONAL,
  serviceSpecificUnit    [6] Unsigned64 OPTIONAL
  -- Above line: Comma (,) is only used with ellipsis notations
  -- Ellipsis notations (...) aren't supported by BSSF
}

UsedServiceUnits ::= SEQUENCE (SIZE (1..3)) OF
  UsedServiceUnit

UserSessionID ::= UTF8String (SIZE (1..255))

END

```

3.6.4.4 update-schema

The update-schema command to update all the asn1 schemas present.

Syntax

Command Deprecated in API v1.1: `beamctl pub-asn-schema update-schema < filename`

Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 update-schema < filename`

Example

Command Deprecated in API v1.1: `beamctl pub-asn-schema update-schema < /var/tmp/asn.zip`



Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 update-schema < /var/tmp/asn.zip`

Note: SI must prepare the zip file with the required ASN.1 schemas included. The same needs to be placed in the path which is given as the input for CLI.

Example 139 updateSchema

```
-- ASN.1 Formal Description

-- schemaNameSpace : "com.ericsson.bss.sys.event.schema"
-- schemaName : "BssfChargingCdrOccAsn1"
-- schemaVersion : "5.0.0"
--
-- Definition of schemaVersion format:
-- Major field = Incompatible changes (e.g. to
-- introduce a new Mandatory parameter)
-- Minor field = Backwards compatible changes (e.g. to
-- introduce a new Optional parameter)
-- Patch field = Backwards compatible bug fixes
-- (i.e. related to existing parameter use)
--
-- Note: Code changes below this line (e.g. assignment of new
-- field Ids) must be coordinated with other documents/files that
-- are making use of the given record type
--

CHARGINGCDR {iso(1) member-body(2) bsi(826) disc(0)
ericsson(1249) mobileDomain (0) charging (5) cCHARGINGCDR (1)
asn1Module (0)}

DEFINITIONS IMPLICIT TAGS ::=

BEGIN

EXPORTS ChargingDataOutputRecord;
ChargingDataOutputRecord ::= CHOICE {
-- Reserved, not used [0]-[5]
    onlineCreditControlRecord [6] OnlineCreditControlRecord
-- Above line: Comma (,) is only used with ellipsis notations
-- Reserved, not used [7]
-- Ellipsis notations (...) aren't supported by BSSF
}

OnlineCreditControlRecord ::= SEQUENCE {
    resultCode [0] Unsigned32,
    resultCodeExtension [1] Unsigned32 OPTIONAL,
    triggerTime [2] TimeStamp,
    nodeName [3] UTF8String (SIZE(1..255)),
    serviceContextID [4] UTF8String,
```



```

chargingContextID      [5] ChargingContext,
serviceSessionID       [6] UTF8String,
recordIdentificationNumber [7] RecordIdentificationNumber,
partialSequenceNumber  [8] INTEGER OPTIONAL,
lastPartialOutput      [9] BOOLEAN OPTIONAL,
servedSubscriptionID   [10] SEQUENCE OF SubscriptionID
                        OPTIONAL,
correlationID          [11] EXPLICIT CorrelationID OPTIONAL,
servingElement         [12] EXPLICIT ServingElement OPTIONAL,
creditControlRecords   [13] SEQUENCE OF SDPCreditControlRecord
                        OPTIONAL,
usedUnchargedServiceUnits [14] UsedServiceUnits OPTIONAL,
-- Reserved, not used   [15]-[24]
imsChargingIdentifier   [25] IMSChargingIdentifier OPTIONAL,
outgoingSessionId      [26] OutgoingSessionId OPTIONAL,
userSessionID          [27] UserSessionID OPTIONAL,
causeCode              [28] CauseCode OPTIONAL,
creditControlFailureHandling [29] CreditControlFailureHandling OPTIONAL,
-- Reserved, not used   [30]-[32]
relatedICIDs           [33] SEQUENCE OF RelatedICID OPTIONAL,
-- Reserved, not used   [34]-[44]
supplementaryService    [45] SEQUENCE OF SupplementaryServices OPTIONAL →
L,
-- Reserved, not used   [46]-[51]
nodeFunctionality       [52] NodeFunctionality OPTIONAL,
reasonHeader            [53] SEQUENCE OF ReasonHeader OPTIONAL,
accessNetworkInformation [54] SEQUENCE OF AccessNetworkInformation OPTI →
ONAL,
reportingReason         [55] ReportingReason OPTIONAL,
-- Reserved, not used   [56]-[64]
sessionStartTime        [65] TimeStamp OPTIONAL,
sessionStopTime         [66] TimeStamp OPTIONAL,
fUIRedirectAddress      [67] UTF8String OPTIONAL,
retargetAddress         [68] UTF8String OPTIONAL,
failOpenRecord          [69] BOOLEAN OPTIONAL,
failOpenChargingContextSpecific [70] SEQUENCE OF ContextParameter
                        OPTIONAL
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

-----
--
--   ONLINE CREDIT CONTROL RECORD DATA TYPES
--
-----

AccessNetworkInformation ::= OCTET STRING (SIZE (1..255))

AcctMultiSessionID ::= OCTET STRING (SIZE (1..255))

AddressString ::= OCTET STRING (SIZE (1..20))

```



```
AssociatedPartyAddress ::= UTF8String (SIZE (1..255))

CauseCode ::= Integer32

ChangeTime ::= TimeStamp

ChargingContext ::= UTF8String (SIZE (1..255))

ContextParameter ::= SEQUENCE
{
    parameterID          [0] Integer64,
    parameterValue       [1] ContextParameterValueType
}

ContextParameterValueType ::= CHOICE
{
    boolean              [0] BOOLEAN,
    integer32            [1] Integer32,
    -- Reserved, not used [2]
    unsigned32           [3] Unsigned32,
    unsigned64           [4] Unsigned64,
    -- Reserved, not used [5]
    octetString          [6] OCTET STRING,
    string               [7] UTF8String,
    time                 [8] Time,
    partyInformation     [9] PartyInformation,
    group                [10] Group,
    -- Reserved, not used [11]
    strings              [12] Strings,
    groups               [13] Groups,
    -- Reserved, not used [14]-[15]
    unsigned32s          [16] Unsigned32s
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

CorrelationID ::= CHOICE {
    -- Reserved, not used [0]-[2]
    dccCorrelationId     [3] DccCorrelationID
}

CreditControlFailureHandling ::= ENUMERATED {
    terminate (0),
    continue (1)
}

CreditControlRecord ::= SEQUENCE
{
    -- Reserved, used      [1]-[28] ASN.1 BER encoded blob data,
    --                      no decoding required
    -- Ellipsis notations (...) aren't supported by BSSF
}
```



```

}

CUGInformation ::= OCTET STRING (SIZE (1..255))

DccCorrelationID ::= SEQUENCE {
    acctMultiSessionID    [0] AcctMultiSessionID OPTIONAL,
    ccCorrelationId       [1] OCTET STRING OPTIONAL,
    sessionId             [2] UTF8String OPTIONAL,
    ccRequestNumber       [3] Unsigned32 OPTIONAL
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

Group ::= SEQUENCE OF ContextParameter

Groups ::= SEQUENCE OF Group

GSNAddress ::= IPBinaryAddress

IMSChargingIdentifier ::= UTF8String (SIZE (1..255))

Integer8 ::= INTEGER (-128..127)

Integer32 ::= INTEGER (-2147483648..2147483647)

Integer64 ::= INTEGER
            (-9223372036854775808..9223372036854775807)

IPBinaryAddress ::= CHOICE {
    iPBinV4Address        [0] OCTET STRING (SIZE(4)),
    iPBinV6Address        [1] OCTET STRING (SIZE(16))
}

MonetaryUnits ::= SEQUENCE
{
    amount                [0] Integer64,
    decimals              [1] Integer32,
    currency              [2] Unsigned32
}

NodeFunctionality ::= ENUMERATED {
sCSCF                (0),
pCSCF                (1),
iCSCF                (2),
mRFC                 (3),
mGCF                 (4),
bGCF                 (5),
aS                   (6),
iBCF                 (7),
sGW                  (8),
pGW                  (9),
hSGW                 (10),

```



```
eCSCF          (11),
mME            (12),
tRF            (13),
tF             (14),
aTCF           (15),
proxyFunction  (16),
ePDG           (17),
tDF            (18),
tWAG           (19)
}

NumberOfDiversions ::= Unsigned32

NumberOfParticipants ::= Unsigned32

OriginHost ::= OCTET STRING (SIZE (1..255))

OriginInfo ::= SEQUENCE {
    originRealm          [0] OriginRealm OPTIONAL,
    originHost           [1] OriginHost OPTIONAL
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

OriginRealm ::= OCTET STRING (SIZE (1..255))

OutgoingSessionId ::= UTF8String (SIZE (1..255))

ParticipantActionType ::= ENUMERATED {
    createConf          (0),
    joinConf            (1),
    inviteIntoConf      (2),
    quitConf            (3)
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

PartyInformation ::= CHOICE
{
    msisdn              [0] AddressString,
    imsi                [1] AddressString,
    sipUri              [2] UTF8String,
    nai                 [3] UTF8String
    -- Above line: Comma (,) is only used with ellipsis notations
    -- Ellipsis notations (...) aren't supported by BSSF
}

ReasonHeader ::= UTF8String (SIZE (1..255))

RecordIdentificationNumber ::= OCTET STRING (SIZE(1..4))

RelatedICID ::= UTF8String (SIZE (1..255))
```




```

ReportingReason ::= ENUMERATED {
    threshold                (0),
    qht                      (1),
    final                    (2),
    quotaExhausted           (3),
    validityTime             (4),
    otherQuotaType           (5),
    ratingConditionChanged   (6),
    forcedReAuthorisation    (7),
    poolExhausted            (8)
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

SDPCreditControlRecord ::= CHOICE
{
    creditControlRecord      [0] CreditControlRecord
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

ServiceID ::= UTF8String (SIZE (1..255))

ServiceMode ::= Unsigned32

ServiceType ::= Unsigned32

ServingElement ::= CHOICE {
    originInfo                [0] OriginInfo,
    mSCAddress                [1] AddressString,
    ggsnAddress                [2] GSNAddress,
    sgsnAddress                [3] GSNAddress
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

Strings ::= SEQUENCE OF UTF8String

SubscriptionID ::= SEQUENCE {
    subscriptionIDType        [0] SubscriptionIDType,
    subscriptionIDValue       [1] SubscriptionIDData
}

SubscriptionIDData ::= UTF8String (SIZE (1..255))

SubscriptionIDType ::= ENUMERATED {
    endUserE164                (0),
    endUserIMSI                (1),
--    Reserved, not used        (2),
    endUserNAI                 (3)

```



```
-- Reserved, not used          (4)
-- Above line: Comma (,) is only used with ellipsis notations
-- Ellipsis notations (...) aren't supported by BSSF
}

SupplementaryServices ::= SEQUENCE {
  serviceType           [0] ServiceType OPTIONAL,
  serviceMode           [1] ServiceMode OPTIONAL,
  numberOfDiversions    [2] NumberOfDiversions OPTIONAL,
  associatedPartyAddress [3] AssociatedPartyAddress OPTIONAL,
  serviceID             [4] ServiceID OPTIONAL,
  changeTime           [5] ChangeTime OPTIONAL,
  numberOfParticipants  [6] NumberOfParticipants OPTIONAL,
  participantActionType [7] ParticipantActionType OPTIONAL,
  cUGInformation        [8] CUGInformation OPTIONAL
  -- Above line: Comma (,) is only used with ellipsis notations
  -- Ellipsis notations (...) aren't supported by BSSF
}

TariffChangeUsage ::= ENUMERATED
{
  before(0),
  after(1),
  indeterminate(2)
  -- Above line: Comma (,) is only used with ellipsis notations
  -- Ellipsis notations (...) aren't supported by BSSF
}

Time ::= SEQUENCE
{
  sec      [0] Integer32,
  tz       [1] Integer8
}

TimeStamp ::= OCTET STRING (SIZE(9))

Unsigned32 ::= INTEGER (0..4294967295)

Unsigned32s ::= SEQUENCE OF Unsigned32

Unsigned64 ::= INTEGER (0..18446744073709551615)

UsedServiceUnit ::= SEQUENCE
{
  tariffChangeUsage      [0] TariffChangeUsage OPTIONAL,
  timeUnit               [1] Unsigned32 OPTIONAL,
  moneyUnit              [2] MonetaryUnits OPTIONAL,
  totalOctetsUnit        [3] Unsigned64 OPTIONAL,
  uplinkOctetsUnit        [4] Unsigned64 OPTIONAL,
  downlinkOctetsUnit      [5] Unsigned64 OPTIONAL,
  serviceSpecificUnit     [6] Unsigned64 OPTIONAL
  -- Above line: Comma (,) is only used with ellipsis notations
```



```
-- Ellipsis notations (...) aren't supported by BSSF
}

UsedServiceUnits ::= SEQUENCE (SIZE (1..3)) OF
    UsedServiceUnit

UserSessionID ::= UTF8String (SIZE (1..255))

END
```

3.6.4.5 get-schema

The `get-schema` command is used to export the specified ASN1 schema available in the system. Schema name is a mandatory flag.

Syntax

Command Deprecated in API v1.1: `beamctl pub-asn-schema get-schema --schema-name <asn1 schema name>`

Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 get-schema --schema-name <asn1 schema name>`

Arguments

The command has no arguments.

Flags

Table 180 Flags

| Name | Description |
|-------------|--------------------------------|
| schema-name | To export the ASN schema name. |

Example 140

Command Deprecated in API v1.1:

```
beamctl pub-asn-schema get-schema --schema-name
com.ericsson.bss.sys.event.schema.BssfChfCdrAsn1_1.0.0
```

Command Introduced in API v2.0:

```
beamctl pub-asn-schema-v2 get-schema --schema-name
com.ericsson.bss.sys.event.schema.BssfChfCdrAsn1_1.0.0
```



3.6.4.6 **get-all-schemas**

The `get-all-schemas` command is to export all the ASN schemas available in the system.

Syntax

Command Deprecated in API v1.1: `beamctl pub-asn-schema get-all-schemas`

Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 get-all-schemas`

Arguments

The command has no arguments.

Example 141

Command Deprecated in API v1.1: `beamctl pub-asn-schema get-all-schemas`

Command Introduced in API v2.0: `beamctl pub-asn-schema-v2 get-all-schemas`

3.6.5 **Publishing Application Configuration**

3.6.5.1 **get**

The `get` command retrieves the information for all Event Data Management app-configurations.

Ensure to include the query parameter to retrieve the information for a specific key of Event Data Management app-configuration.

Syntax

Command Deprecated in API v1.1: `beamctl pub-appconfig get --key <key>`

Command Introduced in API v2.0: `beamctl pub-appconfig-v2 get --key <key>`

Flag

Table 181 Flags

| Name | Description |
|------|--------------------------------|
| key | Application configuration name |



Example

Command Deprecated in API v1.1: `beamctl pub-appconfig get --key eventRecoveryFrequency`

Command Introduced in API v2.0: `beamctl pub-appconfig-v2 get --key eventRecoveryFrequency`

3.6.5.2

update

The `update` command updates the existing Event Data Management specific configuration.

The command requires a JSON object from stdin.

Syntax

Command Deprecated in API v1.1: `beamctl pub-appconfig update < JsonFile`

Command Introduced in API v2.0: `beamctl pub-appconfig-v2 update < JsonFile`

Example

Command Deprecated in API v1.1: `beamctl pub-appconfig update < config.json`

Command Introduced in API v2.0: `beamctl pub-appconfig-v2 update < config.json`

Example: Config JSON

```
{
  "key": "eventPublishingEndPoint",
  "value": "SFTP"
}
```

3.6.6

Snapshot Publisher Application Configuration

3.6.6.1

get

The `get` command retrieves all Snapshot Publisher application configurations. Use the `--key` flag to get a specific configuration.

Syntax

`beamctl snapshot-pub-appconfig get --key <key>`



Flag

Table 182 Flags

| Name | Description |
|------|--------------------------------|
| Key | Application configuration name |

Example

```
beamctl snapshot-pub-appconfig get --key  
enableSFTPKeyBasedAuthentication
```

3.6.6.2

update

The update command modifies existing application configurations using a JSON input from stdin.

Syntax

```
beamctl snapshot-pub-appconfig update < JsonFile
```

Example

```
beamctl snapshot-pub-appconfig update < config.json
```

Example: Config JSON

```
{  
  "key": "noOfEventsFetchPerPoll",  
  "value": "2000"  
}
```

3.6.7

Publishing Data Format Configuration

3.6.7.1

get

The get command gives a list of all the configured publishing dataformat configurations.

Syntax

Command Deprecated in API v1.1: `beamctl pub-dataformatconfig get`

Command Introduced in API v2.0: `beamctl pub-dataformatconfig-v2 get`



Export the Current Configuration to a File

Command Deprecated in API v1.1: `beamctl pub-dataformatconfig get > <filename.json>`

Command Introduced in API v2.0: `beamctl pub-dataformatconfig-v2 get > <filename.json>`

Example 142

Command Deprecated in API v1.1: `beamctl pub-dataformatconfig get`

Command Introduced in API v2.0: `beamctl pub-dataformatconfig-v2 get`

Export the Current Configuration to a File

Command Deprecated in API v1.1: `beamctl pub-dataformatconfig get > /var/tmp/Pubdataoutputconfig.json`

Command Introduced in API v2.0: `beamctl pub-dataformatconfig-v2 get > /var/tmp/Pubdataoutputconfig.json`

Example 143 Pubdataoutputconfig JSON

```
[
  {
    "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.CpmMassResourceIdChan →
geEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.UserChangeEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.RefillBlockEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.rm.coba.core.eventservice.ReferenceDataListCh →
angeEvent",
    "dataFormat": "COMPRESSEDJSON",
    "supportedFormats": [
      "COMPRESSEDJSON"
    ]
  },
  {
    "eventName": "com.ericsson.bss.fin.event.schema.PaymentInstructionEvent",
    "dataFormat": "AVROJSON",
```



```
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.CpmCustomerChangeEvent" →
  },
  {
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.PurchaseHistoryEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.BillingCostCalculationEvent" →
  },
  {
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.rmca.eventservice.BusinessConfigurationChange" →
    "Event",
    "dataFormat": "COMPRESSEDJSON",
    "supportedFormats": [
      "COMPRESSEDJSON"
    ]
  },
  {
    "eventName": "com.ericsson.bss.inv.event.schema.SettlementRunStatisticsEvent" →
  },
  {
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.ChargingInformationEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.IndividualUsageChargeSuccess" →
    "Event",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.PurchaseChargeEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.BalanceCreationEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  }
]
```




```

    },
    {
      "eventName": "com.ericsson.bss.inv.event.schema.SettlementNoteGenerationEven →
t",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.NetworkPolicyCounterStatusEv →
ent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.ChargingFunctionEvent",
      "dataFormat": "ASN1BER",
      "supportedFormats": [
        "ASN1BER",
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.ActionNotification",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.IndividualUsageChargeFailure →
Event",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.ProviderUsageChargeEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.ReplicateChargingConflictEve →
nt",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.CustomerInformationUpdateEve →
nt",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.ContractBalanceSnapshotEvent →
",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {

```



```

    "eventName": "com.ericsson.bss.inv.event.schema.BillRunStatisticsEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.PartyRoleChangeEvent" →
    ,
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.CreditControlEvent",
    "dataFormat": "ASN1BER",
    "supportedFormats": [
      "ASN1BER",
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.RecurringEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.NetworkAuthorizationEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.ProductAdministrativeChargeE →
event",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cil.cdal.ntf.service.domain.UserRoleHistoryDo →
main.EndUserNotificationHistoryEventForUserRole",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cil.cdal.ntf.service.domain.PartyRoleHistoryD →
omain.PartyRoleNotificationHistoryEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.ChargingNetworkFunctionEvent →
",
    "dataFormat": "ASN1BER",
    "supportedFormats": [
      "ASN1BER",
      "COMPRESSEDJSON"
    ]
  },
  {
    "eventName": "com.ericsson.bss.fin.colhub.event.schema.DunningCaseInfoEvent" →

```



```

    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.ServiceChargeEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.BucketMaintenanceEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.PromotionEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.inv.event.schema.InvoiceGenerationEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cil.cdal.ntf.service.domain.HistoryDomain.End →
UserNotificationHistoryEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.UsageChargeEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.RefundEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.ConsolidatedUsageChargeEvent →
",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cpm.event.schema.BusinessLogicProcessingError →
Event",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  }

```



```

    },
    {
      "eventName": "com.ericsson.bss.inv.event.schema.ExternalCalculatedChargeEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.PartnerSettlementEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.fin.event.schema.FinanceEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.RefillEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.OrganizationChangeEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.BalanceAdjustmentEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.TransferProductDataEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.RecurrenceManagementStatusEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    }
  ]
}

```

3.6.7.2 update

The update command is used to upload the expected publishing dataformat.



The command requires a JSON object from stdin.

Syntax

Command Deprecated in API v1.1: `beamctl pub-dataformatconfig update`
<path of the input json file>

Command Introduced in API v2.0: `beamctl pub-dataformatconfig-v2`
`update` *<path of the input json file>*

Example 144

Command Deprecated in API v1.1: `beamctl pub-dataformatconfig update`
</var/tmp/Pubdatainputconfig.json>

Command Introduced in API v2.0: `beamctl pub-dataformatconfig-v2`
`update` *< /var/tmp/Pubdatainputconfig.json>*

Example 145 Pubdatainputconfig JSON

```
[
  {
    "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.CpmMassResourceIdChan →
geEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.UserChangeEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.RefillBlockEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.rm.coba.core.eventservice.ReferenceDataListCh →
angeEvent",
    "dataFormat": "COMPRESSEDJSON",
    "supportedFormats": [
      "COMPRESSEDJSON"
    ]
  },
  {
    "eventName": "com.ericsson.bss.fin.event.schema.PaymentInstructionEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.CpmCustomerChangeEven →
t",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  }
]
```



```
]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.PurchaseHistoryEvent",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.BillingCostCalculationEvent" →
,
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.rmca.eventservice.BusinessConfigurationChange
Event",
  "dataFormat": "COMPRESSEDJSON",
  "supportedFormats": [
    "COMPRESSEDJSON"
  ]
},
{
  "eventName": "com.ericsson.bss.inv.event.schema.SettlementRunStatisticsEvent" →
,
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.ChargingInformationEvent",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.IndividualUsageChargeSuccess
Event",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.PurchaseChargeEvent",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.BalanceCreationEvent",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.inv.event.schema.SettlementNoteGenerationEven
t",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.NetworkPolicyCounterStatusEv" →
```



```

ent"
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.ChargingFunctionEvent",
  "dataFormat": "ASN1BER",
  "supportedFormats": [
    "ASN1BER",
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.ActionNotification",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.IndividualUsageChargeFailure
Event",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.ProviderUsageChargeEvent",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.ReplicateChargingConflictEve
nt",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.CustomerInformationUpdateEve
nt",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.cha.event.schema.ContractBalanceSnapshotEvent
",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "com.ericsson.bss.inv.event.schema.BillRunStatisticsEvent",
  "dataFormat": "AVROJSON",
  "supportedFormats": [
    "AVROJSON",
    "CELSerializedAVROBlob"
  ]
},
{
  "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.PartyRoleChangeEvent"
,
  "dataFormat": "AVROJSON",
  "supportedFormats": [

```



```
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.CreditControlEvent",
      "dataFormat": "ASN1BER",
      "supportedFormats": [
        "ASN1BER",
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.RecurringEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.NetworkAuthorizationEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.ProductAdministrativeChargeEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cil.cdal.ntf.service.domain.UserRoleHistoryDomain.EndUserNotificationHistoryEventForUserRole",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cil.cdal.ntf.service.domain.PartyRoleHistoryDomain.PartyRoleNotificationHistoryEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.ChargingNetworkFunctionEvent",
      "dataFormat": "ASN1BER",
      "supportedFormats": [
        "ASN1BER",
        "COMPRESSEDJSON"
      ]
    },
    {
      "eventName": "com.ericsson.bss.fin.colhub.event.schema.DunningCaseInfoEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.ServiceChargeEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    }
  ],
  "eventNames": [
    "com.ericsson.bss.cha.event.schema.CreditControlEvent",
    "com.ericsson.bss.cha.event.schema.RecurringEvent",
    "com.ericsson.bss.cha.event.schema.NetworkAuthorizationEvent",
    "com.ericsson.bss.cha.event.schema.ProductAdministrativeChargeEvent",
    "com.ericsson.bss.cil.cdal.ntf.service.domain.UserRoleHistoryDomain.EndUserNotificationHistoryEventForUserRole",
    "com.ericsson.bss.cil.cdal.ntf.service.domain.PartyRoleHistoryDomain.PartyRoleNotificationHistoryEvent",
    "com.ericsson.bss.cha.event.schema.ChargingNetworkFunctionEvent",
    "com.ericsson.bss.fin.colhub.event.schema.DunningCaseInfoEvent",
    "com.ericsson.bss.cha.event.schema.ServiceChargeEvent"
  ]
}
```




```

    ],
    {
      "eventName": "com.ericsson.bss.cha.event.schema.BucketMaintenanceEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.PromotionEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.inv.event.schema.InvoiceGenerationEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cil.cdal.ntf.service.domain.HistoryDomain.End →
UserNotificationHistoryEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.UsageChargeEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.RefundEvent",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.ConsolidatedUsageChargeEvent →
",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cpm.event.schema.BusinessLogicProcessingError →
Event",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.inv.event.schema.ExternalCalculatedChargeEven →
t",
      "dataFormat": "AVROJSON",
      "supportedFormats": [
        "AVROJSON",
        "CELSerializedAVROBlob"
      ]
    },
    {
      "eventName": "com.ericsson.bss.cha.event.schema.PartnerSettlementEvent",

```



```

    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.fin.event.schema.FinanceEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.RefillEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "CPMCDM.com.ericsson.bss.cpm.event.schema.OrganizationChangeEve →
nt",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.BalanceAdjustmentEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.TransferProductDataEvent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  },
  {
    "eventName": "com.ericsson.bss.cha.event.schema.RecurrenceManagementStatusEv →
ent",
    "dataFormat": "AVROJSON",
    "supportedFormats": [
      "AVROJSON",
      "CELSerializedAVROBlob"
    ]
  }
]

```

3.6.8 Publishing Variation Point Configuration

3.6.8.1 list-all-schemas

The `list-all-schemas` command gives a list of variation point schemas present in the table.

Syntax

```
beamctl pub-variationpoint-config list-all-schemas
```



Example

```
beamctl pub-variationpoint-config list-all-schemas
```

3.6.8.2

get-schema

The `get-schema` command is used to get the variation point schema metadata for the given schema name and schema version.

Syntax

```
beamctl pub-variationpoint-config get-schema --schema-name
<schema-name> --schema-version <schema-version>
```

Flag

Table 183 Flags

| Name | Description |
|----------------|--|
| schema-name | schema name of the variation point metadata |
| schema-version | schema version of the variation point metadata |

Example

```
beamctl pub-variationpoint-config get-schema --schema-name
edmeventpublishingcobaenrichment --schema-version 1.3.0
```

```
beamctl pub-variationpoint-config get-schema --schema-name
edmeventpublishingfilter --schema-version 1.5.0
```

3.6.8.3

add-schema

The `add-schema` command is used to add a variation point metadata into the System.

Syntax

```
beamctl pub-variationpoint-config add-schema < <path of the input
metadata file to add> --schema-name <schema-name> --schema-version
<schema-version>
```

Example

```
beamctl pub-variationpoint-config add-schema
< newenrichmentschema.csv --schema-name
edmeventpublishingcobaenrichment --schema-version 1.3.1
```



```
beamctl pub-variationpoint-config add-schema <
newfilterschema.csv --schema-name edmeventpublishingfilter --
schema-version 1.5.1
```

CSV Format for Event Publishing Filter Configuration

The Event Publishing Filter configuration is provided using a CSV file.

- The CSV file must contain a header named FilteredEvents.
- Event names to be filtered must be listed under this header, with one event per line.

Example: CSV Format

FilteredEvents

RecurrenceManagementStatusEvent

OverdueReceivableEvent

AggregateMappingEvent

ExternalCalculatedChargeEvent

3.6.8.4

activate-schema

The `activate-schema` command is used to activate the variation point metadata for the given schema details.

Syntax

```
beamctl pub-variationpoint-config activate-schema --schema-name
<schema-name> --schema-version <schema-version>
```

Flag

Table 184 Flags

| Name | Description |
|----------------|--|
| schema-name | schema name of the variation point metadata |
| schema-version | schema version of the variation point metadata |

Example

```
beamctl pub-variationpoint-config activate-schema --schema-name
edmeventpublishingcobaenrichment --schema-version 1.3.1
```



```
beamctl pub-variationpoint-config activate-schema --schema-name
edmeventpublishingfilter --schema-version 1.5.1
```

3.6.8.5

delete-schema

The `delete-schema` command is used to delete the variation point metadata for the given schema details.

Syntax

```
beamctl pub-variationpoint-config delete-schema --schema-name
<schema-name> --schema-version <schema-version>
```

Flag

Table 185 Flags

| Name | Description |
|----------------|--|
| schema-name | schema name of the variation point metadata |
| schema-version | schema version of the variation point metadata |

Example

```
beamctl pub-variationpoint-config delete-schema --schema-name
edmeventpublishingcobaenrichment --schema-version 1.3.1
```

```
beamctl pub-variationpoint-config delete-schema --schema-name
edmeventpublishingfilter --schema-version 1.5.1
```

3.6.8.6

validate-schema

The `validate-schema` command is used to validate the given variation point metadata.

Syntax

```
beamctl pub-variationpoint-config validate-schema < <path of the
metadata file to validate> --schema-name <schema-name>
```

Flag

Table 186 Flags

| Name | Description |
|-------------|---|
| schema-name | schema name of the variation point metadata |



Example

```
beamctl pub-variationpoint-config validate-schema  
< newenrichmentschema.csv --schema-name  
edmeventpublishingcobaenrichment
```

```
beamctl pub-variationpoint-config validate-schema <  
newfilterschema.csv --schema-name edmeventpublishingfilter
```

3.6.9 Grouping Stale Removal

When the events stored in grouping table have no reporting time entry in the reporting table, the events become stale. This can happen when the start or end session events are missed to be stored in the grouping table or published as individual events. In such case, grouping stale removal commands can be used to publish and remove the stale entries.

3.6.9.1 remove

The remove command is to remove the stale entries from grouping table. Ensure to mention the customer partition id to remove the stale entries for the required partitions. Based on the provided customer partition id, stale entries are removed.

Note: This command must be run only during the maintenance window. Use remove command only if the events are corrupted, and events are not needed anymore. It is always recommended to use **publish** command to get the stale events published and removed.

Syntax

Command Introduced in API: `beamctl pub-grouping-stale-entry remove <customer-partition>`

```
beamctl pub-grouping-stale-entry remove <customer-partition> --  
customer-id <customer-id> --session-id <session-id> --operation-  
number <operation-number> --count <count> --force <force run>
```

Argument

Table 187 Mandatory Arguments

| Argument | Description |
|--------------------|--|
| customer-partition | To remove the stale session for the given customer partition id. |



| Argument | Description |
|----------|---|
| | customer-partition takes a single valid customer partition value or Comma-Separated Values. |

Flags

Table 188 Flags

| Flag | Type | Description |
|------------------|---------|---|
| customer-id | string | To remove the stale session for the given customerId. It is optional flag and value can be valid customer id. |
| session-id | string | To remove the stale session for the given sessionId. It is optional and value can be valid session id. This should always be used in conjunction with customer-id. |
| operation-number | int | To remove the stale session for the given operationNumber. It is optional and value can be valid number. This should always be used in conjunction with customer-id and session-id |
| count | int | To remove the number of stale session for the given count. It is optional and value can be valid number and should be greater than zero and less than or equal to 100k.If not provided, the default count will be 2000. For partitionId specific stale removal, the count value cannot be guaranteed as the stale removal job runs across edm pods and deletion depends on the stale sessions present in the segments assigned to that edm pod. |
| force | boolean | To remove the stale session forcefully even though any stale session removal job already running. Use this option if the last command |



| Flag | Type | Description |
|------|------|---|
| | | run is terminated abruptly in middle of the execution due to any interruptions. It is optional and value can be either true or false. |

Example 146

The following command removes the stale entries for the customer partition 1.

Command Introduced in API: **beamctl pub-grouping-stale-entry remove <customer-partition>**

```
beamctl pub-grouping-stale-entry remove 1
```

Note: *<customer-partition>* takes a single valid customer partition value or Comma-Separated Values. Comma separated partition id is applicable only for command with partition id option. For commands with options customer id, session id, and operation number combinations, only single valid customer partition is allowed.

Example 147

The following command removes the stale entries for the customer partition 1 and customer id DFE6F4CA4E1F40F38F9E09DA5F261299.

Command Introduced in API: **beamctl pub-grouping-stale-entry remove <customer-partition> --customer-id <customer-id>**

```
beamctl pub-grouping-stale-entry remove 1 --customer-id
DFE6F4CA4E1F40F38F9E09DA5F261299
```

Example 148

The following command removes the stale entries for the customer partition 1, customer id DFE6F4CA4E1F40F38F9E09DA5F261299, and session id 29b70aab-9672-4ff8-898f-67d49c837564

Command Introduced in API: **beamctl pub-grouping-stale-entry remove <customer-partition> --customer-id <customer-id> --session-id <session-id>**

```
beamctl pub-grouping-stale-entry remove 1 --
customer-id DFE6F4CA4E1F40F38F9E09DA5F261299 --session-id
29b70aab-9672-4ff8-898f-67d49c837564
```

Example 149

The following command removes the stale entries for the customer partition 1, customer id DFE6F4CA4E1F40F38F9E09DA5F261299, session id



29b70aab-9672-4ff8-898f-67d49c837564, and session operation number 1 (INTERM events)

Command Introduced in API: **beamctl pub-grouping-stale-entry remove** *<customer-partition>* --customer-id *<customer-id>* --session-id *<session-id>* --operation-number *<operation-number>*

```
beamctl pub-grouping-stale-entry remove 1 --
customer-id DFE6F4CA4E1F40F38F9E09DA5F261299 --session-id
29b70aab-9672-4ff8-898f-67d49c837564 --operation-number 1
```

Example 150

The following command removes the stale entries for the customer partition 1, customer id DFE6F4CA4E1F40F38F9E09DA5F261299, session id 29b70aab-9672-4ff8-898f-67d49c837564, session operation number 1 (INTERM events), and limits the removal to 500 sessions.

Command Introduced in API: **beamctl pub-grouping-stale-entry remove** *<customer-partition>* --customer-id *<customer-id>* --session-id *<session-id>* --operation-number *<operation-number>* --count *<count>*

```
beamctl pub-grouping-stale-entry remove 1 --
customer-id DFE6F4CA4E1F40F38F9E09DA5F261299 --session-id
29b70aab-9672-4ff8-898f-67d49c837564 --operation-number 1 --count
500
```

Example 151

The following command removes the stale entries for the customer partition 1, customer id DFE6F4CA4E1F40F38F9E09DA5F261299, session id 29b70aab-9672-4ff8-898f-67d49c837564, session operation number 1 (INTERM events), and limits the removal to 500 sessions. Using force option, the command removes the stale entries irrespective of any stale handling job which is running already.

Command Introduced in API: **beamctl pub-grouping-stale-entry remove** *<customer-partition>* --customer-id *<customer-id>* --session-id *<session-id>* --operation-number *<operation-number>* --count *<count>* --force *<boolean>*

```
beamctl pub-grouping-stale-entry remove 1 --
customer-id DFE6F4CA4E1F40F38F9E09DA5F261299 --session-id
29b70aab-9672-4ff8-898f-67d49c837564 --operation-number 1 --count
500 --force true
```



3.6.9.2 publish

The publish command is to publish and remove the stale entries from grouping table. Ensure to mention the customer-partition id to publish and remove the stale entries for the required partitions. Based on the customer-partition id provided, stale entries are published and removed. Stale events that are published are sent to error directory with error extension.

Note: This command must be run only during the maintenance window. It is recommended to use publish command to get the stale events published and removed.

Syntax

Command Introduced in API: `beamctl pub-grouping-stale-entry publish <customer-partition>`

```
beamctl pub-grouping-stale-entry publish <customer-partition> --customer-id <customer-id> --session-id <session-id> --operation-number <operation-number> --count <count> --force <force run>
```

Argument

Table 189 Mandatory Arguments

| Argument | Description |
|--------------------|---|
| customer-partition | To remove the stale session for the given customer partition id.

customer-partition takes a single valid customer partition value or Comma-Separated Values. |

Flags

Table 190 Flags

| Flag | Type | Description |
|-------------|--------|--|
| customer-id | string | To publish and remove the stale session for the given customerId. It is optional and value can be valid customer id. |
| session-id | string | To publish and remove the stale session for the given sessionid. It is optional and value can be valid session id. |



| Flag | Type | Description |
|------------------|---------|--|
| | | This must always be used in conjunction with the customer-id. |
| operation-number | int | To publish and remove the stale session for the given operationNumber. It is optional and value can be valid number. This must always be used in conjunction with customer-id and session-id |
| count | int | To publish and remove the number of stale session for the given count. It is optional and value can be valid number and must be greater than zero and less than or equal to 100k. If not provided, the default count is 2000. For partitionId specific stale removal, the count value cannot be guaranteed as the stale removal job runs across edm pods, and the deletion depends on the stale sessions available in the segments which are assigned to that edm pod. |
| force | boolean | To publish and remove the stale session forcefully even though any stale session removal job is running already. Use this option, if the last command run is terminated abruptly in middle of the execution due to any interruptions. It is optional and value can be either true or false. |

Example 152

The following command removes the stale entries for the customer partition 1.

Command Introduced in API: **beamctl pub-grouping-stale-entry publish <customer-partition>**

beamctl pub-grouping-stale-entry publish 1



Note: *<customer-partition>* takes a single valid customer partition value or Comma-Separated Values. Comma separated partition id is applicable only for command with partition id option, for commands with options customer id, session id, and operation number combinations, only single valid customer partition is allowed.

Example 153

The following command publishes and removes the stale entries for the customer partition 1 and customer id DFE6F4CA4E1F40F38F9E09DA5F261299.

Command Introduced in API: **beamctl pub-grouping-stale-entry publish <customer-partition> --customer-id <customer-id>**

```
beamctl pub-grouping-stale-entry publish 1 --customer-id
DFE6F4CA4E1F40F38F9E09DA5F261299
```

Example 154

The following command publishes and removes the stale entries for the customer partition 1, customer id DFE6F4CA4E1F40F38F9E09DA5F261299, and session id 29b70aab-9672-4ff8-898f-67d49c837564

Command Introduced in API: **beamctl pub-grouping-stale-entry publish <customer-partition> --customer-id <customer-id> --session-id <session-id>**

```
beamctl pub-grouping-stale-entry publish 1 --
customer-id DFE6F4CA4E1F40F38F9E09DA5F261299 --session-id
29b70aab-9672-4ff8-898f-67d49c837564
```

Example 155

The following command publishes and removes the stale entries for the customer partition 1, customer id DFE6F4CA4E1F40F38F9E09DA5F261299, session id 29b70aab-9672-4ff8-898f-67d49c837564, and session operation number 1 (INTERM events)

Command Introduced in API: **beamctl pub-grouping-stale-entry publish <customer-partition> --customer-id <customer-id> --session-id <session-id> --operation-number <operation-number>**

```
beamctl pub-grouping-stale-entry publish 1 --
customer-id DFE6F4CA4E1F40F38F9E09DA5F261299 --session-id
29b70aab-9672-4ff8-898f-67d49c837564 --operation-number 1
```

Example 156

The following command publishes and removes the stale entries for the customer partition 1, customer id DFE6F4CA4E1F40F38F9E09DA5F261299, session id 29b70aab-9672-4ff8-898f-67d49c837564, session operation number 1 (INTERM events), and limits the removal to 500 sessions.



Command Introduced in API: **beamctl pub-grouping-stale-entry publish** *<customer-partition>* --customer-id *<customer-id>* --session-id *<session-id>* --operation-number *<operation-number>* --count *<count>*

```
beamctl pub-grouping-stale-entry publish 1 --
customer-id DFE6F4CA4E1F40F38F9E09DA5F261299 --session-id
29b70aab-9672-4ff8-898f-67d49c837564 --operation-number 1 --count
500
```

Example 157

The following command publishes and removes the stale entries for the customer partition 1, customer id DFE6F4CA4E1F40F38F9E09DA5F261299, session id 29b70aab-9672-4ff8-898f-67d49c837564, session operation number 1 (INTERM events), and limits the removal to 500 sessions. Using **force** option, the command removes the stale entries irrespective of any stale handling job that is already running.

Command Introduced in API: **beamctl pub-grouping-stale-entry publish** *<customer-partition>* --customer-id *<customer-id>* --session-id *<session-id>* --operation-number *<operation-number>* --count *<count>* --force *<boolean>*

```
beamctl pub-grouping-stale-entry publish 1 --
customer-id DFE6F4CA4E1F40F38F9E09DA5F261299 --session-id
29b70aab-9672-4ff8-898f-67d49c837564 --operation-number 1 --count
500 --force true
```

3.6.9.3

status

The status command fetches the status of the stale entry removal job.

Syntax

Command Introduced in API: **beamctl pub-grouping-stale-entry status**

```
beamctl pub-grouping-stale-entry status
```

Argument

The command has no arguments.

Example 158

Command Introduced in API: **beamctl pub-grouping-stale-entry status**



3.7 FIN Configuration Commands

3.7.1 FIN Application Configuration

The `fin-access-config` command line option can be used to read and update values present in the FIN Application Configuration.

Syntax

```
beamctl fin-access-config
```

Available Commands

Table 191 Available Sub Commands

| Command | Description |
|---------|---|
| get | Gets the app configuration parameter. |
| get-all | Gets application configuration values for all contexts or for a specific context with all attributes. |
| set | Sets application configuration values for a given context and attribute(s). |

For a list of application configuration contexts, see [FIN Application Configuration Function Guide](#).

Note: The ConfigMap `eric-bss-fin-processor-overdue-receivable-properties` and its parameters are kept in the system for future use.

3.7.1.1 get

The `get` command retrieves application configuration values for a given context and attribute.

Syntax

```
beamctl fin-access-config get <context> <attribute>
```

Arguments

Table 192 Mandatory Arguments

| Argument | Description |
|----------|--|
| context | The application configuration context. |



| Argument | Description |
|-----------|--|
| attribute | The name of the attribute from within the context. |

Table 193 JSON Parameters

| Parameter | Parameter Type | Description |
|---------------------|----------------|---|
| dataExportSlackTime | Mandatory | This parameter sets the slack time for Data Export exportTask in seconds. |

Example 159

```
beamctl fin-access-config get eric-bss-fin-processor-data-export-properties dataExportSlackTime
```

Sample Response:

```
{
  "dataExportSlackTime": 3605
}
```

3.7.1.2

get-all

The `get-all` command lists all application configuration values for all contexts and their attributes.

Syntax

```
beamctl fin-access-config get-all
```

Arguments

The command has no arguments.

Parameters

Table 194 Parameters

| Parameter | Parameter Type | Description |
|--------------------------------|----------------|--|
| name | | |
| data | | |
| adminOperationLookupTimeToLive | Optional | <p>The Time to Live interval for adminOperationLookuptable in seconds.</p> <p>Note: — The TTL value must be greater than 0.</p> |



| Parameter | Parameter Type | Description |
|--|----------------|--|
| precisionValue | Mandatory | Default value for the Currency Precision under FIN. |
| invoiceQueuePollingInterval | Mandatory | The time interval in milliseconds that defines the delay, FIN applies when polling for new messages on the Invoice queue. |
| numberOfTopicPartitonsForExportData | Mandatory | Number of topic partitions for Export Data topic. |
| numberOfTopicPartitionsForFinCustomerEvent | Mandatory | Number of topic partitions for FIN Customer Event topic. |
| numberOfTopicPartitionsForFinCustomerInvoiceEvent | Mandatory | Number of topic partitions for Fin Customer Invoice Event topic. |
| numberOfTopicPartitionsForOverdueReceivableDetection | Mandatory | Number of topic partitions for Overdue Receivable Detection topic. |
| numberOfTopicPartitionsForPartyRoleTasks | Mandatory | Number of topic partitions for Party Role Tasks topic. This is not over-writable. |
| numberOfTopicPartitionsForPaymentInitiation | Mandatory | Number of topic partitions for Payment Initiation topic. |
| maxNumberOfFinancialEntriesToDisplay | Optional | The maximum number of FEs to be retrieved for display purposes in the Financial details view of the FIN Accounting Center. Only applicable for the FIN internal GUI REST. Use value '-1' to turn off the FE limit. Values less than '-1' are not accepted. A value of zero means that no FEs are displayed at all. |
| enableRequestValidation | Mandatory | Switching on or off JSON schema validation. If set to "true", the request payload is validated against the JSON schema meaning the request is rejected if it does not match the schema. If set to "false", schema validation is skipped. It is expected that schema validation is switched off in production systems and used only in test environments because of the performance impact brought by the validation process. |
| logDirectoryPath | Mandatory | Used to set the log directory path for monitoring. |
| logDirectoryUsageThreshold | Mandatory | Used to set the log directory usage threshold before raising an alarm. |
| lingerMs | Mandatory | The amount of time (in milliseconds), a producer is willing to wait before sending out a batch. |
| maxBlockMs | Mandatory | The amount of time (in milliseconds), a producer is used to block buffer time and serialization time. |
| maxMessageBytes | Optional | The largest record batch size (in bytes) allowed by Kafka after compression. |
| maxRequestSize | Optional | The maximum size of a request is in bytes. |
| ttlInterval | Optional | The Time to Live interval for duplicate requests or events, in seconds. |



| Parameter | Parameter Type | Description |
|--------------------------------------|----------------|--|
| finSensitiveDataEncryptionEnabled | Mandatory | Application-level flag for sensitive data encryption. |
| finTransientDataEncryptionEnabled | Mandatory | Application-level flag for transient data encryption. |
| actorCreationTimeout | Optional | Time out used to define the actor creation. |
| forkJoinExecutorParallelismFactor | Optional | The factor of the Fork Join Executor Parallelism. |
| forkJoinExecutorParallelismMax | Optional | The maximum number of the Fork Join Executor Parallelism. |
| forkJoinExecutorParallelismMin | Optional | The minimum number of the Fork Join Executor Parallelism. |
| dataExportMessageBlockSize | Optional | This parameter sets the block size of ExportData entities. |
| dataExportSlackTime | Mandatory | This parameter sets the slack time for Data Export exportTask in seconds. |
| dataExportTimeToLive | | |
| defaultDataExportPeriodDuration | | |
| financeEvent | Mandatory | Used to generate an external event Finance Event towards EDM if the parameter value is set to "enabled". |
| overdueReceivableEvent | Mandatory | Used to generate external event Overdue Receivable Event towards EDM if the parameter value is set to "enabled". |
| paymentInstructionEvent | Mandatory | Used to generate an external event Payment Instruction Event towards EDM if the parameter value is set to "enabled". |
| noLoadIncrementSleepDuration | Mandatory | Incremental sleep duration in ms, which is accumulated over base sleep duration. |
| noLoadMaxIncrementSleepDurationCount | Mandatory | Max accumulation count for increment sleep duration addition over base sleep duration. |
| fetchMaxBytes | Mandatory | Maximum amount of data (in bytes) which the server shall return for a fetch request. The consumer fetches the records in batches and if the first record batch in the first non-empty partition of the fetch is larger than this value, the record batch is still returned to ensure that the consumer advances. |
| maxPartitionFetchBytes | Mandatory | Maximum amount of data (in bytes) which the server returns per partition. The consumer fetches the records in batches and if the first record batch in the first non-empty partition of the fetch is larger than this value, the batch is still returned to ensure that the consumer advances. |
| maxPollIntervalBatchTasks | Mandatory | Maximum delay time (in minutes) between invocation of poll for the Batch tasks. |
| maxPollIntervalFinTasks | Mandatory | Maximum delay time (in hours) between invocation of poll for the Financial tasks. It shall be set |



| Parameter | Parameter Type | Description |
|--------------------------------------|----------------|--|
| | | to one hour to cover the long-running transactions. |
| noOfPollBatchTasks | Mandatory | The maximum number of records returned in a single call to poll. |
| noOfPollFinTasks | Mandatory | The maximum number of records returned in a single call to poll. |
| partitionAssignmentStrategy | Mandatory | It is used for the consumer assignment strategy. If all the consumers of a consumer group are assigned to one topic, the range and Round Robin are the same. Round Robin is better balanced if the consumer is assigned to several topics. |
| acks | | |
| batchSize | Optional | The maximum batch size (in bytes) for message batching optimization. |
| bootstrapServers | | |
| bufferMemory | | |
| deliveryTimeoutMs | | |
| maxInFlightRequestsPerConnection | | |
| overdueReceivableTaskSlackTime | | |
| requestTimeoutMs | | |
| retries | | |
| retryBackoffMs | | |
| securityProtocol | | |
| sslEnabledProtocols | | |
| sslKeystoreLocation | | |
| sslKeystoreType | | |
| sslProvider | | |
| sslTruststoreLocation | | |
| sslTruststoreType | | |
| topicName | | |
| paymentInitiationCutOffDueDateOffset | Mandatory | Sets a default value (in days) for CutOffDueDateOffset in case Business Scheduler does not send it at the scheduled payload. |
| paymentInitiationReportTimeToLive | Mandatory | Sets the Time to Live (in seconds) for Payment Initiation Reports. |
| paymentInitiationTaskSlackTime | Mandatory | Sets the slack time (in milliseconds) to process invoices at the Payment Initiation Task. |
| timeoutForBulkTaskProcessing | Optional | Sets the time out for bulk tasks processing (in milliseconds). The maximum time limit per message in a bulk is 3 seconds (3000 milliseconds). Each bulk has approximately 100 messages (with an assumption of no more than five extra messages). |
| dataExportJournalPurgePeriod | Mandatory | To configure the purging period (in days) for data export Journal entries. |



| Parameter | Parameter Type | Description |
|------------------------------------|----------------|--|
| | | Note: The TTL value must be greater than 0. |
| encryptionPurgePeriod | Mandatory | Purging period (in days) for Data Maintenance task status in encryptionJournal table. |
| fhPurgePeriodForCustomer | Mandatory | <p>To configure the purging period (in days) for Financial Headers belonging to customer party role.</p> <p>Note:</p> <ul style="list-style-type: none"> — The TTL value must be greater than or equal to the billing cycle. — The TTL value must be greater than 0. — Financial Header Purge Period must be greater than the financial transaction purge period. |
| fhPurgePeriodForPaymentResponsible | Mandatory | <p>To configure the purging period (in days) for Financial Headers belonging to payment responsible party role.</p> <p>Note:</p> <ul style="list-style-type: none"> — The TTL value must be greater than or equal to the billing cycle. — The TTL value must be greater than 0. — Financial Header Purge Period must be greater than the financial transaction purge period. |
| ftxPurgePeriodForCustomer | Mandatory | To configure the purging period (in days) for Financial Transactions belonging to customer party role. |



| Parameter | Parameter Type | Description |
|--------------------------------------|----------------|--|
| | | Note: <ul style="list-style-type: none">— The TTL value must be greater than or equal to the billing cycle.— The TTL value must be greater than 0.— Financial Header Purge Period must be greater than the financial transaction purge period. |
| ftxPurgePeriodForPaymentResponsible | Mandatory | To configure the purging period (in days) for financial transactions belonging to payment responsible party role.
Note: <ul style="list-style-type: none">— The TTL value must be greater than or equal to the billing cycle.— The TTL value must be greater than 0.— Financial Header Purge Period must be greater than the financial transaction purge period. |
| paymentInstructionJournalPurgePeriod | Mandatory | To configure the purging period (in days) for data export Journal entries.
Note: The TTL value must be greater than 0. |
| paymentInstructionsPurgePeriod | Mandatory | To configure the purging period (in days) for payment instructions.
Note: The TTL value must be greater than 0. |

Example 160

beamctl fin-access-config get-all

Sample Response:

```
{
  "configmaps": [
    {
      "data": {
```



```

      "adminOperationLookupTimeToLive": 86400
    },
    {
      "name": "eric-bss-fin-access-admin-operation-lookup-properties"
    },
    {
      "data": {
        "precisionValue": 6
      },
      "name": "eric-bss-fin-access-base-precision-value-properties"
    },
    {
      "data": {
        "invoiceQueuePollingInterval": 1000,
        "numberOfTopicPartitionsForExportData": 20,
        "numberOfTopicPartitionsForFinCustomerEvent": 20,
        "numberOfTopicPartitionsForFinCustomerInvoiceEvent": 20,
        "numberOfTopicPartitionsForOverdueReceivableDetection": 20,
        "numberOfTopicPartitionsForPartyRoleTasks": 20,
        "numberOfTopicPartitionsForPaymentInitiation": 20
      },
      "name": "eric-bss-fin-access-finance-messaging-properties"
    },
    {
      "data": {
        "maxNumberOfFinancialEntriesToDisplay": 50
      },
      "name": "eric-bss-fin-access-internal-gui-properties"
    },
    {
      "data": {
        "enableRequestValidation": true
      },
      "name": "eric-bss-fin-access-json-validator-properties"
    },
    {
      "data": {
        "logDirectoryPath": "/var/log",
        "logDirectoryUsageThreshold": 80
      },
      "name": "eric-bss-fin-access-log-monitoring-config-properties"
    },
    {
      "data": {
        "lingerMs": 16,
        "maxBlockMs": 500,
        "maxMessageBytes": 1048588,
        "maxRequestSize": 1048576
      },
      "name": "eric-bss-fin-access-msg-producer-config-properties"
    },
    {
      "data": {
        "ttlInterval": 604800
      },
      "name": "eric-bss-fin-access-ttl-properties"
    },
    {
      "data": {
        "finSensitiveDataEncryptionEnabled": "N",
        "finTransientDataEncryptionEnabled": "N"
      },
      "name": "eric-bss-fin-config-encryption-properties"
    },
    {
      "data": {
        "adminOperationLookupTimeToLive": 86400
      },
      "name": "eric-bss-fin-processor-admin-operation-lookup-properties"
    },
    {
      "data": {
        "actorCreationTimeout": "20s",
        "forkJoinExecutorParallelismFactor": 10,
        "forkJoinExecutorParallelismMax": 128,
        "forkJoinExecutorParallelismMin": 8
      },
      "name": "eric-bss-fin-processor-akka-properties"
    },
    {

```



```
"data": {
  "dataExportMessageBlockSize": 12000000,
  "dataExportSlackTime": 3600,
  "dataExportTimeToLive": 259200,
  "defaultDataExportPeriodDuration": 1
},
"name": "eric-bss-fin-processor-data-export-properties"
},
{
  "data": {
    "financeEvent": "enabled",
    "overdueReceivableEvent": "enabled",
    "paymentInstructionEvent": "enabled"
  },
  "name": "eric-bss-fin-processor-event-generate-properties"
},
{
  "data": {
    "invoiceQueuePollingInterval": 1000,
    "numberOfTopicPartitionsForExportData": 20,
    "numberOfTopicPartitionsForFinCustomerEvent": 20,
    "numberOfTopicPartitionsForFinCustomerInvoiceEvent": 20,
    "numberOfTopicPartitionsForOverdueReceivableDetection": 20,
    "numberOfTopicPartitionsForPartyRoleTasks": 20,
    "numberOfTopicPartitionsForPaymentInitiation": 20
  },
  "name": "eric-bss-fin-processor-finance-messaging-properties"
},
{
  "data": {
    "noLoadIncrementSleepDuration": 1000,
    "noLoadMaxIncrementSleepDurationCount": 1
  },
  "name": "eric-bss-fin-processor-finance-queue-message-properties"
},
{
  "data": {
    "logDirectoryPath": "/var/log",
    "logDirectoryUsageThreshold": 80
  },
  "name": "eric-bss-fin-processor-log-monitoring-config-properties"
},
{
  "data": {
    "fetchMaxBytes": 52428800,
    "maxPartitionFetchBytes": 1048576,
    "maxPollIntervalBatchTasks": 300000,
    "maxPollIntervalFinTasks": 3600000,
    "noOfPollBatchTasks": 5,
    "noOfPollFinTasks": 1000,
    "partitionAssignmentStrategy": "org.apache.kafka.clients.consumer.RangeA →
ssignor"
  },
  "name": "eric-bss-fin-processor-msg-consumer-config-properties"
},
{
  "data": {
    "lingerMs": 16,
    "maxBlockMs": 500,
    "maxMessageBytes": 1048588,
    "maxRequestSize": 1048576
  },
  "name": "eric-bss-fin-processor-msg-producer-config-properties"
},
{
  "data": {
    "acks": "all",
    "batchSize": 16384,
    "bootstrapServers": "localhost:9092",
    "bufferMemory": 33554432,
    "deliveryTimeoutMs": 120000,
    "lingerMs": 0,
    "maxBlockMs": 500,
    "maxInFlightRequestsPerConnection": 5,
    "overdueReceivableTaskSlackTime": 3600000,
    "requestTimeoutMs": 30000,
    "retries": 2147483647,
    "retryBackoffMs": 100,
    "securityProtocol": "PLAINTEXT",
```



```

      "sslEnabledProtocols": "n/a",
      "sslKeystoreLocation": "n/a",
      "sslKeystoreType": "n/a",
      "sslProvider": "n/a",
      "sslTruststoreLocation": "n/a",
      "sslTruststoreType": "n/a",
      "topicName": "overdue_receivable_event"
    },
    {
      "name": "eric-bss-fin-processor-overdue-receivable-properties"
    },
    {
      "data": {
        "paymentInitiationCutOffDueDateOffset": 5,
        "paymentInitiationReportTimeToLive": 259200,
        "paymentInitiationTaskSlackTime": 3600000
      },
      "name": "eric-bss-fin-processor-payment-initiation-properties"
    },
    {
      "data": {
        "timeoutForBulkTaskProcessing": 315000
      },
      "name": "eric-bss-fin-processor-processor-service-config-properties"
    },
    {
      "data": {
        "dataExportJournalPurgePeriod": 30,
        "encryptionPurgePeriod": 10,
        "fhPurgePeriodForCustomer": 365,
        "fhPurgePeriodForPaymentResponsible": 365,
        "ftxPurgePeriodForCustomer": 180,
        "ftxPurgePeriodForPaymentResponsible": 180,
        "paymentInstructionJournalPurgePeriod": 30,
        "paymentInstructionsPurgePeriod": 30
      },
      "name": "eric-bss-fin-processor-purge-period-properties"
    }
  ]
}

```

3.7.1.2.1 get-all --context

The `get-all --context` command allows the user to limit the result to a specific context and its attributes.

Syntax

```
beamctl fin-access-config get-all --context <context>
```

Arguments

Table 195 Mandatory Arguments

| Argument | Description |
|----------|-------------------------------|
| context | The context of the ConfigMap. |



Parameters

Table 196 Parameters

| Parameter | Parameter Type | Description |
|-------------------------------|----------------|--|
| dataExportMessageBlockSize | Optional | This parameter sets the block size of ExportData entities. |
| dataExportSlackTime | Mandatory | This parameter sets the slack time for Data Export exportTask in seconds. |
| defaultExportPeriodDuration | Mandatory | This parameter sets the default exportPeriodDuration, used if this information is not sent by Business Scheduler. This is a positive number and its unit is [day]. |
| generatedExportDataTimeToLive | Optional | This parameter sets the Time to Live of ExportData generated entities in seconds. |

Example 161

```
beamctl fin-access-config get-all --context eric-bss-fin-processor-data-export-properties
```

Sample Response:

```
{
  "configmaps": [
    {
      "data": {
        "dataExportMessageBlockSize": 12000000,
        "dataExportSlackTime": 3600,
        "defaultExportPeriodDuration": 1,
        "generatedExportDataTimeToLive": 604800
      },
      "name": "eric-bss-fin-processor-data-export-properties"
    }
  ]
}
```

3.7.1.3

set

The set command updates single or multiple application configuration values for a given context and its attributes.

Syntax

```
beamctl fin-access-config set <context> [<|<<|<<<] <RedirectedInput>
```

```
beamctl fin-access-config set <context> <attribute>:<value>
[, <anotherattribute>:<anothervalue>]
```




Arguments

Table 197 Mandatory Arguments

| Arguments | Description |
|-----------------------|---|
| context | The context of the ConfigMap. |
| attribute | Key in ConfigMap. |
| value | Value of the key. |
| < <RedirectedInput> | Input from a JSON file. |
| << <RedirectedInput> | Here document, which tells the CLI to continue to read input until a specified delimiter is found, usually EOF. |
| <<< <RedirectedInput> | Here string, which is like here document but it only takes one word as input without a delimiter. |

Parameters

Table 198 Parameters

| Parameter | Parameter Type | Description |
|---------------------|----------------|---|
| dataExportSlackTime | Mandatory | This parameter sets the slack time for Data Export exportTask in seconds. |
| | | |
| | | |

Example 162

Using set context <<< <<< '{"property1":"new value","property2":"new value"}':

```
beamctl fin-access-config set eric-bss-fin-processor-data-export-properties <<< '{"dataExportSlackTime":3605}'
```

Example 163

Creating a payload JSON file containing the properties and new values to update and then use it with the set command:

```
beamctl fin-access-config set eric-bss-fin-processor-data-export-properties < dataexport.json
```



Example 164

```
beamctl fin-access-config set eric-bss-fin-processor-data-export-properties "dataExportSlackTime":3601
```

Example 165

```
beamctl fin-access-config set eric-bss-fin-processor-data-export-properties << EOF

{
  "dataExportSlackTime": 3606
}
EOF
```

3.8 NTF Configuration Commands

3.8.1 NTF APP Application Configuration

NTF Application Configuration REST Interface has a command `ntf-config`, which provides the options to manage (Read and Update) a resource.

Note: When failover of an AG is done, the O&M services from the failed AG give a response as "Service Unavailable" with error code 503.

Table 199 NTF Config Commands

| Command | Description |
|---------|---|
| get-all | It retrieves all the keys available in the resource. |
| get | It retrieves the value for a key passed as an argument. |
| set | It sets the value of a key, available in three different ways of input. |

Table 200 NTF Valid Context

| Context | Description |
|------------------|--|
| msgkafkaconsumer | MSG Kafka client consumer configuration. |



| Context | Description |
|----------------------------------|--|
| ntfcilrequestconfig | Notif CIL Request Priorities. |
| ntfgenericlookuprestcommonconfig | NTFGenericLookUpRest Global configuration parameters. |
| ntfgenericlookuprestconfig | NTFGenericLookUpRest Client configuration parameters. |
| ntfmrriconfig | NTF Messaging Receiver REST Interface configuration parameters for MRRI Gateway. |
| ntfsmppcommonconfig | SMPP Global configuration parameters. |
| ntfsmppconfig | SMPP client configuration parameters. |
| ntfsmtpcommonconfig | SMTP Global configuration parameters. |
| ntfsmtpconfig | SMTP client configuration parameters. |
| ntfssystemconfig | Default Notifies System configuration. |
| ntfmsgkafkaconfig | MSG Kafka client configuration parameters. |
| ntfmsgkafkatopicconfig | MSG Kafka topic configuration parameters. |
| ntfmultisiteconfig | NTF Multisite configuration parameters. |
| ntfeventcontrolconfig | NTF Event Control configuration parameters. |

3.8.1.1

get-all

Syntax

```
beamctl ntf-config get-all
```

```
beamctl ntf-config get-all --context <context>
```

Arguments

Table 201 Mandatory Arguments

| Argument | Description |
|----------|--|
| context | The value for this must be a string as mentioned in the valid context table. |



| Argument | Description |
|----------|---|
| | If incorrect or a non-exist key is provided, the response is 404 Not Found. |

Parameters

Table 202 JSON Parameters

| Parameter | Parameter Type | Description |
|---------------------|----------------|--|
| topicConfiguration | | |
| topicNames | Optional | Name of the topic |
| retentionInMins | Optional | Retention in Minutes at topic level |
| topicPartition | Optional | Total number of partitions to be for the topic |
| partitionInfo | Optional | Partition id for the corresponding topic |
| topicPartitionCount | Optional | Partition count against the partition Info |

Example 166

```
beamctl ntf-config get-all --context ntFMsgKafkaTopicConfig
```

Response:

```
{
  "ntFMsgKafkaTopicConfig": {
    "topicConfiguration": {
      "topicNames": [
        {
          "retentionInMins": 78,
          "topicName": "ntf_event_customer",
          "topicPartition": [
            {
              "partitionInfo": 1,
              "topicPartitionCount": 28
            },
            {
              "partitionInfo": 2,
              "topicPartitionCount": 28
            },
            {
              "partitionInfo": 3,
              "topicPartitionCount": 28
            },
            {
              "partitionInfo": 4,
              "topicPartitionCount": 28
            }
          ]
        }
      ]
    }
  }
}
```



```

    }
  }
]

```

3.8.1.2

get

Syntax

```
beamctl ntf-config get <context> <key>
```

Arguments

Table 203 Mandatory Arguments

| Argument | Description |
|----------|---|
| context | The value for this must be a string as mentioned in the valid context table.

If incorrect or non exist key is provided, the response is 404 Not Found. |
| key | The value is a string and must exist in the context of a resource.

See get-all on page 351 section on how to get the keys for a specific context.

If incorrect or non-exist key is provided, the response is 404 Not Found. |

Parameters

Table 204 JSON Parameters

| Parameter | Parameter Type | Description |
|--------------------|----------------|---|
| topicConfiguration | | |
| topicNames | Optional | Name of the topic |
| retentionInMins | Optional | Retention in Minutes at topic level |
| topicPartition | Optional | Total number of partitions to be for the topic. |
| partitionInfo | Optional | Partition id for the corresponding topic |



| Parameter | Parameter Type | Description |
|---------------------|----------------|--|
| topicPartitionCount | Optional | Partition count against the partition Info |

Example 167

To view values of a specific key in a context:

```
beamctl ntf-config get ntFMsgKafkaTopicConfig topicConfiguration
```

Response:

```
{
  "topicConfiguration": {
    "topicNames": [
      {
        "retentionInMins": 78,
        "topicName": "ntf_event_customer",
        "topicPartition": [
          {
            "partitionInfo": 1,
            "topicPartitionCount": 28
          },
          {
            "partitionInfo": 2,
            "topicPartitionCount": 28
          },
          {
            "partitionInfo": 3,
            "topicPartitionCount": 28
          },
          {
            "partitionInfo": 4,
            "topicPartitionCount": 28
          }
        ]
      }
    ]
  }
}
```

3.8.1.3

set

The set command updates a single or multiple attributes of an application configuration.

Syntax

```
beamctl ntf-config set <context> <available input options>
```



Arguments

Table 205 Mandatory Arguments

| Argument | Description |
|----------|--|
| context | The value for this must be a String, as mentioned in the valid context table.

If it is incorrect or a non-exist key is provided, the response is <i>404 Not Found</i> . |

Options

Option 1

Using shorthand syntax, key, and value with a colon character `:` delimiter and multiple keys are delimited with a comma character `,`.

Example:

```
beamctl ntf-config set ntfcilrequestconfig
readRestartPriority:6
```

Option 2

Using `<` means input from a file, typically a JSON file.

Example:

```
beamctl ntf-config set ntFCilRequestConfig <
ntfcil.json
```

```
ntfcil.json
{
  "dataCenterSelection": "USE_ANY",
  "numOfSegments": 1200,
  "numOfSegmentsPerThread": 300,
  "readHighPriority": 250,
  "readLowPriority": 32,
  "readRestartPriority": 10,
  "writeHighPriority": 4095,
  "writeLowPriority": 256
}
```

Option 3

Using `<<` operator, which tells the CLI to continue to read input until a specified delimiter is found, usually EOF.

Example:

```
beamctl ntf-config set ntFCilRequestConfig <<
EOF
```

```
{
  "readRestartPriority": 10
}
EOF
```



Option 4 Using <<< operator, which is like here document but it only takes one word as input without a delimiter.

Example:

```
beamctl ntf-config set ntFCilRequestConfig <<<
'{"readRestartPriority":10,"writeHighPriority"
:4095}'
```

Note: BEAM CLI set command operation is not allowed for the context ntfmtsgkafkatopicconfig. If set command operation is done, then the following error response is displayed:

```
HTTP 405:
{"status":405,"code":"InvalidUpdateOperation","message":"U
pdate operation is not allowed.","details":"Update
operation for NTFMsgKafkaTopicConfig is not allowed"}
```

3.9 RMCA Configuration Commands

3.9.1 RMCA-SERVER-CONFIG

The rmca-server-config command provides the options to manage (Read and Update) RMCA Server configuration.

Table 206 RMCA Server Config Commands

| Command | Description |
|---------|---|
| get-all | It retrieves all the keys available in the resource. |
| get | It retrieves the value for a key passed as an argument. |
| set | It sets the value of a key, available in three different ways of input. |

3.9.1.1 get-all

This command retrieves all the keys present in the configmap resource.

Syntax

```
beamctl rmca-server-config get-all
```

Arguments

The command has no arguments.



Parameters

Table 207 JSON Parameters

| Parameter | Parameter Type | Description |
|--|----------------|--|
| businessConfigurationChangeEventGeneration | Mandatory | This parameter is used to control generation of event. |
| businessConfigurationChangeEventVersion | Mandatory | This parameter is used to change event version. |
| This parameter is deprecated.
msgTopicPartitionCount | Mandatory | This parameter value represents the MSG Topic partition count. |
| This parameter is deprecated.
msgTopicReplicationFactor | Mandatory | This parameter value represents the MSG Topic replication factor. |
| This parameter is deprecated.
msgTopicRetentionInMins | Mandatory | This parameter value represents the MSG Topic retention in mins. |
| This parameter is deprecated.
msgTopicRetriesCount | Mandatory | This parameter value represents the MSG Topic retries count. |
| rollbackWindow | Mandatory | This parameter value is used to configure the length of the rollback window in days. |
| topicConfiguration | Optional | This parameter value represents the MSG/CIL Topic Configuration. |

Example 168

Request:

```
beamctl rmca-server-config get-all
```

Response:

```
{
  "businessConfigurationChangeEventGeneration": "enabled"
  "businessConfigurationChangeEventVersion": 0,
  "msgTopicPartitionCount": 16,
  "msgTopicReplicationFactor": 3,
```



```
"msgTopicRetentionInMins": 3,
"msgTopicRetriesCount": 3,
"rollbackWindow": 7
"topicConfiguration": "[{topicNames: [{retentionInMins: 3, top →
icName: rmca_businessconfig_event, topicPartition: [{partitionIn →
fo: 0, topicPartitionCount: 18}}]]}"
}
```

3.9.1.2 **get**

This command retrieves all the keys present in the configmap resource.

Syntax

beamctl rmca-server-config get <existingKey>

Arguments

Table 208 Mandatory Arguments

| Argument | Description |
|-------------|--|
| existingKey | The value must be a string and exist in the configMap resource.

If the given key value is incorrect or non-exist, then the response is 404 Not Found. |

Parameters

Table 209 JSON Parameters

| Parameter | Parameter Type | Description |
|--|----------------|--|
| businessConfigurationChangeEventVersion | Mandatory | This parameter is used to change event version. |
| businessConfigurationChangeEventGeneration | Mandatory | This parameter is used to control generation of event. |
| rollbackWindow | Mandatory | This parameter value is used to configure the length of the rollback window in days. |
| topicConfiguration | Optional | This parameter value represents the MSG/CIL Topic Configuration. |



Example 169

Request:

```
beamctl rmca-server-config get  
businessConfigurationChangeEventVersion
```

Response:

```
{  
  "businessConfigurationChangeEventVersion": 0  
}
```

3.9.1.3

set

This command sets the value for a key passed in the payload with the help of the available options.

Syntax

```
beamctl rmca-server-config set <available input options>
```

Arguments

The command has no arguments.

Input Options

Option 1

Using shorthand syntax, key and value with a colon character : delimiter and multiple keys are delimited with a comma character ,.

Parameters

Table 210 JSON Parameters

| Parameter | Parameter Type | Description |
|---|----------------|---|
| businessConfigurationChangeEventVersion | Mandatory | This parameter is used to change event version. |
| This parameter is deprecated.
msgTopicPartitionCount | Mandatory | This parameter value represents the MSG Topic partition count. |
| This parameter is deprecated. | Mandatory | This parameter value represents the MSG Topic replication factor. |



| Parameter | Parameter Type | Description |
|--|----------------|--|
| msgTopicReplicationFactor | | |
| This parameter is deprecated.
msgTopicRetentionInMins | Mandatory | This parameter value represents the MSG Topic retention in mins. |
| This parameter is deprecated.
msgTopicRetriesCount | Mandatory | This parameter value represents the MSG Topic retries count. |
| rollbackWindow | Mandatory | This parameter value is used to configure the length of the rollback window in days. |

Example 170

Request:

```
beamctl rmca-server-config set  
businessConfigurationChangeEventVersion: 0
```

Response:

```
Service config [businessConfigurationChangeEventVersion] updated  
successfully
```

Example 171

Request:

```
beamctl rmca-server-config set  
businessConfigurationChangeEventVersion: 0,  
msgTopicPartitionCount: 16
```

Response:

```
Service config [businessConfigurationChangeEventVersion,  
msgTopicPartitionCount] updated successfully
```

Option 2

Using < means input from a file, typically a JSON file.

Example 172

Request:

```
beamctl rmca-server-config set < option1.json
```



Response:

```
Service config [businessConfigurationChangeEventVersion,
msgTopicPartitionCount, msgTopicReplicationFactor,
msgTopicRetentionInMins, msgTopicRetriesCount] updated
successfully
```

Option 3

Using << operator, which tells the CLI to continue to read input until a specified delimiter is found, usually EOF.

Example 173

Request:

```
beamctl rmca-server-config set << EOF
```

```
{
    " businessConfigurationChangeEventVersion ": 0
}
EOF
```

Response:

```
Service config [businessConfigurationChangeEventVersion] updated
successfully
```

Example 174

Request:

```
beamctl rmca-server-config set << EOF
```

```
{
  "businessConfigurationChangeEventVersion": 0,
  "msgTopicPartitionCount": 16,
  "msgTopicReplicationFactor": 3
}
EOF
```

Response:

```
Service config [businessConfigurationChangeEventVersion,
msgTopicPartitionCount, msgTopicReplicationFactor] updated
successfully
```

Option 4



Using <<< operator, which is like here document but it only takes one word as input without a delimiter.

Example 175

Request:

```
beamctl rmca-server-config set <<<
'{"businessConfigurationChangeEventVersion": 0}'
```

Response:

Service config [businessConfigurationChangeEventVersion] updated successfully

Example 176

Request:

```
beamctl rmca-server-config set
<<< '{"businessConfigurationChangeEventVersion":
0,"msgTopicPartitionCount": 16}'
```

Response:

Service config [businessConfigurationChangeEventVersion, msgTopicPartitionCount] updated successfully

3.9.2 RMCA-BUSINESSCONFIGURATION-EXPORT-UTILITY

The rmca-businessconfiguration-export-utility command provides the capability to export specifications from RMCA.

Table 211 RMCA Business Configuration Export Utility Command

| Command | Description |
|---------|--|
| export | Export all the specifications specified in the JSON payload request. |

3.9.2.1 export

The command exports the RMCA Specifications.

Syntax

```
beamctl rmca-businessconfiguration-export-utility export <

```



Arguments

The command has no arguments.

Parameters

Table 212 JSON Parameters

| Parameter | Parameter Type | Parameter Description |
|------------|----------------|--|
| entities | Optional | Array of entity objects containing entityType and optional ID fields |
| entityType | Mandatory | The type of entity to export (e.g., PolicyEvent) |
| id | Optional | Array of specific entity IDs to export. If not specified, all entities of the given type are exported. |

Example 177

Request:

```
beamctl rmca-businessconfiguration-export-utility export <
export_request.json > export_file.json
```

Response: Redirected to the export_file.json

The export_request.json file contains an entities field consisting of a list of entity types and optional ID fields. If only the entity type is specified, all entities corresponding to that entity type will be exported. If both the entity type and ID are specified, only the corresponding entity will be exported. If the entire payload is empty, then all entities will be exported.

Example 178

This example describes a request payload for a specific PolicyEvent entity. The PolicyEvent entity with the id d080a409-0306-458f-b246-6c777a38125f is exported.

Request:

```
{
  "entities": [{
    "entityType": "PolicyEvent",
    "id": ["d080a409-0306-458f-b246-6c777a38125f"]
  }]
}
```



```
]
}
```

Response: The redirection operator redirects the output to the `export_file.json`.

Note: The `export_file.json` contains the JSON content with all the requested specifications. It is recommended to use redirection to fetch the output in a file.

3.9.3 Entity

It provides the options to manage (Create, Read and Update) global data related to RMCA. It has below sub-commands:

Table 213 Available Sub Commands

| Command | Description |
|---------------------------|---|
| get | It fetches the entity data. |
| get-key | It fetches the entity data as per public key. |
| get-default-configuration | It retrieves the default value of common business entity configuration. |
| set-default-configuration | It sets the default value of common business entity configuration. |
| set-external-mapping | It sets the external mapping for common business entity configuration. |
| list-versions | It retrieves list versions of the common business entity configuration. |
| update | It updates the common business entity configuration. |

3.9.3.1 get

This command fetches the entity data.

Syntax

`beamctl entity get <entity-name>`



Arguments

Table 214 Mandatory Arguments

| Argument | Description |
|-------------|-------------------------|
| entity-name | The name of the entity. |

Parameters

Each resource of this entity has its own parameters. See, [RMCA Global Data Item Management REST Interface](#).

Example 179

Request:

```
beamctl entity get Country
```

Response:

```
[
  {
    "alpha2Code": "US",
    "alpha3Code": "USA",
    "countryCode": "1",
    "description": "United States",
    "id": "US",
    "lastModified": "2025-04-30T16:07:12.864+02:00",
    "name": "United States",
    "numericCode": "840",
    "status": "released",
    "uri": "rmref://coba/Country/US",
    "validFor": [
      {
        "endDateTime": "9999-12-31T12:00:00.001+00:00",
        "startDateTime": "0001-01-01T12:00:00.001+00:00"
      }
    ],
    "versionNumber": 1746022032864
  }
]
```



3.9.3.2 update

This command updates the existing common business entity configuration of the given entity name.

Syntax

beamctl entity update <entity-name> < <input json file>

Arguments

Table 215 Mandatory Arguments

| Argument | Description |
|-------------|---------------------|
| entity-name | Name of the entity. |

Parameters

Each resource of this entity has its own parameters. See, [RMCA Global Data Item Management REST Interface](#).

Example 180

Request:

```
beamctl entity update Currency < /data/currency.json
[
  {
    "name": "US Dollar",
    "description": "US Dollar",
    "alpha3Code": "USD",
    "numericCode": "840",
    "minorUnit": 2
  },
  {
    "name": "Euro",
    "description": "Euro",
    "alpha3Code": "EUR",
    "numericCode": "978",
    "minorUnit": 2
  },
  {
    "name": "Swedish Krona",
    "description": "Swedish Krona",
    "alpha3Code": "SEK",
    "numericCode": "752",
    "minorUnit": 2
  }
]
```

Response:

Requested action successfully completed



3.9.3.3 get-key

This command retrieves the common business entity configuration of the given entity name and its public key.

Syntax

```
beamctl entity get-key <entity-name> <public-key>
```

Arguments

Table 216 Mandatory Arguments

| Argument | Description |
|-------------|---|
| entity-name | Name of the entity. |
| public-key | Public key of the entity configuration. |

Parameters

Each resource of this entity has its own parameters. See, [RMCA Global Data Item Management REST Interface](#).

Example 181

Request:

```
beamctl entity get-key Country US
```

Response:

```
[
  {
    "alpha2Code": "US",
    "alpha3Code": "USA",
    "countryCode": "1",
    "description": "United States",
    "id": "US",
    "lastModified": "2025-04-30T16:07:12.864+02:00",
    "name": "United States",
    "numericCode": "840",
    "status": "released",
    "uri": "rmref://coba/Country/US",
    "validFor": [
      {
        "endDateTime": "9999-12-31T12:00:00.001+00:00",
        "startDateTime": "0001-01-01T12:00:00.001+00:00"
      }
    ]
  },
]
```



```
    "versionNumber": 1746022032864
  }
]
```

3.9.3.4 **get-default-configuration**

This command retrieves the default value of common business entity configuration.

Syntax

`beamctl entity get-default-configuration <entity-name>`

Arguments

Table 217 Mandatory Arguments

| Argument | Description |
|-------------|---------------------|
| entity-name | Name of the entity. |

Parameters

This entity has the following parameter. For more information, see [RMCA Global Data Item Default Value Management](#).

Table 218 JSON Parameters

| Parameter | Parameter Type | Description |
|--------------|----------------|--|
| defaultValue | Mandatory | Default value of the entity type.
Example: IN |

Example 182

Request:

`beamctl entity get-default-configuration Country`

Response:

```
{
  "defaultValue": ""
}
```



3.9.3.5

set-default-configuration

This command sets the default value of common business entity configuration.

Syntax

```
beamctl entity set-default-configuration < <input json file>
```

Arguments

The command has no arguments.

Parameters

This entity has the following parameters. For more information, see [RMCA Global Data Item Default Value Management](#).

Table 219 JSON Parameters

| Parameter | Parameter Type | Description |
|--------------|----------------|--|
| entityType | Mandatory | Name of the entity type.
Example: Country |
| defaultValue | Mandatory | Default value of the entity type.
Example: IN |

Example 183

Request:

```
beamctl entity set-default-configuration < /data/  
defaultValue.json
```

```
[  
  {  
    "entityType": "Country",  
    "defaultValue": ""  
  }  
]
```

Response:

Requested action successfully completed



3.9.3.6 list-versions

This command retrieves the versions of the common business entity configuration.

Syntax

beamctl entity list-versions <entity-name> <public-key>

Arguments

Table 220 Mandatory Arguments

| Argument | Description |
|-------------|---|
| entity-name | Name of the entity. |
| public-key | Public key of the entity configuration. |

Parameters

Each resource of this entity has its own parameters. See, RMCA Global Data Item Management REST Interface.

Example 184

Request:

beamctl entity list-versions Country US

Response:

```
[
  {
    "alpha2Code": "US",
    "alpha3Code": "USA",
    "countryCode": "1",
    "description": "United States",
    "id": "US",
    "lastModified": "2025-04-30T16:07:12.864+02:00",
    "name": "United States",
    "numericCode": "840",
    "status": "released",
    "uri": "rmref://coba/Country/US",
    "validFor": [
      {
        "endDateTime": "9999-12-31T12:00:00.001+00:00",
        "startDateTime": "0001-01-01T12:00:00.001+00:00"
      }
    ]
  }
]
```



```
    ],  
    "versionNumber": 1746022032864  
  }  
]
```

3.9.3.7

set-external-mapping

This command sets the external mapping for common business entity configuration.

Syntax

beamctl entity set-external-mapping < <input json file>

Arguments

The command has no arguments.

Parameters

This entity has the following parameters. For more information, see [RMCA Global Data Item External Key Management](#)

Table 221 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------|----------------|---|
| entityType | Mandatory | Name of the global data item type or a global list.
Example: Country |
| publicKey | Mandatory | Public key of the record.
Example: IN |
| externalKey | Mandatory | External key of the record.
Example: INDIA |

Example 185

Request:

```
beamctl entity set-external-mapping < /data/external-mapping.json  
[  
  {  
    "entityType": "Country",  
    "publicKey": "US",  
    "externalKey": "A13"  
  }  
]
```



Response:

```
Requested action successfully completed
```

3.9.4 globallist

It provides the options to manage (Create, Read and Update) global list configurations related to RMCA.

Table 222 Available Sub Commands

| Command | Description |
|---------------|--|
| get | It fetches the data lists as per given parameters. |
| get-all | It fetches all data lists as per the given data list specification name. |
| update | It updates the data list as per the given data list specification name. |
| list-versions | It retrieves versions of a data list configuration. |

3.9.4.1 get

This command retrieves the common business global list configurations of the given global list specification name and global list name.

Syntax

beamctl globallist get <globallist-spec-name> <globallist-name>

Arguments

Table 223 Mandatory Arguments

| Argument | Description |
|----------------------|--|
| globallist-spec-name | Name of the global list specification. |
| globallist-name | Name of the global list. |

Parameters

This entity has the following parameters.



Table 224 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------------|----------------|---|
| id | Mandatory | Specifies the unique Id of the global list. |
| globalListSpecification | Mandatory | A global list specification consisting of id and columnDefinition . |
| description | Mandatory | Description of the global list. |
| label | Mandatory | Specifies the unique Id of the global list. |
| lastModified | Mandatory | Specifies creation/ modification time stamp of a record in UTC format. |
| status | Mandatory | Defines the status of the entity. |
| versionIdentifier | Mandatory | A version identifier is unique number of the version of a global list. The version number is generated by the system when a version is created or modified. |
| rmref | Mandatory | Unique identifier of a record. |
| validFor | Mandatory | The period for which this specification is valid. |
| globalListEntry | Mandatory | An array of global list items. It includes columnName and value |

Example 186

Request:

```
beamctl globallist get EquipmentId userEquipmentIds
```

Response:

```
[
  {
    "description": "",
    "globalListEntry": [
      {
        "columnName": "equipmentId",
        "value": "35715705"
      },
    ],
  },
]
```



```
{
  {
    "columnName": "manufacturer",
    "value": "Sony_Mobile_Communications_AB"
  },
  {
    "columnName": "model",
    "value": "Sony_M35c"
  }
],
[
  {
    "columnName": "equipmentId",
    "value": "35712305"
  },
  {
    "columnName": "manufacturer",
    "value": "Sony_Mobile_Communications_AB"
  },
  {
    "columnName": "model",
    "value": "Sony_SGP321"
  }
]
],
"globalListSpecification": {
  "columnDefinition": [
    {
      "canContainRange": false,
      "description": "",
      "name": "equipmentId",
      "overlappingValidator": "",
      "type": "string"
    },
    {
      "canContainRange": false,
      "description": "",
      "name": "manufacturer",
      "overlappingValidator": "",
      "type": "string"
    },
    {
      "canContainRange": false,
      "description": "",
      "name": "model",
      "overlappingValidator": "",
      "type": "string"
    }
  ],
  "id": "EquipmentId"
},
{id": "userEquipmentIds",
"label": "userEquipmentIds",
"lastModified": "2025-05-02T10:08:51.773+02:00",
"rmref": "rmref://coba/globalListSpecification/EquipmentId/globalList/userEquipmentIds",
"status": "released",
"validFor": [
  {
    "endTime": "9999-12-31T12:00:00.001+00:00",
    "startTime": "0001-01-01T12:00:00.001+00:00"
  }
]
},
"versionIdentifier": "1746173331773"
}
```



3.9.4.2

get-all

This command retrieves the common business global list configurations of the given globallist specification name.

Syntax

```
beamctl globallist get-all <globallist-spec-name>
```

Arguments

Table 225 Mandatory Arguments

| Argument | Description |
|----------------------|--|
| globallist-spec-name | Name of the global list specification. |

Parameters

This entity has the following parameters.

Table 226 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------------|----------------|---|
| id | Mandatory | Specifies the unique Id of the global list. |
| globalListSpecification | Mandatory | A global list specification consisting of id and columnDefinition . |
| description | Mandatory | Description of the global list. |
| label | Mandatory | Specifies the unique Id of the global list. |
| lastModified | Mandatory | Specifies creation/ modification time stamp of a record in UTC format. |
| status | Mandatory | Defines the status of the entity. |
| versionIdentifier | Mandatory | A version identifier is unique number of the version of a global list. The version number is generated by the system when a version is created or modified. |



| Parameter | Parameter Type | Description |
|-----------------|----------------|---|
| rmref | Mandatory | Unique identifier of a record. |
| validFor | Mandatory | The period for which this specification is valid. |
| globalListEntry | Mandatory | An array of global list items. It includes columnName and value |

Example 187

Request:

beamctl globallist get-all EquipmentId

Response:

```
[
  {
    "description": "",
    "globalListSpecification": {
      "columnDefinition": [
        {
          "canContainRange": false,
          "description": "",
          "name": "equipmentId",
          "overlappingValidator": "",
          "type": "string"
        },
        {
          "canContainRange": false,
          "description": "",
          "name": "manufacturer",
          "overlappingValidator": "",
          "type": "string"
        },
        {
          "canContainRange": false,
          "description": "",
          "name": "model",
          "overlappingValidator": "",
          "type": "string"
        }
      ]
    },
    "id": "EquipmentId"
  },
  {
    "id": "userEquipmentIds",
    "label": "userEquipmentIds",
    "lastModified": "2025-05-02T10:08:51.773+02:00",
    "rmref": "rmref://coba/globalListSpecification/EquipmentId/globalList/userEquipmentIds",
    "status": "released",
    "validFor": [
      {
        "endTime": "9999-12-31T12:00:00.001+00:00",
        "startTime": "0001-01-01T12:00:00.001+00:00"
      }
    ],
    "versionIdentifier": "1746173331773"
  }
]
```



3.9.4.3

update

This command updates the existing common business globallist configuration of the given globallist specification name.

Syntax

```
beamctl globallist update <globallist-spec-name> < <input json file>
```

Arguments

Table 227 Mandatory Arguments

| Argument | Description |
|----------------------|--|
| globallist-spec-name | Name of the global list specification. |

Parameters

This entity has the following parameters.

Table 228 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------------|----------------|---|
| id | Mandatory | Specifies the unique Id of the global list. |
| description | Mandatory | Description of the global list. |
| status | Mandatory | Defines the status of the entity. |
| validFor | Mandatory | The period for which this specification is valid. |
| globalListEntry | Mandatory | An array of global list items. It includes columnName and value |

Example 188

Request:

```
beamctl globallist update EquipmentId < /data/globalList-
equipmentId.json
```

```
[
  {
    "id": "userEquipmentIds",
    "status": "released",
    "globalListEntry": [
```



```
[
  {
    "columnName": "equipmentId",
    "value": "35715705"
  },
  {
    "columnName": "model",
    "value": "Sony_M35c"
  },
  {
    "columnName": "manufacturer",
    "value": "Sony_Mobile_Communications_AB"
  }
],
[
  {
    "columnName": "equipmentId",
    "value": "35474905"
  },
  {
    "columnName": "model",
    "value": "SONY_SO-01E"
  },
  {
    "columnName": "manufacturer",
    "value": "Sony_Mobile_Communications_AB"
  }
]
],
"validFor": {
  "startDateTime": "2025-01-01T00:00:00.000+00:00",
  "endDateTime": "9999-12-31T23:59:59.000+00:00"
}
}
```

Response:

Global list registered successfully

3.9.4.4 list-versions

This command retrieves versions of the global list configuration of the given global list specification name and global list name.

Syntax

beamctl globallist list-versions <globallistspec-name> <globallist-name>

Arguments

Table 229 Mandatory Arguments

| Argument | Description |
|---------------------|--|
| globallistspec-name | Name of the global list specification. |
| globallist-name | Name of the global list. |



Parameters

This entity has the following parameters.

Table 230 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------------|----------------|---|
| id | Mandatory | Specifies the unique Id of the global list. |
| globalListSpecification | Mandatory | A global list specification consisting of id and columnDefinition . |
| description | Mandatory | Description of the global list. |
| label | Mandatory | Specifies the unique Id of the global list. |
| lastModified | Mandatory | Specifies creation/ modification time stamp of a record in UTC format. |
| status | Mandatory | Defines the status of the entity. |
| versionIdentifier | Mandatory | A version identifier is unique number of the version of a global list. The version number is generated by the system when a version is created or modified. |
| rmref | Mandatory | Unique identifier of a record. |
| validFor | Mandatory | The period for which this specification is valid. |
| globalListEntry | Mandatory | An array of global list items. It includes columnName and value |

Example 189

Request:

```
beamctl globallist list-versions EquipmentId userEquipmentIds
```

Response:

```
[
  {
    "description": "",
```



```

"globalListEntry": [
  [
    {
      "columnName": "equipmentId",
      "value": "35715705"
    },
    {
      "columnName": "manufacturer",
      "value": "Sony_Mobile_Communications_AB"
    },
    {
      "columnName": "model",
      "value": "Sony_M35c"
    }
  ],
  [
    {
      "columnName": "equipmentId",
      "value": "35712305"
    },
    {
      "columnName": "manufacturer",
      "value": "Sony_Mobile_Communications_AB"
    },
    {
      "columnName": "model",
      "value": "Sony_SGP321"
    }
  ]
],
"globalListSpecification": {
  "columnDefinition": [
    {
      "canContainRange": false,
      "description": "",
      "name": "equipmentId",
      "overlappingValidator": "",
      "type": "string"
    },
    {
      "canContainRange": false,
      "description": "",
      "name": "manufacturer",
      "overlappingValidator": "",
      "type": "string"
    },
    {
      "canContainRange": false,
      "description": "",
      "name": "model",
      "overlappingValidator": "",
      "type": "string"
    }
  ]
},
"id": "EquipmentId",
"id": "userEquipmentIds",
"label": "userEquipmentIds",
"lastModified": "2025-05-02T10:08:51.773+02:00",
"rmref": "rmref://coba/globalListSpecification/EquipmentId/globalList/userEquipmentIds",
"status": "released",
"validFor": [
  {
    "endTime": "9999-12-31T12:00:00.001+00:00",
    "startTime": "0001-01-01T12:00:00.001+00:00"
  }
],
"versionIdentifier": "1746173331773"
}
]

```

3.9.5 globallistspec

It provides the options to manage (Create, Read and Update) related to RMCA.



Table 231 Available Sub Commands

| Command | Description |
|---------|---|
| get | It retrieves the common business global list specification as per the given name. |
| get-all | It retrieves the common business global list specifications. |
| update | It updates the existing common business global list specification. |

3.9.5.1

get

This command retrieves the common business global list specification as per the given name.

Syntax

```
beamctl globallistspec get <spec-name>
```

Arguments

Table 232 Mandatory Arguments

| Argument | Description |
|-----------|--|
| spec-name | Name of the global list specification. |

Parameters

This entity has the following parameters.

Table 233 JSON Parameters

| Parameter | Parameter Type | Description |
|------------------|----------------|---|
| id | Mandatory | Identifier of the global list specification. |
| description | Mandatory | Description of the global list specification. |
| columnDefinition | Mandatory | An array of column definition identifier.
It includes the following: <ul style="list-style-type: none">— name - Name of the column (or field) in a global list. |



| Parameter | Parameter Type | Description |
|--------------|----------------|---|
| | | <ul style="list-style-type: none">— type - Supported data type for attribute name in a global list.— description - Description of column in a global list.— canContainRange - Specifies the range value support for a global list as true or false— overlappingValidator - Specifies the range validator with possible values as NO_OVERLAP_WITHIN_LIST, NO_OVERLAP_WITHIN_COLUMN, and "". The value "" specifies that overlapping constraints are not applicable. |
| created | Mandatory | Specifies creation timestamp of a specification in UTC format. |
| lastModified | Mandatory | Specifies modification timestamp of a specification in UTC format. |
| status | Mandatory | Defines the status of the global list specification. |

Example 190

Request:

```
beamctl globallistspec get EquipmentId
```

Response:

```
[  
  {
```



```
"columnDefinition": [
  {
    "canContainRange": false,
    "description": "",
    "name": "equipmentId",
    "overlappingValidator": "",
    "type": "string"
  },
  {
    "canContainRange": false,
    "description": "",
    "name": "manufacturer",
    "overlappingValidator": "",
    "type": "string"
  },
  {
    "canContainRange": false,
    "description": "",
    "name": "model",
    "overlappingValidator": "",
    "type": "string"
  }
],
"created": "",
"description": "",
"id": "EquipmentId",
"lastModified": "",
"status": "released"
}
```

3.9.5.2 get-all

This command retrieves the common business global list specifications. It has no mandatory arguments.

Syntax

```
beamctl globallistspec get-all
```

Arguments

This command has no arguments.

Parameters

This entity has the following parameters.

Table 234 JSON Parameters

| Parameter | Parameter Type | Description |
|------------------|----------------|---|
| id | Mandatory | Identifier of the global list specification. |
| description | Mandatory | Description of the global list specification. |
| columnDefinition | Mandatory | An array of column definition identifier. |



| Parameter | Parameter Type | Description |
|--------------|----------------|---|
| | | <p>It includes the following:</p> <ul style="list-style-type: none">— name - Name of the column (or field) in a global list.— type - Supported data type for attribute name in a global list.— description - Description of column in a global list.— canContainRange - Specifies the range value support for a global list as true or false— overlappingValidator - Specifies the range validator with possible values as NO_OVERLAP_WITHIN_LIST, NO_OVERLAP_WITHIN_COLUMN, and "". The value "" specifies that overlapping constraints are not applicable. |
| created | Mandatory | Specifies creation timestamp of a specification in UTC format. |
| lastModified | Mandatory | Specifies modification timestamp of a specification in UTC format. |
| status | Mandatory | Defines the status of the global list specification. |

Example 191

Request:



beamctl globallistspec get-all

Response:

```
[
  {
    "columnDefinition": [
      {
        "canContainRange": false,
        "description": "",
        "name": "equipmentId",
        "overlappingValidator": "",
        "type": "string"
      },
      {
        "canContainRange": false,
        "description": "",
        "name": "manufacturer",
        "overlappingValidator": "",
        "type": "string"
      },
      {
        "canContainRange": false,
        "description": "",
        "name": "model",
        "overlappingValidator": "",
        "type": "string"
      }
    ],
    "created": "",
    "description": "",
    "id": "EquipmentId",
    "lastModified": "",
    "status": "released"
  },
  {
    "columnDefinition": [
      {
        "canContainRange": false,
        "description": "name",
        "name": "name",
        "overlappingValidator": "",
        "type": "string"
      }
    ],
    "created": "",
    "description": "",
    "id": "ZoneInformation",
    "lastModified": "",
    "status": "released"
  }
]
```

3.9.5.3 update

This command is used to create global list specifications.

Note: This command can only be used to create global list specifications, and cannot support updating an existing global list specification.

Syntax

beamctl globallistspec update < <input json file>



Arguments

The command has no arguments.

Parameters

This entity has the following parameters.

Table 235 JSON Parameters

| Parameter | Parameter Type | Description |
|------------------|----------------|---|
| id | Mandatory | Identifier of the global list specification. |
| description | Mandatory | Description of the global list specification. |
| columnDefinition | Mandatory | <p>An array of column definition identifier.</p> <p>It includes the following:</p> <ul style="list-style-type: none">— name - Name of the column (or field) in a global list.— type - Supported data type for attribute name in a global list.— description - Description of column in a global list.— canContainRange - Specifies the range value support for a global list as true or false.— overlappingValidator - Specifies the range validator with possible values as NO_OVERLAP_WITHIN_LIST, NO_OVERLAP_WITHIN_COLUMN, and "". The value "" specifies that overlapping |



| Parameter | Parameter Type | Description |
|-----------|----------------|---------------------------------|
| | | constraints are not applicable. |

Example 192

Request:

```
beamctl globallistspec update < /data/globalListSpec-productdata →
ils.json
{
  "id": "ProductDetails",
  "description": "Customer Product Details",
  "columnDefinition": [
    {
      "name": "productId",
      "type": "int",
      "description": "Description of the product Id"
    },
    {
      "name": "productDescription",
      "type": "string",
      "description": "Description of the product"
    }
  ]
}
```

Response:

Global list specification registered successfully

3.10 MSG Configuration Commands

3.10.1 Enable Exposure of Kafka Topics

Lists MSG topics and bootstrap URLs.

3.10.1.1 list-topics

The `list-topics` command retrieves a list of topics for each AppGroup in the current site from MSG.

Syntax

```
beamctl msg-topic-management list-topics
```



Arguments

The command has no arguments.

Parameters

Table 236 JSON Parameters

| Parameter | Parameter Type | Description |
|---------------|----------------|---|
| appgroups | Mandatory | Array of Application Group objects.
Each object contains: <ul style="list-style-type: none">— appgroup-name— topics |
| appgroup-name | Mandatory | Name of the AppGroup. It s a string. |
| topics | Mandatory | List of topic names for the Application Group. It is an array of string. |
| location-name | Mandatory | Name of the current site or location. |

Example 193

beamctl msg-topic-management list-topics

Response:

```
{
  "appgroups": [
    {
      "appgroup-name": "msg1",
      "topics": [
        "msg1_test_topic_1"
      ]
    }
  ],
  "location-name": "l1"
}
```

Example 194

beamctl msg-topic-management list-topics

Response:



```
{
  "appgroups": [
    {
      "appgroup-name": "msg1",
      "topics": [
        "msg1_test_topic_1",
        "msg3_test_topic_1"
      ]
    }
  ],
  "location-name": "11"
}
```

Example 195

```
beamctl msg-topic-management list-topics
```

Response:

```
{
  "appgroups": [
    {
      "appgroup-name": "msg3",
      "topics": [
        "msg3_test_topic_1",
        "msg1_test_topic_1"
      ]
    }
  ],
  "location-name": "12"
}
```

3.10.1.2 list-bootstrap-urls

The `list-bootstrap-urls` command retrieves bootstrap URL for each MSG AppGroup across the sites.

Syntax

```
beamctl msg-topic-management list-bootstrap-urls
```

Arguments

The command has no arguments.

Parameters

Table 237 JSON Parameters

| Parameter | Parameter Type | Description |
|----------------|----------------|---------------------------------|
| bootstrap-urls | Mandatory | Array of bootstrap URL objects. |



| Parameter | Parameter Type | Description |
|---------------|----------------|--|
| | | Each object contains: <ul style="list-style-type: none">— appgroup-name— location— url |
| appgroup-name | Mandatory | Name of the AppGroup. It is a string. |
| location | Mandatory | Location or site name for the Application Group. It is a string. |
| url | Mandatory | Bootstrap URL for the Application Group. It is a string. |

Example 196

```
beamctl msg-topic-management list-bootstrap-urls
```

Response:

```
{
  "bootstrap-urls": [
    {
      "appgroup-name": "msg1",
      "location": "l1",
      "url": "eric-bss-msg-msg1-kafka-bootstrap.zone1.domain.se:1892"
    },
    {
      "appgroup-name": "msg3",
      "location": "l2",
      "url": "eric-bss-msg-msg3-kafka-bootstrap.zone2.domain.se:1892"
    }
  ]
}
```

3.10.2 MSG External User Management

Commands allow to create, update, and delete MSG External User for exposing MSG topics to an external consumer and to get the list of available users and associated certificates.

3.10.2.1 create-acl-user

The `create-acl-user` command creates a user for an MSG External User along with associated consumer groups and topics ACLs (Access Control List) for one or more Application Groups.



Syntax

beamctl msg-acl-authorization create-acl-user *<three input options>*

Arguments

The command has no arguments.

Parameters

Table 238 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------------|----------------|---|
| acl-user | Mandatory | Name of the ACL user to create. |
| consumer-groups | Mandatory | List of <i>consumer-group</i> names to create and bind to the ACL user. It is an array of string. |
| appgroups | Mandatory | Array of Application Group objects.
Each object contains:
— appgroup-name
— topics |
| appgroup-name | Mandatory | Name of the AppGroup. It is a string. |
| topics | Mandatory | List of topic names for the Application Group. It is an array of string. |

Example 197 Option 1

beamctl msg-acl-authorization create-acl-user *< inputFile.json*

```
inputFile.json
{
  "acl-user": "foreign-consumer",
  "consumer-groups": [
    "external_foreign-consumer_msg1"
  ],
  "appgroups": [
    {
      "appgroup-name": "msg1",
      "topics": [
        "msg1_test_topic_1"
      ]
    }
  ]
}
```



```
inputFile.json
{
  "acl-user": "foreign-consumer",
  "consumer-groups": [
    "external_foreign-consumer_msg1",
    "external_foreign-consumer_msg3"
  ],
  "appgroups": [
    {
      "appgroup-name": "msg1",
      "topics": [
        "msg1_test_topic_1",
        "msg3_test_topic_1"
      ]
    }
  ]
}
```

```
inputFile.json
{
  "acl-user": "foreign-consumer",
  "consumer-groups": [
    "external_foreign-consumer_msg3",
    "external_foreign-consumer_msg1"
  ],
  "appgroups": [
    {
      "appgroup-name": "msg3",
      "topics": [
        "msg3_test_topic_1",
        "msg1_test_topic_1"
      ]
    }
  ]
}
```

Example 198 Option 2:

```
beamctl msg-acl-authorization create-acl-user << EOF
```

```
{
  "acl-user": "foreign-consumer",
  "consumer-groups": [
    "external_foreign-consumer_msg1"
  ],
  "appgroups": [
    {
      "appgroup-name": "msg1",
      "topics": [
        "msg1_test_topic_1"
      ]
    }
  ]
}
EOF
```

Example 199 Option 3:

```
beamctl msg-acl-authorization create-acl-user <<< '{"acl-  
user":"foreign-consumer","consumer-groups":["external_foreign-
```



```
consumer_msg1"], "appgroups": [{"appgroup-name": "msg1", "topics":
["msg1_test_topic_1"]}]]}'
```

3.10.2.2

list-acl-users

The `list-acl-users` command retrieves a list of MSG External User along with associated consumer groups and topics.

Syntax

```
beamctl msg-acl-authorization list-acl-users
```

Arguments

The command has no arguments.

Parameters

Table 239 JSON Parameters

| Parameter | Parameter Type | Description |
|------------------------------|----------------|---|
| <code>acl-users</code> | Mandatory | Array of ACL user objects.
Each object contains: <ul style="list-style-type: none"> — <code>acl-user</code> — <code>consumer-groups</code> — <code>appgroups</code> |
| <code>acl-user</code> | Mandatory | Name of the ACL user. It is a string. |
| <code>consumer-groups</code> | Mandatory | List of <i>consumer-group</i> names associated with the ACL user. It is an array of string. |
| <code>appgroups</code> | Mandatory | Array of Application Group objects.
Each object contains: <ul style="list-style-type: none"> — <code>appgroup-name</code> — <code>topics</code> |
| <code>appgroup-name</code> | Mandatory | Name of the AppGroup. It is a string. |



| Parameter | Parameter Type | Description |
|-----------|----------------|--|
| topics | Mandatory | List of topic names for the Application Group. It is an array of string. |

Example 200

beamctl msg-acl-authorization list-acl-users

Response:

```
{
  "acl-users": [
    {
      "acl-user": "foreign-consumer ",
      "appgroups": [
        {
          "appgroup-name": "msg1",
          "topics": [
            "msg1_test_topic_1",
            "msg3_test_topic_1"
          ]
        }
      ],
      "consumer-groups": [
        "external_foreign-consumer _msg1",
        "external_foreign-consumer _msg3",
      ]
    }
  ]
}
```

3.10.2.3 delete-acl-user

The delete-acl-user command deletes an existing MSG External User created for an external consumer.

Syntax

beamctl msg-acl-authorization delete-acl-user *<username>*

Argument

Table 240 Mandatory Arguments

| Argument | Description |
|----------|------------------------|
| username | MSG External username. |



Example 201

```
beamctl msg-acl-authorization delete-acl-user foreign-consumer
```

3.10.2.4 update-acl-user

The `update-acl-user` command updates an existing MSG External User created for an external consumer.

Syntax

```
beamctl msg-acl-authorization update-acl-user <three input options>
```

Arguments

The command has no arguments.

Parameters

Table 241 JSON Parameters

| Parameter | Parameter Type | Description |
|------------------------------|----------------|--|
| <code>acl-user</code> | Mandatory | Name of the ACL user to update. It is a string. |
| <code>consumer-groups</code> | Mandatory | List of <i>consumer-group</i> names to associate with the ACL user.

Replace existing <i>consumer-group</i> associations with this list.

It is an array of strings. |
| <code>appgroups</code> | Mandatory | Array of Application Group objects.

Each object contains:
— <code>appgroup-name</code>
— <code>topics</code> |
| <code>appgroup-name</code> | Mandatory | Name of the Application Group. It is a string. |
| <code>topics</code> | Mandatory | List of topic names within the Application |



| Parameter | Parameter Type | Description |
|-----------|----------------|--|
| | | <p>Group to grant access to. It is an array of string.</p> <p>Note: appgroups is required in the request payload, but it might be an empty array [] when the ACL user has no Application Group or topic assignments.</p> |

Example 202 Option 1:

```
beamctl msg-acl-authorization update-acl-user < inputFile.json
```

```
inputFile.json
{
  "acl-user": "foreign-consumer",
  "consumer-groups": [
    "external_foreign-consumer_msg1",
    "external_foreign-consumer_msg3"
  ],
  "appgroups": [
    {
      "appgroup-name": "msg1",
      "topics": [
        "msg1_test_topic_1",
        "msg3_test_topic_1"
      ]
    }
  ]
}
```

Example 203 Option 2:

```
beamctl msg-acl-authorization update-acl-user << EOF
```

```
{
  "acl-user": "foreign-consumer",
  "consumer-groups": [
    "external_foreign-consumer_msg1",
    "external_foreign-consumer_msg3"
  ],
  "appgroups": [
    {
      "appgroup-name": "msg1",
      "topics": [
        "msg1_test_topic_1",
        "msg3_test_topic_1"
      ]
    }
  ]
}
```




```
}
EOF
```

Example 204 Option 3:

```
beamctl msg-acl-authorization update-acl-
user <<< '{"acl-user":"foreign-consumer","consumer-
groups":["external_foreign-consumer_msg1","external_foreign-
consumer_msg3"],"appgroups":[{"appgroup-name":"msg1","topics":
["msg1_test_topic_1","msg3_test_topic_1"]}]]'
```

3.10.2.5

list-acl-user-certificates

This command retrieves certificates associated with a specific MSG External User created for an external consumer that contains password-protected PKCS12 certificates packaged in `tar.gz` format.

The package includes:

- `client.p12` – Client certificate + private key (PKCS12)
- `server-ca.p12` – Server CA certificate (PKCS12)
- `README.md` – Usage instructions

Syntax

```
beamctl msg-acl-authorization list-acl-user-certificates <
<acl_user>.json
```

Arguments

The command has no arguments.

JSON Parameters

Table 242 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|--------------------------|
| username | Mandatory | An existing ACL user |
| password | Mandatory | Base64 encoded password. |



Example 205

```
beamctl msg-acl-authorization list-acl-user-certificates < certificate-request.json
```

certificate-request.json content:

```
{
  "username": "testuser",
  "password": "VGVzdFBhc3N3b3JkMTIzIQ=="
}
```

Response:

```
{"Downloaded filename":"testuser-certificates.tar.gz"}
```

3.11 EML Configuration Commands

3.11.1 EML Event Enquiry Variation Point Configuration

3.11.1.1 list-configs

The `list-configs` command lists the details of variation point configurations available in the Event Enquiry Interface.

Syntax

```
beamctl eml-event-enq-vp list-configs --configtype <configtype_value>
```

Arguments

Table 243 Mandatory Arguments

| Argument | Description |
|---------------------------|---|
| <code>--configtype</code> | Possible values are (case insensitive):
— all
— default |



| Argument | Description |
|----------|--|
| | Note: <ul style="list-style-type: none"> — To list all the configurations available in the system, that includes custom and default profiles, execute the command with <i><config-profile-type></i> as all. — To list the default configurations available in the system, execute the command with <i><config-profile-type></i> as default. |

Example 206

```
beamctl eml-event-enq-vp list-configs --configtype all
```

```
beamctl eml-event-enq-vp list-configs --configtype default
```

3.11.1.2

import-configs

The `import-configs` command imports the config file such as *response config*, *response filter*, *consolidation config*, or all the three, to create the custom profile with the given variation point.

Syntax

```
beamctl eml-event-enq-vp import-configs --configprofile
<config_profile_name_value> --configtype <config_type_value> [--force]
< /absolute/file-path
```

Arguments

Table 244 Mandatory Arguments

| Argument | Description |
|------------------------------|--|
| <code>--configprofile</code> | The name of the config profile to be imported. It must start with alphanumeric characters (except 'default' and 'CP' keywords) and it allows only hyphens and underscores. |
| <code>--configtype</code> | Possible values are (case insensitive): <ul style="list-style-type: none"> — all — responseconfig |



| Argument | Description |
|----------|--|
| | <ul style="list-style-type: none">— responsefilter— consolidationconfig <p>Note:
To import the three, responseconfig, responsefilter, and consolidationconfig, place all the three configs in a single JSON file, and specify the <i><config-profile-type></i> as all.</p> |
| filepath | The path of the JSON file which needs to be imported. |

Table 245 Optional Arguments

| Argument | Description |
|----------|---|
| --force | To override the existing config profile, include this option in the import-configs command. |

Example 207

- To create the CSR1 profile with responseconfig:

```
beamctl eml-event-enq-vp import-configs --configprofile  
CSR1 --configtype responseconfig < /path/to/json/  
responseconfig.json
```

- To create the CSR1 profile with responsefilter:

```
beamctl eml-event-enq-vp import-configs --configprofile  
CSR1 --configtype responsefilter < /path/to/json/  
responsefilter.json
```

- To create the CSR1 profile with consolidationconfig:

```
beamctl eml-event-enq-vp import-configs --configprofile  
CSR1 --configtype consolidationconfig < /path/to/json/  
consolidationconfig.json
```

- To override the CSR1 profile with responseconfig, responsefilter, and consolidationconfig:

```
beamctl eml-event-enq-vp import-configs --configprofile CSR1  
--configtype all --force < /path/to/json/all.json
```



Example 208 JSON Files

JSON file sample containing only the responseconfig:

```
{
  "data": {
    "responseconfig": {
      "BalanceCreationEvent_3": "customerInformation,balanceCreationResult",
      "BalanceCreationEvent_4": "customerInformation,balanceCreationResult",
      "BalanceCreationEvent_5": "customerInformation,balanceCreationResult",
      "UsageChargeEvent_1": "customerInformation,serviceUsageResult",
      "UsageChargeEvent_2": "customerInformation,serviceUsageResult",
      "UsageChargeEvent_3": "customerInformation,serviceReservationResult,serviceUsageResult"
    }
  }
}
```

JSON file sample containing only the responsefilter:

```
{
  "data": {
    "responsefilter": {
      "events": ["BalanceCreationEvent", "UsageChargeEvent"]
    }
  }
}
```

JSON file containing only the consolidationconfig:

```
{
  "data": {
    "consolidationconfig": {
      "name": "consolidationMetaData",
      "description": "MetaData for Consolidation",
      "consolidationMetaData": [
        {
          "UsageChargeEvent_1": {
            "compositekey": {
              "consolidateField": "",
              "consolidateBy": "customerInformation.customerId,customerInformation.contractId",
              "operation": ""
            },
            "eventHeader": {
              "consolidateField": "eventCreationTime,recordUniqueId",
              "consolidateBy": "",
              "operation": "NONE"
            },
            "eventHeader.eventCreationTime": {
              "consolidateField": "",
              "consolidateBy": "",
              "operation": "FIRST"
            },
            "eventHeader.recordUniqueId": {
              "consolidateField": "",
              "consolidateBy": "",
              "operation": "DISTINCT"
            }
          }
        }
      ]
    }
  }
}
```

JSON file sample containing the responseconfig, responsefilter, and consolidationconfig:



```

{
  "data": {
    "responseconfig": {
      "BalanceCreationEvent_3": "customerInformation,balanceCreationResult",
      "BalanceCreationEvent_4": "customerInformation,balanceCreationResult",
      "BalanceCreationEvent_5": "customerInformation,balanceCreationResult",
      "UsageChargeEvent_1": "customerInformation,serviceUsageResult",
      "UsageChargeEvent_2": "customerInformation,serviceUsageResult",
      "UsageChargeEvent_3": "customerInformation,serviceReservationResult,serviceUsageResult"
    },
    "responsefilter": {
      "events": [
        "BalanceCreationEvent",
        "UsageChargeEvent"
      ]
    },
    "consolidationconfig": {
      "name": "consolidationMetaData",
      "description": "MetaData for Consolidation",
      "consolidationMetaData": [
        {
          "UsageChargeEvent_1": {
            "compositekey": {
              "consolidateField": "",
              "consolidateBy": "customerInformation.customerId,customerInformation.contractId",
              "operation": ""
            },
            "eventHeader": {
              "consolidateField": "eventCreationTime,recordUniqueId",
              "consolidateBy": "",
              "operation": "NONE"
            },
            "eventHeader.eventCreationTime": {
              "consolidateField": "",
              "consolidateBy": "",
              "operation": "FIRST"
            },
            "eventHeader.recordUniqueId": {
              "consolidateField": "",
              "consolidateBy": "",
              "operation": "DISTINCT"
            }
          }
        }
      ]
    }
  }
}

```

Note: Make sure that the JSON file provided for import contains the expected configurations, otherwise the import fails.

Example 209

```

beamctl eml-event-enq-vp import-configs --configprofile
CSR1 --configtype responseconfig --force < /path/to/json/
csr1_config.json

```

The `csr1_config.json` file must only have the `responseconfig` content as shown in the following:

```

{
  "data": {
    "responseconfig": {
      "BalanceCreationEvent_3": "customerInformation,balanceCreationResult",
      ...
    }
  }
}

```



```
}
}
```

3.11.1.3

validate-configs

The `validate-configs` command validates the config file that needs to be imported.

Syntax

```
beamctl eml-event-enq-vp validate-configs --configprofile
<configprofile_value> --configtype <configtype_value> < /absolute/
file-path
```

Arguments

Table 246 Mandatory Arguments

| Argument | Description |
|------------------------------|--|
| <code>--configprofile</code> | The name of the config profile to be imported. |
| <code>--configtype</code> | <p>Possible values are (case insensitive):</p> <ul style="list-style-type: none"> — all — responseconfig — responsefilter — consolidationconfig <p>Note:
To import the three, <code>responseconfig</code>, <code>responsefilter</code>, and <code>consolidationconfig</code>, place all the three configs in a single JSON file, and specify the <code><config-profile-type></code> as all.</p> |
| <code>filepath</code> | The path of the JSON file which needs to be imported. |

Example 210

- To validate config profile with `responseconfig` type:

```
beamctl eml-event-enq-vp validate-configs --configprofile
CSR1 --configtype responseconfig < /path/to/json/
responseconfig.json
```



- To validate config profile with `responsefilter` type:

```
beamctl eml-event-enq-vp validate-configs --configprofile
CSR1 --configtype responsefilter < /path/to/json/
responsefilter.json
```

- To validate config profile with `consolidationconfig` type:

```
beamctl eml-event-enq-vp validate-configs --configprofile
CSR1 --configtype consolidationconfig < /path/to/json/
consolidationconfig.json
```

- To validate config profile with the three `responseconfig`, `responsefilter`, and `consolidationconfig` type:

```
beamctl eml-event-enq-vp validate-configs --configprofile CSR1
--configtype all < /path/to/json/all.json
```

3.11.1.4

export-template

The `export-template` command exports default config profile available in the system.

By default the `export-template` prints the current version of the default config profile configured in the system.

Syntax

```
beamctl eml-event-enq-vp export-template [--configprofile
<default_configprofile_value>] [--configtype <configtype_value>] > /
absolute/file_path
```

Arguments

Table 247 Optional Arguments

| Argument | Description |
|------------------------------|--|
| <code>--configprofile</code> | The name of the default config profile to be exported in the following format:
default- <version_number>
Note:
To know the versions available for default profile, use the <code>list-config</code> command with <i>default</i> option. |
| <code>--configtype</code> | Possible values are (case insensitive):
— all |



| Argument | Description |
|----------|---|
| | <ul style="list-style-type: none"> — responseconfig — responsefilter — consolidationconfig <p>Note:
To export the three responseconfig, responsefilter, and consolidationconfig, specify the <i><config-profile-type></i> as all.</p> |
| filepath | The file path where the default configs must be exported. |

Example 211

- To view the current version of the default config profile:

```
beamctl eml-event-enq-vp export-template> /path/to/json/
default_all_1.0.0.json
```

- To view responseconfig, responsefilter, and consolidationconfig of the default config profile with version 1.0.0:

```
beamctl eml-event-enq-vp export-template --configprofile
default-1.0.0 --configtype all > /path/to/json/
default_all_1.0.0.json
```

- To view the responseconfig of the default config profile with version 1.0.0:

```
beamctl eml-event-enq-vp export-template --configprofile
default-1.0.0 --configtype responseconfig > /path/to/json/
default_ responseconfig_1.0.0.json
```

3.11.1.5 export-configs

The export-configs command only exports custom config profiles. No default configs can be exported using this command.

Syntax

```
beamctl eml-event-enq-vp export-configs --configprofile
<configprofile_value> --configtype <configtype_value> > /absolute/
file_path
```



Arguments

Table 248 Mandatory Arguments

| Argument | Description |
|------------------------------|---|
| <code>--configprofile</code> | The name of the config profile to be exported. |
| <code>--configtype</code> | Possible values are (case insensitive): <ul style="list-style-type: none">— all— responseconfig— responsefilter— consolidationconfig Note:
To export the three responseconfig, responsefilter, and consolidationconfig, specify the <i><config-profile-type></i> as all. |
| <code>filepath</code> | The file path where the configs must be exported. |

Example 212

```
beamctl eml-event-enq-vp export-configs --configprofile CSR1 --configtype all > /path/to/json/csr1_all.json
```

```
beamctl eml-event-enq-vp export-configs --configprofile CSR1 --configtype responseconfig > /path/to/json/csr1_responseconfig.json
```

3.11.1.6

delete-config

The `delete-config` command deletes config profiles including the default config profiles except the latest version of the default config profile.

Syntax

```
beamctl eml-event-enq-vp delete-config --configprofile <configprofile_value> --configtype <configtype_value>
```



Arguments

Table 249 Mandatory Arguments

| Argument | Description |
|------------------------------|--|
| <code>--configprofile</code> | <p>The name of the config profile to be exported.</p> <p>Ensure that the name of the default config profile is in the following format:</p> <p>default-<i><version_number></i></p> <p>Note:</p> <p>To know the versions available for default profile, use the <code>list-config</code> command with <i>default</i> option.</p> |
| <code>--configtype</code> | <p>Possible values are (case insensitive):</p> <ul style="list-style-type: none"> — all — responseconfig — responsefilter — consolidationconfig <p>Note:</p> <p>To delete the three <code>responseconfig</code>, <code>responsefilter</code>, and <code>consolidationconfig</code>, specify the <i><configtype></i> as all.</p> |

Example 213

- To delete `responseconfig`, `responsefilter`, and `consolidationconfig` of the config profile:

```
beamctl eml-event-enq-vp delete-config --configprofile CSR1 --configtype all
```

- To delete `responseconfig`, `responsefilter`, and `consolidationconfig` of the Default config profile:

```
beamctl eml-event-enq-vp delete-config --configprofile default-1.0.0 --configtype all
```

- To delete only the `responseconfig` of the config profile:



```
beamctl eml-event-enq-vp delete-config --configprofile CSR1 --  
configtype responseconfig
```

The Configuration Status of a given Profile in Response Data:

When executing the BEAM CLI commands, the following status is shown in the response data:

- **Configured** – This denotes that the configuration is already present in the system, or the configuration is successfully configured.
- **Unconfigured** – This denotes that the configuration is not present in the system, or the configuration is successfully unconfigured.
- **Reconfigured** – This denotes that the configuration is already present in the system for a given config profile and was successfully overridden as per the import-configs command action.
- **Deconfigured** - This denotes that the configuration which was present in the system was now successfully deleted as per the delete-config command action.
- **Valid** – This denotes that the provided configuration JSON content is valid and it is fine to import the configuration into the system.

3.11.2 EML TTL Configuration

3.11.2.1 get-config

The `get-config` command lists the current TTL configurations for the various Supported Events available in the system.

Syntax

```
beamctl eml-ttl-management get-config
```

Arguments

The command has no arguments.

Example 214

```
beamctl eml-ttl-management get-config
```

3.11.2.2 update-config

The `update-config` is used to modify the TTL configurations for one or more events that are supported by the system.



Syntax

```
beamctl eml-ttl-management update-config < <absolute_file_path>
```

Arguments

Table 250 Mandatory Arguments

| Argument | Description |
|--------------------|---|
| absolute_file_path | The absolute path of the file. The file contains the TTL configuration to be updated. |

Example 215

```
beamctl eml-ttl-management update-config < /path/to/json/
input.json
```

Sample input.json file:

```
{
  "ttlConfig": [
    {
      "event": "ActionNotification",
      "ttl": 11
    },
    {
      "event": "BalanceAdjustmentEvent",
      "ttl": 8
    }
  ]
}
```

- Note:**
- The get-config command lists the TTL configuration available in the system for all the supported events. TTL updates can be applied exclusively to these events.
 - Ensure that the input file includes only the events for which a TTL update is necessary. Including events that do not require a TTL update results in validation errors.

3.11.3 EML Snapshot Event Management Configuration

3.11.3.1 get-onboarding-status

The get-onboarding-status command returns the status of snapshot onboarding configuration.

Syntax

```
beamctl eml-snapshot-config get-onboarding-status
```



Arguments

The command has no arguments.

Example 216

```
beamctl eml-snapshot-config get-onboarding-status
```

3.11.3.2 start-onboarding

The `start-onboarding` command starts onboarding of existing customers into EAD by enabling entity snapshot subscription.

Syntax

```
beamctl eml-snapshot-config start-onboarding
```

Arguments

The command has no arguments.

Example 217

```
beamctl eml-snapshot-config start-onboarding
```

3.11.3.3 stop-onboarding

The `stop-onboarding` command stops onboarding of existing customers into EAD by disabling entity snapshot subscription.

Syntax

```
beamctl eml-snapshot-config stop-onboarding
```

Arguments

The command has no arguments.

Example 218

```
beamctl eml-snapshot-config stop-onboarding
```

3.12 EPL Configuration Commands

3.12.1 EPL Web Console Access Command

The Web Console Access command provides a UI for monitoring and managing EPL Application Groups. There is typically one URL per EPL Application Group.



This command can be used to retrieve the login URLs that operators can open in a browser.

Syntax

```
beamctl epl-webconsole <sub-command>
```

Available Sub Commands

Table 251 Sub Command

| Command | Description |
|--------------------------------------|---|
| get-urls on page 411 | It is used to extract the details to access the web console of EPL cluster. |

3.12.1.1

get-urls

The `get-urls` command is used to extract the details to access the web console of EPL cluster. It provides the URL to login to the web console. The number of URLs is equal to the number of EPL application groups.

Syntax

```
beamctl epl-webconsole get-urls
```

Arguments

The command has no arguments.

Example 219

```
beamctl epl-webconsole get-urls
```

Sample Response:

```
{
  "data": {
    "url": "[https://eric-bss-epl-epl1-event-db.beam-cluster01.operator.com:443/
ui/index.html]"
  }
}
```

3.12.2

EPL GeoRed Replication Manager Commands

The EPL GeoRed Replication Manager commands let operators inspect remote connections and replication status, control replication, view and



update replication configuration parameters for replicated buckets, and obtain replication statistics. They are used to monitor and operate cross-site data replication for EPL buckets (for example the Global bucket) and to troubleshoot replication issues.

Note: The following commands work only when the Geographical Redundancy is enabled or at least in a Multisite setup.

In a Multisite setup, the following commands are applicable only for Global bucket. For example, `pause-replication` command pauses only the Global bucket data replication.

Syntax

```
beamctl epl-replication-manager <sub-command>
```

Available Sub Commands

Table 252 Sub Commands

| Sub Command | Description |
|---|---|
| list-configs on page 417 | It fetches configuration parameters for all the replicated buckets between the given EPL source and destination Application Groups. |
| list-connections on page 413 | It displays the remote references established for the EPL Application Groups. |
| list-replications on page 414 | It fetches the replicated buckets associated between the specified source and destination EPL Application Groups. |
| pause-replication on page 415 | It halts replication of the buckets between EPL source to destination Application Groups. |
| remove-connection on page 416 | It removes the replications along with the references existing between EPL source and destination Application Groups. |
| remove-replication on page 416 | It deletes the replications that exist between the specified EPL source and destination Application Groups. |
| replication-details on page 413 | It displays replication information that includes bucket-name, status of the replication, and cluster details. |
| replication-stats on page 420 | It displays the number of documents that are to be copied for each |



| Sub Command | Description |
|--|---|
| | replicated bucket, between the source and destination EPL Application Groups. |
| resume-replication on page 415 | It resumes back the existing buckets replication from EPL source to destination Application Groups. |
| update-configs on page 418 | It updates the replication configuration parameters provided in a JSON file. |

3.12.2.1 list-connections

The `list-connections` command displays the remote references established for the EPL Application Groups.

Syntax

```
beamctl epl-replication-manager list-connections [--appgroup
<EPL_AG_NAME>]
```

Arguments

Table 253 Optional Arguments

| Arguments | Description |
|-------------------------|---|
| <code>--appgroup</code> | The name of the EPL Application Group name within the local site for which to fetch the reference connection details.
By default, it lists all the remote references established for all the Application Groups from the local site. |

Example 220

```
beamctl epl-replication-manager list-connections
beamctl epl-replication-manager list-connections --appgroup ep11
beamctl epl-replication-manager list-connections --appgroup ep12
```

3.12.2.2 replication-details

The `replication-details` command displays replication information that includes bucket-name, status of the replication, and cluster details.



Syntax

```
beamctl epl-replication-manager replication-details [--appgroup
<EPL_AG_NAME>]
```

Arguments

Table 254 Mandatory Arguments

| Arguments | Description |
|------------|---|
| --appgroup | The name of the EPL application group within the local site for which the replication details is to be fetched. |

Example 221

```
beamctl epl-replication-manager replication-details --appgroup
ep11
```

3.12.2.3

list-replications

The list-replications command fetches the replicated buckets associated between the specified source and destination EPL Application Groups.

Syntax

```
beamctl epl-replication-manager list-replications [--appgroup
<EPL_AG_NAME>]
```

Arguments

Table 255 Optional Arguments

| Arguments | Description |
|------------|---|
| --appgroup | The name of the EPL application group name within the local site for which replication is listed. By default, it lists all the replications for all the Application Groups exist in the local site. |

Example 222

```
beamctl epl-replication-manager list-replications
beamctl epl-replication-manager list-replications --appgroup ep11
```



3.12.2.4 pause-replication

The `pause-replication` command temporarily halts replication of the buckets between EPL source to destination Application Groups.

Syntax

```
beamctl epl-replication-manager pause-replication --
source-appgroup <SOURCE_EPL_AG_NAME> --destination-appgroup
<DESTINATION_EPL_AG_NAME>
```

Arguments

Table 256 Mandatory Arguments

| Arguments | Description |
|-------------------------------------|---|
| <code>--source-appgroup</code> | Name of EPL source Application Group within the local site. |
| <code>--destination-appgroup</code> | Name of EPL destination Application Group name. |

Example 223

```
beamctl epl-replication-manager pause-replication --source-
appgroup epl1 --destination-appgroup epl2
```

3.12.2.5 resume-replication

The `resume-replication` command resumes back the existing buckets replication from EPL source to destination Application Groups.

Syntax

```
beamctl epl-replication-manager resume-replication --
source-appgroup <SOURCE_EPL_AG_NAME> --destination-appgroup
<DESTINATION_EPL_AG_NAME>
```

Arguments

Table 257 Mandatory Arguments

| Arguments | Description |
|-------------------------------------|---|
| <code>--source-appgroup</code> | Name of EPL source app group within the local site. |
| <code>--destination-appgroup</code> | Name of EPL destination Application Group name. |



Example 224

```
beamctl epl-replication-manager resume-replication --source-appgroup epl1 --destination-appgroup epl2
```

3.12.2.6

remove-replication

The `remove-replication` command deletes the replications exist between the specified EPL source and destination app groups.

Note: This command is typically not required during normal operations. It is used only in specific cases, such as when a replication is no longer needed due to EPL wanted state changes, or during emergency situations that require immediate intervention.

Syntax

```
beamctl epl-replication-manager remove-replication --source-appgroup <SOURCE_EPL_AG_NAME> --destination-appgroup <DESTINATION_EPL_AG_NAME>
```

Arguments

Table 258 Mandatory Arguments

| Arguments | Description |
|-------------------------------------|---|
| <code>--source-appgroup</code> | Name of EPL source Application Group within the local site. |
| <code>--destination-appgroup</code> | Name of EPL destination Application Group name. |

Example 225

```
beamctl epl-replication-manager remove-replication --source-appgroup epl1 --destination-appgroup epl2
```

3.12.2.7

remove-connection

The `remove-connection` command removes the replications along with the references existing between EPL source and destination app groups.

Note: This command is typically not required during normal operations. It is used only in specific cases, such as when a replication is no longer needed due to EPL wanted state changes, or during emergency situations that require immediate intervention.



Syntax

```
beamctl epl-replication-manager remove-connection --
source-appgroup <SOURCE_EPL_AG_NAME> --destination-appgroup
<DESTINATION_EPL_AG_NAME>
```

Arguments

Table 259 Mandatory Arguments

| Arguments | Description |
|------------------------|---|
| --source-appgroup | Name of EPL source Application Group within the local site. |
| --destination-appgroup | Name of EPL destination Application Group name. |

Example 226

```
beamctl epl-replication-manager remove-connection --source-
appgroup epl1 --destination-appgroup epl2
```

3.12.2.8

list-configs

The `list-configs` command fetches configuration parameters for all the replicated buckets between the given EPL source and destination Application Groups.

Syntax

```
beamctl epl-replication-manager list-configs --source-appgroup
<SOURCE_EPL_AG_NAME> --destination-appgroup <DESTINATION_EPL_AG_NAME>
```

Arguments

Table 260 Mandatory Arguments

| Arguments | Description |
|------------------------|---|
| --source-appgroup | Name of EPL source Application Group within the local site. |
| --destination-appgroup | Name of EPL destination Application Group name. |

Example 227

```
beamctl epl-replication-manager list-configs --source-appgroup
epl1 --destination-appgroup epl2
```



3.12.2.9

update-configs

The `update-configs` command updates the replication configuration parameters provided in the JSON file for the given buckets between the source and destination EPL Application Groups. If no buckets match with the argument, it shows a message indicating the absence.

Syntax

```
beamctl epl-replication-manager update-configs \  
  
--source-appgroup <SOURCE_EPL_AG_NAME> \  
  
--destination-appgroup <DESTINATION_EPL_AG_NAME> \  
  
< <config_params_filepath>.json
```

Arguments

Table 261 Mandatory Arguments

| Arguments | Description |
|------------------------|---|
| --source-appgroup | Name of EPL source Application Group within the local site. |
| --destination-appgroup | Name of EPL destination Application Group name. |
| config_params_file | The JSON file containing the configuration parameters of bucket settings. |

Parameters

Table 262 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------|----------------|---|
| bucketName | Mandatory | Name of the bucket to which the replication settings apply. |
| desiredLatency | Mandatory | Maximum desired latency in milliseconds. |
| docBatchSizeKb | Mandatory | Size, in kB, of each document batch sent per replication operation. |
| networkUsageLimit | Mandatory | Maximum percentage of network it might use for replication. |
| priority | Mandatory | Replication thread priority. It controls resource sharing. |



| Parameter | Parameter Type | Description |
|---------------------|----------------|--|
| | | It can be <i>High</i> , <i>Medium</i> , <i>Low</i> . |
| sourceNozzlePerNode | Mandatory | Number of parallel outgoing replication streams (nozzles) per source node. |
| targetNozzlePerNode | Mandatory | Number of parallel incoming replication streams per target node. |
| workerBatchSize | Mandatory | Number of operations a replication worker processes per batch. |

Example 228

```
beamctl epl-replication-manager update-configs \
--source-appgroup epl1 \
--destination-appgroup epl2 < /tmp/config_params.json
```

Example 229 Config Parameters JSON Content

```
[
  {
    "bucketName": "global",
    "desiredLatency": 51,
    "docBatchSizeKb": 2048,
    "networkUsageLimit": 0,
    "priority": "High",
    "sourceNozzlePerNode": 2,
    "targetNozzlePerNode": 2,
    "workerBatchSize": 500
  },
  {
    "bucketName": "customer_1",
    "desiredLatency": 50,
    "docBatchSizeKb": 2048,
    "networkUsageLimit": 0,
    "priority": "Medium",
    "sourceNozzlePerNode": 2,
    "targetNozzlePerNode": 2,
    "workerBatchSize": 400
  }
]
```



3.12.2.10 replication-stats

The `replication-stats` command displays the number of documents that are to be copied for each replicated bucket, between the source and destination EPL app groups.

Syntax

```
beamctl epl-replication-manager replication-stats \  
  
--source-appgroup <SOURCE_EPL_AG_NAME> \  
  
--destination-appgroup <DESTINATION_EPL_AG_NAME>  
  
[ --mode <live/copy>]
```

Arguments

Table 263 Mandatory Arguments

| Arguments | Description |
|-------------------------------------|---|
| <code>--source-appgroup</code> | Name of EPL source Application group within the local site. |
| <code>--destination-appgroup</code> | Name of EPL destination Application group name. |

Table 264 Optional Arguments

| Arguments | Description |
|---------------------|---|
| <code>--mode</code> | The possible values (case insensitive) are: <i>copy</i> and <i>live</i> . |

Note:

- *copy* shows the number of documents to be copied from source to destination at the time that the replication was created.
- *live* shows the currently running count of documents that are copied, and the pending document which needs to be copied between the given source and destination Application Groups.

Parameters

Table 265 XDCR Settings Configuration Parameters

| Parameter | Parameter Type | Description |
|-----------------------------|----------------|--|
| <code>desiredLatency</code> | Mandatory | Maximum desired latency in milliseconds for replication. Lower values prioritize faster replication, |



| Parameter | Parameter Type | Description |
|---------------------|----------------|--|
| | | but it might increase CPU and network usage.
The default value is 50. |
| docBatchSizeKb | Mandatory | Size, in kB, of the document batch sent per replication operation. Affects memory usage and throughput.
The default value is 2048. |
| networkUsageLimit | Mandatory | The maximum percentage of network bandwidth XDCR is allowed to use.
The default value is 0. |
| priority | Mandatory | Priority of replication threads. It can be <i>High</i> , <i>Medium</i> , <i>Low</i> . It affects how the system resources are shared with other processes.
The default value is <i>High</i> . |
| sourceNozzlePerNode | Mandatory | Number of parallel data streams (nozzles) from each source node. Higher values improve parallelism, but use more CPU and network. |
| targetNozzlePerNode | Mandatory | Number of receiving streams on each target node. Works with source nozzles to control replication parallelism. |
| workerBatchSize | Mandatory | Number of operations each worker processes per batch. Larger batches can improve performance at the cost of more memory use.
The default value is 500. |

Example 230

```
beamctl epl-replication-manager replication-stats \
--source-appgroup epl1 \
--destination-appgroup epl2 --mode copy
```

3.12.3

EPL Extraction Manager Commands

The EPL Extraction Manager commands control whether particular events are extracted in real time, by batch, or both, and let operators view the current configuration and apply updates.

Syntax

```
beamctl epl-extraction-mgr <sub-command>
```



Available Sub Commands

Table 266 Sub Commands

| Command | Description |
|---|---|
| list-config on page 422 | Retrieves the current EPL extraction configuration for all events. |
| update-config on page 422 | Updates the EPL extraction configuration for the specified events using a JSON input. |

3.12.3.1 list-config

The `list-config` command retrieves the current EPL extraction configuration for all events, detailing whether each event is enabled for real-time extraction, batch extraction, or both.

Syntax

```
beamctl epl-extraction-mgr list-config
```

Arguments

The command has no arguments.

Example 231

beamctl epl-extraction-mgr list-config

3.12.3.2 update-config

The `update-config` command updates the EPL extraction configuration for the specified events using JSON input.

Syntax

```
beamctl epl-extraction-mgr update-config < <absolute_file_path>
```

Arguments

Table 267 Mandatory Arguments

| Argument | Description |
|---------------------------------|--------------------------------|
| <code>absolute_file_path</code> | The absolute path of the file. |



| Argument | Description |
|----------|---|
| | The file contains the EPL extraction configuration to be updated. |

- Note:**
- The EPL Extraction Configuration can only be updated for the existing events that are listed through the `list-config` command.
 - While updating the EPL Extraction Configuration, both real time and batch values must be provided for each event in the input JSON.
 - If the extraction configuration is modified on one site using the `update-config` command, the same change must be applied to all other sites using the `update-config` command.

Example 232

```
beamctl epl-extraction-mgr update-config < /path/to/json/
input.json
```

Example 233 JSON Input File

```
{
  "eventExtractionConfigs": {
    "ActionNotification": {
      "batch": true,
      "realtime": false
    },
    "BalanceAdjustmentEvent": {
      "batch": false,
      "realtime": true
    }
  }
}
```

3.12.4 EPL Analytics Manager Configuration Commands

The EPL Analytics Manager Configuration Commands provide the information for managing and operating EPL analytics components. It includes MART artifacts, batch and nrtr jobs, job runs and tasks, Spark configurations, external storage resources, analytics datasets and views, reruns, import errors, and BI integration details.

Syntax

```
beamctl epl-analytics-mgr
```



Available Sub Commands

Table 268 Available Sub Commands

| Command | Description |
|--|---|
| add-batch-job-rerun on page 445 | Creates a new rerun operation for scheduled DAGs and non-scheduled batch-job. |
| add-ext-resource on page 446 | Adds a new external variable which holds the configuration for the AWS connection. |
| batch-job-import-err on page 446 | Retrieves import errors for batch jobs that failed to load properly. |
| batch-job-sqltsk-ls on page 428 | Retrieves all SQL tasks for a batch job. |
| batch-job-sqltsk-upd on page 429 | Updates SQL task configurations from a JSON file. |
| crt-analytics-view on page 452 | Creates new analytics view from a JSON configuration file. |
| delete-nrtr-job on page 433 | Deletes a nrtr job. |
| delete-ext-resource on page 448 | Deletes an existing external variable which holds the configuration for the AWS connection. |
| desc-batch-job on page 437 | Retrieves detailed information about a specific batch job. |
| desc-batch-job-rerun on page 444 | Retrieves information about a specific rerun created for a scheduled batch job. |
| desc-batch-job-run on page 439 | Retrieves detailed information about a specific batch job run. |
| desc-batch-job-task on page 442 | Retrieves detailed information about a specific task within a batch job. |
| edit-nrtr-config on page 435 | Updates retention in database and spark-conf in the config.json file. |
| import-batch-job on page 425 | Imports a ZIP file containing batch MART artifacts and converts them into Airflow jobs. |
| import-nrtr-mart on page 430 | Imports a ZIP file containing nrtr MART artifacts. |
| list-all-batch-job on page 436 | Retrieves a list of all available batch jobs. |
| list-batch-job on page 455 | Retrieves spark configurations for all batch MARTs. |



| Command | Description |
|--|--|
| list-batch-job-rerun on page 443 | Retrieves a list of rerun operations for scheduled batch jobs. |
| list-batch-job-run on page 438 | Retrieves a list of runs for a specific batch job. |
| list-batch-job-task on page 440 | Retrieves a list of tasks for a specific batch job. |
| list-ext-resource on page 449 | Retrieves a list of all external variables created. |
| list-nrtr-jobs on page 431 | Lists all the nrtr jobs. |
| ls-analytics-dataset on page 450 | Lists all available analytics datasets. |
| ls-analytics-view on page 453 | Lists all available analytics views. |
| preview-dataset on page 451 | Retrieves the record from the specified analytics dataset for preview purposes. |
| start-nrtr-job on page 432 | Starts a nrtr job by job name. |
| stop-nrtr-job on page 433 | Stops a running nrtr job by creating a stop signal file. |
| show-bi-details on page 454 | Retrieves BI connection details. |
| show-nrtr-config on page 434 | Shows the <code>config.json</code> content for a specific nrtr job with retention from database. |
| update-batch-job on page 456 | Updates spark configurations for all batch MARTs and jobs. |
| update-ext-resource on page 447 | Updates an existing external variable which holds the configuration for the AWS connection. |
| list-all-batch-marts on page 457 | Lists all the active and inactive batch marts in the system. |
| update-batch-mart-status on page 458 | This command enables/disables batch marts in the system. |

3.12.4.1 **import-batch-job**

The `import-batch-job` command imports a ZIP file containing batch MART artifacts, such as DAG files, SQL files, and task-level configuration files, and converts them into Airflow jobs.

Syntax

```
beamctl epl-analytics-mgr import-batch-job < <zip_file>
```



Arguments

Table 269 Mandatory Arguments

| Argument | Description |
|----------|---|
| zip file | It is a zip file containing batch MART artifacts. |

Structure of a ZIP File

FileName.zip

- config.json
- CustomDags.py
- sql
- Sql1.txt
- Sql2.txt

Parameters

Table 270 JSON Parameters

| Parameter | Parameter Type | Description |
|------------------------|--|---|
| JobConfig.name | Mandatory | Job name. |
| Taskname.sql-file | Mandatory | Path to the SQL file. |
| Taskname.id | Optional | Task identifier. |
| Taskname.source-events | Mandatory | List of the source events. Acceptable values are event names listed by the list-config on page 422 command under EPL Extraction Manager Commands on page 421 or it should be specified as EXTERNAL for external resources. |
| Taskname.destination | Mandatory | Destination type.
It can be <i>fs</i> or <i>table</i> . |
| Taskname.mart.enabled | Mandatory when the destination is <i>table</i> . | Enable the MART functionality. |



| Parameter | Parameter Type | Description |
|--------------------|--|-------------------------------|
| Taskname.mart.name | Mandatory when the destination is <i>table</i> . | MART name. |
| Taskname.conf | Optional | Spark configuration settings. |

Example 234

```
beamctl epl-analytics-mgr import-batch-job <
batch_marts_artifact.zip
```

Example 235 config.json Example

```
{
  "JobConfig": {
    "name": "DailyDetail"
  },
  "EndUserNotificationStatusSummary": {
    "sql-file": "DailyDetail/sql/EndUserNotificationStatusSummary.txt",
    "id": "DateId, CustomerId, ContractId, DeliveryChannel, NotificationStatus",
    "source-events": [
      "EndUserNotificationHistoryEvent"
    ],
    "destination": "fs"
  },
  "ProductFeeSummary": {
    "sql-file": "DailyDetail/sql/ProductFeeSummary.txt",
    "id": "DateId, Location, PartitionId, CustomerId, ProductOfferingID, BillingAccountID, BillingAccountSpecificationId, BillingAccountBucketId, BillingAccountBucketSpecificationId, UoM",
    "destination": "table",
    "source-events": [
      "RecurringEvent"
    ],
    "mart": {
      "enabled": true,
      "name": "ProductFeeSummary"
    }
  }
}
```

Example 236 config.json Example for external resources

```
{
  "JobConfig": {
    "name": "MyCustomDAG"
  },
  "task2": {
    "sql-file": "MyCustomDAG/sql/task2.sql",
    "destination": "table",
    "source-events": [
      "EXTERNAL"
    ],
    "mart": {
      "enabled": true,
      "name": "external_mart"
    },
    "external-source": {
      "type": "s3",
      "connection-ref": "s3_connection"
    }
  }
}
```



```
} }
```

3.12.4.2 batch-job-sqltsk-ls

The `batch-job-sqltsk-ls` command retrieves all SQL tasks for a batch job. SQL tasks are data processing units that configure MART operations and retention policies within the job.

Syntax

`beamctl epl-analytics-mgr batch-job-sqltsk-ls <job_name>`

Arguments

Table 271 Mandatory Arguments

| Argument | Description |
|----------|----------------------------|
| job_name | The name of the batch job. |

Example 237

`beamctl epl-analytics-mgr batch-job-sqltsk-ls eric-bss-epl-sb-DailyDetail`

Sample Response:

```
{
  "tasks": [
    {
      "task-name": "ProductFeeSummary",
      "enabled": true,
      "mart-name": "ProductFeeSummary",
      "retention": {
        "table": 30
      }
    },
    {
      "task-name": "EndUserNotificationStatusSummary",
      "enabled": true,
      "mart-name": "EndUserNotificationStatusSummary",
      "retention": {
        "table": 30
      }
    },
    {
      "task-name": "CustomerStateTransition",
      "enabled": true,
      "mart-name": "CustomerStateTransition",
      "retention": {
        "table": 30
      }
    }
  ]
}
```




3.12.4.3 batch-job-sqltsk-upd

The `batch-job-sqltsk-upd` command updates SQL task configurations from a JSON file. Only retention values can be changed for all tasks. Enable and disable can be changed using command `update-batch-mart-status`.

Syntax

```
beamctl epl-analytics-mgr batch-job-sqltsk-upd <job_name> <
config_file>
```

Arguments

Table 272 Mandatory Arguments

| Argument | Description |
|----------|----------------------------|
| job_name | The name of the batch job. |

Parameters

This command requires a JSON configuration file from `stdin` containing SQL tasks update request.

Table 273 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|------------------------------------|---|
| task-name | Mandatory | Name of the SQL task to update. |
| enabled | Optional | Boolean flag to enable or disable the task.
Only supported for tasks that write to table or MART destinations. |
| mart-name | Option | Name of the MART (table destination) associated with the task.
It is required when enabling or disabling a task that writes to a MART. |
| retention | Mandatory if changing retention | Object specifying retention settings to update for the task. |
| table | Mandatory if retention is provided | Retention value for table destination. Only retention values might be changed for all tasks. |

Example 238 Required JSON Structure

```
{
  "tasks": [
    {
      "task-name": "ProductFeeSummary",
      "enabled": true,
      "mart-name": "ProductFeeSummary",
      "retention": {
```



```
    "table": 40
  }
}
]
```

Note: Only the parameters mentioned above will be considered for update. Any other parameters will be ignored and not processed.

Example 239

```
beamctl epl-analytics-mgr batch-job-sqltsk-upd eric-bss-epl-sb-DailyDetail < sql_tasks_config.json
```

3.12.4.4 import-nrtr-mart

The `import-nrtr-mart` command imports a zip file containing `nrtr` MART artifacts such as `sql` files.

Syntax

```
beamctl epl-analytics-mgr import-nrtr-mart < zip_file
```

Arguments

Table 274 Mandatory Arguments

| Argument | Description |
|----------|---|
| zip file | It is a zip file containing <code>nrtr</code> MART artifacts. |

Structure of a ZIP File

Filename.zip

- config.json
- sql-file.txt

Parameters

Table 275 Config JSON Parameters

| Parameter | Parameter Type | Description |
|------------|----------------|--------------------|
| event-name | Mandatory | Name of the event. |



| Parameter | Parameter Type | Description |
|------------|----------------|-------------------------------|
| job-name | Mandatory | Name of the MART. |
| spark-conf | Optional | Spark configuration settings. |

Example 240

```
beamctl epl-analytics-mgr import-nrtr-mart <
nrtr_marts_artifact.zip
```

Example 241 config.json Example

```
{
  "event-name": "CpmCustomerChangeEvent",
  "job-name": "ContractMetaStateTransition",
  "spark-conf": {
    "spark.network.timeout": "800s"
  }
}
```

3.12.4.5 list-nrtr-jobs

The `list-nrtr-jobs` command lists all the `nrtr` jobs in a table format.

Syntax

```
beamctl epl-analytics-mgr list-nrtr-jobs
```

Arguments

The command has no arguments.

Example 242

```
beamctl epl-analytics-mgr list-nrtr-jobs
```

Table 276 Sample Response:

| Job Name | Status | Introduced Version | Current Version |
|--|----------|--------------------|-----------------|
| eric-bss-epl-sr-contractmetastatettransition | Complete | 0.1.0-8 | 0.1.0-9 |



| Job Name | Status | Introduced Version | Current Version |
|--|---------|--------------------|-----------------|
| eric-bss-epl-sr-cpmcustomerchangeevent-fileconsolidation | Running | 0.1.0-8 | 0.1.0-9 |
| eric-bss-epl-cr-balanceadjustmentevent-fileconsolidation | New | 0.1.0-8 | 0.1.0-9 |

Where:

- *Job Name* is the name of the MART or consolidation job.
- *Status* is the status of the job.
- *Introduced Version* is the helm chart version of the job that is created.
- *Current Version* is the version of the helm chart presently deployed.

3.12.4.6

start-nrtr-job

The `start-nrtr-job` command starts an `nrtr` job by job name. It also validates the existing job in a database that is not running before starting.

Syntax

```
beamctl epl-analytics-mgr start-nrtr-job <job_name>
```

Arguments

Table 277 Mandatory Arguments

| Argument | Description |
|----------|---|
| job_name | The name of the <code>nrtr</code> job to start. |

Example 243

```
beamctl epl-analytics-mgr start-nrtr-job eric-bss-epl-sr-contractmetastatetransition
```



3.12.4.7 stop-nrtr-job

The `stop-nrtr-job` command stops a running `nrtr` job by creating a stop signal file. It also validates an existing job that is currently running before stopping.

Syntax

```
beamctl epl-analytics-mgr stop-nrtr-job <job_name>
```

Arguments

Table 278 Mandatory Arguments

| Argument | Description |
|----------|--|
| job_name | Name of the <code>nrtr</code> job to stop. |

Example 244

```
beamctl epl-analytics-mgr stop-nrtr-job eric-bss-epl-sr-  
contractmetastatetransition
```

3.12.4.8 delete-nrtr-job

The `delete-nrtr-job` command deletes a `nrtr` job including artifacts, configuration, and extracted files. It validates if the job is not running before deletion.

Note: Standard `nrtr` jobs cannot be deleted.

Syntax

```
beamctl epl-analytics-mgr delete-nrtr-job <job_name>
```

Arguments

Table 279 Mandatory Arguments

| Argument | Description |
|----------|--|
| job_name | The name of the <code>nrtr</code> job to be deleted. |



Example 245

```
beamctl epl-analytics-mgr delete-nrtr-job eric-bss-epl-cr-  
balanceadjustmentevent-fileconsolidation
```

3.12.4.9

show-nrtr-config

The `show-nrtr-config` command shows the `config.json` content for a specific `nrtr` job with retention from database.

Syntax

```
beamctl epl-analytics-mgr show-nrtr-config <job_name>
```

Arguments

Table 280 Mandatory Arguments

| Argument | Description |
|----------|--|
| job_name | The name of the <code>nrtr</code> job to show the JSON config for. |

Parameters

Table 281 Config JSON Parameters

| Parameter | Parameter Type | Description |
|------------|----------------|---------------------------------------|
| event-name | Mandatory | Name of the event. |
| job-name | Mandatory | Name of the <code>nrtr</code> config. |
| spark-conf | Optional | Spark configuration settings. |

Example 246

```
beamctl epl-analytics-mgr show-nrtr-job eric-bss-epl-sr-  
contractmetastatetransition
```

Sample Response:

```
{  
  "event-name": "CpmCustomerChangeEvent",  
  "job-name": "ContractMetaStateTransition",  
  "retention": 3,  
  "spark-conf": {  
    "spark.network.timeout": "800s"  
  }  
}
```



3.12.4.10 edit-nrtr-config

The `edit-nrtr-config` command updates retention in database and `spark-conf` in `config.json` file. It rezips and updates artifact database if the `config.json` is modified.

Syntax

```
beamctl epl-analytics-mgr edit-nrtr-config <job_name> <config_file>
```

Arguments

Table 282 Mandatory Arguments

| Argument | Description |
|-----------------------|--|
| <code>job_name</code> | The name of the <code>nrtr</code> job to edit. |

The command also requires a JSON configuration file from `stdin` containing retention and `spark-conf`. See an example of a `nrtr-config.json` below.

Parameters

Table 283 Config JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------------|----------------|---------------------------------------|
| <code>event-name</code> | Mandatory | Name of the event. |
| <code>job-name</code> | Mandatory | Name of the <code>nrtr</code> config. |
| <code>spark-conf</code> | Optional | Spark configuration settings. |

Example 247

```
beamctl epl-analytics-mgr edit-nrtr-config eric-bss-epl-sr-
contractmetastatettransition
```

Example 248 Examples of a `nrtr-config.json`

```
{
  "event-name": "CpmCustomerChangeEvent",
  "job-name": "ContractMetaStateTransition",
  "retention": 40,
  "spark-conf": {
    "spark.network.timeout": "900s"
  }
}
```



3.12.4.11 list-all-batch-job

The `list-all-batch-job` command retrieves a list of all available batch jobs in a table format.

Syntax

```
beamctl epl-analytics-mgr list-all-batch-job
```

Arguments

This command has no arguments.

Example 249

```
beamctl epl-analytics-mgr list-all-batch-job
```

Table 284 Sample Response:

| Job Name | Description | Paused | Import Error | Next Run Date | Job Type | Scheduled Job | Scheduled Frequency | Introduced Version | Current Version |
|-------------------------------------|-------------------------------------|--------|--------------|----------------------|-------------|---------------|---------------------|--------------------|-----------------|
| eric-bss-epl-sb-HourlySummary | eric-bss-epl-sb-HourlySummary | No | No | N/A | Standard | No | N/A | 1.4.0-7 | 1.4.0-8 |
| eric-bss-epl-sb-WeeklySummary | eric-bss-epl-sb-WeeklySummary | No | No | N/A | Standard | No | N/A | 1.4.0-7 | 1.4.0-8 |
| eric-bss-epl-sb-MonthlySummary | eric-bss-epl-sb-MonthlySummary | No | No | N/A | Standard | No | N/A | 1.4.0-7 | 1.4.0-8 |
| eric-bss-epl-sb-FileConsolidation | eric-bss-epl-sb-FileConsolidation | No | No | 2025-11-03T13:00:00Z | Airflow Job | Yes | 0 * * * * | N/A | N/A |
| eric-bss-epl-mdextration | eric-bss-epl-mdextration | No | No | 2025-11-04T02:00:00Z | Airflow Job | Yes | 0 2 * * * | N/A | N/A |
| eric-bss-epl-sb-DailyTransformation | eric-bss-epl-sb-DailyTransformation | No | No | 2025-11-04T04:00:00Z | Standard | Yes | 0 4 * * * | 1.4.0-7 | 1.4.0-8 |
| eric-bss-epl-sb-DailySummary | eric-bss-epl-sb-DailySummary | No | No | N/A | Standard | No | N/A | 1.4.0-7 | 1.4.0-8 |



| Job Name | Description | Paused | Import Error | Next Run Date | Job Type | Scheduled Job | Scheduled Frequency | Introduced Version | Current Version |
|------------------------------|------------------------------|--------|--------------|---------------|----------|---------------|---------------------|--------------------|-----------------|
| eric-bss-epl-sb-DailyDe tail | eric-bss-epl-sb-DailyDe tail | No | No | N/A | Standard | No | N/A | 1.4.0-7 | 1.4.0-8 |

3.12.4.12 desc-batch-job

The `desc-batch-job` command retrieves detailed information about a specific batch job.

Syntax

```
beamctl epl-analytics-mgr desc-batch-job <job_name>
```

Arguments

Table 285 Mandatory Arguments

| Argument | Description |
|----------|---|
| job_name | The unique identifier of the batch job. |

Example 250

```
beamctl epl-analytics-mgr desc-batch-job eric-bss-epl-sb-FileConsolidation
```

```
{
  "bundle_name": "dags-folder",
  "bundle_version": null,
  "description": null,
  "file_token": ".eJyrVkoqzUvJSY3PS8xNVbJSSk1ML9ZNY89JSS1S0lEqSs...",
  "fileloc": "/mnt/data/airflow/dags/FileConsolidation/Consolidation.py",
  "has_import_errors": false,
  "has_task_concurrency_limits": false,
  "is_paused": false,
  "is_stale": false,
  "job_display_name": "eric-bss-epl-sb-FileConsolidation",
  "job_name": "eric-bss-epl-sb-FileConsolidation",
  "last_expired": null,
  "last_parsed_time": "2025-10-08T10:43:42.998759Z",
  "max_active_runs": 16,
  "max_active_tasks": 16,
  "max_consecutive_failed_dag_runs": 0,
  "next_dagrun_data_interval_end": "2025-10-08T11:00:00Z",
  "next_dagrun_data_interval_start": "2025-10-08T11:00:00Z",
  "next_dagrun_logical_date": "2025-10-08T11:00:00Z",
  "next_dagrun_after": "2025-10-08T11:00:00Z",
  "owners": [
    "airflow"
  ],
  "relative_fileloc": "FileConsolidation/Consolidation.py",
  "tags": [
    {

```



```
{
  "job_name": "eric-bss-epl-sb-FileConsolidation",
  "name": "spark"
},
{
  "job_name": "eric-bss-epl-sb-FileConsolidation",
  "name": "Hourly"
},
{
  "job_name": "eric-bss-epl-sb-FileConsolidation",
  "name": "batch"
},
{
  "job_name": "eric-bss-epl-sb-FileConsolidation",
  "name": "Consolidation"
}
],
"timetable_description": "Every hour",
"timetable_summary": "0 * * * *"
}
```

3.12.4.13 list-batch-job-run

The `list-batch-job-run` command retrieves a list of runs for a specific batch job in table format.

Syntax

```
beamctl epl-analytics-mgr list-batch-job-run <job_name> [--
start_date <start_date>] [--end_date <end_date>] [--state <state>]
```

Arguments

Table 286 Mandatory Arguments

| Argument | Description |
|----------|---|
| job_name | The unique identifier of the batch job. |

Table 287 Optional Arguments

| Argument | Description |
|------------|---|
| start_date | Start date filter used in UTC format. |
| end_date | End date filter used in UTC format. |
| state | State filter, such as success, failed, running or queued.

These are the status of the batch-job. |

Example 251

```
beamctl epl-analytics-mgr list-batch-job-run eric-bss-epl-sb-
FileConsolidation --state success
```



```
beamctl epl-analytics-mgr list-batch-job-run eric-bss-
epl-sb-FileConsolidation --state success --start_date
2025-09-01T14:30:45Z --end_date 2025-09-02T14:30:45Z
```

```
beamctl epl-analytics-mgr list-batch-job-run eric-bss-epl-sb-
FileConsolidation --start_date 2025-09-01T14:30:45Z --end_date
2025-09-02T14:30:45Z
```

Sample response:

```
beamctl epl-analytics-mgr list-batch-job-run eric-
bss-epl-sb-FileConsolidation --state success --
start_date 2025-10-08T00:00:00.533892Z --end_date
2025-10-08T02:00:00.533892Z
```

Table 288 Sample Response:

| Run ID | Job Name | Start Date | End Date | Run Type | State |
|--------------------------------------|-----------------------------------|-----------------------------|-----------------------------|-----------|---------|
| scheduled__2025-10-08T01:00:00+00:00 | eric-bss-epl-sb-FileConsolidation | 2025-10-08T01:00:00.721506Z | 2025-10-08T01:01:58.074925Z | Scheduled | Success |
| scheduled__2025-10-08T00:00:00+00:00 | eric-bss-epl-sb-FileConsolidation | 2025-10-08T00:00:00.533892Z | 2025-10-08T00:02:02.928867Z | Scheduled | Success |

3.12.4.14

desc-batch-job-run

The `desc-batch-job-run` command retrieves detailed information about a specific batch job run (Standard Marts or Custom Marts).

Syntax

```
beamctl epl-analytics-mgr desc-batch-job-run <job_name> <job_run_id>
```

Arguments

Table 289 Mandatory Arguments

| Argument | Description |
|------------|--|
| job_name | The unique identifier of the batch job. |
| job_run_id | The unique identifier of the job run, which can be taken from the <code>list-batch-job-run</code> command. |



Example 252

```
beamctl epl-analytics-mgr desc-batch-job-run eric-bss-epl-sb-FileConsolidation scheduled__2025-10-08T01:00:00+00:00
```

Sample Response:

```
{
  "bundle_version": null,
  "conf": {},
  "data_interval_end": "2025-10-08T01:00:00Z",
  "data_interval_start": "2025-10-08T01:00:00Z",
  "end_date": "2025-10-08T01:01:58.074925Z",
  "job_name": "eric-bss-epl-sb-FileConsolidation",
  "job_run_id": "scheduled__2025-10-08T01:00:00+00:00",
  "job_versions": [
    {
      "bundle_name": "dags-folder",
      "bundle_url": null,
      "bundle_version": null,
      "created_at": "2025-10-08T00:06:10.177562Z",
      "id": "0199c123-c201-711b-bf4a-99862aa72d8e",
      "job_name": "eric-bss-epl-sb-FileConsolidation",
      "version_number": 242
    }
  ],
  "last_scheduling_decision": "2025-10-08T01:01:58.067553Z",
  "logical_date": "2025-10-08T01:00:00Z",
  "note": null,
  "queued_at": "2025-10-08T01:00:00.631886Z",
  "run_after": "2025-10-08T01:00:00Z",
  "run_type": "scheduled",
  "start_date": "2025-10-08T01:00:00.721506Z",
  "state": "success",
  "triggered_by": "timetable"
}
```

3.12.4.15 list-batch-job-task

The `list-batch-job-task` command retrieves a list of tasks for a specific batch job in a table format.

The users with the roles *Business Config Admin* and *Assistance* can only execute this command.

Syntax

```
beamctl epl-analytics-mgr list-batch-job-task <job_name>
```

Arguments

Table 290 Mandatory Arguments

| Argument | Description |
|----------|---|
| job_name | The unique identifier of the batch job. |



Example 253

```
beamctl epl-analytics-mgr list-batch-job-task eric-bss-epl-sb-FileConsolidation
```

Table 291 Sample Response:

| Task Name | Description | Operator | Trigger Rule |
|---|---|----------------------|--------------|
| ContractMetaStateTransition | ContractMetaStateTransition | SparkSubmitOperator | all_success |
| ProductUsageSummary | ProductUsageSummary | SparkSubmitOperator | all_success |
| ServiceUsageSummary | ServiceUsageSummary | SparkSubmitOperator | all_success |
| TransactionTypeProduct.TransactionTypeProductSummaryBalAdj | TransactionTypeProduct.TransactionTypeProductSummaryBalAdj | SparkSubmitOperator | all_success |
| TransactionTypeProduct.TransactionTypeProductSummaryRecurring | TransactionTypeProduct.TransactionTypeProductSummaryRecurring | SparkSubmitOperator | all_success |
| TransactionTypeProduct.TransactionTypeProductSummaryUsage | TransactionTypeProduct.TransactionTypeProductSummaryUsage | SparkSubmitOperator | all_success |
| TransactionTypeProduct.transaction_product_decision | TransactionTypeProduct.transaction_product_decision | BranchPythonOperator | all_success |
| config_loader | config_loader | HttpOperator | all_success |
| contract_meta_state_transition_decision | contract_meta_state_transition_decision | BranchPythonOperator | all_success |
| end | end | EmptyOperator | all_success |
| join | join | EmptyOperator | all_success |
| product_usage_summary_decision | product_usage_summary_decision | BranchPythonOperator | all_success |
| service_usage_summary_decision | service_usage_summary_decision | BranchPythonOperator | all_success |
| start | start | EmptyOperator | all_success |



3.12.4.16 desc-batch-job-task

The `desc-batch-job-task` command retrieves detailed information about a specific task within a batch job.

The users with the roles *Business Config Admin* and *Assistance* can only execute this command.

Syntax

```
beamctl epl-analytics-mgr desc-batch-job-task <job_name> <task_name>
```

Arguments

Table 292 Mandatory Arguments

| Argument | Description |
|-----------|---|
| job_name | The unique identifier of the batch job. |
| task_name | The unique identifier of the task. |

Example 254

```
beamctl epl-analytics-mgr desc-batch-job-task eric-bss-epl-sb-FileConsolidation ContractMetaStateTransition
```

Sample Response:

```
{
  "class_ref": {
    "class_name": "SparkSubmitOperator",
    "module_path": "airflow.providers.apache.spark.operators.spark_submit"
  },
  "depends_on_past": false,
  "description": "ContractMetaStateTransition",
  "doc_md": null,
  "downstream_task_ids": [
    "join"
  ],
  "end_date": null,
  "execution_timeout": null,
  "extra_links": [],
  "is_mapped": false,
  "operator_name": "SparkSubmitOperator",
  "owner": "airflow",
  "params": {},
  "pool": "default_pool",
  "pool_slots": 1,
  "priority_weight": 1,
  "queue": "default",
  "retries": 1,
  "retry_delay": {
    "type": "TimeDelta",
    "days": 0,
    "microseconds": 0,
    "seconds": 300
  },
  "retry_exponential_backoff": false,
  "start_date": "2025-10-08T16:22:13.295691Z",
  "task_name": "ContractMetaStateTransition",
  "template_fields": [
    "application",
```



```

    "conf",
    "files",
    "py_files",
    "jars",
    "driver_class_path",
    "packages",
    "exclude_packages",
    "keytab",
    "principal",
    "proxy_user",
    "name",
    "application_args",
    "env_vars",
    "properties_file",
    "application_args"
  ],
  "trigger_rule": "all_success",
  "ui_color": "#FF9933",
  "ui_fgcolor": "#000",
  "wait_for_downstream": false,
  "weight_rule": "downstream"
}

```

3.12.4.17 list-batch-job-rerun

The `list-batch-job-rerun` command retrieves a list of rerun operations for scheduled batch jobs in a table format.

The users with the roles *Business Config Admin* and *Assistance* can only execute this command.

Syntax

```
beamctl epl-analytics-mgr list-batch-job-rerun <job_name>
```

Arguments

Table 293 Mandatory Arguments

| Argument | Description |
|----------|---|
| job_name | The unique identifier of the batch job. |

Example 255

```
beamctl epl-analytics-mgr list-batch-job-rerun eric-bss-epl-sb-FileConsolidation
```



Table 294 Sample Response

| Id | Job Name | Start Date | End Date |
|----|-----------------------------------|----------------------|----------------------|
| 4 | eric-bss-epl-sb-FileConsolidation | 2025-10-01T05:00:00Z | 2025-10-01T05:30:00Z |
| 3 | eric-bss-epl-sb-FileConsolidation | 2025-09-30T23:30:00Z | 2025-10-01T00:00:00Z |

3.12.4.18

desc-batch-job-rerun

The `desc-batch-job-rerun` command retrieves detailed information about a specific rerun created for scheduled batch job.

Syntax

```
beamctl epl-analytics-mgr desc-batch-job-rerun <job_name>
<rerun_id>
```

Arguments

Table 295 Mandatory Arguments

| Argument | Description |
|----------|---|
| job_name | The unique identifier of the batch job. |
| rerun_id | The unique identifier of the rerun. It can be fetched from the <code>list-batch-job-rerun</code> command. |

Example 256

```
beamctl epl-analytics-mgr desc-batch-job-rerun eric-bss-epl-sb-
DailyTransformation 4
```

Sample Response:

```
{
  "completed_at": "2025-10-01T05:14:42.602293Z",
  "created_at": "2025-10-01T05:12:02.997538Z",
  "from_date": "2025-10-01T05:00:00Z",
  "id": 4,
  "is_paused": false,
  "job_name": "eric-bss-epl-sb-DailyTransformation",
  "job_run_conf": {},
}
```




```

    "max_active_runs": 1,
    "reprocess_behavior": "completed",
    "to_date": "2025-10-01T05:30:00Z",
    "updated_at": "2025-10-01T05:14:42.608560Z"
  }

```

3.12.4.19 add-batch-job-rerun

The `add-batch-job-rerun` command creates a new rerun operation for scheduled DAGs and non-scheduled batch-job. Which is basically to rerun all the jobs within the mentioned time frame.

Rerun works differently for scheduled batch-job and non-scheduled batch-job:

- For scheduled batch-job, the rerun creates a set of rerun with the timeframe.
- For non-scheduled batch-job, the rerun triggers the specific batch-job only once.

Syntax

```
beamctl epl-analytics-mgr add-batch-job-rerun <job_name> --
start_date <start_date> --end_date <end_date> --run_date <run_date>
```

Arguments

Table 296 Mandatory Arguments

| Argument | Description |
|----------|---|
| job_name | The unique identifier of the batch job. |

Table 297 Optional Arguments

| Argument | Description |
|------------|---|
| end_date | End date for scheduled jobs, in the UTC format. |
| run_date | Run date for non-scheduled jobs, in the UTC format. |
| start_date | Start date for scheduled jobs, in the UTC format. |



Example 257

```
beamctl epl-analytics-mgr add-batch-job-rerun eric-bss-epl-sb-FileConsolidation --start_date 2025-10-01T05:00:00Z --end_date 2025-10-01T05:30:00Z
```

```
beamctl epl-analytics-mgr add-batch-job-rerun eric-bss-epl-sb-WeeklySummary --run_date 2025-10-01T05:00:00Z
```

Note:

- Scheduled and Non-Schedule jobs information is available from the `list-all-batch-job` command. Refer to [list-all-batch-job](#) on page 436.
- Adding milliseconds to the time stamp is not allowed, for example: `2025-10-01T05:00:00.489381Z`.

3.12.4.20

batch-job-import-err

The `batch-job-import-err` command retrieves import errors for batch jobs that failed to load properly. Import errors are basically any errors that are found in the implementation of the batch-job.

Syntax

```
beamctl epl-analytics-mgr batch-job-import-err
```

Arguments

This command has no arguments.

Example 258

```
beamctl epl-analytics-mgr batch-job-import-err
```

Sample Response:

```
{
  "import_errors": [],
  "total_entries": 0
}
```

3.12.4.21

add-ext-resource

The `add-ext-resource` command adds a new external variable which holds the configuration for AWS connection.



Syntax

```
beamctl epl-analytics-mgr add-ext-resource < <file_path>
```

Arguments

Table 298 Mandatory Arguments

| Argument | Description |
|-----------|--|
| file_path | The JSON file containing AWS connection configuration. |

Example 259

```
beamctl epl-analytics-mgr add-ext-resource < /path/to/config.json
```

Sample Content of the file /path/to/updated-config.json file:

```
{
  "key": "minio-connection",
  "value": {
    "AWS_S3_ENDPOINT": "https://minio.minio.svc.cluster.local:9000",
    "AWS_DEFAULT_REGION": "us-east-1",
    "AWS_ACCESS_KEY_ID": "admin",
    "AWS_SECRET_ACCESS_KEY": "XXXX",
    "S3_SSL_ENABLED": "true"
  },
  "description": "MinIO S3-compatible storage connection details with ssl"
}
```

Note: S3_SSL_ENABLED must be enabled for S3 compliant backends configured with private CAs.

S3_SSL_ENABLED is optional for backends, such as AWS S3, configured with well-known CAs.

For configuring trusted private CAs, refer to section *Injection of Trusted CA Certificate for Standalone S3-compliant Backend* in EPL Custom Data Marts Function Guide.

3.12.4.22 update-ext-resource

The update-ext-resource command updates an existing external variable which holds the configuration for AWS connection.

Syntax

```
beamctl epl-analytics-mgr update-ext-resource < <file_path>
```



Arguments

Table 299 Mandatory Arguments

| Argument | Description |
|-----------|--|
| file_path | The JSON file containing AWS connection configuration. |

Example 260

```
beamctl epl-analytics-mgr update-ext-resource < /path/to/updated-config.json
```

Sample Content of the /path/to/updated-config.json file:

```
{
  "key": "minio-connection",
  "value": {
    "AWS_S3_ENDPOINT": "https://minio.minio.svc.cluster.local:9000",
    "AWS_DEFAULT_REGION": "us-east-1",
    "AWS_ACCESS_KEY_ID": "admin",
    "AWS_SECRET_ACCESS_KEY": "XXXX",
    "S3_SSL_ENABLED": "true"
  },
  "description": "MinIO S3-compatible storage connection details with ssl"
}
```

Note: S3_SSL_ENABLED must be enabled for S3 compliant backends configured with private CAs.

S3_SSL_ENABLED is optional for backends, such as AWS S3, configured with well-known CAs.

For configuring trusted private CAs, refer to section *Injection of Trusted CA Certificate for Standalone S3-compliant Backend* in EPL Custom Data Marts Function Guide.

3.12.4.23 delete-ext-resource

The delete-ext-resource command deletes an existing external variable which holds the configuration for AWS connection.

Syntax

```
beamctl epl-analytics-mgr delete-ext-resource variable-name
```



Arguments

Table 300 Mandatory Arguments

| Argument | Description |
|---------------|---|
| variable-name | The variable name is the given key at the time of creation. |

Flags

Table 301 Optional Flags

| Flags | Type | Description |
|-------------------------|---------|---|
| dry-run | boolean | Dry-run the command. |
| o, output-format string | string | It provides the output format [JSON, YAML]. Default is JSON format. |
| q, query string | string | Filter/project results using JMESPath |
| raw | boolean | Output result of query as raw rather than a JSON string or list. |
| server string | string | Override server URL |
| show-headers | boolean | Output predefined response headers |
| test-env | boolean | Run in developer test environment |
| verbose | boolean | Enable verbose log output |

Example 261

```
beamctl epl-analytics-mgr delete-ext-resource minio-connection
```

3.12.4.24 list-ext-resource

The `list-ext-resource` command retrieves a list of all external variables created.

Syntax

```
beamctl epl-analytics-mgr list-ext-resource
```



Arguments

This command has no arguments.

Example 262

```
beamctl epl-analytics-mgr list-ext-resource
```

Sample Response:

```
{
  "total_entries": 1,
  "variables": [
    {
      "description": "MinIO S3-compatible storage connection details",
      "is_encrypted": true,
      "key": "minio-connection",
      "value": {
        "AWS_S3_ENDPOINT": "https://minio.minio.svc.cluster.local:9000",
        "AWS_DEFAULT_REGION": "us-east-1",
        "AWS_ACCESS_KEY_ID": "admin",
        "AWS_SECRET_ACCESS_KEY": "XXXX",
        "S3_SSL_ENABLED": "true"
      }
    }
  ]
}
```

3.12.4.25

ls-analytics-dataset

The `ls-analytics-dataset` command lists all available analytics datasets in table format with their available clusters.

Syntax

```
beamctl epl-analytics-mgr ls-analytics-dataset [--analytics-type
<analytics-type>]
```

Arguments

Table 302

| Argument | Description |
|------------------|--|
| <analytics-type> | Type of analytics (optional, default is primary) |

Example 263

```
beamctl epl-analytics-mgr ls-analytics-dataset
```

```
beamctl epl-analytics-mgr ls-analytics-dataset --analytics-type
primary
```



```
beamctl epl-analytics-mgr ls-analytics-dataset --analytics-type
secondary
```

Sample Response

Table 303

| Dataset Name | Available clusters |
|---------------------|--------------------|
| ServiceUsageSummary | epl1,epl2 |

3.12.4.26

preview-dataset

The `preview-dataset` command retrieves the record from the specified analytics dataset for preview purposes. This helps to know the columns available in dataset such that the same columns could be used for view creation.

Syntax

```
beamctl epl-analytics-mgr preview-dataset <DATASET_NAME> [--
analytics-type <analytics-type>]
```

Arguments

Table 304

| Argument | Description |
|------------------|---|
| <DATASET_NAME> | Mandatory. Name of the dataset to preview. |
| <analytics-type> | Type of analytics (optional, default is primary). |

Example 264

```
beamctl epl-analytics-mgr preview-dataset ServiceUsageSummary
```

```
beamctl epl-analytics-mgr preview-dataset ServiceUsageSummary --
analytics-type primary
```

```
beamctl epl-analytics-mgr preview-dataset ServiceUsageSummary --
analytics-type secondary
```

Sample Response

```
{
  "RFSSpecificationId": "ID766443",
  "SumOfChargedUnits": "4.5077608488E8",
  "UnitUoM": "SMS",
  "PartitionId": "1",
  "SumOfRatedUnits": "4.5077608488E8",
```



```
"Location": "l1",
"DateId": "20251111"
}
```

3.12.4.27 crt-analytics-view

The `crt-analytics-view` command creates new analytics view from a JSON configuration file.

Syntax

```
beamctl epl-analytics-mgr crt-analytics-view < <config_file>
```

Arguments

Table 305 Mandatory Arguments

| Argument | Description |
|-------------------------|--|
| JSON configuration file | It is a JSON configuration file from stdin containing the view definition. |

Parameters

Table 306 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------------|----------------|--|
| view_name | Mandatory | Name of the analytics views to create. |
| view_definition | Mandatory | SQL statement with <code><View_Name></code> placeholder.
It must start with <i>CREATE</i> or <i>REPLACE</i> . |
| view_source | Mandatory | SQL <code>SELECT</code> statement with <code><dataset_name></code> placeholder. |
| dataset_name | Mandatory | Base name of the dataset. |

The required JSON structure is:

```
{
  "view_name": "view_name",
  "view_definition": "CREATE OR REPLACE ANALYTICS VIEW <View_Name> (column1 TYPE, column2 TYPE) DE →
  FAULT NULL PRIMARY KEY (column1) NOT ENFORCED",
  "view_source": "SELECT column1, column2 FROM <dataset_name>",
}
```




```
{
  "dataset_name": "dataset_name"
}
```

Note: It is important to provide data type to each column to create analytics views. The following is the list of the supported data types:

- BOOLEAN
- STRING
- BIGINT (synonym INT)
- DOUBLE
- DATE (includes date only, no timezone)
- TIME (includes time only, no timezone)
- DATETIME (includes date and time, no timezone)

Example 265

```
beamctl epl-analytics-mgr crt-analytics-view < view_config.json
```

Example 266 view_config.json Example

```
{
  "view_name": "service_usage_summary_view",
  "dataset_name": "ServiceUsageSummary",
  "view_definition": "CREATE OR REPLACE ANALYTICS VIEW <View_Name> (RFSSpecificationId STRING, Sum →
OfChargedUnits STRING, UnitUoM STRING, PartitionId STRING, SumOfRatedUnits STRING, Location STRING →
, DateId STRING) DEFAULT NULL PRIMARY KEY (RFSSpecificationId) NOT ENFORCED",
  "view_source": "SELECT RFSSpecificationId, SumOfChargedUnits, UnitUoM, PartitionId, SumOfRatedUn →
its, Location, DateId FROM <dataset_name>"
}
```

Sample Response:

```
Analytics view 'service_usage_summary_view' created successfully
```

3.12.4.28 ls-analytics-view

The `ls-analytics-view` command lists all available analytics views with their associated dataset names.

Syntax

```
beamctl epl-analytics-mgr ls-analytics-view [--analytics-type
<analytics-type>]
```



Arguments

Table 307 Arguments

| Arguments | Description |
|------------------|--|
| <analytics-type> | Type of analytics (optional, default is primary) |

Example 267

```
beamctl epl-analytics-mgr ls-analytics-view

beamctl epl-analytics-mgr ls-analytics-view --analytics-type
primary

beamctl epl-analytics-mgr ls-analytics-view --analytics-type
secondary
```

Table 308 Sample Response

| View Name | Dataset Name |
|----------------------------|---|
| service_usage_summary_view | ServiceUsageSummary_epl1,
ServiceUsageSummary_epl2 |

3.12.4.29 show-bi-details

The show-bi-details command retrieves BI connection details including appgroup endpoints, role, data service URLs, and certificates required for BI tools integration.

Syntax

```
beamctl epl-analytics-mgr show-bi-details < password.txt
```

Arguments

Table 309 Arguments

| Arguments | Description |
|---------------|--|
| Password File | Create a text file (example, password.txt) containing your password string (example, myTestPassword123) and use it with the command. |



Example 268

```
beamctl epl-analytics-mgr show-bi-details < password.txt
```

Sample Response

```
json
{
  "servers": [
    {
      "appgroup": "epl1",
      "endpoint": "eric-bss-epl-epl1-event-db-0000.test-bi-integration.com",
      "role": "primary"
    },
    {
      "appgroup": "epl3",
      "endpoint": "eric-bss-epl-epl3-event-db-0000.test-bi-integration.com",
      "role": "secondary"
    }
  ],
  "server-ca": "<ENCRYPTED_BASE64_ENCODED_STRING>"
}
```

3.12.4.30 list-batch-job

The `list-batch-job` command retrieves spark configurations for all batch marts, showing both standard and custom job configurations.

Syntax

```
beamctl epl-analytics-mgr list-batch-job
```

Arguments

The command has no arguments.

Example 269

```
beamctl epl-analytics-mgr list-batch-job
```

Sample Response:

```
json
{
  "jobs": [
    {
      "jobName": "eric-epl-bss-cb-CustomEvents",
      "type": "custom",
      "marts": [
        {
          "martName": "CustomUsageSummary",
          "conf": {
            "spark.executor.memory": "50g",
            "spark.driver.memory": "20g",
            "spark.executor.cores": "32"
          }
        }
      ]
    },
    {
      "jobName": "eric-epl-bss-sb-HourlyDetail",
```



```
"type": "standard",
"mars": [
  {
    "martName": "HourlyDetailMart",
    "conf": {
      "spark.executor.memory": "40g",
      "spark.driver.memory": "15g"
    }
  }
]
}
```

3.12.4.31 update-batch-job

The update-batch-job command updates spark configurations for all batch mars and jobs. Custom jobs update both file system and EPL storage.

Syntax

beamctl epl-analytics-mgr update-batch-job < <config_file.json>

Arguments

Table 310 Mandatory Arguments

| Argument | Description |
|------------------|---|
| config_file.json | JSON configuration file from stdin, containing job configurations with Mart Spark settings. |

Example 270

```
beamctl epl-analytics-mgr update-batch-job < spark-config-
update.json
```

Example 271 Examples of spark-config-update.json

```
json
{
  {
    "jobName": "eric-epl-bss-cb-CustomEvents",
    "mars": [
      {
        "martName": "CustomUsageSummary",
        "conf": {
          "spark.executor.memory": "50g",
          "spark.driver.memory": "20g",
          "spark.executor.cores": "32"
        }
      }
    ]
  }
}
```



```
] } ]
```

Sample Response:

```
json
{
  "message": "Spark configurations updated successfully",
  "results": [
    {
      "jobName": "eric-epl-bss-cb-CustomEvents",
      "marts": [
        {
          "martName": "CustomUsageSummary",
          "status": "updated",
          "type": "custom"
        }
      ]
    }
  ],
  "status": "success"
}
```

Note: Only the configurations listed by the `list-batch-job` command can be updated using this command. For any issues or additional support, contact your local Ericsson representative.

3.12.4.32

list-all-batch-marts

The `list-all-batch-marts` command lists all the active and inactive batch marts in the system.

Syntax

```
beamctl epl-analytics-mgr list-all-batch-marts
```

Arguments

This command has no arguments.

Example 272

```
beamctl epl-analytics-mgr list-all-batch-marts
```

Table 311 Sample Response

| Mart Name | Status |
|----------------------------------|--------|
| ContractMetaStateTransition | false |
| ContractStateTransition | false |
| CustomerStateTransition | false |
| EndUserNotificationStatusSummary | false |
| FinancialEntrySummary | false |



| Mart Name | Status |
|----------------------------------|--------|
| InvoiceBillRunStatistics | false |
| InvoiceGenerationSummary | false |
| OverdueReceivableSummary | false |
| ProductFeeSummary | false |
| ProductUsageSummary | false |
| ServiceUsageSummary | false |
| TransactionTypeBillingAccSummary | false |
| TransactionTypeProductSummary | false |

3.12.4.33 update-batch-mart-status

The update-batch-mart-status command is to enable/disable batch marts in the system.

Syntax

```
beamctl epl-analytics-mgr update-batch-mart-status
<batch_mart_name> <enabled>
```

Arguments

Table 312 Optional Arguments

| Arguments | Description |
|-----------------|--|
| batch_mart_name | The unique identifier of the batch mart. It can be taken from list-all-batch-mart command. |
| enabled | To enable/disable the mart using true/false. |

Example 273

```
beamctl epl-analytics-mgr update-batch-mart-status
TransactionTypeProductSummary true

beamctl epl-analytics-mgr update-batch-mart-status
TransactionTypeProductSummary false
```

Sample Response:

```
beamctl epl-analytics-mgr update-batch-mart-status TransactionTypeProductSummary true
```



```
{"Mart-name":"TransactionTypeProductSummary","Message":"Updated mart 'TransactionTypeProductSummary' status to enabled"} →
```

3.12.5 EPL Master Dimension Configuration Manager Commands

The EPL Master Dimension Configuration Manager commands control which master-entity types are considered by the extraction job that populates the master dimension data in the database.

Syntax

```
beamctl epl-md-config <sub-command>
```

Available Sub Commands

Table 313 Sub Commands

| Command | Description |
|--|---|
| list-configs on page 459 | Displays the list of entity names and their status in the system. |
| update-configs on page 460 | Updates the status of the entity. |

3.12.5.1 list-configs

The `list-configs` displays the list of entity names and their status in the system.

Syntax

```
beamctl epl-md-config list-configs
```

Arguments

This command has no arguments.

Example 274

```
beamctl epl-md-config list-configs
```

Table 314 Sample Response:

| Entity Name | Status |
|-----------------------------|--------|
| productoffering | true |
| billingAccountSpecification | true |



| Entity Name | Status |
|------------------------------------|--------|
| customerFacingServiceSpecification | false |
| characteristicSetSpecification | false |

3.12.5.2

update-configs

The `update-configs` command updates the status of the entity. Depending on the updated status, the corresponding entity is either bypassed or considered for the master dimension extraction job that stores data for the entity in the database.

The update command takes entity names, with comma separated, such that multiple entities can be either disabled or enabled at a time. If an invalid entity name is specified, the corresponding error is displayed.

Syntax

```
beamctl epl-md-config update-configs
```

Arguments

Table 315 Mandatory Arguments

| Arguments | Description |
|-------------|---|
| entity-name | Name of the entity. This argument intakes multiple entity names with comma separated. |
| enable | It is a boolean value: <i>true</i> or <i>false</i> .
Value is set against an entity or entities. |

Note: Entity or entities that are enabled or disabled, are reflected only in the next schedule of the master dimension extraction job.

Example 275

```
beamctl epl-md-config update-configs productoffering true
beamctl epl-md-config update-configs
productoffering,customerFacingServiceSpecification,billingAccount
Specification true
```




3.13 Orchestrator

The Orchestrator command provides the functionality to perform different operations towards the Orchestrator service.

Syntax

```
beamctl orchestrator
```

Sub Commands

Table 316 Available Sub Commands

| Subcommand | Description |
|---|--|
| decommission-partition-for-datatype on page 462 | It performs a decommission of a partition for a particular application type. For example: db-user, db-provider, or msg-provider. |
| failback on page 462 | It performs the failback of a partition. |
| failback-appgroup on page 463 | It performs the failback of an app group. |
| failover on page 464 | It performs the failover of a partition. |
| failover-appgroup on page 464 | It performs the failover for an app group. |
| get-msg-partition on page 465 | It provides MSG partitionconfiguration for a partitionid. |
| get-msg-partition-for-datatype on page 466 | It provides MSG partitionconfiguration for partitionid and datatype. |
| get-original-state-partition-config on page 467 | It provides the current original state partition configuration in JSON format. |
| get-partition on page 483 | It provides information about a specific partition. |
| get-partition-for-datatype on page 485 | It provides information about a specific partition and datatype. |
| get-status on page 486 | It provides information about the currently ongoing operation, it can be <i>failover</i> or <i>failback</i> . |
| replace-original-state-partition-config on page 487 | It takes the given original state partition configuration and replaces the current original state partition configuration. |



| Subcommand | Description |
|--|--|
| update-original-state-partition-config on page 488 | It updates the original state partition configuration. |

3.13.1 decommission-partition-for-datatype

The `decommission-partition-for-datatype` command performs a decommission of a partition for a particular application type (db-provider, db-user).

Syntax

```
beamctl orchestrator decommission-partition-for-datatype  
  <partitionId> <partition> <datatype> <apptype>
```

Argument

Table 317 Mandatory Arguments

| Argument | Description |
|-------------|---|
| partitionid | Partition ID: it is a number. |
| partition | Partition type. |
| datatype | Data type for the partition. |
| apptype | Application type (db-user and db-provider). |

Example 276

```
beamctl orchestrator decommission-partition-for-datatype 1  
customer cha db-user
```

```
beamctl orchestrator decommission-partition-for-datatype 1  
customer cha db-provider
```

3.13.2 failback

The `failback` command performs failback of a partition.

Syntax

```
beamctl orchestrator failback <appgroup> <partitionid> <partition>  
  <datatype>
```



Arguments

Table 318 Mandatory Arguments

| Argument | Description |
|-------------|--|
| appgroup | Application group: it is an application name followed by a number. |
| partitionid | Partition ID: it is a number. |
| partition | Partition type. |
| datatype | Data type for the partition. |

Example 277

```
beamctl orchestrator failback cha1 0 global cha
beamctl orchestrator failback cil3 3 customer edm
```

3.13.3

failback-appgroup

The `failback-appgroup` command performs the failback of an app group.

Syntax

```
beamctl orchestrator failback-appgroup <appgroup>
```

Arguments

Table 319 Mandatory Arguments

| Argument | Description |
|----------|--|
| appgroup | Application group: it is an application name followed by a number. |

Example 278

```
beamctl orchestrator failback-appgroup cha1
```



3.13.4 failover

The `failover` command performs the failover of a partition.

Syntax

```
beamctl orchestrator failover <appgroup> <partitionid> <partition>
<datatype>
```

Table 320 Mandatory Arguments

| Argument | Description |
|-------------|---|
| appgroup | Application group: it is the application name followed by a number. |
| partitionid | Partition ID: it is a number. |
| partition | Partition type. |
| datatype | Data type for the partition. |

Example 279

```
beamctl orchestrator failover cha1 0 global cha
beamctl orchestrator failover cil3 3 customer edm
```

3.13.5 failover-appgroup

The `failover-appgroup` command performs failover of an app group.

Syntax

```
beamctl orchestrator failover-appgroup <appgroup>
```

Arguments

Table 321 Mandatory Arguments

| Argument | Description |
|----------|---|
| appgroup | Application group: this is the application name followed by a number. |

**Example 280**

```
beamctl orchestrator failover-appgroup cha1
```

3.13.6 get-msg-partition

The `get-msg-partition` command provides the information about a specific partition for `msg-provider`.

Syntax

```
beamctl orchestrator get-msg-partition <partitionId> <partition>
```

Arguments

Table 322 Mandatory Arguments

| Argument | Description |
|-------------|-------------------------------|
| partitionId | Partition ID: it is a number. |
| partition | Partition type. |

Parameters

Table 323 Parameters

| Parameter | Parameter Type | Description |
|---------------------------------|----------------|--|
| partition | Mandatory | The partition name. |
| partitionId | Mandatory | The partition ID. |
| partitionType | Mandatory | Type of the partition. |
| cacheNotificationConfig | Mandatory | Cache notification enabled flag. |
| replicationAndStatesPerDataType | Mandatory | Array of replication and states per data type. |

Example 281

```
beamctl orchestrator get-msg-partition 2 customer
```

Sample Response:

```
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "customer",
  "partitionId": 2,
```



```
"partitionType": "PARTITIONED",
"replicationAndStatesPerDataType": [
  {
    "dataType": "msg",
    "states": {
      "activemsgProviderAgs": [
        "msg2",
        "msg1"
      ],
      "passivemsgProviderAgs": [],
      "suspendedmsgProviderAgs": []
    }
  }
]
```

3.13.7 get-msg-partition-for-datatype

The `get-msg-partition-for-datatype` command provides the information about a specific partition and datatype for `msg-provider`.

Syntax

```
beamctl orchestrator get-msg-partition-for-datatype <partitionId>
<partition> <datatype>
```

Arguments

Table 324 Mandatory Arguments

| Argument | Description |
|-------------|-------------------------------|
| partitionid | Partition ID: it is a number. |
| partition | Partition type. |
| datatype | Datatype for msg-provider. |

Parameters

Table 325 Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|--------------------------|
| dataType | Mandatory | Data type identifier. |
| states | Mandatory | Message provider states. |

Example 282

```
beamctl orchestrator get-msg-partition-for-datatype 1 customer
msg
```



Sample Response:

```
{
  "dataType": "msg",
  "states": {
    "activemsgProviderAgs": [
      "msg1",
      "msg2"
    ],
    "passivemsgProviderAgs": [],
    "suspendedmsgProviderAgs": []
  }
}
```

3.13.8 get-original-state-partition-config

The `get-original-state-partition-config` command provides the current original state partition configuration in JSON format.

Syntax

```
beamctl orchestrator get-original-state-partition-config
```

Arguments

The command has no arguments.

Parameters

Table 326 Parameters

| Parameter | Parameter Type | Description |
|--------------------------------|----------------|---|
| OriginalStateDbProviderConfig | Mandatory | Array of DB provider configurations. |
| OriginalStateDbUserConfig | Mandatory | Array of DB user configurations. |
| OriginalStateMsgProviderConfig | Mandatory | Array of message provider configurations. |
| OriginalStateStandAloneConfig | Mandatory | Array of standalone configurations. |
| version | Mandatory | Orchestrator version. |

Example 283

```
beamctl orchestrator get-original-state-partition-config
```

Sample Response:

```
{
  "OriginalStateStandAloneConfig": [
    {
      "applicationName": "cusapp",
```



```
"standAloneAG": {
  "activeStandAloneAgs": [
    "cusapp1",
    "cusapp2"
  ],
  "stoppedStandAloneAgs": []
},
"stateKey": "Original",
"targetMap": "eric-bss-originalstate-configmap",
"version": "2.0.0"
}
],
"OriginalStatedbProviderConfig": [
{
  "applicationName": "cil",
  "partitions": [
    {
      "cacheNotificationConfig": {
        "isEnabled": true
      },
      "partition": "global",
      "partitionId": 0,
      "partitionType": "GLOBAL",
      "replicationAndStatesPerDataType": [
        {
          "dataType": "bae",
          "replication": [
            {
              "datacenterName": "cil1",
              "replicationFactor": 3
            },
            {
              "datacenterName": "cil2",
              "replicationFactor": 3
            }
          ],
          "states": {
            "activedbProviderDCs": [
              "cil1",
              "cil2"
            ],
            "passivedbProviderDCs": [],
            "suspendeddbProviderDCs": []
          }
        },
        {
          "dataType": "bsr",
          "replication": [
            {
              "datacenterName": "cil1",
              "replicationFactor": 3
            },
            {
              "datacenterName": "cil2",
              "replicationFactor": 3
            }
          ],
          "states": {
            "activedbProviderDCs": [
              "cil1",
              "cil2"
            ],
            "passivedbProviderDCs": [],
            "suspendeddbProviderDCs": []
          }
        },
        {
          "dataType": "cha",
          "replication": [
            {
              "datacenterName": "cil1",
              "replicationFactor": 3
            },
            {
              "datacenterName": "cil2",
              "replicationFactor": 3
            }
          ],
          "states": {
            "activedbProviderDCs": [
```




```

        "cil1",
        "cil2"
    ],
    "passivedbProviderDCs": [],
    "suspendeddbProviderDCs": []
  },
  {
    "dataType": "coba",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil1",
        "cil2"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  },
  {
    "dataType": "cpm",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil1",
        "cil2"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  },
  {
    "dataType": "edm",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil1",
        "cil2"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  },
  {
    "dataType": "ntf",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil1",
        "cil2"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  }
]

```



```
    },
    "states": {
      "activedbProviderDCs": [
        "cil1",
        "cil2",
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  },
  {
    "dataType": "rmca",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil1",
        "cil2",
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  }
],
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "region",
  "partitionId": 1,
  "partitionType": "PARTITIONED",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "cha",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil1",
          "cil2",
        ],
        "passivedbProviderDCs": [],
        "suspendeddbProviderDCs": []
      }
    }
  ]
},
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "index",
  "partitionId": 0,
  "partitionType": "GLOBAL",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "cpm",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        }
      ],
    }
  ]
}
```



```

        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil1",
          "cil2"
        ],
        "passivedbProviderDCs": [],
        "suspendeddbProviderDCs": []
      }
    }
  ],
},
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "site",
  "partitionId": 1,
  "partitionType": "PARTITIONED",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "bae",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil1",
          "cil2"
        ],
        "passivedbProviderDCs": [],
        "suspendeddbProviderDCs": []
      }
    }
  ]
},
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "site",
  "partitionId": 2,
  "partitionType": "PARTITIONED",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "bae",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil2",
          "cil1"
        ],
        "passivedbProviderDCs": [],
        "suspendeddbProviderDCs": []
      }
    }
  ]
},
{
  "cacheNotificationConfig": {

```



```
    "isEnabled": false
  },
  "partition": "customer",
  "partitionId": 1,
  "partitionType": "PARTITIONED",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "bae",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil1",
          "cil2"
        ],
        "passivedbProviderDCs": [],
        "suspendeddbProviderDCs": []
      }
    },
    {
      "dataType": "bsr",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil1",
          "cil2"
        ],
        "passivedbProviderDCs": [],
        "suspendeddbProviderDCs": []
      }
    },
    {
      "dataType": "cha",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil1",
          "cil2"
        ],
        "passivedbProviderDCs": [],
        "suspendeddbProviderDCs": []
      }
    },
    {
      "dataType": "cpm",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ]
    }
  ]
}
```



```

    ],
    "states": {
      "activedbProviderDCs": [
        "cil1",
        "cil2"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  },
  {
    "dataType": "edm",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil1",
        "cil2"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  },
  {
    "dataType": "ntf",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil1",
        "cil2"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  }
]
},
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "customer",
  "partitionId": 2,
  "partitionType": "PARTITIONED",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "bae",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil2",
          "cil1"
        ],
        "passivedbProviderDCs": [],

```



```
    "suspendeddbProviderDCs": []
  },
  {
    "dataType": "bsr",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil2",
        "cil1"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  },
  {
    "dataType": "cha",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil2",
        "cil1"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  },
  {
    "dataType": "cpm",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil2",
        "cil1"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  },
  {
    "dataType": "edm",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
```



```

        "cil2",
        "cil1"
    ],
    "passivedbProviderDCs": [],
    "suspendeddbProviderDCs": []
  },
  {
    "dataType": "ntf",
    "replication": [
      {
        "datacenterName": "cil1",
        "replicationFactor": 3
      },
      {
        "datacenterName": "cil2",
        "replicationFactor": 3
      }
    ],
    "states": {
      "activedbProviderDCs": [
        "cil2",
        "cil1"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  }
],
},
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "customer",
  "partitionId": 3,
  "partitionType": "PARTITIONED",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "cha",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil1",
          "cil2"
        ],
        "passivedbProviderDCs": [],
        "suspendeddbProviderDCs": []
      }
    }
  ]
},
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "customer",
  "partitionId": 4,
  "partitionType": "PARTITIONED",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "cha",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ]
    }
  ]
}
]
}

```



```
    ],
    "states": {
      "activedbProviderDCs": [
        "cil2",
        "cil1"
      ],
      "passivedbProviderDCs": [],
      "suspendeddbProviderDCs": []
    }
  ]
},
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "customer",
  "partitionId": 5,
  "partitionType": "PARTITIONED",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "cha",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ],
      "states": {
        "activedbProviderDCs": [
          "cil1",
          "cil2"
        ],
        "passivedbProviderDCs": [],
        "suspendeddbProviderDCs": []
      }
    }
  ],
  "stateKey": "Original",
  "targetMap": "eric-bss-originalstate-configmap",
  "version": "2.0.0"
},
]
```




```

"OriginalStatedbUserConfig": [
  {
    "applicationName": "bae",
    "partitions": [
      {
        "dataType": "bae",
        "partition": "global",
        "partitionId": 0,
        "states": {
          "activedbUserAgs": [
            "bae1",
            "bae2"
          ],
          "passivedbUserAgs": [],
          "stoppeddbUserAgs": []
        }
      },
      {
        "dataType": "bae",
        "partition": "site",
        "partitionId": 1,
        "states": {
          "activedbUserAgs": [
            "bae1"
          ],
          "passivedbUserAgs": [
            "bae2"
          ],
          "stoppeddbUserAgs": []
        }
      },
      {
        "dataType": "bae",
        "partition": "site",
        "partitionId": 2,
        "states": {
          "activedbUserAgs": [
            "bae2"
          ],
          "passivedbUserAgs": [
            "bae1"
          ],
          "stoppeddbUserAgs": []
        }
      },
      {
        "dataType": "bae",
        "partition": "customer",
        "partitionId": 1,
        "states": {
          "activedbUserAgs": [
            "bae1"
          ],
          "passivedbUserAgs": [
            "bae2"
          ],
          "stoppeddbUserAgs": []
        }
      },
      {
        "dataType": "bae",
        "partition": "customer",
        "partitionId": 2,
        "states": {
          "activedbUserAgs": [
            "bae2"
          ],
          "passivedbUserAgs": [
            "bae1"
          ],
          "stoppeddbUserAgs": []
        }
      }
    ],
    "stateKey": "Original",
    "targetMap": "eric-bss-originalstate-configmap",
    "version": "2.0.0"
  },
  {
    "applicationName": "bsr",

```



```
"partitions": [
  {
    "dataType": "bsr",
    "partition": "global",
    "partitionId": 0,
    "states": {
      "activedbUserAgs": [
        "bsr1",
        "bsr2"
      ],
      "passivedbUserAgs": [],
      "stoppeddbUserAgs": []
    }
  },
  {
    "dataType": "bsr",
    "partition": "customer",
    "partitionId": 1,
    "states": {
      "activedbUserAgs": [
        "bsr1"
      ],
      "passivedbUserAgs": [
        "bsr2"
      ],
      "stoppeddbUserAgs": []
    }
  },
  {
    "dataType": "bsr",
    "partition": "customer",
    "partitionId": 2,
    "states": {
      "activedbUserAgs": [
        "bsr2"
      ],
      "passivedbUserAgs": [
        "bsr1"
      ],
      "stoppeddbUserAgs": []
    }
  }
],
"stateKey": "Original",
"targetMap": "eric-bss-originalstate-configmap",
"version": "2.0.0"
},
{
  "applicationName": "cha",
  "partitions": [
    {
      "dataType": "cha",
      "partition": "global",
      "partitionId": 0,
      "states": {
        "activedbUserAgs": [
          "cha1",
          "cha2",
          "cha11",
          "cha12",
          "cha21",
          "cha22"
        ],
        "passivedbUserAgs": [],
        "stoppeddbUserAgs": []
      }
    },
    {
      "dataType": "cha",
      "partition": "region",
      "partitionId": 1,
      "states": {
        "activedbUserAgs": [
          "cha1",
          "cha2",
          "cha11",
          "cha12",
          "cha21",
          "cha22"
        ],

```



```

        "passivedbUserAgs": [],
        "stoppeddbUserAgs": []
    },
    {
        "dataType": "cha",
        "partition": "customer",
        "partitionId": 1,
        "states": {
            "activedbUserAgs": [
                "cha1"
            ],
            "passivedbUserAgs": [
                "cha2"
            ],
            "stoppeddbUserAgs": []
        }
    },
    {
        "dataType": "cha",
        "partition": "customer",
        "partitionId": 2,
        "states": {
            "activedbUserAgs": [
                "cha2"
            ],
            "passivedbUserAgs": [
                "cha1"
            ],
            "stoppeddbUserAgs": []
        }
    },
    {
        "dataType": "cha",
        "partition": "customer",
        "partitionId": 3,
        "states": {
            "activedbUserAgs": [
                "cha11"
            ],
            "passivedbUserAgs": [
                "cha21"
            ],
            "stoppeddbUserAgs": []
        }
    },
    {
        "dataType": "cha",
        "partition": "customer",
        "partitionId": 4,
        "states": {
            "activedbUserAgs": [
                "cha21"
            ],
            "passivedbUserAgs": [
                "cha11"
            ],
            "stoppeddbUserAgs": []
        }
    },
    {
        "dataType": "cha",
        "partition": "customer",
        "partitionId": 5,
        "states": {
            "activedbUserAgs": [
                "cha12"
            ],
            "passivedbUserAgs": [
                "cha22"
            ],
            "stoppeddbUserAgs": []
        }
    },
    {
        "dataType": "cha",
        "partition": "customer",
        "partitionId": 6,
        "states": {
            "activedbUserAgs": [

```



```
        "cha22"
      ],
      "passivedbUserAgs": [
        "cha12"
      ],
      "stoppeddbUserAgs": []
    }
  ],
  "stateKey": "Original",
  "targetMap": "eric-bss-originalstate-configmap",
  "version": "2.0.0"
},
{
  "applicationName": "coba",
  "partitions": [
    {
      "dataType": "coba",
      "partition": "global",
      "partitionId": 0,
      "states": {
        "activedbUserAgs": [
          "coba1"
        ],
        "passivedbUserAgs": [
          "coba2"
        ],
        "stoppeddbUserAgs": []
      }
    }
  ],
  "stateKey": "Original",
  "targetMap": "eric-bss-originalstate-configmap",
  "version": "2.0.0"
},
{
  "applicationName": "cpm",
  "partitions": [
    {
      "dataType": "cpm",
      "partition": "global",
      "partitionId": 0,
      "states": {
        "activedbUserAgs": [
          "cpm1",
          "cpm2"
        ],
        "passivedbUserAgs": [],
        "stoppeddbUserAgs": []
      }
    },
    {
      "dataType": "cpm",
      "partition": "index",
      "partitionId": 0,
      "states": {
        "activedbUserAgs": [
          "cpm1",
          "cpm2"
        ],
        "passivedbUserAgs": [],
        "stoppeddbUserAgs": []
      }
    }
  ],
  {
    "dataType": "cpm",
    "partition": "customer",
    "partitionId": 1,
    "states": {
      "activedbUserAgs": [
        "cpm1"
      ],
      "passivedbUserAgs": [
        "cpm2"
      ],
      "stoppeddbUserAgs": []
    }
  },
  {
    "dataType": "cpm",
```



```

        "partition": "customer",
        "partitionId": 2,
        "states": {
          "activedbUserAgs": [
            "cpm2"
          ],
          "passivedbUserAgs": [
            "cpm1"
          ],
          "stoppeddbUserAgs": []
        }
      ],
      "stateKey": "Original",
      "targetMap": "eric-bss-originalstate-configmap",
      "version": "2.0.0"
    },
    {
      "applicationName": "edm",
      "partitions": [
        {
          "dataType": "edm",
          "partition": "global",
          "partitionId": 0,
          "states": {
            "activedbUserAgs": [
              "edm1",
              "edm2"
            ],
            "passivedbUserAgs": [],
            "stoppeddbUserAgs": []
          }
        },
        {
          "dataType": "edm",
          "partition": "customer",
          "partitionId": 1,
          "states": {
            "activedbUserAgs": [
              "edm1"
            ],
            "passivedbUserAgs": [
              "edm2"
            ],
            "stoppeddbUserAgs": []
          }
        },
        {
          "dataType": "edm",
          "partition": "customer",
          "partitionId": 2,
          "states": {
            "activedbUserAgs": [
              "edm2"
            ],
            "passivedbUserAgs": [
              "edm1"
            ],
            "stoppeddbUserAgs": []
          }
        }
      ],
      "stateKey": "Original",
      "targetMap": "eric-bss-originalstate-configmap",
      "version": "2.0.0"
    },
    {
      "applicationName": "ntf",
      "partitions": [
        {
          "dataType": "ntf",
          "partition": "global",
          "partitionId": 0,
          "states": {
            "activedbUserAgs": [
              "ntf1",
              "ntf2"
            ],
            "passivedbUserAgs": [],
            "stoppeddbUserAgs": []
          }
        }
      ]
    }
  ]
}

```



```
    },
    {
      "dataType": "ntf",
      "partition": "customer",
      "partitionId": 1,
      "states": {
        "activedbUserAgs": [
          "ntf1"
        ],
        "passivedbUserAgs": [
          "ntf2"
        ],
        "stoppeddbUserAgs": []
      }
    },
    {
      "dataType": "ntf",
      "partition": "customer",
      "partitionId": 2,
      "states": {
        "activedbUserAgs": [
          "ntf2"
        ],
        "passivedbUserAgs": [
          "ntf1"
        ],
        "stoppeddbUserAgs": []
      }
    }
  ],
  "stateKey": "Original",
  "targetMap": "eric-bss-originalstate-configmap",
  "version": "2.0.0"
},
{
  "applicationName": "rmca",
  "partitions": [
    {
      "dataType": "rmca",
      "partition": "global",
      "partitionId": 0,
      "states": {
        "activedbUserAgs": [
          "rmca1"
        ],
        "passivedbUserAgs": [
          "rmca2"
        ],
        "stoppeddbUserAgs": []
      }
    }
  ],
  "stateKey": "Original",
  "targetMap": "eric-bss-originalstate-configmap",
  "version": "2.0.0"
}
],
"OriginalStatemsgProviderConfig": [
  {
    "applicationName": "msg",
    "partitions": [
      {
        "cacheNotificationConfig": {
          "isEnabled": true
        },
        "partition": "global",
        "partitionId": 0,
        "partitionType": "GLOBAL",
        "replicationAndStatesPerDataType": [
          {
            "dataType": "msg",
            "states": {
              "activemsgProviderAgs": [
                "msg1",
                "msg2"
              ],
              "passivemsgProviderAgs": [],
              "suspendedmsgProviderAgs": []
            }
          }
        ]
      }
    ]
  }
]
```



```

    }
  ],
  {
    "cacheNotificationConfig": {
      "isEnabled": false
    },
    "partition": "customer",
    "partitionId": 1,
    "partitionType": "PARTITIONED",
    "replicationAndStatesPerDataType": [
      {
        "dataType": "msg",
        "states": {
          "activemsgProviderAgs": [
            "msg1",
            "msg2"
          ],
          "passivemsgProviderAgs": [],
          "suspendedmsgProviderAgs": []
        }
      }
    ]
  },
  {
    "cacheNotificationConfig": {
      "isEnabled": false
    },
    "partition": "customer",
    "partitionId": 2,
    "partitionType": "PARTITIONED",
    "replicationAndStatesPerDataType": [
      {
        "dataType": "msg",
        "states": {
          "activemsgProviderAgs": [
            "msg2",
            "msg1"
          ],
          "passivemsgProviderAgs": [],
          "suspendedmsgProviderAgs": []
        }
      }
    ]
  }
],
"stateKey": "Original",
"targetMap": "eric-bss-originalstate-configmap",
"version": "2.0.0"
},
"version": "2.0.0"
}

```

3.13.9 get-partition

The `get-partition` command provides the information about a specific partition.

Syntax

```
beamctl orchestrator get-partition <partitionId> <partition>
```



Arguments

Table 327 Mandatory Arguments

| Argument | Description |
|-------------|-------------------------------|
| partitionid | Partition ID: it is a number. |
| partition | Partition type. |

Parameters

Table 328 Parameters

| Parameter | Parameter Type | Description |
|---------------------------------|----------------|--|
| partition | Mandatory | The partition name. |
| partitionId | Mandatory | The partition ID. |
| partitionType | Mandatory | Type of the partition. |
| cacheNotificationConfig | Mandatory | Cache notification enabled flag. |
| replicationAndStatesPerDataType | Mandatory | Array of replication and states per data type. |

Example 284

beamctl orchestrator get-partition 3 customer

Sample Response:

```
{
  "cacheNotificationConfig": {
    "isEnabled": false
  },
  "partition": "customer",
  "partitionId": 3,
  "partitionType": "PARTITIONED",
  "replicationAndStatesPerDataType": [
    {
      "dataType": "cha",
      "replication": [
        {
          "datacenterName": "cil1",
          "replicationFactor": 3
        },
        {
          "datacenterName": "cil2",
          "replicationFactor": 3
        }
      ]
    },
    {
      "states": {
        "activedbProviderDCs": [
          "cil1",
          "cil2"
        ],
        "activedbUserAgs": [
          "cha11"
        ],
        "passivedbProviderDCs": [],
        "passivedbUserAgs": [
          "cha21"
        ],
        "stoppeddbUserAgs": [],
      }
    }
  ]
}
```




```

    "suspendeddbProviderDCs": []
  }
}

```

3.13.10 get-partition-for-datatype

The `get-partition-for-datatype` command provides the information about a specific partition and datatype.

Syntax

```
beamctl orchestrator get-partition-for-datatype <partitionId>
<partition> <datatype>
```

Arguments

Table 329 Mandatory Arguments

| Argument | Description |
|-------------|-------------------------------|
| partitionid | Partition ID: it is a number. |
| partition | Partition type. |
| datatype | Data type for the partition. |

Parameters

Table 330 Parameters

| Parameter | Parameter Type | Description |
|-------------|----------------|--------------------------------------|
| states | Mandatory | DB user and provider states. |
| replication | Mandatory | Array of replication configurations. |

Example 285

```
beamctl orchestrator get-partition-for-datatype 0 global edm
```

Sample Response:

```

{
  "dataType": "edm",
  "replication": [
    {
      "datacenterName": "cil1",
      "replicationFactor": 3
    },
    {
      "datacenterName": "cil2",

```



```
    "replicationFactor": 3
  },
  "states": {
    "activedbProviderDCs": [
      "cil1",
      "cil2"
    ],
    "activedbUserAgs": [
      "edm2"
    ],
    "passivedbProviderDCs": [],
    "passivedbUserAgs": [],
    "stoppeddbUserAgs": [
      "edm1"
    ],
    "suspendeddbProviderDCs": []
  }
}
```

3.13.11 get-status

The `get-status` command provides the information about currently ongoing operation (failover or failback).

Syntax

```
beamctl orchestrator get-status
```

Arguments

The command has no arguments.

Parameters

Table 331 Parameters

| Parameter | Parameter Type | Description |
|----------------------|----------------|--------------------------------|
| operation | Mandatory | Operation type. |
| affectedApplications | Mandatory | Affected applications details. |

Example 286

```
beamctl orchestrator get-status
```

Sample Response:

```
{
  "affectedApplications": {},
  "operation": ""
}
```



Note: The Failover or Failback process completes only when the current state ConfigMap matches the desired state in the wanted state ConfigMap. If the node fails to update its current state ConfigMap, the orchestrator might not be able to mark the ongoing operation as complete.

3.13.12 replace-original-state-partition-config

The `replace-original-state-partition-config` command takes the given original state partition configuration and replaces the current original state partition configuration.

Note: The replace command completely replaces the existing partition configuration data, so this command must be used carefully. Because of the potential impact of this action, it is necessary to follow this recommended sequence before using the replace command with the `--apply` flag:

- Retrieve the existing partition configuration data.
- Make the desired additions or modifications.
- Validate the schema and ensure that the data is accurate.
- Perform a dry run of the command (without the `--apply` flag).

Syntax

```
beamctl orchestrator replace-original-state-partition-config <
<partition-config-file>
```

Arguments

Table 332 Mandatory Arguments

| Argument | Description |
|-----------------------|--|
| partition-config-file | Modified original state config file in JSON format with timestamp added. |

Flags

Table 333 Optional Flags

| Flag | Type | Description |
|-------|---------|---|
| apply | boolean | This flag must be set if the <code>replace-original-state-partition-</code> |



| Flag | Type | Description |
|------|------|--|
| | | config command shall be applied.

If the --apply flag is not set, no changes are applied, and the command only simulates the update. |

Example 287

```
beamctl orchestrator replace-original-state-partition-config <
file.json

beamctl orchestrator replace-original-state-partition-config --
apply < file.json
```

3.13.13 update-original-state-partition-config

The update-original-state-partition-config command updates the original state partition configuration.

This command does not delete or rename any settings in the original state partition configuration. The new settings are merged into the original state partition configuration. It means that partial configurations can be added to the configuration.

Syntax

```
beamctl orchestrator update-original-state-partition-config <
<partition-config-file>
```

Arguments

Table 334 Mandatory Arguments

| Argument | Description |
|-----------------------|--|
| partition-config-file | Modified original state config file in JSON format with timestamp added. |



Flags

Table 335 Optional Flags

| Flag | Type | Description |
|-------|---------|--|
| apply | boolean | This flag must be set if the <code>update-original-state-partition-config</code> command shall be applied. If the <code>--apply</code> flag is not set, no changes are applied, and the command only simulates the update. |

Example 288

```
beamctl orchestrator update-original-state-partition-config <
file.json
```

```
beamctl orchestrator update-original-state-partition-config --
apply < file.json
```

3.14 Supervisor

3.14.1 get-clusters

The `get-clusters` command retrieves the node status and roles within a supervisor cluster.

Syntax

```
beamctl supervisor get-clusters
```

Arguments

The command has no mandatory arguments.

Parameters

Table 336 JSON Parameters

| Parameter | Parameter Type | Description |
|-----------|----------------|------------------------------------|
| clusterId | Mandatory | Cluster Id of a supervisor cluster |
| leaderId | Mandatory | Leader node identifier |



| Parameter | Parameter Type | Description |
|-------------------|----------------|--|
| nodes | Mandatory | Collection of supervisor cluster nodes |
| node-id | Mandatory | Node identifier |
| hexFormatNodeID | Mandatory | Hexadecimal format node ID |
| podName | Mandatory | Pod name |
| protocol | Mandatory | Communication protocol |
| raftAddress | Mandatory | Raft consensus address |
| httpAddress | Mandatory | HTTP address |
| role | Mandatory | Node role |
| isLeader | Mandatory | Leader status |
| lastUpdated | Mandatory | Last update timestamp |
| siteName | Mandatory | Site name |
| lastClusterUpdate | Mandatory | Last cluster update timestamp |

Example 289

beamctl supervisor get-clusters

Sample Response:

```
{
  "clusterId": <cluster-id>,
  "leaderId": <leader-node-id>,
  "nodes": {
    "<node-id>": {
      "id": "<node-id>",
      "hexFormatNodeID": "<hex-format-node-id>",
      "podName": "<pod-name>",
      "protocol": "<protocol>",
      "raftAddress": "<raft-address>",
      "httpAddress": "<http-address>",
      "role": "<role>",
      "isLeader": <true-or-false>,
      "lastUpdated": "<timestamp>",
      "siteName": "<site-name>"
    }
  },
  "lastClusterUpdate": "<timestamp>"
}
```



3.14.2

get-health-monitor-config

The `get-health-monitor-config` command retrieves the supervisor health monitor config map data.

Syntax

```
beamctl supervisor get-health-monitor-config
```

Parameters

Table 337 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------------|----------------|--------------------------------------|
| supervisor | Mandatory | Supervisor configuration data |
| maintenanceMode | Mandatory | List of sites in maintenance mode |
| monitoringStatus | Mandatory | Monitoring status (enabled/disabled) |
| siteFailureDetection | Mandatory | Site failure detection status |
| defaultMonitoringConfig | Mandatory | Default monitoring configuration |
| podHealthChecks | Mandatory | Pod health check settings |
| minAvailable | Mandatory | Minimum available pods |
| overloadThreshold | Mandatory | Overload threshold percentage |
| monitoringDelay | Mandatory | Monitoring delay in seconds |
| monitoringFailureCount | Mandatory | Failure count threshold |

Arguments

The command has no arguments.

Example 290

```
beamctl supervisor get-health-monitor-config
```

Sample Response:

```
{
  "supervisor": {
    "maintenanceMode": [ ],
    "monitoringStatus": "disabled",
    "siteFailureDetection": "disabled",
    "defaultMonitoringConfig": [
      {
        "podHealthChecks": {
          "minAvailable": "30%",
          "overloadThreshold": 90,
          "monitoringDelay": 2,
          "monitoringFailureCount": 3
        }
      }
    ]
  }
}
```



```
    }  
  ]  
}
```

3.14.3 get-health-monitor-metrics

The `get-health-monitor-metrics` command fetches the Supervisor health monitor metrics (such as the number of pods monitored, memory usage, and processing time) for an app group on a Supervisor node.

Syntax

```
beamctl supervisor get-health-monitor-metrics
```

Arguments

The command has no arguments.

Parameters

Table 338 JSON Parameters

| Parameter | Parameter Type | Description |
|----------------|----------------|---------------------------|
| service-name | Mandatory | Service name |
| totalPods | Mandatory | Total number of pods |
| healthyPods | Mandatory | Number of healthy pods |
| unhealthyPods | Mandatory | Number of unhealthy pods |
| overloadedPods | Mandatory | Number of overloaded pods |
| memoryUsageMB | Mandatory | Memory usage in MB |
| processingTime | Mandatory | Processing time |

Example 291

```
beamctl supervisor get-health-monitor-metrics
```

Sample Response:

```
{  
  "healthMonitorMetrics": {  
    "<app-group>": {  
      "<service-name>": {  
        "totalPods": <number of total-pods>,  
        "healthyPods": <number of healthy-pods>,  
        "unhealthyPods": <number of unhealthy-pods>,  
        "overloadedPods": <number of overloaded-pods>,  
        "memoryUsageMB": <memory-usage-mb>,  
        "processingTime": <processing-time>  
      }  
    }  
  },  
}
```




```
"timestamp": "<timestamp>"
}
```

3.14.4

get-health-monitor-status

The `get-health-monitor-status` command fetches the Supervisor health monitor status (such as the service name, path, port, monitoring status etc) for an app group on a Supervisor node.

Syntax

```
beamctl supervisor get-health-monitor-status
```

Arguments

The command has no arguments.

Parameters

Table 339 JSON Parameters

| Parameter | Parameter Type | Description |
|----------------------|----------------|-------------------------------------|
| site-id | Mandatory | Site identifier |
| node-id | Mandatory | Node identifier |
| app-group-name | Mandatory | Application group name |
| monitoringStatus | Mandatory | Monitoring status |
| services | Mandatory | List of services |
| appServiceName | Mandatory | Application service name |
| path | Mandatory | Health check endpoint |
| port | Mandatory | Port number |
| scheme | Mandatory | Protocol scheme (http/https) |
| replicaSetName | Mandatory | Application service replicaset name |
| serviceMonitorStatus | Mandatory | Service monitor status |
| status | Mandatory | Status |

Example 292

```
beamctl supervisor get-health-monitor-status
```

Sample Response:

```
{
  "<site-id>": {
    "<node-id>": {
      "<app-group-name>": {
        "monitoringStatus": "<enabled>",
        "services": [
          {
            "appServiceName": "<application-service-name>",
```



```
        "path": "<health-check-path>",
        "port": "<port-number>",
        "scheme": "<http|https>",
        "replicaSetName": "<k8s-replicaset-name>",
        "serviceMonitorStatus": "<enabled|disabled>"
      },
      "status": "<status-string>"
    }
  }
}
```

3.14.5 transfer-leader

The `transfer-leader` command transfers leadership to a specified node only if the current role is "follower".

Syntax

```
beamctl supervisor transfer-leader <clusterId> < input.json
```

Arguments

Table 340 Mandatory Arguments

| Argument | Description |
|------------|--|
| clusterId | Cluster Id of a supervisor cluster |
| input.json | Input from a file will be a json file. |

Note: After a successful leadership transfer, the former leader node is restarted to rejoin the cluster under the new leader.

Parameters

Table 341 JSON Parameters

| Parameter | Parameter Type | Description |
|----------------|----------------|---|
| target_node_id | Mandatory | Target follower node ID for leadership transfer |

Example 293

```
beamctl supervisor transfer-leader
```

Sample Response:

```
{
  "target_node_id": <follower_node_id>
}
```



3.14.6 update-health-monitor-config

The `update-health-monitor-config` command updates the supervisor health monitor configuration.

Syntax

```
beamctl supervisor update-health-monitor-config < input.json
```

Arguments

This command has no arguments.

Parameters

Table 342 JSON Parameters

| Parameter | Parameter Type | Description |
|-------------------------------------|----------------|--|
| <code>maintenanceMode</code> | Mandatory | List of locations currently undergoing maintenance. |
| <code>minAvailable</code> | Mandatory | The minimum percentage, or number, of pods that must be healthy. |
| <code>monitoringDelay</code> | Mandatory | Delay between monitoring intervals, in seconds. |
| <code>monitoringFailureCount</code> | Mandatory | Number of consecutive failures required before triggering failover logic. |
| <code>monitoringStatus</code> | Mandatory | Flag to enable or disable supervision. |
| <code>overloadThreshold</code> | Mandatory | CPU memory usage percentage threshold that marks a pod as overloaded. |
| <code>siteFailureDetection</code> | Mandatory | Flag to enable or disable site failure detection, which governs the site failover. |

Example 294

Example of `input.json` where site `/1` is configured as `maintenanceMode`:

```
{
  "supervisor": {
    "maintenanceMode": [
      "11"
    ],
    "monitoringStatus": "disabled",
    "siteFailureDetection": "disabled",
    "defaultMonitoringConfig": [
      {
        "podHealthChecks": {
          "minAvailable": "30%",
          "overloadThreshold": 90,
          "monitoringDelay": 2,
          "monitoringFailureCount": 3
        }
      }
    ]
  }
}
```



```
} }
```

Sample Response:

This command must be executed on the primary CMAgent site. The primary site can be identified using the `bamctl bam-cm-agent get-config` command via BAMCLI.

Note: The `maintenanceMode` field in the input JSON specifies the list of sites, for example `"I1"`, `"I2"`, and so on.. It must be placed under maintenance mode.

Maintenance mode must be applied incrementally, one site at a time, to ensure high availability and avoid supervision gaps.

Recommended Procedure:

1. Update the health monitor configuration to include the intended site (e.g., `"I1"`) in `maintenanceMode`.
2. After applying maintenance mode, ensure that health monitoring is no longer active for the site by verifying with the `get-health-monitor-status` command.
3. Perform maintenance operations on the specified site, for example `"I1"`.
4. If you plan to add another site to the maintenance mode, for example `"I2"`, update the configuration to include it along with previously completed ones, for example `"I1"`, `"I2"`.
5. Repeat steps 2–4 for each additional site as required.

This approach ensures that only sites intended for maintenance are excluded from health monitoring, while all other sites continue to be supervised as expected.

3.15 Global Config Manager

The `global-config-manager` commands allow user to read and to update global ConfigMap attributes.

3.15.1 list-config-map

The `list-config-map` operation displays all the existing values for the given global ConfigMap.



Syntax

```
beamctl global-config-manager list-config-map <configmap name>
```

Arguments

Table 343 Mandatory Arguments

| Argument | Description |
|----------------|---|
| configmap name | The name of the ConfigMap values to be displayed. |

Example 295

```
beamctl global-config-manager list-config-map eric-bss-beam-privacy-data-definitions
```

```
beamctl global-config-manager list-config-map eric-bss-beam-global-system-configuration
```

```
beamctl global-config-manager list-config-map eric-bss-beam-local-system-configuration
```

3.15.2

update-config-map

The `update-config-map` operation modifies the value of an existing attribute in the specified configuration map.

Syntax

```
beamctl global-config-manager update-config-map <configmap name>
[<|<<] <RedirectedInput>
```

Arguments

Table 344 Arguments

| Argument | Description |
|---------------------|--|
| configmap name | The name of the ConfigMap to be updated. |
| <<RedirectedInput> | Input from a file. It is a JSON file. |
| <<<RedirectedInput> | This argument tells the CLI to continue to read input until a specified delimiter is found, usually EOF. |



| Argument | Description |
|----------|---------------------|
| | It is a JSON input. |

- Note:**
- When updating the `eric-bss-beam-privacy-data-definitions` ConfigMap, only the following fields can be included in the input:
 - `sensitiveType`
 - `logHidingScheme`
 - `databaseHidingScheme`
 - `exposedInterfaceHidingScheme`
 - Currently, only one `sensitiveType` attribute can be updated at a time for the `eric-bss-beam-privacy-data-definitions` ConfigMap. Including multiple `sensitiveType` details in the input is not supported.

Example 296 Update-config-map of `eric-bss-beam-privacy-data-definitions`

```
beamctl global-config-manager update-config-map eric-bss-beam-privacy-data-definitions < input.json
```

Example of an `input.json`:

```
{
  "definition": [
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "1a",
      "sensitiveType": "MSISDN",
      "sensitiveTypeDescription": "MSISDN numbers of the users"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "1b",
      "sensitiveType": "IP-SOURCE",
      "sensitiveTypeDescription": "IP source information for internet users"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "1c",
      "sensitiveType": "IP-DESTINATION",
      "sensitiveTypeDescription": "IP destination information for internet users"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "1d",
      "sensitiveType": "IP-PROXY",
      "sensitiveTypeDescription": "IP proxy information for internet users"
    }
  ],
  {}
}
```



```

    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1e",
    "sensitiveType": "URI",
    "sensitiveTypeDescription": "URI, URLs , SIP URIs"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1f",
    "sensitiveType": "IMEI",
    "sensitiveTypeDescription": "IMEI numbers"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1g",
    "sensitiveType": "PHY-EQUIPMENT-ADDRESS",
    "sensitiveTypeDescription": "Physical equipment address like MAC address or others"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1h",
    "sensitiveType": "MEDIA-CONTENT",
    "sensitiveTypeDescription": "MMS , SMS , and other media content of subscribers"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "ENCRYPT",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1i",
    "sensitiveType": "IDENTITY",
    "sensitiveTypeDescription": "Personal Identity"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1j",
    "sensitiveType": "EMAIL",
    "sensitiveTypeDescription": "End user Email Address"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1k",
    "sensitiveType": "COUNTRY",
    "sensitiveTypeDescription": "End user Country information"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1l",
    "sensitiveType": "POSTAL-CODE",
    "sensitiveTypeDescription": "Postal code information"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1m",
    "sensitiveType": "USER-ID",
    "sensitiveTypeDescription": "User id information"
  },

```



```
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1n",
  "sensitiveType": "IP-ADDRESS",
  "sensitiveTypeDescription": "IP address of subscriber"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1o",
  "sensitiveType": "IMSI",
  "sensitiveTypeDescription": "IMSI or TIMSI number information"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1p",
  "sensitiveType": "COMMUNICATION-ID",
  "sensitiveTypeDescription": "Any end user related network identifier which is generic and could be either of MSISDN, IMSI, IP Address, EMAIL or any"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2",
  "sensitiveType": "CALL-HISTORY",
  "sensitiveTypeDescription": "Call history information"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2a",
  "sensitiveType": "TIME-OF-CALL",
  "sensitiveTypeDescription": "Time of call"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2b",
  "sensitiveType": "CONNECTION-TYPE",
  "sensitiveTypeDescription": "Connection type details of mobile \u0026amp; fixed connection services"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2c",
  "sensitiveType": "CALL-DURATION",
  "sensitiveTypeDescription": "Information of call duration"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2d",
  "sensitiveType": "LBS-DATA",
  "sensitiveTypeDescription": "Subscriber's location based service data"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2e",
```




```

    "sensitiveType": "LOCATION-LAC",
    "sensitiveTypeDescription": "LAC CID information"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "2f",
    "sensitiveType": "LOCATION-OTHER",
    "sensitiveTypeDescription": "subscriber's location (LBS) or any other related to subscriber
location info"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "2g",
    "sensitiveType": "ROAMING-DETAILS",
    "sensitiveTypeDescription": "Information of subscriber's connections in the roaming (includi
ng country, information regarding identifying of the roaming partner)"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "2h",
    "sensitiveType": "SUBSCRIBER-SERVICES-INFO",
    "sensitiveTypeDescription": "The status and the information of services delivered or receive
d by the subscriber, including information regarding tariff plans, additional numbers"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "ENCRYPT",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "3a",
    "sensitiveType": "ICC",
    "sensitiveTypeDescription": "ICC id of the SIM card"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "ENCRYPT",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "3b",
    "sensitiveType": "KI",
    "sensitiveTypeDescription": "KI number information or eKI"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "ENCRYPT",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "3c",
    "sensitiveType": "PIN",
    "sensitiveTypeDescription": "SIM card PIN number information"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "ENCRYPT",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "3d",
    "sensitiveType": "PUK",
    "sensitiveTypeDescription": "SIM card PUK number information(Personal Unlocking Key)"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "4",
    "sensitiveType": "TRANSACTION-INFO",
    "sensitiveTypeDescription": "Subscriber transaction information like usage, amount, balance
, date, time and other transaction details"
  },
  {

```



```
"configurable": true,
"databaseHidingScheme": "NONE",
"exposedInterfaceHidingScheme": "NONE",
"logHidingScheme": "MASK",
"sensitiveTag": "4a",
"sensitiveType": "BALANCE-INFO",
"sensitiveTypeDescription": "Subscriber balance information"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "4b",
  "sensitiveType": "TRANSACTION-AMOUNT",
  "sensitiveTypeDescription": "Transaction amount"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "4c",
  "sensitiveType": "VOLUME",
  "sensitiveTypeDescription": "Volume information"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "4d",
  "sensitiveType": "SUBSCRIBER-DISCOUNT-DATE",
  "sensitiveTypeDescription": "Discount date when subscriber got discount"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "4e",
  "sensitiveType": "TRANSACTION-PAYMENT-DATE",
  "sensitiveTypeDescription": "Date of a financial transaction (e.g., Payment/Adjustment/Write
-off...)"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "4f",
  "sensitiveType": "INVOICE-AMOUNT",
  "sensitiveTypeDescription": "Amount associated with invoice (e.g., Invoice/Tax/Dispute, etc.
)"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "4g",
  "sensitiveType": "INVOICE-ADDRESS",
  "sensitiveTypeDescription": "Invoice address"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "5",
  "sensitiveType": "NAME",
  "sensitiveTypeDescription": "Name of an individual"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "5a",
  "sensitiveType": "FIRST-NAME",
```



```

    "sensitiveTypeDescription": "First name"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "5b",
    "sensitiveType": "MIDDLE-NAME",
    "sensitiveTypeDescription": "Middle name"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "5c",
    "sensitiveType": "LAST-NAME",
    "sensitiveTypeDescription": "Last name"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "5d",
    "sensitiveType": "DATE-OF-BIRTH",
    "sensitiveTypeDescription": "End user date of birth (year or date)"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "5e",
    "sensitiveType": "BIRTH-PLACE",
    "sensitiveTypeDescription": "End user birth place"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "5f",
    "sensitiveType": "ADDRESS",
    "sensitiveTypeDescription": "Legal address of individuals, addresses of the messages deliver →
y"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "5g",
    "sensitiveType": "TAX-ID",
    "sensitiveTypeDescription": "Tax information identity"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "ENCRYPT",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "6",
    "sensitiveType": "SUBS-SPECIAL-PERSONAL-DATA",
    "sensitiveTypeDescription": "Any additional special personal information of a subscriber e.g →
. religious beliefs etc."
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "7a",
    "sensitiveType": "USER-ACTIVITY",
    "sensitiveTypeDescription": "Metadata showing user activity e.g browsing or connection histo →
ry"
  },
  {
    "configurable": false,
    "databaseHidingScheme": "NONE",

```




```

    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "18a",
    "sensitiveType": "CUSTOMER-TAX-CATEGORY",
    "sensitiveTypeDescription": "Tax category of customer"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "18b",
    "sensitiveType": "TAX-JURISDICTION-CODE",
    "sensitiveTypeDescription": "TAX Jurisdiction code"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "19",
    "sensitiveType": "INVOICE-IDENTIFIER",
    "sensitiveTypeDescription": "Identifier for external invoice id and document id"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "MASK",
    "logHidingScheme": "MASK",
    "sensitiveTag": "19a",
    "sensitiveType": "OTHER-PARTY-NUMBER-FOR-CONSUMER",
    "sensitiveTypeDescription": "Other party number for the transaction done by consumer"
  }
]
}

```

Example 297 Update-config-map of eric-bss-beam-privacy-data-definitions << EOF

beamctl global-config-manager update-config-map eric-bss-beam-privacy-data-definitions << EOF

```

{
  "definition": [
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "1a",
      "sensitiveType": "MSISDN",
      "sensitiveTypeDescription": "MSISDN numbers of the users"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "1b",
      "sensitiveType": "IP-SOURCE",
      "sensitiveTypeDescription": "IP source information for internet users"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "1c",
      "sensitiveType": "IP-DESTINATION",
      "sensitiveTypeDescription": "IP destination information for internet users"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",

```



```
"logHidingScheme": "MASK",
"sensitiveTag": "1d",
"sensitiveType": "IP-PROXY",
"sensitiveTypeDescription": "IP proxy information for internet users"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1e",
  "sensitiveType": "URI",
  "sensitiveTypeDescription": "URI, URLs , SIP URIs"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1f",
  "sensitiveType": "IMEI",
  "sensitiveTypeDescription": "IMEI numbers"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1g",
  "sensitiveType": "PHY-EQUIPMENT-ADDRESS",
  "sensitiveTypeDescription": "Physical equipment address like MAC address or others"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1h",
  "sensitiveType": "MEDIA-CONTENT",
  "sensitiveTypeDescription": "MMS , SMS , and other media content of subscribers"
},
{
  "configurable": true,
  "databaseHidingScheme": "ENCRYPT",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1i",
  "sensitiveType": "IDENTITY",
  "sensitiveTypeDescription": "Personal Identity"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1j",
  "sensitiveType": "EMAIL",
  "sensitiveTypeDescription": "End user Email Address"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1k",
  "sensitiveType": "COUNTRY",
  "sensitiveTypeDescription": "End user Country information"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "1l",
  "sensitiveType": "POSTAL-CODE",
  "sensitiveTypeDescription": "Postal code information"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
```



```

    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1m",
    "sensitiveType": "USER-ID",
    "sensitiveTypeDescription": "User id information"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1n",
    "sensitiveType": "IP-ADDRESS",
    "sensitiveTypeDescription": "IP address of subscriber"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1o",
    "sensitiveType": "IMSI",
    "sensitiveTypeDescription": "IMSI or TIMSI number information"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "1p",
    "sensitiveType": "COMMUNICATION-ID",
    "sensitiveTypeDescription": "Any end user related network identifier which is generic and could be either of MSISDN, IMSI, IP Address, EMAIL or any" →
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "2",
    "sensitiveType": "CALL-HISTORY",
    "sensitiveTypeDescription": "Call history information"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "2a",
    "sensitiveType": "TIME-OF-CALL",
    "sensitiveTypeDescription": "Time of call"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "2b",
    "sensitiveType": "CONNECTION-TYPE",
    "sensitiveTypeDescription": "Connection type details of mobile \u0026amp; fixed connection services" →
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "2c",
    "sensitiveType": "CALL-DURATION",
    "sensitiveTypeDescription": "Information of call duration"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "2d",
    "sensitiveType": "LBS-DATA",
    "sensitiveTypeDescription": "Subscriber's location based service data"
  },

```



```
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2e",
  "sensitiveType": "LOCATION-LAC",
  "sensitiveTypeDescription": "LAC CID information"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2f",
  "sensitiveType": "LOCATION-OTHER",
  "sensitiveTypeDescription": "subscriber's location (LBS) or any other related to subscriber
location info"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2g",
  "sensitiveType": "ROAMING-DETAILS",
  "sensitiveTypeDescription": "Information of subscriber's connections in the roaming (includi
ng country, information regarding identifying of the roaming partner)"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "2h",
  "sensitiveType": "SUBSCRIBER-SERVICES-INFO",
  "sensitiveTypeDescription": "The status and the information of services delivered or receive
d by the subscriber, including information regarding tariff plans, additional numbers"
},
{
  "configurable": true,
  "databaseHidingScheme": "ENCRYPT",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "3a",
  "sensitiveType": "ICC",
  "sensitiveTypeDescription": "ICC id of the SIM card"
},
{
  "configurable": true,
  "databaseHidingScheme": "ENCRYPT",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "3b",
  "sensitiveType": "KI",
  "sensitiveTypeDescription": "KI number information or eKI"
},
{
  "configurable": true,
  "databaseHidingScheme": "ENCRYPT",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "3c",
  "sensitiveType": "PIN",
  "sensitiveTypeDescription": "SIM card PIN number information"
},
{
  "configurable": true,
  "databaseHidingScheme": "ENCRYPT",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "3d",
  "sensitiveType": "PUK",
  "sensitiveTypeDescription": "SIM card PUK number information(Personal Unlocking Key)"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
```




```

    "sensitiveTag": "4",
    "sensitiveType": "TRANSACTION-INFO",
    "sensitiveTypeDescription": "Subscriber transaction information like usage, amount, balance
, date, time and other transaction details" →
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "4a",
    "sensitiveType": "BALANCE-INFO",
    "sensitiveTypeDescription": "Subscriber balance information"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "4b",
    "sensitiveType": "TRANSACTION-AMOUNT",
    "sensitiveTypeDescription": "Transaction amount"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "4c",
    "sensitiveType": "VOLUME",
    "sensitiveTypeDescription": "Volume information"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "4d",
    "sensitiveType": "SUBSCRIBER-DISCOUNT-DATE",
    "sensitiveTypeDescription": "Discount date when subscriber got discount"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "4e",
    "sensitiveType": "TRANSACTION-PAYMENT-DATE",
    "sensitiveTypeDescription": "Date of a financial transaction (e.g., Payment/Adjustment/Write
-off...)" →
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "4f",
    "sensitiveType": "INVOICE-AMOUNT",
    "sensitiveTypeDescription": "Amount associated with invoice (e.g., Invoice/Tax/Dispute, etc. →
)"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "4g",
    "sensitiveType": "INVOICE-ADDRESS",
    "sensitiveTypeDescription": "Invoice address"
  },
  {
    "configurable": true,
    "databaseHidingScheme": "NONE",
    "exposedInterfaceHidingScheme": "NONE",
    "logHidingScheme": "MASK",
    "sensitiveTag": "5",
    "sensitiveType": "NAME",
    "sensitiveTypeDescription": "Name of an individual"
  },
  {

```



```
"configurable": true,
"databaseHidingScheme": "NONE",
"exposedInterfaceHidingScheme": "NONE",
"logHidingScheme": "MASK",
"sensitiveTag": "5a",
"sensitiveType": "FIRST-NAME",
"sensitiveTypeDescription": "First name"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "5b",
  "sensitiveType": "MIDDLE-NAME",
  "sensitiveTypeDescription": "Middle name"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "5c",
  "sensitiveType": "LAST-NAME",
  "sensitiveTypeDescription": "Last name"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "5d",
  "sensitiveType": "DATE-OF-BIRTH",
  "sensitiveTypeDescription": "End user date of birth (year or date)"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "5e",
  "sensitiveType": "BIRTH-PLACE",
  "sensitiveTypeDescription": "End user birth place"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "5f",
  "sensitiveType": "ADDRESS",
  "sensitiveTypeDescription": "Legal address of individuals, addresses of the messages deliver →
y"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "5g",
  "sensitiveType": "TAX-ID",
  "sensitiveTypeDescription": "Tax information identity"
},
{
  "configurable": true,
  "databaseHidingScheme": "ENCRYPT",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "6",
  "sensitiveType": "SUBS-SPECIAL-PERSONAL-DATA",
  "sensitiveTypeDescription": "Any additional special personal information of a subscriber e.g →
. religious beliefs etc."
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "7a",
  "sensitiveType": "USER-ACTIVITY",
```



```

ry"    "sensitiveTypeDescription": "Metadata showing user activity e.g browsing or connection histo →
    },
    {
      "configurable": false,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "8",
      "sensitiveType": "POTENTIAL",
      "sensitiveTypeDescription": "Information which could be potentially sensitive"
    },
    {
      "configurable": false,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "9",
      "sensitiveType": "EVENT-POTENTIAL",
      "sensitiveTypeDescription": "Information which could be potentially sensitive in EDM events"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "10",
      "sensitiveType": "FINANCIAL-HEADER-EXTERNAL-ID",
      "sensitiveTypeDescription": "The external Id of financial headers"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "11",
      "sensitiveType": "BI-DATA",
      "sensitiveTypeDescription": "BusinessInteraction Data"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "12",
      "sensitiveType": "NUM-RESOURCE",
      "sensitiveTypeDescription": "Any number which is generic and could be either of MSISDN,IMSI →
, IP Address,EMAIL or any used as resource String in NUM"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "13",
      "sensitiveType": "NOTIFICATION",
      "sensitiveTypeDescription": "Notification data"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "14",
      "sensitiveType": "BUSINESS-LOGIC-DATA",
      "sensitiveTypeDescription": "Business Rules defined for execution of business logic"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",
      "logHidingScheme": "MASK",
      "sensitiveTag": "16",
      "sensitiveType": "ACTIVATION-CODES",
      "sensitiveTypeDescription": "Business Rules defined for execution of business logic"
    },
    {
      "configurable": true,
      "databaseHidingScheme": "NONE",
      "exposedInterfaceHidingScheme": "NONE",

```



```
"logHidingScheme": "MASK",
"sensitiveTag": "17",
"sensitiveType": "OTHER-SENSITIVE",
"sensitiveTypeDescription": "Other sensitive type"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "18a",
  "sensitiveType": "CUSTOMER-TAX-CATEGORY",
  "sensitiveTypeDescription": "Tax category of customer"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "18b",
  "sensitiveType": "TAX-JURISDICTION-CODE",
  "sensitiveTypeDescription": "TAX Jurisdiction code"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "NONE",
  "logHidingScheme": "MASK",
  "sensitiveTag": "19",
  "sensitiveType": "INVOICE-IDENTIFIER",
  "sensitiveTypeDescription": "Identifier for external invoice id and document id"
},
{
  "configurable": true,
  "databaseHidingScheme": "NONE",
  "exposedInterfaceHidingScheme": "MASK",
  "logHidingScheme": "MASK",
  "sensitiveTag": "19a",
  "sensitiveType": "OTHER-PARTY-NUMBER-FOR-CONSUMER",
  "sensitiveTypeDescription": "Other party number for the transaction done by consumer"
}
]
EOF
```

Example 298

```
beamctl global-config-manager update-config-map eric-bss-beam-
global-system-configuration < input.json
```

Example of an input.json:

```
{
  "sensitiveDataEncryptionEnabled": "false"
}
```

Example 299

```
beamctl global-config-manager update-config-map eric-bss-beam-
global-system-configuration << EOF
```

```
{
  "sensitiveDataEncryptionEnabled": "false"
}
```



```
}
EOF
```

Example 300

```
beamctl global-config-manager update-config-map eric-bss-beam-  
local-system-configuration < input.json
```

```
{
  "schedule": "0 3 * * *",
  "eventCategory": {
    "billingAccountBucketBalance": "true",
    "productBucketBalance": "true",
    "settlementAccountBucketBalance": "true",
    "customerEntity": "true",
    "contractEntity": "true",
    "partyEntity": "true",
    "organizationEntity": "true",
    "genericPartyRoleEntity": "true",
    "agreementRealizationEntity": "true",
    "settlementAccountEntity": "true",
    "settlementContractEntity": "true",
    "customerListEntity": "true",
    "userEntity": "true"
  }
}
```

Example 301

```
beamctl global-config-manager update-config-map eric-bss-beam-  
local-system-configuration << EOF
```

```
{
  "schedule": "0 3 * * *",
  "eventCategory": {
    "billingAccountBucketBalance": "true",
    "productBucketBalance": "true",
    "settlementAccountBucketBalance": "true",
    "customerEntity": "true",
    "contractEntity": "true",
    "partyEntity": "true",
    "organizationEntity": "true",
    "genericPartyRoleEntity": "true",
    "agreementRealizationEntity": "true",
    "settlementAccountEntity": "true",
    "settlementContractEntity": "true",
    "customerListEntity": "true",
    "userEntity": "true"
  }
}
EOF
```

3.16 Config

The `config` command provides the functionality to add configurations. Added configuration is persisted in `config.yaml` file in `config-directory` location.



The default is: \$HOME/.beamctl.

Commands

Table 345 Available Commands

| Command | Description |
|-------------|-----------------------------------|
| get-config | Gets configuration for key. |
| list-config | Lists all configuration. |
| set-config | Sets configuration for a command. |

Syntax

beamctl config list-config

Example 302

beamctl config list-config

Response:

```
alarms:
server: https://eric-fh-alarm-handler-rbac-proxy
login:
client_id: AuthorizationClient
token_url: https://eric-sec-access-mgmt/auth/realms/master/protocol/openid-connect/token →
```

Example 303

beamctl config get-config --key login.token_url

3.17

Help

The help command displays a list of commands and a help for a given command.

If called without a parameter, the help command lists all available commands. When called with a command supported by the CLI, a detailed description of the command is displayed.

Syntax

beamctl [command] --help

Example 304

beamctl config --help

Response:



```
Available Commands:
  get-config  Get configuration for key
  list-config List all configuration
  set-config  Set configuration for a command
```

3.18 Help-Config

The `help-config` command displays information about CLI configuration help.

The configuration for the CLI comes from the following places:

- Command options
- Environment variables
- Configuration files

Global Command Option

Command options are passed when invoking the command.

Table 346 Global Commands

| Name | Type | Description |
|----------------------------|--------|---|
| <code>output-format</code> | string | Output format [JSON, YAML]. |
| <code>query</code> | string | Filter / project results using JMESPath. |
| <code>raw</code> | bool | Output result of the query as raw rather than an escaped JSON string or list. |
| <code>server</code> | string | Override the server URL. |
| <code>verbose</code> | bool | Enable verbose log output.

Note: <code>--verbose</code> configures the CLI to run with additional output for debugging. |

Environment Variables

Environment variables must be capitalized, prefixed with `BEAMCTL`, and words are separated by an underscore rather than a dash.

Example 305

Setting `BEAMCTL_VERBOSE=1` is equivalent to passing `--verbose` to the command.



Configuration Files

Configuration files can be used to configure the CLI and can be written using JSON, YAML, or TOML.

The CLI searches in your home directory first, for example `'/home/ebatsum/.beamctl/config.json'` and on Mac/Linux also looks in `'/etc/beamctl/config.json'`

Example 306

The following is equivalent to passing `'--verbose'` to the command:

```
'''json
{
  "verbose": true
},,
```

3.19 Help-Input

Input to the CLI is handled via parameters, arguments, and standard input. The help for an individual command shows the available optional parameters and required arguments.

For requests that require a body, standard input and a CLI shorthand can complement each other to supply the request data.

Standard Input

Standard input allows you to send in whatever data is required to make a successful request against the API.

Example 307

```
my-cli command <input.json' or 'echo '{"hello": "world"}' I
my-cli command'
```

CLI Shorthand Syntax

Any argument beyond those that are required for a command is treated as CLI shorthand and used to generate structured data for requests. Shorthand objects are specified as key and value pairs. They complement standard input so can be used to override or to add additional fields as needed.



Example 308

```
my-cli command <input.json field: value, other: value2
```

3.20 Login

The `login` command authenticates the user with Keycloak API and persists the refresh token from the response output which is used to get identity token during the pre-run phase of other commands. Based on an identity token, an user is authorized to access different services with-in cluster. It also displays a notice to user and administrator which is fetched from the BAM CLI server.

Syntax

```
beamctl login --username <username> --password-file <path-to-  
password-file> -t <token-url> [-n <notice-url> | -s]
```

Example 309

```
beamctl login -u <user> -p <path-to-password-file>  
-t https://eric-sec-access-mgmt:8443/auth/realms/master/protocol/  
openid-connect/token -n https://eric-bss-bam-cli-proxy:8443/  
notice | -s
```

- Note:**
- `--token-url` is an optional parameter. If it is not provided, it uses <https://eric-sec-access-mgmt/auth/realms/master/protocol/openid-connect/token>
 - `--notice-url` is an optional parameter. If it is not provided, it uses <https://eric-bss-bam-cli-proxy/notice>
 - If `--silent` is set, or BAM CLI server is not accessible, a notice to user and administrator is skipped.

3.21 Logout

The `logout` command provides the functionality to logout from CLI. It removes the saved user token and CA certificates.

Example 310

```
beamctl logout
```

Response:

```
Logout Successful
```